



CS EXECUTIVE

FINANCIAL MANAGEMENT



Comprehensive Curriculum Coverage

Covers every Concept as per the latest ICSI Syllabus



Strictly Based on ICSI Modules

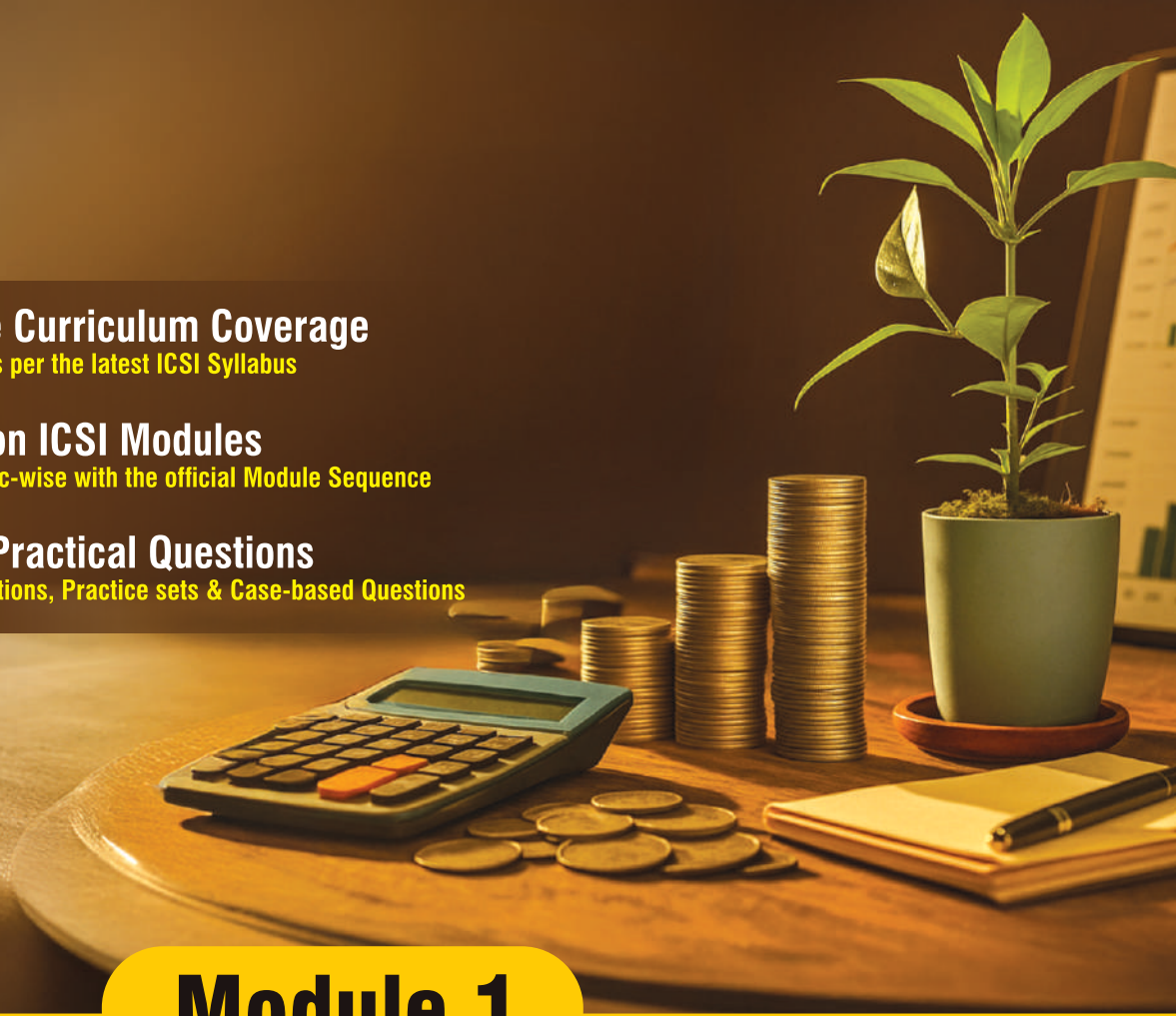
All chapters aligned topic-wise with the official Module Sequence



Theoretical + Practical Questions

Includes Past Year Questions, Practice sets & Case-based Questions

Module 1





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PREFACE

Financial Management is an indispensable component of the CS Executive curriculum, equipping students with the knowledge and skills required to understand corporate finance, financial planning, and strategic decision-making in today's dynamic business environment. For a Company Secretary, expertise in financial management is vital, as it supports advisory functions on capital structuring, investment decisions, dividend policy, and overall financial governance.

This book has been developed with the specific objective of facilitating students in mastering the **Financial Management** syllabus prescribed by the Institute of Company Secretaries of India (ICSI). The subject matter is presented in a logical sequence, covering fundamental as well as advanced concepts such as time value of money, cost of capital, capital budgeting, leverage, working capital management, dividend decisions, and financial analysis.

Each chapter has been crafted with precision to balance conceptual clarity with practical application. Numerical problems, illustrations, and case-based examples have been provided to enhance problem-solving and analytical skills. The pedagogical approach emphasizes not only the assimilation of theoretical knowledge but also its application in decision-making, thereby preparing students for both examination success and professional responsibilities.

We are of the view that comprehensive and well-structured study material is central to effective learning. Accordingly, this book is designed to function both as a detailed study resource and as a concise revision aid, ensuring that students are well-prepared and confident in approaching their examinations.

We sincerely acknowledge the contributions of academic experts, reviewers, and the editorial team in shaping this volume. It is our earnest belief that this book will serve as a valuable academic resource, helping students build proficiency in Financial Management and excel in their professional journey as future Company Secretaries.

ABOUT THE AUTHOR

Sir, **Dr. Debasish Goswami** is a multiple degree holder and qualified exams of Company Secretary, Cost Accountancy and holder of MBA degree from Chandigarh University, MA (Economics) from Aligarh Muslim University, LL.B, from Kalyan Law College and B.com. With over 6 years of experience in field of commerce and expertise in Law, Finance, Tax and Economics. Sir has cleared 4 papers of CA Final as well. Out of curiosity sir has also learnt Mandarin, Sanskrit and Shorthand.

His 36 students have scored All India Rank by 2025 and 1000+ students have scored exemptions over the years, having mentored over 30,000 students both online and offline.

Sir has been visiting faculty for ICMAI and won multiple awards including National winner of Debate competition, National Moot Court competition of ICSI and also National Conference Winner of ICAI.

Dr. Debasish Sir is passionate and professional teacher giving students diverse and deep knowledge with fun loving examples and case studies so that students can retain and excel each and every topic and perform well in exam. He understands that every student is different and should be treated differently according to his/her needs.

His mission is to inspire more and more students towards excellence and aim as high as possible because once you dream it, you can achieve it.

ACKNOWLEDGEMENT

We extend our sincere gratitude to the subject experts, illustrators, and content development team whose dedication and insight have shaped the quality and depth of this book. Their contributions have been instrumental in crafting a resource that balances academic rigor with practical clarity. We also thank our proofreaders, reviewers, and publishing professionals for their meticulous attention to detail, ensuring that the final work upholds the highest standards of accuracy, coherence, and accessibility.

A special word of thanks to **Miss Tanuja Baghel (Content Professor)** for going above and beyond in contributing her subject expertise, which has added immense value to this book.

This book is the result of a collaborative effort to empower learners and professionals in their pursuit of legal and compliance excellence. It reflects our shared commitment to supporting aspirants as they navigate complex statutory frameworks with confidence and clarity. May this guide serve as a meaningful step toward your success—not only in examinations and practice, but in shaping a more informed and inclusive legal ecosystem.

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INTRODUCTION TO FINANCIAL MANAGEMENT

A. INTRODUCTION TO FINANCIAL MANAGEMENT

Starting a business? Whether it's a tech startup, a retail shop, or a manufacturing unit, every aspiring entrepreneur must answer these **three crucial questions**:

What to invest in?

Think land, machines, R&D, IT systems — what capital assets are needed?

Where will the money come from?

Will it be **Equity**, **Debt**, or a mix of both? Your capital structure matters.

How to manage daily money matters?

Collecting payments, paying suppliers, and managing working capital - these daily operations keep the business running.

Understanding these helps build a **strong financial foundation**, and that's where **Financial Management** plays a hero's role

EVOLUTION OF FINANCIAL MANAGEMENT

Financial Management didn't emerge overnight. It evolved through **3 key phases**:

1. Traditional Phase (1900s–1940s)

- (a) Focus: Rare events — mergers, expansions, bankruptcies.
- (b) Approach: Legalistic and descriptive.
- (c) Viewpoint: Mainly from outsiders like bankers and investors.
- (d) Classic Work: The Financial Policy of Corporations by Arthur S. Dewing.

2. Transitional Phase (1940s–1950s)

- (a) Shift in focus to **daily finance issues**: cash flows, planning, control.
- (b) Highlight: **Working Capital Management**.
- (c) Key Work: Essays on Business Finance by Wilford J. Eiteman et al.

3. Modern Phase (1950s–Present)

- (a) **Goal**: Maximize shareholder wealth with smart fund allocation.
- (b) **Tools**: Data, analysis, economic theory.
- (c) **Modern concepts include**:
 - (i) Capital Budgeting
 - (ii) Asset Pricing & Valuation
 - (iii) Capital Structure Theory
 - (iv) Option & Agency Theory
 - (v) Working Capital & Behavioural Finance

Finance today is **dynamic, data-driven, and decision-oriented** — a perfect blend of logic and numbers, offering endless challenges and opportunities!

B. MEANING OF FINANCE – THE LIFELINE OF BUSINESS

Finance is like the **heartbeat** of any business — without it, nothing moves!

What is Finance?

Finance is the **art & science** of managing **money**—from raising funds to using them efficiently. It enables businesses to define their **scope of operations**, shape their **strategy**, and drive **growth**.

Think of finance as

- ◆ **Oil** for machines (Production)
- ◆ **Blood** for the body (Business operations)

No **startup**, no **growth**, and no **development** is possible without **finance**.

a. ORIGIN & INTERPRETATION

1. Derived from Latin word '**finis**' = end/finish
2. Refers to **funds, capital, investments, money**
3. Covers both the **creation** and **management** of money in various forms – banking, credit, securities, and more.

b. WEBSTER'S DICTIONARY

“Finance is the science or study of the management of funds — including money circulation, credit granting, investments, and banking services.”

c. KEY CONCEPT: TIME VALUE OF MONEY

Finance deeply connects with:

1. Time value of money
2. Interest rates
3. Future cash flows

It's about making **smart choices today** that pay off **tomorrow**!

C. TYPES OF FINANCE (3 BROAD CATEGORIES)

a. PUBLIC FINANCE

1. Managed by **Central, State & Local governments**
2. **Sources:** Taxes, fees, borrowings
3. **Purpose:** **Welfare**, not profit
4. **Focus:** Policies for **social & economic upliftment**
Study of government revenues, expenditures & financial systems

b. CORPORATE FINANCE

1. Concerned with **companies**
2. **Focus:** Raising capital, investment decisions, financial planning
3. **Objectives:** Maximize shareholder value
4. **Activities include:**
 - (a) Managing cash flows
 - (b) Structuring capital
 - (c) Preparing statements
 - (d) Issuing stocks/bonds

C. PERSONAL FINANCE

1. Managing **individual or family finances**
2. Involves:
 - (a) Budgeting
 - (b) Savings & Investments
 - (c) Insurance
 - (d) Retirement & Tax planning
 - (e) **Goal:** Achieve **financial freedom** & make **smart decisions**

Finance is **everywhere** — in your wallet, your business, and your government. The more you know about it, the better you can manage, grow, and secure your future.

D. DEFINITION OF FINANCIAL MANAGEMENT – THE BRAIN BEHIND BUSINESS MONEY

What is Financial Management?

Financial Management is like the **GPS system** for any business's money matters — guiding it through **planning, raising, allocating, and utilizing** funds efficiently.

It's a **core part of overall management**, ensuring that **every rupee works hard** for the business!

EXPERT DEFINITIONS

Let's break down what top scholars have said:

1. Core Focus: Efficient Use of Capital

- ◆ **Solomon:**
“Efficient use of capital funds – a vital economic resource.”
- ◆ **Kuchal:**
“Procurement and effective use of funds.”

2. Decision-Making & Strategic Role

- ◆ **Howard & Upton:**
“Applying general management to financial decisions.”
- ◆ **Weston & Brigham:**
“Balancing personal motives with enterprise goals.”

3. Operations & Activities

- ◆ **Joseph & Massie:**
“Operational activity of obtaining and using funds effectively.”
- ◆ **Guthman & Dougal:**
“Planning, raising, controlling, and administering funds.”

4. Capital Selection & Goal Alignment

- ◆ **J.F. Brandley:**
“Judicious use of capital to guide spending towards goals.”

5. In Simple Terms

- ◆ **Khan & Jain:**
“Finance is the **art and science** of managing money.”
- ◆ **Wheeler:**
“Acquiring & converting capital to meet financial needs.”

6. Business Finance Angle

Let's understand what **Business Finance** scholars have to say:

- ◆ **E.W. Walker:**
“Planning, coordinating, and implementing financial activities.”
- ◆ **Henry Hoagland:**
“Decisions about financing & investments to achieve corporate goals.”
- ◆ **Gloss & Baker:**
“Finding sources and ensuring proper use of money.”
- ◆ **Parhter & Wert:**
“Focuses on raising and using funds in non-financial businesses.”

E. NATURE OF FINANCIAL MANAGEMENT

- a. **Lifeblood of business** – Every activity needs funds.
- b. **Continuous & strategic** – Involves planning, execution & control.
- c. **Universal** – Applies to all organizations (business, NGO, schools, etc.).

F. SCOPE OF FINANCIAL MANAGEMENT

- a. **Planning** – Estimating financial needs & setting goals.
- b. **Procurement** – Raising funds via equity, debt, loans.
- c. **Allocation** – Investing wisely for better returns.
- d. **Utilization** – Ensuring efficient use of funds.
- e. **Control** – Monitoring budgets, costs & profits.
- f. **Decision Support** – Helps in financial strategies & investments.

G. OBJECTIVES OF FINANCIAL MANAGEMENT

- a. Ensure regular & sufficient funds
- b. Use funds efficiently
- c. Maximize shareholder wealth
- d. Maintain balanced capital structure
- e. Maintain financial discipline
- f. Plan for future needs

Financial Management = Smart use of money to ensure **growth, stability, and wealth creation!**

H. TYPES OF FINANCIAL DECISIONS

a. INVESTMENT DECISIONS

An **investment decision** involves allocating money to projects/assets expected to yield **future returns**. It's all about **value creation**, often linked to **capital budgeting**.

1. Why it matters?

- (a) High risk
- (b) Long-term impact
- (c) Affects growth, survival, and profitability

2. Categories of Investment Decisions

- (a) **Inventory Management** – Investment in stock/materials
- (b) **Strategic Investment** – Long-term market power
- (c) **Modernisation** – New tech for cost reduction
- (d) **New Business** – Diversification or startups
- (e) **Replacement** – Replace old assets
- (f) **Expansion** – Increase production (capital widening)

3. Key Concepts in Investment Decisions

- (a) **Capital Budgeting:** Evaluate long-term investments like new products, R&D, acquisition, etc.
- (b) **Main Tools:**
 - (i) Net Present Value (NPV)
 - (ii) Internal Rate of Return (IRR)
 - (iii) Payback Period
 - (iv) Profitability Index (PI)
 - (v) Sensitivity Analysis

Real-life Example

Forrest Gump made \$650M globally. Yet the studio reported a **loss** of \$62M and paid the author **nothing** — why? Because of **creative accounting**—cash flow $\neq$ earnings.

Conclusion

Investment decisions are **critical** to a firm's long-term success. They must be taken with careful financial analysis, projections, and **rational capital allocation** to ensure **maximum value creation**.

4. Checklist: Cash Flow from Investing Activities (CFIA)

- (a) **Source of Investment Funds**
 - (i) Is investment financed via:
 - (1) Operating cash flow
 - (2) Retained earnings
 - (3) Borrowed funds/debt/equity?
 - (ii) Return on Investment (ROI) must be **greater than** cost of capital.
- (b) **Nature & Quality of Investments**
 - (i) Are funds used for **core business** assets (e.g., machinery) or for **external securities**?
 - (ii) Analyze industry trends, firm's expansion strategies, and management commentary for insights.
 - (iii) Persistent large outflows = probable **business expansion** or **capex-heavy model**.
- (c) **Capital Expenditure Requirements**
 - (i) Use **CapEx-to-Sales ratio** to assess capital intensity.
 - (ii) Distinguish between:
 - (1) **Maintenance Capital Expenditure** (to sustain current assets)
 - (2) **Growth Capital Expenditure** (to expand capacity)
- (d) **Capitalisation of Operating Expenses**
 - (i) Be wary of companies **inflating cash flow from operations** by shifting operating expenses to investing.
 - (ii) Example: WorldCom classified regular expenses as asset purchases to fake profits.
- (e) **Inorganic Expansion (M&A)**

- (i) Check acquisitions for:
 - (1) Strategic fit
 - (2) Value creation
 - (3) Past success of M&As
- (ii) Watch out for unrelated or failed deals (e.g., Kodak–Sterling Drugs).
- (f) **Sale of Assets or Business Units**
 - (i) Look at **churn** in business (divestitures).
 - (ii) Assess:
 - (1) Strategy behind asset sale
 - (2) Impact on shareholder value
 - (3) Compliance with legal provisions (esp. related-party transactions)
- (g) **R&D and In-House Expenditures**
 - (i) Only R&D that **creates an asset** is shown in CFIA.
 - (ii) Pharma & tech firms require extra scrutiny.
 - (iii) Promotional/advertising expenses are **not** included.

5. Understanding Cash Flow from Investing Activities

Is Negative Cash Flow Always Bad?

Not necessarily.

A company may show **negative overall cash flow** in a quarter, and it might still be okay **if** the negative cash is due to **investing for future growth** (like buying assets), not because of poor core operations.

(a) What Belongs in Investing Activities?

Cash flows are classified under **Investing Activities** if they relate to:

- (i) **Buying or selling investments** (stocks, bonds, subsidiaries)
- (ii) **Capital Expenditure (CAPEX)** – buying/selling:
 - (1) Property
 - (2) Plant
 - (3) Equipment

(b) Common Examples of Investing Cash Flows

- (i) Cash Outflows (Money Spent)
 - (1) Purchase of property, plant, and equipment (PPE)
 - (2) Buying shares or debt instruments of other entities
 - (3) Acquiring a subsidiary or business
 - (4) Capitalised R&D and software development
- (ii) Cash Inflows (Money Received)
 - (1) Sale of PPE
 - (2) Selling shares or debt instruments of other entities
 - (3) Proceeds from disposal of business segments
 - (4) Interest or dividends received (if classified under investing)

(c) Key Insight

A **healthy investing cash outflow** may indicate that a company is:

- (i) Building assets for future growth
- (ii) Restructuring to realign strategy

But **persistent outflows without returns** or when funded by debt might raise red flags.

Inflows from Investing Activities	Outflows from Investing Activities
Proceeds from disposal of property, plant, and equipment	Payments for acquisition of property, plant, and equipment
Cash receipts from the disposal of debt instruments of other entities	Payments for purchase of debt instruments of other entities
Receipts from sale-of-equity instruments of other entities	Payments for purchase of equity instruments of other entities
	Sales/maturities of investments
	Purchasing and selling long-term assets and other investments

b. FINANCING DECISIONS

Financing decisions involve choosing the right mix of **equity and debt** to fund the company's activities after investment decisions are made. The main goal is to build an **optimal capital structure**—a combination that maximizes shareholder wealth while minimizing cost and risk. A company's **balance sheet** reflects this mix through equity share capital, preference capital, reserves, debentures, loans, etc.

Capital structure is essentially about how a firm divides its cash flow between fixed returns to debt holders and residual income for equity shareholders. Debt is cheaper but riskier, while equity is costlier but less risky. Hence, choosing the right proportion is key. These decisions are strategic and influence profitability, risk level, and overall firm value.

Modern financing decisions are deeply integrated with capital budgeting, long-term planning, and performance measurement. They now rely heavily on financial analysis and technology tools for better decision-making.

1. Factors Affecting Financing Decisions

Several factors influence financing decisions.

- (a) **Cost** – Choose the cheapest source of funds.
- (b) **Risk** – Debt increases financial risk; equity offers more safety.
- (c) **Cash Flow Position** – Firms with strong, steady cash flow can afford more debt.
- (d) **Control** – To retain control, promoters may prefer debt over issuing new equity.
- (e) **Floatation Cost** – Lower costs of raising funds are preferred (e.g., fewer fees, commissions).

2. Cash Flow from Financing Activities (CFFA)

Cash flow from financing activities in the cash flow statement shows how a company raises and repays capital. It tracks cash movements between the firm and its investors or creditors. **Positive cash flow** indicates raising money (through loans or equity), while **negative cash flow** shows repayments (like loan repayments, dividend payouts, or share buybacks).

Common examples include cash received from issuing shares or debt, repayment of loans, dividend payments, and proceeds from stock options or hybrid securities like convertible debt.

In essence, financing decisions determine how much, from where, and at what cost capital is raised. Cash flow from financing activities reflects these decisions in action—whether the firm is growing by raising capital or returning profits to investors.

c. DIVIDEND DECISIONS – DISTRIBUTE OR RETAIN?

A **dividend decision** involves choosing how much of the firm's profit should be:

- ◆ **Distributed to shareholders** as dividends, or
- ◆ **Retained** for reinvestment in business.

Objective: Maximize shareholder wealth while maintaining sufficient funds for future growth.

1. Linked with Other Financial Decisions

- (a) **Investment Decision** – Profitable use of funds
- (b) **Financing Decision** – How to raise the required funds
- (c) **Dividend Decision** – Whether to pay or retain profits

All three are interconnected and affect firm's long-term value.

2. Factors Affecting Dividend Policy

- (a) **Legal Requirements**
Must follow rules: Net profit rule, Capital impairment rule, Insolvency rule.
- (b) **Liquidity Position**
Even with profits, no cash = no dividends.
- (c) **Repayment Needs**
Firm may retain earnings to repay debts.
- (d) **Expected Return on Investment**
Higher return projects = more retention, less dividend.
- (e) **Stability of Earnings**
Stable income = stable/larger dividends; fluctuating income = cautious payout.

3. Dividend Decision Matrix

Factors	FCFE > Dividends	FCFE < Dividends
ROE > Cost of Equity	(a) Good Projects (b) Cash flow surplus (c) No Change	(a) Good Projects Decrease (b) Dividends Invest in Projects
ROE < Cost of Equity	(a) Poor Projects (b) Cash flow surplus (c) Increase Dividends (d) Reduce Investment	(a) Poor Projects (b) Cash flow Deficit (c) Decrease Dividends (d) Reduce Investment

I. STABLE DIVIDEND POLICY – SMOOTHING OVER TIME

A **Stable Dividend Policy** means companies aim to keep **dividend payouts consistent** over time, even if earnings (EPS) fluctuate.

a. KEY INSIGHT – LINTNER'S OBSERVATION (1956)

Companies **prefer gradual increases** in dividends and **avoid cutting them**, leading to:

1. Less Volatile Dividends (DPS)
2. More Volatile Earnings (EPS)

b. WHY DO COMPANIES DO THIS?

1. **Build investor confidence**
2. **Predictability** for income-focused shareholders
3. **Avoid negative market signals** from dividend cuts
4. **Maintain reserves** for future stability

c. EPS VS. DPS TREND

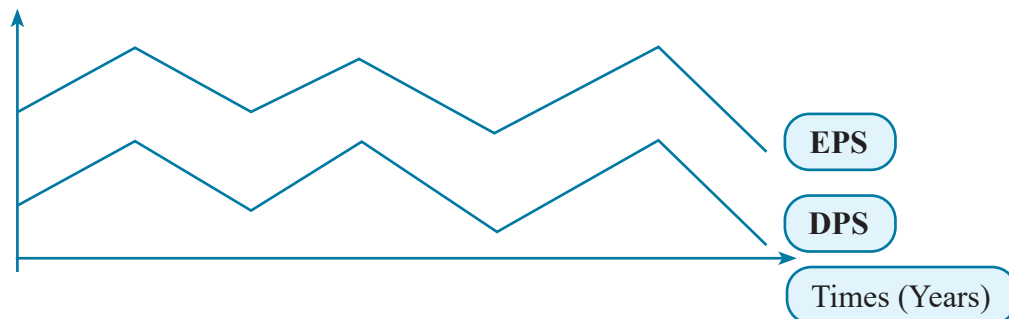
1. If **EPS falls** → DPS may **slightly fall** or stay the same
2. If **EPS rises** → DPS increases **slowly** (lagging behind)

This gradual adjustment is called **dividend smoothing**

d. FIRM MAY ADAPT ANY OF THE FOLLOWING STABLE DIVIDEND POLICIES:

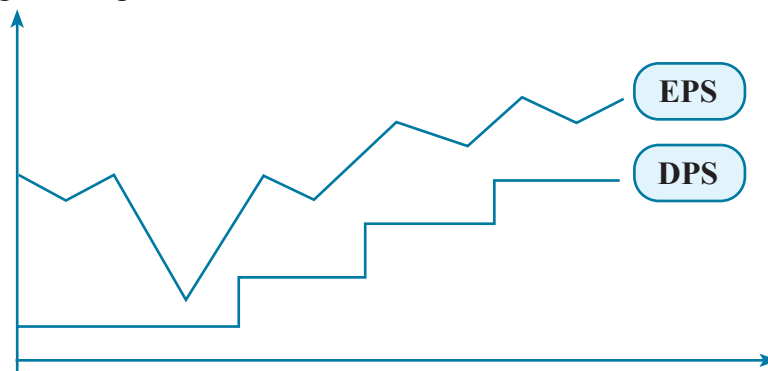
1. Stable dividend payout ratio
2. Stable dividends per share
3. A regular plus extra dividend policy

- 1. Stable Dividend Payout Ratio:** As per this policy the percentage of dividends paid out of earnings remains constant.



Example: if a company adopts a 30% payout ratio and if EPS is Rs 100, then shareholder having 10 shares will receive Rs.300 as dividend under this policy.

- 2. Stable Dividends Per Share:** According to this policy, the firm pays a certain fixed amount of dividend per share every year. Annual dividend per share is increased only when the company reaches a new level of earnings and expects to maintain it.



- 3. A Regular Plus Extra Dividend Policy:** According to this policy a certain fixed percentage or a minimum amount of dividend is paid every year, which is referred to as regular dividend. The firm pays 'additional' or 'extra' dividend if earnings are higher than normal in any year.

e. RATIONALE FOR STABLE DIVIDEND POLICY

Companies prefer a stable dividend policy because it builds investor trust and ensures a steady income for shareholders, especially retirees. It reduces uncertainty, which can lead to lower capital costs and higher stock prices. Institutional investors also favor firms with consistent dividend records. Such stability boosts investor confidence and helps in raising external funds more easily. However, once adopted, this policy reduces corporate flexibility—cutting dividends later may hurt the company's image and shake investor confidence.

J. DECISION CRITERIA IN FINANCIAL MANAGEMENT

Decision-making in finance revolves around selecting the best project or financial option to **maximize returns** and **minimize costs**. A sound decision criterion helps distinguish between acceptable and non-viable proposals. Two guiding principles are: “**Bigger and Better**” (choose higher benefits) and “**Bird in Hand**” (prefer earlier returns).

a. KEY INVESTMENT DECISION CRITERIA

1. **Urgency** – Non-quantifiable but useful when immediate action is needed for critical projects or goal alignment.
2. **Payback Period** – Measures how quickly the investment is recovered. Shorter payback = better. However, it **ignores cash flows after recovery**, violating both decision principles.
3. **Rate of Return (ROR)** – Shows percentage return over time. It helps compare profitability but may vary depending on income or asset treatment.
4. **Benefit-Cost Ratio (Undiscounted)** – Compares total benefits with costs. Useful but gives equal weight to future and present benefits, ignoring the time value of money.
5. **Discounted Benefit-Cost Ratio** – More reliable as it applies present value calculations. It follows both financial principles and adjusts for timing of cash flows.
6. **Net Present Value (NPV)** – Evaluates total present value of future cash inflows against the investment cost. Positive NPV = project is viable. It reflects the **time value of money** and helps in selecting between mutually exclusive projects.
7. **Internal Rate of Return (IRR)** – The discount rate at which NPV becomes zero. It considers both timing and size of cash flows. IRR is widely used and aligns with both “bigger” and “earlier” return preferences.

K. CAPITAL STRUCTURE – THE FINANCIAL FRAMEWORK

Capital structure is the **mix of equity (owner’s funds)** and **debt (borrowed funds)** used by a company to finance its operations and assets.

a. KEY COMPONENTS

1. **Equity** = Shareholders’ capital
2. **Debt** = Loans, debentures, bonds

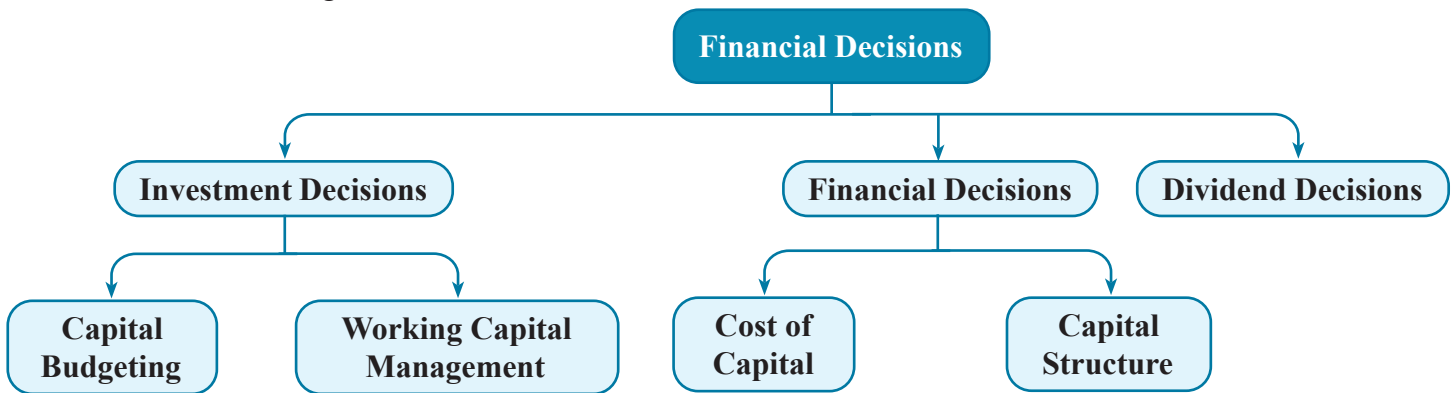
b. WHY INCLUDE DEBT?

1. **Tax savings** (interest is tax-deductible)
2. **Lower cost** vs. equity
3. **Reduces flotation cost**
4. **Leverage advantage** (boosts returns)
5. **No equity dilution** (ownership/control retained)
6. **Refinancing tool** for short-term liabilities
7. **Improves financial ratios** in inflationary times

c. ROLE OF EQUITY

1. Acts as a **cushion** for debt
2. Strong equity base = easier to raise debt

Please refer the following exhibit on forms of financial decisions.



L. VALUE OF FIRM – RISK & RETURN

Every **financial decision** (investment, financing, or dividend) involves a **degree of risk** and aims to generate **returns**. These two always go **hand in hand**:

- ◆ **Higher Risk = Higher Expected Return**
- ◆ **Lower Risk = Lower Expected Return**

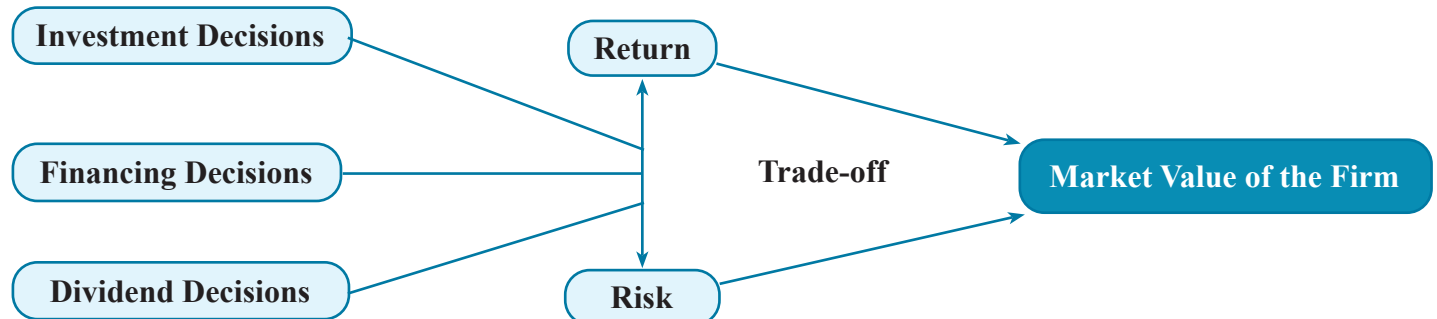
Investor Example

- ◆ **Govt Bonds** = Low risk, fixed returns
- ◆ **Shares** = High risk, uncertain but potentially higher returns

M. FINANCE MANAGER'S ROLE

The **goal** is to **maximize firm value** (market price of shares) by finding the **right balance between risk and return**—known as the **Risk-Return Trade-Off**.

A finance manager cannot avoid risk but must **optimize** it. Decisions should not chase return alone, but also evaluate the **associated risks** to protect shareholder value.



N. LIQUIDITY – THE BUSINESS LIFELINE

Liquidity refers to a company's **ability to meet short-term obligations** using its **current or near-cash assets**. It shows how quickly a firm can convert its assets into cash to pay off its bills.

a. WHY IT MATTERS?

According to Ezra Solomon, liquidity ensures a company can handle:

1. Operating losses
2. Asset expansion
3. Liability reduction

It improves **cash availability** and enhances **creditworthiness** in the market.

b. KEY LIQUIDITY RATIOS

1. **Current Ratio** = Current Assets / Current Liabilities
Higher = Strong short-term solvency
Crude, ignores asset quality (e.g., slow-moving inventory)
2. **Quick Ratio (Acid Test)** = (Cash + Marketable Securities + Receivables) / Current Liabilities
Excludes inventory for a sharper liquidity view
3. **Net Working Capital** = Current Assets – Current Liabilities
Shows surplus funds available for day-to-day operations

c. RECEIVABLES & INVENTORY LIQUIDITY

1. **Average Collection Period (ACP)**: Days it takes to collect receivables
 - (a) Low ACP = strict credit policy
 - (b) High ACP = risk of bad debts
2. **Receivables Turnover Ratio** = Sales / Avg. Receivables
Measures efficiency in collections
3. **Inventory Turnover Ratio** = Cost of Goods Sold / Avg. Inventory
Higher ratio = better inventory management

d. IMPACT ON FINANCIAL DECISIONS

Liquidity analysis helps in

1. Managing cash & securities
2. Formulating credit policies
3. Controlling inventory
4. Administering fixed assets
5. Using current assets cost-effectively
6. Avoiding bankruptcy by planning debt repayments

O. PROFITABILITY – A CORE DECISION CRITERION

Profitability measures a company's ability to generate profits from its resources. It's essential for **long-term survival** and **business growth**, while also serving as a tool for **performance evaluation** and **decision-making** across various financial areas.

a. WHY PROFITABILITY MATTERS?

1. **Creditors** look for profitability to judge repayment capacity.
2. **Investors** check it to assess returns on equity.
3. **Managers** use it to improve pricing, cost control, and capital efficiency.

b. KEY PROFITABILITY RATIOS

1. **Net Profit Margin (NPM)** = Net Profit after Tax/Sales
Shows how much of each rupee of sales turns into profit after all expenses.
2. **Gross Profit Margin** = (Sales – Cost of Goods Sold)/Sales
Reflects **production efficiency** and product pricing.
3. **Return on Investment (ROI)** = EBIT/Capital Employed
Measures return on total resources used.
High ROI = efficient asset use
Used to compare divisions or evaluate investments

4. **Profit on Equity** = Net Income/Total Equity
Reflects returns earned for shareholders
5. **Times Interest Earned (TIE)** = EBIT/Interest Expense
Indicates company's ability to cover interest obligations. A ratio of **5–6x** is considered healthy.

c. PROFITABILITY VS. LIQUIDITY

1. **Liquidity** ensures short-term survival (cash availability)
2. **Profitability** ensures long-term growth

A **balance** between the two is vital. Excess focus on one can hurt the other.

P. COSTING AND RISK – BALANCING CAPITAL & CONSEQUENCES

In financial management, **costing** refers to calculating the **cost of capital** from different sources like **equity**, **preference shares**, **debentures**, and **loans**.

a. EQUITY VS. DEBT

1. **Equity** = Owner's funds
2. **Debt** = Borrowed funds with fixed interest

Debt is cheaper (due to tax benefits) but **adds risk**, especially if the firm can't repay interest or principal on time. Still, using debt can increase returns on equity through **leverage/gearing**.

b. GEARING/LEVERAGE

Using fixed-cost funds (debt) is called **financial gearing** (UK) or **leverage** (US).

1. In **good times**, high-g geared companies show **higher returns** for shareholders
2. In **bad times**, the same gearing **amplifies losses**
3. Higher gearing = **higher financial risk**

c. FINANCIAL RISK VS. BUSINESS RISK

1. **Financial Risk**: Linked to debt; risk of **default or insolvency** due to **interest obligations**
2. **Business Risk**: Variability in **EBIT** due to internal/external business factors
 - (a) **EBIT** – reflects business risk
 - (b) **EAIT** – reflects financial risk

d. CAPITAL STRUCTURE THEORIES

1. **Traditional View**: Optimal mix of debt and equity minimizes cost of capital
2. **Modigliani & Miller (M&M) Theory**: Under ideal conditions, **capital structure doesn't affect** overall firm value. Investors can **arbitrage** between firms to equalize value.

e. TYPES OF RISK

1. **Systematic Risk**
 - (a) Market-wide (e.g., inflation, war, recession)
 - (b) **Cannot be diversified**
 - (c) Examples of Systematic Risk (Undiversifiable Risk)
 - (i) Changes in laws/regulations
 - (ii) Tax reforms
 - (iii) Interest rate hikes

- (iv) Natural disasters (earthquakes, floods etc.)
- (v) Political instability and flight of capital
- (vi) Changes in foreign policy
- (vii) Volatility in currency values
- (viii) Failure of banks (e.g. 2008 mortgage crisis)
- (ix) Economic recessions

2. Unsystematic Risk

- (a) Company-specific (e.g., strikes, lawsuits)
- (b) **Can be diversified** by investing in multiple firms
- (c) Examples of Unsystematic (Diversifiable Risk)
 - (i) The entry of a new competitor into a market
 - (ii) A company is forced to recall one of its products
 - (iii) A company is found to have prepared fraudulent financial statements
 - (iv) A union targets a company for an employee walkout
 - (v) A foreign government expropriates the assets of a specific company

Q. OTHER TYPES OF INVESTMENT-RELATED RISKS

These risks are **not tied to cost of funds**, but to the **outcome and safety of investments**:

- a. Purchasing Power Risk (Inflation Risk)

Risk that inflation erodes the **real value** of future cash flows, especially harmful for **fixed-income** investments.
- b. Market Risk

Risk of loss due to **overall market movements** (e.g., stock market crashes), not tied to any one company.
- c. Interest Rate Risk

If **interest rates rise**, the value of bonds or fixed-income investments **falls**, affecting long-term investors.
- d. Social Risk

Includes **labor issues**, **human rights violations**, or **corruption** that damage community trust and business sustainability.
- e. Regulatory Risk

Changes in **laws or government regulations** may affect industry operations or a company's ability to function profitably.
- f. Reputation Risk

Damage to a company's **public image** due to:

 - (i) Its own actions
 - (ii) Employee behavior
 - (iii) Third-party partnerships
- g. Operational Risk

Unexpected events like **natural disasters**, **fire**, or **internal failures** that may disrupt or halt operations.
- h. Competition Risk

Failure to **adapt** to market trends or competitor innovations can cause **loss of customers** and market share.

R. OBJECTIVES OF A FIRM IN FINANCIAL MANAGEMENT

Every firm sets goals to guide its **financial decisions**—these are primarily:

a. PROFIT MAXIMISATION

Traditionally, this was seen as the **main objective**, where the firm aims to:

1. Maximise **output** from given inputs
2. Or minimise **cost** for given output
3. Achieve **efficiency** in resource allocation under perfect competition

b. WHY IT'S CRITICIZED TODAY

1. **Vague** concept
2. Ignores **timing** of returns
3. Ignores **risk**
4. Focuses on **short-term gains**
5. May reduce **share value**
6. Ignores **stakeholder interests**

c. SHAREHOLDER WEALTH MAXIMISATION

This modern approach focuses on **maximising the Net Present Value (NPV)** of all business decisions.

♦ NPV Formula:

$$NPV = \frac{R_t}{(1 + i)^t}$$

Where,

NPV = Net Present Value

R_t = Net cash flow at time t

i = Discount

t = time value of the cash flow

d. WHY WEALTH MAXIMISATION IS PREFERRED

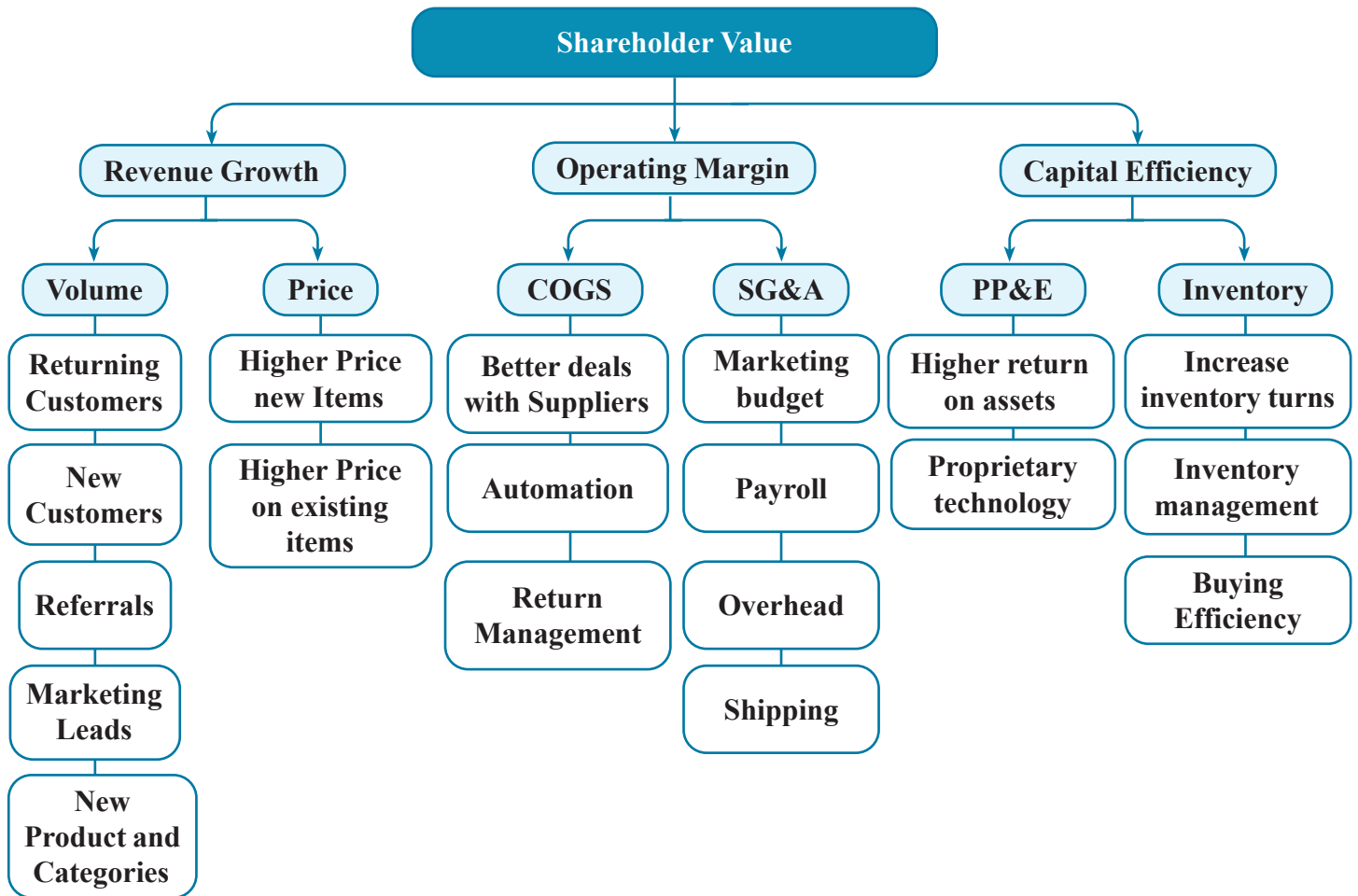
1. Long-term focus
2. Considers **risk & time value of money**
3. Improves **share price**
4. Reflects **stakeholder welfare**
5. Ensures consistent **dividend and earnings growth**

e. PROFIT VS WEALTH MAXIMISATION

Profit Maximisation	Wealth Maximisation
Short-term focus	Long-term growth
Ignores risk	Considers risk
Focus on profits only	Focus on value creation
Vague & outdated	Clear, practical goal

S. HOW TO CREATE SHAREHOLDER VALUE

In order to maximize shareholder value, there are three main strategies for driving profitability in a company:



a. REVENUE GROWTH

1. Increase Sales Volume

- Retain existing customers
- Attract new customers through:
 - Referrals
 - Promotions
 - New offerings & markets

2. Raise Sales Prices

- Gradual or one-time price hikes
- Launch premium products with higher margins
- Ideally, grow **both volume and price** for max impact

b. IMPROVE OPERATING MARGIN

Reduce costs to boost profit margins.

1. Cost of Goods Sold (COGS)

- Negotiate with suppliers for better rates
- Use automation for cost savings & quality control
- Streamline **returns** to lower wastage and costs

2. SG&A Expenses

- (a) Control marketing budget
- (b) Cut overheads and unnecessary labor
- (c) Efficient **shipping & return systems** lower logistics cost

c. INCREASE CAPITAL EFFICIENCY

Measured by **ROCE = EBIT/Capital Employed**

1. Optimize PP&E (Assets)

- (a) Use assets productively = higher ROA
- (b) Invest in **proprietary tech** for competitive edge
- (c) License tech to generate extra income

2. Manage Inventory

- (a) Boost **inventory turnover** = more revenue per item sold
- (b) Reduce **holding costs** (space, theft, waste)
- (c) Use **Just-in-Time (JIT)** system to avoid excess stock

T. ETHICS OF SHAREHOLDER WEALTH MAXIMIZATION

Maximizing shareholder wealth doesn't mean ignoring **ethics or social responsibility**.

a. WHAT WENT WRONG (2008 CRISIS)

- 1. Banks focused on toxic investments instead of responsible lending
- 2. **Outcome:** Crashed portfolios, failing banks, and plummeting share prices

b. DOING IT RIGHT (GM CASE)

- 1. GM recovered post-crisis by:
 - (a) Restructuring operations
 - (b) Embracing **sustainability** (green vehicles)
- 2. **Outcome:** Share price increased

A company **can be ethical and profitable** — good governance & social responsibility **enhance long-term shareholder value**.

U. PROFIT MAXIMISATION VS. SHAREHOLDER WEALTH MAXIMISATION

These are two fundamental, yet often **conflicting objectives** in financial management:

a. PROFIT MAXIMISATION

Goal: Maximise **immediate earnings**

Focus: **Short-term** (single-period focus)

How: Maximise output or minimise cost to earn highest possible profit

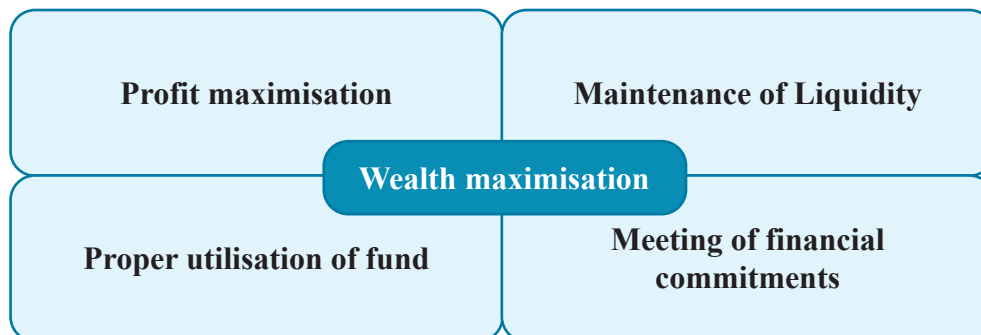
Used in: Traditional business models

1. Advantages Of Profit Maximisation

- (a) Easy to calculate and measure
- (b) Predicts business behavior effectively
- (c) Reflects how competitive firms operate (especially small ones)
- (d) Supports mathematical modelling
- (e) Keeps analysis simple and focused

2. Disadvantages Of Profit Maximisation

- (a) Vague definition (net, total, or rate of profit?)
- (b) Ignores multiple stakeholder interests
- (c) Not suitable for monopolists or non-competitive firms
- (d) Ownership-control separation reduces profit focus
- (e) Preference for power over profits (small firms resist expansion)
- (f) Sometimes **efficiency** or **reputation** matters more than profits
- (g) Businesses avoid certain expansions due to focus limitations
- (h) **Conspicuous consumption** harms investment strategy
- (i) Legal restrictions (esp. in public sector or regulated industries)



b. SHAREHOLDER WEALTH MAXIMISATION (SWM)

Goal: Maximise **market value of shares**

Focus: **Long-term** growth & sustainability

How: Improve NPV, earnings per share, and dividend flows to increase stock price

1. Advantages Of Wealth Maximisation

- (a) Cash-flow based (more reliable than profit)
- (b) Less prone to manipulation
- (c) Focuses on **long-term growth**
- (d) Considers **risk and time value of money**
- (e) Attracts sustained investment and shareholder confidence

2. Disadvantages Of Wealth Maximisation

- (a) Concept is **abstract**, hard to define clearly
- (b) Goals may not be **immediately measurable**
- (c) Dependent on long-term profitability
- (d) Cash flow-based, not profit-based (may overlook accounting realities)

c. KEY DIFFERENCES TABLE

Criteria	Profit Maximisation	Shareholder Wealth Maximisation
Time Horizon	Short-term	Long-term
Risk Consideration	Ignored	Considered

Time Value of Money	Ignored	Recognized
Decision Focus	Immediate profit	Future cash flows, EPS, dividends
Strategic Vision	Narrow	Broad & sustainable

V. ECONOMIC VALUE ADDED (EVA) – GAUGING SHAREHOLDER VALUE

EVA is a modern financial performance metric that evaluates **real value creation** for shareholders. It corrects a major flaw in traditional accounting by factoring in the **cost of capital**, including equity.

◆ EVA Formula

$$\text{EVA} = \text{NOPAT} - (\text{Cost of Capital} \times \text{Capital Employed})$$

Where:

NOPAT = Net Operating Profit After Tax

Cost of Capital = WACC (Weighted Average Cost of Capital)

Capital Employed = Total capital invested in the business

a. WHY EVA MATTERS

Unlike **accounting profit**, EVA deducts **all capital costs** (both debt and equity). It reflects **true economic profit**—i.e., what's left after compensating all capital providers.

b. COMMON MISTAKES IN EVA IMPLEMENTATION

1. Misstating capital cost intentionally
2. Using EVA **only at top levels**
3. Inadequate training for employees
4. Poor communication = poor adoption

c. TWO KEY FUNCTIONS OF EVA

1. **EVA as a Performance Measure**
 - (a) Evaluates true residual income
 - (b) Encourages long-term, value-generating decisions
 - (c) Helps link **manager compensation** to value creation
2. **EVA as a Corporate Philosophy**
 - (a) Adopted **across all decision levels**
 - (b) Shifts mindset from profit-only to **value-based thinking**
 - (c) Focuses on **efficient capital use** and reducing waste

Examples

Problem 1: XYZ Ltd.

- ◆ $\text{NOPAT} = ₹20 \text{ Cr}$, $\text{Capital} = ₹150 \text{ Cr}$, $\text{WACC} = 12\%$
- ◆ $\text{EVA} = 20 - (12\% \text{ of } 150) = ₹2 \text{ Cr}$

Problem 2: Small Business

Investment = ₹20,000, NOPAT = ₹10,000, Cost of Capital = 10%

$$\text{EVA} = 10,000 - (10\% \text{ of } 20,000) = ₹8,000$$

d. WHEN EVA INCREASES

1. NOPAT grows **without increasing capital**
2. Capital is invested in **high-return projects**
3. **Idle/unproductive capital is removed**

e. IMPLEMENTATION CHECKLIST

1. Train & align **top management**
2. Educate **key employees** across functions
3. Embed EVA in **decision-making & performance reviews**

f. ADVANTAGES OF EVA

1. **Focus on Shareholder Value**
Helps managers align decisions with the goal of increasing **shareholders' wealth**.
2. **Investor-Friendly Tool**
Provides a **true economic picture** by considering both **operating and capital costs** – better than EBIT, net profit, or EPS.
3. **Segment Performance Evaluation**
Useful for **monitoring divisions/units** in decentralized firms. Highlights problem areas and supports corrective action.
4. **Enhances Corporate Governance**
With EVA-linked bonuses, managers are motivated to perform **ethically and efficiently**.
5. **Managerial Benchmarking**
Helps identify the most **capable individuals** to lead and manage divisions effectively.

g. DISADVANTAGES OF EVA

1. **Size Distortion**
EVA does not adjust for size—**larger units** may show higher EVA even if less efficient.
2. **Potential Misuse**
Managers might manipulate inputs or decisions for **personal gain**, not firm-wide benefit.
3. **Short-Term Bias**

May push focus toward **immediate results** rather than sustainable long-term growth.

h. INTERPRETING EVA RESULTS

1. **Positive EVA = Value Created**
The firm/division is generating returns above the cost of capital.
2. **Trend Over Time**
Is EVA **rising, stable, or declining**? Even if declining, positive EVA still indicates value creation.
3. **Consider Project Lifecycle**
Early project years may show low EVA due to **high capital charges**; later years may improve as asset value depreciates.

W. MARKET VALUE ADDED (MVA) – GAUGING TRUE WEALTH CREATION

Market Value Added (MVA) shows the **wealth created by a company** for its investors (both equity & debt holders) **beyond the capital invested**.

Formula:

$$\text{MVA} = V - K$$

Where:

V = Market value of the firm (equity + debt)

K = Capital invested by all investors (debt + equity, including retained earnings)

A **positive MVA** means the company has **created wealth**

A **negative MVA** means the company has **destroyed wealth**

Example 1

- ◆ Shares issued = 20,000,000
- ◆ Treasury stock = 1,100,000
- ◆ Current share price = ₹35.5
- ◆ Invested capital + retained earnings = ₹453,503,000
- ◆ Cost of treasury stock = ₹39,050,000

Step 1:

$$\text{Outstanding shares} = 20,000,000 - 1,100,000 = 18,900,000$$

$$\text{Market capitalization} = 18,900,000 \times ₹35.5 = ₹670,950,000$$

Step 2:

$$\text{Shareholder's invested capital} = ₹453,503,000 - ₹39,050,000 = ₹414,453,000$$

$$\text{MVA} = ₹670,950,000 - ₹414,453,000 = ₹256,497,000$$

Example 2

- ◆ Shareholders' Equity = ₹750,000
- ◆ Common shares = 100,000 × ₹12.5 = ₹1,250,000
- ◆ Preferred shares = 5,000 × ₹100 = ₹500,000
- ◆ Total Market Value = ₹1,750,000

$$\text{MVA} = ₹1,750,000 - ₹750,000 = ₹1,000,000$$

a. BENEFITS OF MVA

1. Shows long-term **value creation** beyond accounting profits
2. Indicates company performance from **investor's perspective**
3. Useful for benchmarking firm value over time or against competitors

b. LIMITATIONS

1. Market values fluctuate; may not reflect long-term fundamentals
2. Negative MVA can demotivate even if short-term losses are strategic

c. FINANCIAL DISTRESS AND INSOLVENCY

What is Financial Distress?

A condition when a firm cannot meet its financial obligations:

1. **Reasons:** Low sales, high fixed costs, excess debt, inadequate cash inflows.
2. **Effects:** Creditors lose confidence, share price drops, distress asset sales at below-market value.

d. TYPES OF FAILURE

1. **Technical Failure:** Temporary inability to meet current liabilities.
2. **Insolvency (Bankruptcy):** When total liabilities > assets = **negative net worth**.

e. KEY RATIOS TO DETECT DISTRESS

1. **Liquidity Ratios:** Current ratio, Quick ratio
2. **Solvency Ratios:** Debt-to-Equity, Interest Coverage, Debt-to-Funds
3. **Trend Analysis:** Over 3–5 years to detect early warning signs.

X. FINANCIAL MANAGEMENT – A SCIENCE & AN ART

a. AS A SCIENCE

1. Based on tested **principles, tools, and equations**
2. Uses **statistics, OR, econometrics**, and computers for:
 - (a) Budgeting
 - (b) Investment analysis
 - (c) Financial control

b. AS AN ART

- ◆ Involves **judgement, experience, and intuition**
- ◆ Most financial decisions (like pricing, capital structuring) need:
 - ☞ Human insight
 - ☞ Risk-taking ability
 - ☞ Interpretation beyond formulas

c. WESTON'S VIEW

1. In **planning, organizing, and controlling**, both scientific tools and artistic judgement play vital roles.
2. It is best to say **Financial Management is both a science and an art**.

Y. EMERGING ROLES OF FINANCIAL MANAGER

With financial management evolving from record-keeping to strategic decision-making, the modern **financial manager** handles a broader set of functions:

a. KEY DUTIES

1. **Forecasting Cash Flow**
Match inflows vs outflows to ensure liquidity.
2. **Raising Funds**
Identify sources for short, medium, and long-term financing.

3. **Internal Funds Management**

Utilize surplus funds efficiently across bank accounts.

4. **Cost Control Facilitation**

Monitor variances; suggest corrective measures.

5. **Pricing Assistance**

Provide costing and profit margin insights to set competitive prices.

6. **Profit Forecasting**

Collect data to project future earnings.

7. **Return Measurement**

Determine required rate of return to evaluate investment proposals.

8. **Asset Management**

Decide asset mix; improve usage efficiency and reduce waste.

9. **Fund Management**

Ensure availability of liquid funds for daily operations, debt, asset acquisition, and dividend payments.

b. DUAL ROLE

1. **Decision-maker:** In investment, asset allocation
2. **Staff role:** Advises top management on fund planning & risk

Z. FINANCE AND ITS RELATIONSHIP WITH ECONOMICS & ACCOUNTING

Financial management draws heavily from both **economics** and **accounting**, but each discipline plays a different role in shaping financial decisions.

a. RELATIONSHIP OF FINANCE WITH ECONOMICS

Finance is essentially **applied microeconomics**. It draws insights from both:

1. Macroeconomics

Defines the **external environment** in which a firm operates:

- (a) GDP growth rate
- (b) Inflation and interest rates
- (c) Taxation & fiscal policy
- (d) Monetary policy & savings rate
- (e) Government spending and FDI

Why it matters?

A financial manager must be aware of these factors to identify **risks, threats, and opportunities** in the economy.

2. Microeconomics

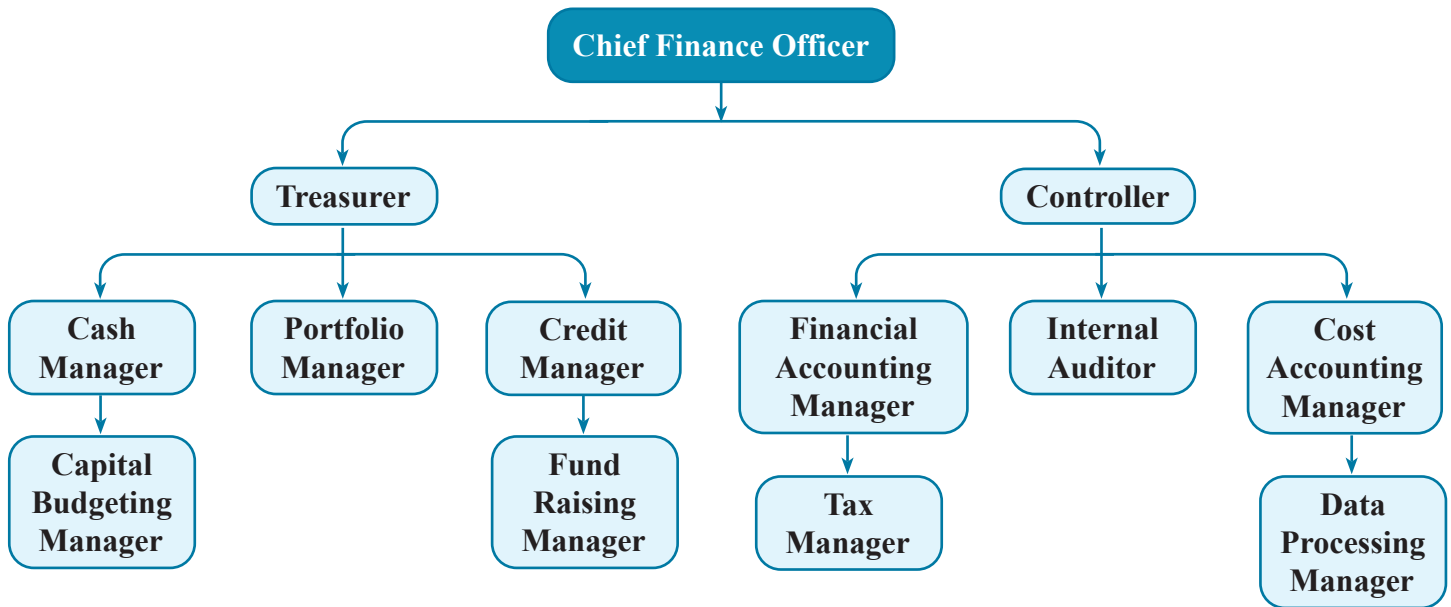
Sharpens financial decision-making with tools like:

- (a) **Marginal analysis**
- (b) Demand-supply principles
- (c) Utility vs. cost

Application:

Used in capital budgeting, pricing, and cost-benefit analysis.

Organisation of Finance Function



b. RELATIONSHIP OF FINANCE WITH ACCOUNTING

Though often confused, finance and accounting are different but deeply interconnected:

Feature	Accounting	Finance
Core Objective	Score-keeping & performance review.	Value maximisation for shareholders.
Focus Area	Past performance & reporting.	Future projections & decisions.
Method	Accrual-based reporting.	Cash flow analysis.
Nature	More objective, fact-based.	More subjective, decision-oriented.
Time Frame	Records the past.	Plans for the future.

TEST YOURSELF

1. What is meant by profit maximisation in financial management?
2. Define shareholder wealth maximisation.
3. What are the main functions of a financial manager?
4. State any two key differences between accounting and finance.
5. What is financial distress? Give one cause.
6. Explain the concept of Economic Value Added (EVA) and how it is calculated.
7. How is financial management both a science and an art? Illustrate with examples.
8. Differentiate between Market Value Added (MVA) and EVA.
9. Describe the relationship of finance with microeconomics and macroeconomics.
10. Highlight the limitations of profit maximisation as an objective of a firm.

TIME VALUE OF MONEY

A. TIME VALUE OF MONEY – A CORE FINANCIAL PRINCIPLE

A rupee in hand today is worth more than the same rupee received later. This is due to its earning potential, inflation effects, and consumption preferences.

a. WHY MONEY HAS TIME VALUE

1. **Preference for Present Consumption:** People generally value immediate use over future use.
2. **Productive Use of Capital:** Money can be invested to earn returns – ₹1 today becomes ₹(1 + r) in the future.
3. **Inflation Impact:** The purchasing power of money erodes over time.

b. APPLICATIONS IN FINANCE

Time Value of Money (TVM) underpins:

1. Security valuation
2. Project appraisal
3. Loan amortisation
4. Lease vs. buy decisions
5. Company Valuation
6. Capital Budgeting

To compare and aggregate cash flows at different periods, we use **compounding and discounting techniques**.

c. KEY TAKEAWAYS

1. **Value Now > Value Later** – due to potential earnings.
2. **Delay = Lost Opportunity** – returns come only through timely investment.
3. **TVM Formula** – involves principal, rate, time, and compounding frequency.
4. **Savings Impacted by Frequency** – more compounding, more growth.
5. **Inflation Diminishes Value** – higher prices reduce future purchasing power.

B. CONCEPTS OF TIME VALUE OF MONEY

a. SIMPLE VS. COMPOUND INTEREST

Simple Interest: Interest is earned only on the principal.

Formula:

$$\text{Future Value (FV)} = \text{Principal} \times (1 + n \times r)$$

Example: ₹1,000 at 12% for 5 years:

$$1,000 \times (1 + 5 \times 0.12) = 1,600$$

Compound Interest: Interest is reinvested and earns further interest.

Growth is exponential rather than linear.

b. POWER OF COMPOUNDING - A COMPARISON

Year	Simple Interest (₹)	Compound Interest (₹)
1	1,100	1,100
5	1,600	1,600
10	2,000	2,594
20	3,000	6,728
50	6,000	117,390
100	11,000	13,780,612

Under simple interest, growth is steady and **linear**.

Under compound interest, growth is **exponential**.

c. PRESENT VALUE OF A SINGLE AMOUNT

Present value (PV) is the value today of a future sum of money, considering the time value of money. It answers:

"How much should I invest today to receive a specific amount in future?"

◆ **Example:** To receive \$15,000 in 5 years at 12%, how much should be invested today?

◆ **Formula:**

$$PV = \frac{FV}{(1+i)^n}$$

where:

PV = Present Value

FV = Future Value

i = interest rate (decimal)

n = number of periods

The higher the interest rate and longer the period, the lower the present value.

d. PRESENT VALUE OF AN UNEVEN SERIES

In real-world financial decisions, such as capital budgeting or equity valuation, **cash inflows are rarely uniform**. Cash flow patterns may vary due to project phases, market cycles, or business performance. In such scenarios, calculating the **present value (PV)** of **each individual cash flow** becomes essential.

◆ **General Formula**

For an **uneven cash flow stream**, the present value is the **sum of discounted values** of individual cash flows:

$$PV = \frac{CF_0}{(1+r)^0} + \frac{CF_1}{(1+r)^1} + \dots + \frac{CF_N}{(1+r)^N}$$

Or simply,

$$PV = \sum \frac{CF_n}{(1+r)^n}$$

where:

CF Cash flow in year n

r = Discount rate (usually opportunity cost of capital)

n = Year number

Illustration: Uneven Cash Flow Stream at 12% Discount Rate

Year	Cash Flow (₹)	PVIF @12%	Discounted Value (₹)
1	1,000	0.893	893
2	2,000	0.797	1,594
3	2,000	0.712	1,424
4	3,000	0.636	1,908
5	3,000	0.567	1,701
6	4,000	0.507	2,028
7	4,000	0.452	1,808
8	5,000	0.404	2,020
			Total PV = ₹13,376

◆ Key Insights

- ☛ Uneven cash flows require year-by-year discounting.
- ☛ The **discount rate** should reflect the best alternative use of funds (opportunity cost).
- ☛ This method is fundamental in **DCF analysis, capital budgeting, and firm valuation**.

e. PRESENT VALUE OF AN ANNUITY

The **Present Value of an Annuity (PVA)** refers to the **current worth** of a stream of equal payments (annuity) made at regular intervals, **discounted at a specific rate of return**.

1. Annuities are common in

- Loan repayments
- Lease rentals
- Pension payouts
- Insurance benefits

2. Why It Matters

- Helps evaluate whether to choose a **lump sum** now or **fixed payments** over time.
- Applies the **time value of money principle**, acknowledging that **₹1 today > ₹1 tomorrow**.
- A **higher discount rate** results in a **lower present value**, due to greater opportunity cost.

3. Key Points Recap

- PVA estimates **how much is needed today** to meet fixed future payments.
- The **time value of money** underpins this – earlier money has greater utility.
- Useful in comparing **lump sum vs. annuity** options in investment and finance decisions.

◆ Formula: Present Value of an Ordinary Annuity

$$P = PMT \times \left[1 - \frac{1}{(1+r)^n} \right] \div r$$

Where:

P = Present value of the annuity

PMT = Equal annuity payment per period

r = Discount rate per period

n = Number of periods

Illustrative Example

If you are to receive ₹10,000 per year for 5 years, and the discount rate is 10%, then:

$$P = 10,000 \times \left[1 - \frac{1}{(1.10)^5} \right] \div 0.10$$

$$P \approx 10,000 \times 3.7908 = ₹37,908$$

4. Applications of the Present Value of an Annuity

The **Present Value of an Annuity (PVA)** is a versatile tool for personal finance, corporate finance, and investment decision-making. Below are key real-world applications:

(a) Borrowing Capacity for Fixed Installments

Scenario: You can afford ₹12,000/month for 3 years. The interest rate is 1.5% per month.

Objective: How much can you borrow?

Solution:

◆ Use PVIFA (1.5%, 36) = 27.70

◆ $PV = ₹12,000 \times 27.70 = ₹3,32,400$

This is the **maximum loan** you can take for the mentioned terms.

(b) Loan Maturity Period (Amortisation Tenure)

Scenario: You want to borrow ₹10,80,000 at 12.5% annual interest. You can pay ₹1,80,000 annually.

Objective: What should be the **loan period**?

Solution:

◆ Set: $₹1,80,000 \times PVIFA(12.5\%, n) = ₹10,80,000$

◆ Solving gives $PVIFA \approx 6$

◆ From tables, this corresponds to **n ≈ 12 years**

So, request a **12-year loan tenure**.

(c) Loan Amortisation Schedule

Scenario: Borrow ₹10,00,000 at 15% for 5 years in equal annual instalments.

Step 1: Find A using $PVIFA(15\%, 5) = 3.3522$

$$A = ₹10,00,000 / 3.3522 = ₹2,98,312$$

Step 2: Prepare an amortisation schedule

Year	Start Balance	Instalment	Interest	Principal	End Balance
1	10,00,000	2,98,312	1,50,000	1,48,312	8,51,688
2	8,51,688	2,98,312	1,27,753	1,70,559	6,81,129
3	—	—	—	—	—
5	—	2,98,312	38,913	2,59,399	23

(d) Periodic Withdrawals from a Lump Sum

Scenario: ₹3,00,000 deposited at 10%. How much can be withdrawn annually for 10 years?

Solution:

◆ $PVIFA(10\%, 10) = 6.145$

$$A = ₹3,00,000 / 6.145 \approx ₹48,819$$

(e) **Finding the Interest Rate**

Scenario: Deposit ₹10,000, receive ₹2,500/year for 6 years.

Objective: What's the interest rate?

Step 1: $PVIFA = ₹10,000/₹2,500 = 4$

Step 2: Use table:

☛ $PVIFA(12\%, 6) = 4.111$

☛ $PVIFA(14\%, 6) = 3.889$

Approx. rate = 13

(f) **Valuing Infrequent Annuities**

Scenario: Receive ₹50,000 every 2 years for 30 years. Annual rate = 8%

Effective 2-year rate:

$$(1.08)^2 - 1 = 16.64$$

PV for 15 periods at 16.64%:

$$PV = ₹50,000 \times \left[\frac{1 - (1/1.1664)^{15}}{0.1664} \right] = ₹2,70,620$$

(g) **Equating Present Value of Two Annuities (Education Planning)**

Scenario: Raj wants to fund ₹1,00,000/year for 4 years of college education starting in 16 years.

Objective: How much should he deposit today?

Approach:

- (i) Calculate **present value of the annuity** needed 16 years later using $PVIFA(8\%, 4)$.
- (ii) Discount this back to **today's value** using $PVIF(8\%, 16)$.

5. Summary of Use Cases

Use Case	Purpose
Car/House Loan	Determine borrowing amount
Loan Tenure	Estimate repayment period
Amortisation Schedule	Break down principal and interest
Periodic Withdrawals	Systematic withdrawals post-retirement
Finding Interest Rate	Evaluate yield on uneven cash flows
Irregular/Deferred Annuities	Value long-term infrequent payments
Educational/Goal-based Planning	Determine required current investments

F. PRESENT VALUE OF PERPETUITY (PVP)

A **perpetuity** is a stream of **identical cash flows that continues forever**. Although the cash flows are infinite, their present value is **finite** when discounted appropriately.

- ◆ The concept is rooted in the **time value of money**.
- ◆ Common in the **valuation of preferred shares, real estate, and retirement/endowment plans**.

1. Basic Formula (Non-Growing Perpetuity)

$$PV \text{ of Perpetuity} = \frac{CF}{r}$$



Where: CF = Constant cash flow per period r = Discount rate (expressed as a decimal)

This formula assumes:

- (a) **Fixed payments**
- (b) No growth over time
- (c) First payment occurs **one period from today**

2. Growing Perpetuity FormulaWhen the cash flow grows at a constant rate (g) each period:

$$PV = \frac{CF}{r - g}$$

Where: g = Constant growth rate of cash flowsCondition: $r > g$ (otherwise PV becomes infinite)This is derived from the **Gordon Growth Model**, often used in equity valuation.**Illustrative Examples****1. Non-Growing Perpetuity:**☛ Annual $CF = ₹10,000$ ☛ $r = 8\%$

$$PV = \frac{10,000}{0.08} = ₹1,25,000$$

2. Growing Perpetuity:☛ $CF = ₹10,000$, $r = 10\%$, $g = 2\%$

$$PV = \frac{10,000}{0.10 - 0.02} = ₹1,25,000$$

3. Key Applications

Domain	Application
Preferred Shares	Valuing stocks with fixed dividends paid indefinitely.
Real Estate	Estimating value of properties with stable rental incomes .
Retirement Planning	Calculating lifetime annuity values or pension entitlements .
Endowment Schemes	Designing plans offering a fixed income stream for indefinite periods.
Valuation Models	Forms basis for terminal value calculations in Discounted Cash Flow (DCF) models.

4. Why It Matters

- ◆ Converts **infinite cash flows into a finite value** for decision-making.
- ◆ Assists in assessing **income-generating assets** that are expected to persist indefinitely.
- ◆ Crucial for **long-term financial planning**, where predictability and sustainability of income are key.

g. FUTURE VALUE OF A SINGLE AMOUNT (FVSA)

The **Future Value (FV)** of a single amount represents the **worth of a lump sum investment at a specific future date**, given a fixed interest rate and compounding over time.

- ◆ It reflects **how much a present investment will grow**, assuming reinvestment of interest.
- ◆ Critical in making long-term investment, retirement, and capital budgeting decisions.

1. Formula for Future Value

$$FV = I \times (1 + R)^T$$

Where:

I = Initial investment (present value)

R = Interest rate per period

T = Number of compounding periods

Note: In your provided version, the formula is simplified as $FV = I \times (1 + R \times T)$, which applies **only under simple interest**. For compound interest, use the exponential form shown above.

Illustrative Example

Scenario: Invest \$10,000 at 12% annual interest, compounded yearly, for 3 years.

Year	Opening Balance (\$)	Interest @12% (\$)	Ending Balance (\$)
1	10,000.00	1,200.00	11,200.00
2	11,200.00	1,344.00	12,544.00
3	12,544.00	1,505.28	14,049.28

Thus, in 3 years, the investment grows to **\$14,049.28** under **compound interest**.

2. Key Interpretations

- (a) **Compounding** plays a crucial role in accelerating growth.
- (b) The longer the investment period and the higher the interest rate, the **greater the future value**.
- (c) **No intermediate cash flows** are involved—only one-time investment left to grow.

3. Applications of FVSA

Area	Use Case
Investment Planning	Projecting future value of lump-sum investments
Retirement Planning	Estimating corpus from a one-time deposit
Business Capital Budgeting	Evaluating returns on retained profits
Loan Decision-Making	Comparing value of future lump sum inflows

4. Caution: Inflation Impact

Although FV shows **nominal growth**, **inflation can erode real value**. Thus, **real FV** must be adjusted using expected inflation rates for meaningful decision-making.

h. FUTURE VALUE OF AN ANNUITY (FVA)

The **Future Value of an Annuity** refers to the total value of a series of **equal payments** made at regular intervals, **compounded** at a specific rate, **evaluated at a future date**.

It answers the question:

“If I invest the same amount periodically, how much will it grow into by a future point in time”

1. Key Features

- (a) **FVA** calculates how much a **series of payments** will accumulate over time.
- (b) It assumes reinvestment of each payment at a given interest rate.
- (c) It is the counterpart to the **Present Value of an Annuity**, which discounts future payments to today.

(d) **Two types of annuities:**

- (i) **Ordinary Annuity:** Payments are made at the **end** of each period.
- (ii) **Annuity Due:** Payments are made at the **beginning** of each period.

2. Formula for Future Value of an Ordinary Annuity

$$FV = PMT \times \left[\frac{(1+r)^n - 1}{r} \right]$$

Where:

FV = Future Value

PMT = Periodic payment

r = Interest rate per period

n = Total number of periods

The formula assumes **payments at the end of each period** (ordinary annuity).

Illustrative Example

Scenario: You invest ₹10,000 annually for 5 years at 10% p.a.

$$FV = 10,000 \times \left[\frac{(1 + 0.10)^5 - 1}{0.10} \right]$$
$$FV = 10,000 \times \left[\frac{1.61051 - 1}{0.10} \right] = 10,000 \times 6.1051 = ₹61,051$$

3. Interpretation

- ◆ **Each annual investment grows**, and earlier payments compound longer.
- ◆ The **final value** reflects both principal and **accumulated interest**.
- ◆ **Higher rate or longer term = higher future value.**

4. Applications

Domain	Application
Retirement Planning	Estimating corpus from annual investments
Education Funds	Projecting future savings for children's education
Corporate Finance	Estimating future value of reserve allocations
Systematic Investment Plans (SIPs)	Calculating maturity value of regular investments

5. Note on Annuity Due

If payments are made at the beginning of each period (annuity due), the formula becomes:

$$FV_{\text{annuity due}} = FV_{\text{ordinary annuity}} \times (1 + r)$$

This adjustment reflects **one extra period of compounding** for each payment.

C. ANNUITY DUE VS. ORDINARY ANNUITY

a. ANNUITY DUE VS. ORDINARY ANNUITY

Basis	Ordinary Annuity	Annuity Due
Payment Timing	End of each period	Beginning of each period
Examples	Loan EMIs, bond interest, dividend payments	Rent, lease, insurance premiums

Present Value (PV)	Lower (each payment is discounted over more periods)	Higher (payments are received/paid earlier)
Benefit to Receiver	Less advantageous-money comes later	More advantageous-money is received sooner
Mathematical Impact	PV or FV formulas don't change, but period shift adds one compounding/discounting factor	Each payment gets one more compounding period

b. KEY INSIGHT

All else equal, an **annuity due** will always have a **higher present value and future value** than an ordinary annuity due to the **early timing of cash flows**.

D. DOUBLING PERIOD

The **doubling period** refers to the **time required for an investment to double in value** at a given rate of interest. Rather than complex logarithmic calculations, investors often use **simple rules of thumb**, such as the **Rule of 72** and **Rule of 69**, to estimate this time efficiently.

a. RULE OF 72 – DOUBLING PERIOD ESTIMATION

Historical Note

The **Rule of 72** dates back to **1494**, first referenced by **Luca Pacioli** in his mathematical text *Summa de Arithmetica*. Although Pacioli didn't explain its derivation, the rule is believed to **predate his work** and is accepted as a **financial heuristic**.

Used for estimating doubling time under **periodic compounding**.

$$\text{Doubling Period (in years)} = \frac{72}{\text{Interest Rate (\%)}}$$

Example:

At 8% interest:

$$\frac{72}{8} = 9 \text{ years}$$

1. Key Takeaways

- (a) A **simplified formula** to calculate the time it takes for an investment to **double** at a given **compound interest rate**.
- (b) Works best within the 6%-10% interest range for **periodic compounding**.
- (c) Versatile applies to investments, inflation, GDP growth, and even fees impact.
- (d) Can be **rearranged** to estimate the **required rate of return**:

$$\text{Required Rate} = \frac{72}{\text{Doubling Period}}$$

- (e) It's a mental math tool, not a precise financial formula-but very effective for rough calculations.

2. When to Use Alternative Rules

Use Rule of 69 or Rule of 70/73:

- (a) For **continuous compounding** (e.g., equity returns).
- (b) For rates **outside the 6%-10% band**.
- (c) When higher **accuracy** is needed.



b. RULE OF 69 - A CONTINUOUS COMPOUNDING ESTIMATOR

More accurate when interest is compounded continuously (e.g., equities).

$$\text{Doubling Period (in years)} = 0.35 + \frac{69}{\text{Interest Rate (\%)}}$$

Example:

At 15% interest:

$$0.35 + \frac{69}{15} = 0.35 + 4.60 = 4.95 \text{ years}$$

1. Key Benefits

- (a) Assumes **continuous compounding**—more accurate for **equity, stocks**, etc.
- (b) Closer to values from a **financial calculator**.
- (c) Encourages **quick decision-making** without complex tools.
- (d) Easy to apply—even for **non-finance users**.
- (e) Acts as a **thought tool** for estimating investment cycles.

2. Limitations

- (a) **Origin of “69”** not easily explained—conceptually less intuitive.
- (b) Accuracy reduces with **very low interest rates** (e.g., 2–3%).
- (c) **Not suitable** for securities that **compound periodically**.
- (d) Does **not account for dividends**, which can alter actual doubling.
- (e) Large projects may need **exact financial models**, not approximations.

c. RULE OF 70 – A GENERAL GROWTH ESTIMATOR

A general-purpose rule used for estimating doubling time of any exponential growth variable (like GDP, inflation, investments).

$$\text{Doubling Period (in years)} = \frac{70}{\text{Growth Rate (\%)}}$$

Example:

At 10% growth:

$$\frac{70}{10} = 7 \text{ years}$$

d. SUMMARY COMPARISON

Aspect	Rule of 69	Rule of 70
Formula	$0.35 + (69 \div \text{Interest Rate})$	$70 \div \text{Growth/Interest Rate}$
Best For	Continuous compounding assets	General exponential growth trends
Accuracy	High (for continuously compounding)	Moderate (less precise than Rule 69)
Simplicity	Moderate	High (easy to use and remember)
Common Use	Equity returns, advanced planning	Inflation, GDP, long-term projections

TEST YOURSELF

1. Suppose a company expects to receive ₹8,000 after 5 years. Calculate the **present value** of this sum if the current market interest rate is **12% per annum**, compounded annually.

Solution:

Step 1:

Write down the formula for Present Value (PV):

$$PV = \frac{FV}{(1+i)^n}$$

Where:

FV = Future Value = ₹8,000

i = Interest rate per annum = 12% = 0.12

n = Number of years = 5

Step 2:

Substitute the values into the formula:

$$PV = \frac{₹8,000}{(1+0.12)^5}$$

Final Answer

The present value of ₹8,000 receivable after 5 years at a 12% interest rate is approximately ₹4,539.

2. What is the **present value** of ₹1,000 received in two years if the interest rate is:

- (a) 12% per year, **compounded annually**
- (b) 12% per year, **compounded semi-annually**

Solution:

- (a) 12% per year, **compounded annually**

Step 1:

Use the formula:

$$PV = \frac{FV}{(1+i)^n}$$

Where:

FV = ₹1,000

i = 12% = 0.12 (annually)

n = 2 years

Step 2:

$$\frac{₹1,000}{(1+0.12)^2} = \frac{₹1,000}{1.2544} = ₹797.19$$

- (b) 12% per year, **compounded semi-annually**

Step 1:

Use the same formula, but divide the rate and multiply the time:

☛ $i = (12\%)/2 = 6\% = 0.06$

☛ $n = 2 \times 2 = 4$ periods

Step 2:

$$\frac{₹1,000}{(1+0.06)^4} = \frac{₹1,000}{1.2669} = ₹792.09$$

Final Answers

Compounding Type	Present Value (₹)
(a) Annual	₹797.19
(b) Semi-Annual	₹792.09

3. ₹7,000 to be received 10 years from now at 7% annual interest is worth how much today?

Solution:

$$PV \frac{₹7,000}{(1+0.17)^{10}} = \frac{₹7,000}{1.9672} = ₹3,558.45$$

4. What is the present value of ₹84,253 to be received or paid in 5 years, discounted at 11% using the table factor?

Solution:

Step 1: Use the Present Value formula with PVIF:

$$PV = FV \times PVIF (11\%, 5)$$

Where:

$$FV = ₹84,253$$

$$PVIF (11\%, 5) = 0.5935$$

Step 2: Multiply the values

$$PV = ₹84,253 \times 0.5935 = ₹50,000$$

So, **unless PVIF is 1**, the present value **cannot be ₹84,253**.

5. Mr. Praveen owes a total of ₹3,060, which includes 12% interest for 3 years. How much did he originally borrow?

Solution:

$$PV = \frac{₹3,060}{(1+0.12)^3} = \frac{₹3,060}{1.4049} = ₹2,179.86$$

6. If Ramesh wants ₹2,000 three years from now, and the compounded interest rate is 8%, how much should he invest today?

Solution:

$$PV = \frac{₹2,000}{(1+0.08)^3} = \frac{₹2,000}{1.2597} = ₹1,587.66$$

7. What is the present value of an offer of ₹14,000 two years from now, if the opportunity cost of capital is 17% per year, compounded annually?

Solution:

$$PV = \frac{₹14,000}{(1+0.17)^2} = \frac{₹14,000}{1.3689} = ₹10,227.19$$

8. If you invested ₹50,000 and received ₹80,000 after 10 years, what was the annual compounded growth rate?

Solution:

$$r = \left(\frac{₹80,000}{₹50,000} \right)^{\frac{1}{10}} - 1$$

$$= (1.6)^{0.1} - 1 = 0.0481 \text{ or } 4.81\%$$

9. How much would you have to deposit today to have ₹10,000 in 5 years at 6% annual interest compounded quarterly?

Solution:

- ◆ $i = 0.06/4 = 0.015$
- ◆ $n = 5 \times 4 = 20$

$$PV = \frac{₹10,000}{(1 + 0.015)^{20}}$$

$$= \frac{₹10,000}{1.3469} = ₹7,424.46$$

10. What is the present value of ₹15,000 one year from now if the annual interest rate is 12%, compounded monthly?

Solution:

- ◆ $i = 0.12/12 = 0.01$
- ◆ $n = 1 \times 12 = 12$

$$PV = \frac{₹15,000}{(1 + 0.01)^{12}}$$

$$= \frac{₹15,000}{1.1268} = ₹13,311.74$$

11. Assume a person has the opportunity to receive an **ordinary annuity** that pays **₹50,000 per year for the next 25 years**, with a **6% annual discount rate**, or take a **₹650,000 lump-sum payment today**.

Which is the better option?

Solution:

We will compare the present value of the annuity to the lump-sum amount of ₹650,000. To calculate the present value (PV) of an ordinary annuity, we use the formula:

$$PV = PMT \times \left[\frac{1 - (1 + i)^{-n}}{i} \right]$$

Where:

PMT = Annual payment = ₹50,000

i = Interest rate = 6% = 0.06

n = Number of years = 25

Step-by-Step Calculation

Step 1:

$$PV = ₹50,000 \times \left[\frac{1 - (1 / (1 + 0.06)^{25})}{0.06} \right]$$

Step 2:

$$= ₹50,000 \times \left[\frac{1 - (1 / 4.2919)}{0.06} \right] = ₹50,000 \times \left[\frac{1 - 0.2331}{0.06} \right]$$

Step 3:

$$= ₹50,000 \times \left[\frac{0.7669}{0.06} \right] = ₹50,000 \times 12.7817 = ₹639,085$$

(Near to the originally cited value of ₹639,168, depending on rounding.)

Conclusion

◆ Present value of annuity = ₹639,168 (approx.)

◆ Lump-sum offered = ₹650,000

Since ₹650,000 > ₹639,168, the lump-sum payment is the better option financially, by about ₹10,832.

12. Magnificent Limited pays ₹2 in dividends annually and estimates that it will pay the dividends indefinitely.

How much are investors willing to pay for the stock if their required rate of return is 5% per annum?

Solution:

Since the dividend is paid forever (in perpetuity), we use the Perpetuity Present Value Formula:

$$PV = \frac{D}{r}$$

Where:

D = Annual dividend = ₹2

r = Required rate of return = 5% = 0.05

Step-by-Step Calculation

$$PV = \frac{₹2}{0.05} = ₹40$$

Conclusion

An investor will be willing to pay ₹40 or less for the stock if their required return is 5% per year.

If the stock price in the market is more than ₹40, the investor may not consider it a good deal based on their return expectations.

13. You have ₹9,000 to deposit.

☞ Jupiter Bank offers 12% per annum compounded monthly

☞ Saturn Bank offers 12% per annum compounded annually

What will your investment be worth in 10 years at each bank, and what is the difference in the final amounts?

Solution:

We use the compound interest formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

Where:

A = Future Value

P = Principal = ₹9,000

r = Annual interest rate = 12% = 0.12

t = Time = 10 years

n = Number of times compounded per year

Jupiter Bank: Compounded Monthly

◆ $r = 0.12$

◆ $n = 12$ monthly)

◆ $t = 10$

$$A = 9,000 \times \left(1 + \frac{0.12}{12} \right)^{12 \times 10} = 9,000 \times (1 + 0.01)^{120} = 9,000 \times (1.01)^{120} = 9,000 \times 3.3004 = ₹29,703.48$$

Saturn Bank: Compounded Annually

◆ $r = 0.12$

◆ $n = 1$ (annually)

◆ $t = 10$

$$A = 9,000 \times (1 + 0.12)^{10} = 9,000 \times (1.12)^{10} = 9,000 \times 3.106 = ₹27,952.63$$

Variance Between the Two Banks

$$₹29,703.48 - ₹27,952.63 = ₹1,750.85$$

Conclusion

Bank	Final Value (₹)
Jupiter Bank	₹29,703,48
Saturn Bank	₹27,952,63
Difference	₹1,750,85

Jupiter Bank gives a higher return because it **compounds monthly**, leading to more frequent interest accumulation.

14. Scenario:

Due to the large capital required to set up a **factory and warehouse for coffee machines**, Akshay turns to **private investors** for funding.

He meets **Jacob**, a high net-worth individual, who is willing to contribute **₹1,000,000** to Akshay's company.

However, Jacob agrees to invest this amount **only if** he is assured of a **12% annual return, compounded yearly**, and wants to know:

How long will it take for my investment to double in value?"

Solution:

We apply the simple financial shortcut known as the **Rule of 72**:

$$\text{Doubling Time (in years)} = \frac{72}{\text{Annual Rate of Return (\%)}}$$

In this case:

$$\frac{72}{12} = 6 \text{ years}$$

Conclusion

Jacob's investment of **₹1,000,000** will approximately **double to ₹2,000,000** in **6 years**, if the annual return is **12% compounded annually**.

CAPITAL BUDGETING

A. INTRODUCTION

Capital budgeting is a decision-making situation where large funds are committed (invested) in the initial stages of a project, and the returns are expected over a long period. These decisions are related to the allocation of investible funds to different long-term assets.

Capital budgeting is a continuous process, and it is carried out by different functional areas of management, such as production, marketing, engineering, financial management, etc.

B. FEATURES OF CAPITAL BUDGETING

- Capital budgeting decisions have long-term implications.
- These decisions involve a substantial commitment of funds.
- These decisions are irreversible and require analysis of minute details.
- These decisions determine and affect the future growth of the firm.

C. CAPITAL BUDGETING DECISION INVOLVES THREE STEPS

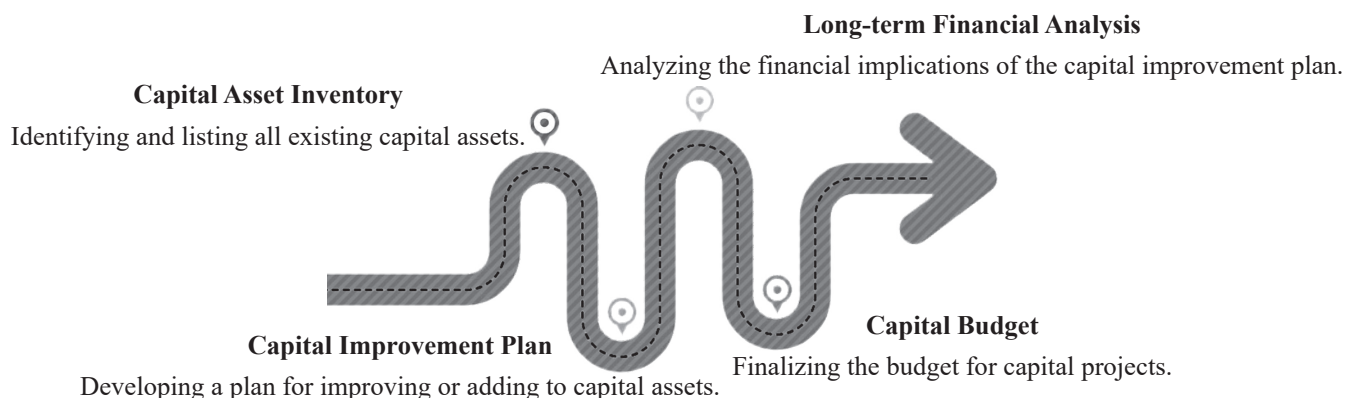
- Estimation of the costs and benefits of a proposal or each alternative.
- Estimation of the required rate of return, i.e., the cost of capital.
- Selection and application of the decision criterion.

D. PROCESS OF CAPITAL BUDGETING

The process is designed to help policy makers: –

- In the selection of a few capital projects from many alternatives.
- From an inventory of capital projects to a capital budget.
- In the timing of the expenditure to be incurred by the projects selected.
- In fitting the selected capital projects into the overall financial program of the government unit.

Steps to Capital Budgeting



CAPITAL ASSET INVENTORY

An inventory and assessment of the existing Capital **assets** includes:

- ◆ Age
- ◆ Condition
- ◆ Degree of use
- ◆ Capacity/Level of Service (LOS)
- ◆ Replacement cost

The inventory of capital assets helps to determine whether the existing assets are to be renewed, replaced, expanded, or retired. It also helps to identify repair and maintenance needs, along with estimated costs.

E. ESTIMATION OF CASH FLOWS

The costs and benefits for a capital budgeting decision situation are measured in terms of cash flows.

An important point is that all cash flows are considered on an after-tax basis. The rule is that all financial decisions are subservient to tax laws. The cash flow from the project is compared with the cost of acquiring the project.

The cash flows may be grouped into relevant and irrelevant cash flows as follows:

Relevant Cash Flows	Irrelevant Cash Flows
Cost of new project	Sunk cost
Scrap value of old/new plant	Allocated overheads
Trade-in-value of old plant	Financial cash flows
Cost reduction/savings	—
Effect on tax liability	—
Incremental repairs	—
Working capital flows	—
Revenue from new proposal	—
Tax benefit of incremental depreciation	—

Calculation of Different Cash Flows May be Summarized as Follows

a. INITIAL CASH OUTFLOW

Particulars	Treatment
Cost of new plant	Add (+)
Installation expenses	Add (+)
Other capital expenditure	Add (+)
Additional working capital required	Add (+)
Tax benefit on capital loss (from sale of old plant, if any)	Less (–)
Salvage value of old plant	Less (–)
Tax liability on capital gain (from sale of old plant, if any)	Add (+)

b. SUBSEQUENT ANNUAL INFLOWS

Particulars	Treatment
Profit after tax	Add (+)
Depreciation	Add (+) (Non-cash)

Financial charges $\times (1 - \text{tax rate})$	Add (+)
Repairs (if any)	Less (–)
Capital expenditure (if any)	Less (–)

c. TERMINAL CASH FLOW

Particulars	Treatment
Final year annual cash inflow	Add (+)
Working capital released	Add (+)
Scrap value of the plant	Add (+)

F. DECISION CRITERIA

TECHNIQUES OF EVALUATION

Category	Methods
Traditional/Non-discounting	1. Payback Period 2. Accounting Rate of Return (ARR)
Time-adjusted / Discounted Cash Flows	1. Net Present Value (NPV) 2. Profitability Index (PI) 3. Internal Rate of Return (IRR)

G. TRADITIONAL OR NON-DISCOUNTING TECHNIQUES

a. PAYBACK PERIOD

1. The payback period is defined as **“the number of years required for the proposal’s cumulative cash inflows to be equal to its cash outflows.”**
2. The payback period is the length of time required to recover the initial cost of the project.
3. The payback period may be suitable if the firm has limited funds available and has no ability or willingness to raise additional funds.

b. ACCOUNTING RATE OF RETURN (ARR)

(Also known as Average Rate of Return)

“The annualized net income earned on the average investment in a project.”

It expresses the annual return from a project as a percentage of the average investment made in the project.

Formula to Compute ARR:

$$\text{ARR} = \left(\frac{\text{Average Annual Profit (after tax)}}{\text{Average Investment}} \right) \times 100$$

H. DISCOUNTED CASH FLOWS OR TIME ADJUSTED TECHNIQUES

These are based upon the fact that the cash flows occurring at different points in time do not have the same economic worth.

a. NET PRESENT VALUE (NPV) METHOD

The NPV of an investment proposal may be defined as the sum of the present values of all the cash inflows less the sum of the present values of all the cash outflows associated with the proposal.

The decision rule is “Accept the proposal if its NPV is positive and reject the proposal if the NPV is negative”.

b. PROFITABILITY INDEX METHOD

This technique is a variant of the NPV technique and is also known as BENEFIT-COST RATIO or PRESENT VALUE INDEX.

Formula:

$$PI = \frac{\text{Total Present Value of Cash Inflows}}{\text{Total Present Value of Cash Outflows}}$$

Interpretation:

- ◆ $PI > 1 \rightarrow$ Project is **profitable**.
- ◆ $PI = 1 \rightarrow$ Project is **break-even**.
- ◆ $PI < 1 \rightarrow$ Project is **not profitable**.

c. INTERNAL RATE OF RETURN (IRR) METHOD

The IRR of a proposal is defined as the discount rate that produces a zero NPV, i.e., the IRR is the discount rate that will equate the present value of cash inflows with the present value of cash outflows.

- ◆ The IRR is also known as the Marginal Rate of Return or the Time Adjusted Rate of Return.
- ◆ The time-schedule of occurrence of future cash flows is known, but the rate of discount is not.
- ◆ The discount rate calculated will equate the present value of cash inflows with the present value of cash outflows.

I. CAPITAL BUDGETING PRACTICES IN INDIA

- a. Capital Budgeting decisions are undertaken at the top management level and are planned. The corporations follow a mostly top-down approach in this regard.
- b. Discounted cash flow techniques are more popular now.
- c. High-growth firms use IRR more frequently, whereas the Payback period is more widely used by small firms.
- d. The Profitability Index technique is used more by public sector units than by private sector units.

Capital budgeting decisions are of paramount importance as they affect the profitability of a firm and are the major determinants of its efficiency and competitive power.

Test Yourself

1. Debasish Ltd. is considering an investment project with the following details:

☛ **Initial Investment** = ₹1,00,000

☛ **Expected Future Cash Inflows:**

Year 1 = ₹20,000

Year 2 = ₹40,000

Year 3 = ₹60,000

Year 4 = ₹70,000

You are required to **calculate the Payback Period** for the project.

Solution:**Payback Period Calculation**

Year	Cash Inflows (₹)	Cumulative Cash Inflows (₹)
1.	20,000	20,000
2.	60,000	40,000
3.	60,000	1,20,000
4.	70,000	1,90,000

- ◆ Initial Investment = ₹1,00,000
- ◆ Investment is recovered between 2nd and 3rd year.

Payback Period Formula:

$$\begin{aligned}
 \text{Payback Period} &= \text{Completed Years} + \left(\frac{\text{Remaining Investment to Recover}}{\text{Cash Inflow of the Next Year}} \times 12 \right) \\
 &= 2 + \left(\frac{1,00,000 - 60,000}{60,000} \times 12 \right) \\
 &= 2 + \left(\frac{40,000}{60,000} \times 12 \right) = 2 + 8 = 2 \text{ year 8 months}
 \end{aligned}$$

Final Answer: Payback Period = 2 year 8 months

2. Tanuja Ltd. is planning to purchase a machine to increase its installed capacity. The company is evaluating **four different machines** - M1, M2, M3, and M4. The relevant financial details including initial investment, estimated annual sales, costs, and economic life of the machines are given below. All sales are for cash. The applicable **Corporate Tax Rate is 33.99%** (inclusive of surcharge and cess).

You are required to **calculate the Payback Period** for each machine and identify the most profitable one.

Given Data:

Particulars	M1	M2	M3	M4
Initial Investment (₹in lacs)	30.00	30.00	40.00	35.00
Estimated Annual Sales (₹in lacs)	50.00	40.00	45.00	48.00
Cost of Production (₹in lacs)	18.00	14.00	16.70	21.00
Economic Life (Years)	2	3	3	4
Scrap Value (₹in lacs)	4.00	2.50	3.00	5.00

Solution:**Statement Showing Payback Period for Four Machines**

Particulars	M1	M2	M3	M4
(1) Initial Investment (₹in lacs)	30.00	30.00	40.00	35.00
(2) Estimated Annual Sales (₹in lacs)	50.00	40.00	45.00	48.00
(3) Estimated Cost of Production (₹lacs)	18.00	14.00	16.70	21.00
(4) Depreciation* (₹in lacs)	13.00	9.17	12.33	7.50
(5) Profit Before Tax (PBT) [2–3–4]	19.00	16.83	15.97	19.50

*Depreciation = (Initial Investment – Scrap Value)/Economic Life

(6) Tax @ 33.99% (₹in lacs)	6.4581	5.721	5.428	6.628
(7) Profit After Tax (PAT) [5–6]	12.5419	11.109	10.542	12.872
(8) Net Cash Flow = PAT + Depreciation	25.5419	20.279	22.872	20.372

Payback Period Calculation:

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Net Annual Cash Flow}}$$

Machine	Payback Period (Years)
M1	$30.00 \div 25.5419 = 1.17$
M2	$30.00 \div 20.279 = 1.48$
M3	$40.00 \div 22.872 = 1.74$
M4	$35.00 \div 20.372 = 1.72$

Conclusion/Analysis:

Machine M1 is the most profitable option as it has the lowest payback period of 1.17 years among all the alternatives considered.

3. Sharda Ltd. is evaluating a new project which requires an initial investment of ₹10,00,000. The project is expected to generate the following **EBITD (Earnings Before Depreciation, Interest and Taxes)** over the next five years:

Year	EBITD (₹)
1	2,50,000
2	3,00,000
3	3,50,000
4	4,00,000
5	5,00,000

- Depreciation is charged at **30% per annum** on the **Written Down Value (WDV)** method.
- Corporate Tax Rate is **33.99%**.
- You are required to **calculate the Accounting Rate of Return (ARR)** for the project.

Solution:

Computation of Project ARR

Step 1: Calculate EBIT and PAT

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5	Average
EBITD (₹)	2,50,000	3,00,000	3,50,000	4,00,000	5,00,000	3,60,000
Less: Depreciation (₹)	3,00,000	2,10,000	1,47,000	1,02,900	72,030	1,66,386
EBIT (₹)	(50,000)	90,000	2,03,000	2,97,100	4,27,970	1,93,614
Less: Tax (₹)	—	13,596	69,000	1,00,984	1,45,467	65,809
Profit After Tax (₹)	(50,000)	76,404	1,34,000	1,96,116	2,82,503	1,27,805

Note: Unabsorbed depreciation of Year 1 is carried forward and adjusted against Year 2's profit before tax.

Tax in Year 2 = 33.99% of (90,000 – 50,000) = 13,596

Step 2: Book Value of Investment

Year	Opening (₹)	Closing (₹)	Average (₹)
1	10,00,000	7,00,000	8,50,000
2	7,00,000	4,90,000	5,95,000
3	4,90,000	3,43,000	4,16,500
4	3,43,000	2,40,100	2,91,550
5	2,40,100	1,68,070	2,04,085
			Average 4,71,427

Step 3: ARR Calculation

$$\text{ARR} = \left(\frac{\text{Average Annual PAT}}{\text{Average Investment}} \right) \times 100 = \left(\frac{1,27,805}{4,71,427} \right) \times 100 = 27.11\%$$

Conclusion/Analysis:

The Accounting Rate of Return (ARR) for the project is 27.11%. This indicates the average return the company expects to earn annually from its investment in this project.

4. Ridhima Ltd. is considering two **mutually exclusive projects** – **Project A** and **Project B**, both requiring an initial investment of **₹60 lakhs**.

- Project A has a life of **4 years**, and
- Project B has a life of **3 years**
- Salvage Value is Nil** for both projects.
- Corporate Tax Rate: 33.99%**
- Cost of Capital: 15%**

You are required to **compute the Net Present Value (NPV)** of both projects using the given net cash inflows and discount factors.

Given Data: Net Cash Inflows and PV Factors

Year	Project A (Lakhs)	Project B (Lakhs)	PV Factor @ 15%
1	60.00	100.00	0.870
2	110.00	130.00	0.756
3	120.00	50.00	0.685
4	50.00	—	0.572

Solution:

Computation of NPV

Project A

Particulars	Year 1	Year 2	Year 3	Year 4
Net Cash Inflow (₹Lacs)	60.00	110.00	120.00	50.00
Depreciation (₹Lacs)	15.00	15.00	15.00	15.00
PBT (1-2)	45.00	95.00	105.00	35.00
Tax @ 33.99%	15.30	32.29	35.70	11.90
PAT	29.70	62.71	69.30	23.10
Net Cash Flow (PAT+Dep.)	44.70	77.71	84.30	38.10
PV Factor @ 15%	0.870	0.756	0.685	0.572
PV of Cash Flows.	38.89	58.75	57.75	21.79

- ◆ **Total PV of Cash Inflows** = ₹177.18 lakhs
- ◆ **Initial Investment** = ₹60.00 lakhs
- ◆ **NPV** = 177.18 – 60.00 = ₹117.18 lakhs

Project B

Particulars	Year 1	Year 2	Year 3
Net Cash Inflow (Lacs)	100.00	130.00	50.00
Depreciation (Lacs)	20.00	20.00	20.00
PBT	80.00	110.00	30.00
Tax @ 33.99%	27.19	37.39	10.20
PAT	52.81	72.61	19.80
Net Cash Flow (PAT+Dep.)	72.81	92.61	39.80
PV Factor @ 15%	0.870	0.756	0.685
PV of Cash Flows	63.345	70.013	27.263

- ◆ **Total PV of Cash Inflows** = ₹160.621 lakhs
- ◆ **Initial Investment** = ₹60.00 lakhs
- ◆ **NPV** = 160.621 – 60.00 = ₹100.621 lakhs

Conclusion/Analysis:

Both projects have positive NPVs, but Project A is more financially viable as it offers a higher Net Present Value of ₹117.18 lakhs compared to Project B's NPV of ₹100.621 lakhs. Therefore, Project A should be selected.

5. Nakshatra Ltd. is evaluating a project with the following details:

👉 **Project Cost (Initial Investment)** = ₹1,10,000

👉 **Estimated Cash Inflows over 4 years:**

Year	Cash Inflow (₹)
1	60,000
2	20,000
3	10,000
4	50,000

You are required to **calculate the Internal Rate of Return (IRR)**.

Solution:

Since the **cash flows are not uniform**, IRR will be calculated using the trial-and-error method.

Step 1: Calculate the Factor

$$\text{Factor} = \frac{\text{Initial Investment}}{\text{Average Annual Cash Inflow}} = \frac{1,10,000}{\left(\frac{60,000 + 20,000 + 10,000 + 50,000}{4} \right)} = \frac{1,10,000}{35,000} = 3.14$$

From the **Present Value Annuity Table**, a factor of 3.14 over 4 years lies between 10% and 12%.

So we test both 10% and 12% to calculate NPVs.

Step 2: NPV Calculation at 10% and 12%

Year	Cash Inflows (₹)	PVF @10%	DCF @10%	PVF @12%	DCF @12%
1	60,000	0.909	54,540	0.893	53,580
2	20,000	0.826	16,520	0.797	15,940
3	10,000	0.751	7,510	0.712	7,120
4	50,000	0.683	34,150	0.636	31,800
			1,12,720		1,08,440

◆ **Initial Investment = ₹1,10,000**

Rate	NPV
10%	₹2,720 (+)
12%	₹(1,560) (-)

Step 3: Apply IRR Formula

Let:

- ◆ A = Lower Rate (10%)
- ◆ B = Higher Rate (12%)
- ◆ NPV at A = +2,720
- ◆ NPV at B = -1,560
- ◆ Difference in NPVs = 4,280

Using Forward Method:

$$\text{IRR} = 10\% + \left(\frac{2,720}{4,280} \right) \times 2\% = 10\% + 1.27\% = 11.27\%$$

Using Backward Method:

$$\text{IRR} = 12\% - \left(\frac{1,560}{4,280} \right) \times 2\% = 12\% - 0.73\% = 11.27\%$$

Conclusion/Analysis:

The Internal Rate of Return (IRR) for the project is 11.27%.

Since the IRR is lower than the cost of capital (15%), the project should not be accepted under the IRR rule.

6. Harshit Ltd. is evaluating a project with the following details:

- ◆ **Initial Investment:** ₹40,000
- ◆ **Project Life:** 4 years
- ◆ **Annual Cash Inflows:** ₹25,000 (for each of the 4 years)
- ◆ **Cost of Capital:** 10% p.a.
- ◆ **Expected Reinvestment Rate for Cash Inflows:** 8% p.a. for each year-end

You are required to compute the Net Present Value (NPV) using the **Terminal Value Approach**.

Solution: Step-by-Step Calculation

Step 1: Compute Terminal Value (Future Value) of Cash Inflows

Year	Cash Inflow (₹)	Years of Compounding at 8%	Future Value Factor (8%)	Future Value (₹)
1	25,000	3	1.260	31,500
2	25,000	2	1.166	29,150
3	25,000	1	1.080	27,000
4	25,000	0	1.000	25,000
				₹Total 1,12,650

Step 2: Discount Terminal Value to Present Value

$$\text{Present Value (PV)} = \frac{\text{Future Value}}{(1+i)^n} = \frac{1,12,650}{(1.10)^4}$$

From the table:

$$\text{PV} = 1,12,650 \times 0.683 = ₹76,940$$

Step 3: Net Present Value

NPV = Present Value of Future Cash Inflows - Initial Investment

$$\text{NPV} = ₹76,940 - ₹40,000 = ₹36,940$$

Conclusion/Analysis:

Since the Present Value of reinvested cash inflows (₹76,940) is greater than the original investment (₹40,000), the project should be accepted as per the Terminal Value criterion.

7. Bijli Ltd. is evaluating a project with the following details:

- ☛ Initial Investment = ₹20 lakhs
- ☛ Expected Annual Cash Inflows = ₹5 lakhs
- ☛ Project Life = 10 years
- ☛ Cost of Capital = 15%

You are required to calculate the Profitability Index (PI) and interpret the decision.

Solution:

Step 1: Calculate Present Value (PV) of Inflows

Given:

- ☛ Annual cash inflow = ₹5 lakhs
- ☛ PV Annuity Factor at 15% for 10 years = **5.019**

$$\text{PV of Inflows} = 5 \times 5.019 = ₹25.095 \text{ lakhs}$$

Step 2: Present Value of Outflows

$$\text{PV of Outflows} = ₹20 \text{ lakhs}$$

Step 3: Calculate Profitability Index (PI)

$$\text{Profitability Index} = \frac{\text{PV of Inflows}}{\text{PV of Outflows}} = \frac{25.095}{20} = 1.255$$

Decision:

Since the Profitability Index is greater than 1 (i.e., 1.255), the project is financially viable and should be accepted.

Additional Note - Discounted Payback Period:

Unlike the traditional payback period, the **discounted payback period** accounts for the **time value of money**. It is calculated by discounting the future cash flows at the **cost of capital (15%)** and then computing the time required to recover the initial investment.

8. Rio Ltd. is evaluating a project with the following details:

- ☛ Initial Investment = ₹1,00,000
- ☛ Cost of Capital = 12% p.a.
- ☛ Expected Cash Inflows:

Year	Cash Inflows (₹)
1	25,000
2	50,000
3	75,000
4	1,00,000
5	1,50,000

You are required to **calculate the Discounted Payback Period**.

Solution:

Step 1: Discount the Cash Flows at 12%

Year	Cash Inflows (₹)	Discount Factor @12%	Discounted Cash Flows (₹)	Cumulative DCF (₹)
1	25,000	0.8929	22,323	22,323
2	50,000	0.7972	39,860	62,183
3	75,000	0.7117	53,378	1,15,561
4	1,00,000	0.6355	63,550	1,79,111
5	1,50,000	0.5674	85,110	2,64,221

Step 2: Determine When Investment is Recovered

- ◆ Initial Investment = ₹1,00,000
- ◆ Cumulative discounted inflow at end of Year 2 = ₹62,183
- ◆ Cumulative discounted inflow at end of Year 3 = ₹1,15,561
- 👉 Recovery occurs **between Year 2 and Year 3**

Step 3: Calculate Discounted Payback Period

$$\text{Remaining Amount to Recover} = 1,00,000 - 62,183 = 37,817$$

$$\text{Discounted Inflow in Year 3} = 53,378$$

$$\text{Fraction of Year 3} = \frac{37,817}{53,378} = 0.708 = 8.5 \text{ months}$$

$$\text{Discounted Payback Period} = 2 \text{ years} + (0.708 \times 12) = 2 \text{ years and } 8.5 \text{ months}$$

Conclusion/Analysis:

The Discounted Payback Period for the project is approximately 2 years and 8.5 months.

Since the project recovers its initial investment within its life, it can be considered acceptable from a payback perspective.

9. Tanuja Ltd. is considering an investment in a project costing ₹20 lakhs.

The project has a **useful life of 5 years**, and the **salvage value is nil**.

Depreciation is charged using the **Straight-Line Method**. The **tax rate is 50%**.

The estimated **cash flows before depreciation and tax** for each year are as follows:

Year	1	2	3	4	5
Cash Flow before Depreciation & Tax (lakhs)	4	6	8	8	10

You are required to compute the following:

- 1. Payback Period**
- 2. Average Rate of Return (ARR)**
- 3. Net Present Value (NPV) at 10% cost of capital**
- 4. Benefit-Cost Ratio (Profitability Index)**

Solution:**Step 1: Annual Cash Flows After Tax**

☛ Depreciation per year = ₹20 lakhs ÷ 5 = ₹4 lakhs

Year	Cash Flow Before Depreciation & Tax	Depreciation	EBT	Tax @50%	EAT	Cash Flow After Tax (EAT + Depreciation)
1	4	4	0	0	0	4
2	6	4	2	1	1	5
3	8	4	4	2	2	6
4	8	4	4	2	2	6
5	10	4	6	3	3	7

1. Payback Period

Year	Cash Flow After Tax	Cumulative Cash Flow
1	4	4
2	5	9
3	6	15
4	6	21
5	7	28

☛ Initial investment = ₹20 lakhs

☛ Recovered by end of 3rd year = ₹15 lakhs

☛ Additional 5 lakhs recovered in year 4, where inflow = ₹6 lakhs

Payback Period 3 years + $\left(\frac{5}{6} \times 12\right)$ months = 3 years and 10 months

2. Average Rate of Return (ARR)

☛ Total Net Profit (EAT) over 5 years = 0 + 1 + 2 + 2 + 3 = ₹8 lakhs

☛ Average Annual Profit = 8 lakhs ÷ 5 = ₹1.6 lakhs

☛ Average Investment = 20 lakhs ÷ 2 = ₹10 lakhs

$$\text{ARR} = \frac{16}{10} \times 100 = 16\%$$

3. Net Present Value (NPV) @10%

Year	Cash Flow After Tax (₹lakhs)	Discount Factor @10%	Present Value (₹lakhs)
1	4	0.909	3.636
2	5	0.826	4.130
3	6	0.751	4.506
4	6	0.683	4.098
5	7	0.621	4.347

☛ Total PV of Cash Inflows = 3.636 + 4.130 + 4.506 + 4.098 + 4.347 = ₹20.717 lakhs

☛ Initial Investment = ₹20 lakhs

$$\text{NPV} = ₹20.717 - ₹20 = ₹0.717 \text{ lakhs}$$

4. Benefit-Cost Ratio (Profitability Index)

$$\text{BCR} = \frac{\text{PV of Inflows}}{\text{PV of Outflows}} = \frac{20.717}{20} = 1.036$$

Conclusion/Analysis:

1. **Payback Period: 3 years 10 months**
2. **ARR: 16%**
3. **NPV: ₹0.717 lakhs (positive, hence acceptable)**
4. **Benefit-Cost Ratio: 1.036 (>1, hence acceptable)**

Thus, the project is financially viable and can be accepted based on all four evaluation methods.

10. Harshita Ltd. is evaluating two alternative internal transportation systems. The details for each system are as follows:

Particulars	System 1	System 2
Initial Investment (Million)	6	4
Annual Operating Costs (Million)	1	0.9
Useful Life (Years)	6	4
Salvage Value at the end (Million)	2	1.5

The **cost of capital** is **6%**.

You are required to:

- Compute the **Present Value of cash outflows** for each system.
- Compute the **Equivalent Annual Cost (EAC)**.
- **Recommend** which system should be adopted and justify the answer with proper analysis.

Solution:

Present Value Factors:

- ◆ PV of Annuity @ 6% for 6 years = 4.917
- ◆ PV of Annuity @ 6% for 4 years = 3.465
- ◆ PV of ₹1 @ 6% for 6th year = 0.705
- ◆ PV of ₹1 @ 6% for 4th year = 0.792

System 1: Present Value of Cash Outflows

$$\text{Initial Investment} = 6.000$$

$$\text{PV of Annual Operating Costs} = 1 \times 4.917 = 4.917$$

$$\text{Total (Before Salvage)} = 6.000 + 4.917 = 10.917$$

$$\text{Less: PV of Salvage Value} = 2 \times 0.705 = 1.410$$

$$\text{Net Present Value of Cash Outflows} = 10.917 - 1.410 = ₹9.507 \text{ million}$$

System 2: Present Value of Cash Outflows

$$\text{Initial Investment} = 4.000$$

$$\text{PV of Annual Operating Costs} = 0.9 \times 3.465 = 3.1185$$

$$\text{Total (Before Salvage)} = 4.000 + 3.1185 = 7.1185$$

$$\text{Less: PV of Salvage Value} = 1.5 \times 0.792 = 1.188$$

$$\text{Net Present Value of Cash Outflows} = 7.1185 - 1.188 = ₹5.9305 \text{ million}$$

Equivalent Annual Cost (EAC):

$$\text{EAC (System 1)} = \frac{5.9305}{3.465} = 1.93 \text{ million}$$

$$\text{EAC (System 2)} = \frac{5.9305}{3.465} = 1.71 \text{ million}$$

Conclusion/Recommendation:

Since the Equivalent Annual Cost of System 2 (₹1.71 million) is lower than that of System 1 (₹1.93 million), System 2 should be selected as it is more cost-effective over its useful life.

11. Sharda Ltd. is considering two **mutually exclusive projects**, each requiring an **initial investment of ₹200 lakhs**. The company's **cost of capital is 10%**. The Finance Director prefers the project with the **higher Net Present Value (NPV)**, while the Managing Director supports the project with the **higher Internal Rate of Return (IRR)**.

The **net after-tax cash flows** of the two projects are given below:

Cash Flows (₹Lakhs):

Year	Project X	Project Y
0	(200)	(200)
1	35	218
2	80	10
3	90	10
4	75	4
5	20	3

Discount Factors:

Year	@10%	@20%
0	1.00	1.00
1	0.91	0.83
2	0.83	0.69
3	0.75	0.58
4	0.68	0.48
5	0.62	0.41

Solution:

a. NPV and IRR Calculations

Project X:

NPV at 10%:

$$\begin{aligned}\text{NPV} &= -200 + (35 \times 0.91) + (80 \times 0.83) + (90 \times 0.75) + (75 \times 0.68) + (20 \times 0.62) \\ &= -200 + 31.85 + 66.40 + 67.50 + 51.00 + 12.40 = ₹29.15 \text{ lakhs (Positive)}\end{aligned}$$

NPV at 20%:

$$-200 + 29.05 + 55.20 + 52.20 + 36.00 + 8.20 = ₹19.35 \text{ lakhs (Negative)}$$

IRR Calculation:

$$\text{IRR} = 10\% + \left(\frac{29.15}{29.15 + 19.35} \right) \times 10 = 10\% + \left(\frac{29.15}{48.50} \right) \times 10 = 16.01\%$$

Project Y:

NPV at 10%:

$$\begin{aligned}&= -200 + (218 \times 0.91) + (10 \times 0.83) + (10 \times 0.75) + (4 \times 0.68) + (3 \times 0.62) \\ &= -200 + 198.38 + 8.30 + 7.50 + 2.72 + 1.86 = ₹18.76 \text{ lakhs}\end{aligned}$$

NPV at 20%:

$$-200 + 180.94 + 6.90 + 5.80 + 1.92 + 1.23 = - ₹3.21 \text{ lakhs}$$

IRR Calculation:

$$\left(\frac{18.76}{18.76 + 3.21} \right) \times 10 = 10\% + \left(\frac{18.76}{21.97} \right) \times 10 = 18.54\%$$

b. Recommendation

1. **Both projects** are financially viable since both have **positive NPVs at 10%**.
2. However, **Project X has a higher NPV** (₹29.15 lakhs vs. ₹18.76 lakhs), meaning it adds more **absolute value** to the firm.
3. Though **Project Y has a higher IRR** (18.54% vs. 16.01%), when **NPV and IRR rankings differ**, preference should be given to the **project with the higher NPV**, as it maximizes shareholder wealth.

Recommendation: Select Project X.

c. Reason for Inconsistency in Ranking

The inconsistency between NPV and IRR arises due to the timing and size of cash flows:

1. **Project Y yields large early cash inflows** (front-loaded), which boost IRR.
2. **Project X gives higher overall cash inflows**, but more evenly spread across its life.
3. IRR may favor earlier cash flows due to **reinvestment assumptions and non-linear returns**, while NPV considers the absolute rupee benefit discounted at a consistent cost of capital.

Conclusion

The inconsistency in the ranking of the two projects arises due to differences in the timing and pattern of cash flows. Project Y generates a large portion of its cash flows in the initial year, which results in a higher IRR. In contrast, Project X has more evenly distributed cash flows over its life, leading to a higher NPV.

When such conflicts arise between NPV and IRR, the Net Present Value method should be preferred. NPV provides a better measure of the absolute increase in wealth and uses a consistent reinvestment rate (the cost of capital), making it more reliable for decision-making.

Therefore, Project X should be selected as it adds greater value to the company.

- 12.** Tanuja Ltd. is evaluating two alternative internal transportation machines, Machine A and Machine B. The company wishes to assess which investment proposal is more profitable using the following methods:
- a. Payback Period
 - b. Net Present Value (NPV)
 - c. Profitability Index (PI)
 - d. Average Rate of Return (ARR)

The projected cash inflows for both machines over 5 years are given below:

Cash Inflows (in lakhs):

Year	Machine A	Machine B
1	1.5	0.5
2	2.0	1.5
3	2.5	2.0
4	1.5	3.0
5	1.0	2.0

Initial Investment for both machines = ₹5 lakhs

Discounting factors at 10%:

Year	DF @10%
1	0.909
2	0.826
3	0.751
4	0.683
5	0.621

Solution:**a. Payback Period****Machine A:**

- ☞ Year 1: ₹1.50
- ☞ Year 2: ₹2.00 (Cumulative: ₹3.50)
- ☞ Year 3: ₹1.50 more needed
 - ☛ Payback within 7.2 months ($1.5/2.5 \times 12$)

Machine B:

- ☞ Year 1 to Year 3: ₹4.00
- ☞ Year 4: ₹1.00 needed
 - ☛ Payback within 4 months ($1/3 \times 12$)

Payback Period = 3 years 4 months**Ranking**

- ◆ Machine A: I
- ◆ Machine B: II

b. Net Present Value (NPV)

Year	Machine A (Inflow $\times$ DF)	Machine B (Inflow $\times$ DF)
1	$1.5 \times 0.909 = 1.36$	$0.5 \times 0.909 = 0.45$
2	$2.0 \times 0.826 = 1.65$	$1.5 \times 0.826 = 1.24$
3	$2.5 \times 0.751 = 1.88$	$2.0 \times 0.751 = 1.50$
4	$1.5 \times 0.683 = 1.02$	$3.0 \times 0.683 = 2.05$
5	$1.0 \times 0.621 = 0.62$	$2.0 \times 0.621 = 1.24$
Total PV	6.53	6.48

Initial Investment = ₹5 lakhs

- a. NPV of Machine A = ₹6.53 – 5.00 = ₹1.53 lakhs
- b. NPV of Machine B = ₹6.48 – ₹5.00 = ₹1.48 lakhs

Ranking:

- ◆ Machine A: I
- ◆ Machine B: II

c. Profitability Index (PI)

- ☞ Machine A: $6.53/5.00 = 1.306$
- ☞ Machine B: $6.48/5.00 = 1.296$

Ranking:

- ◆ Machine A: I
- ◆ Machine B: II

d. Average Rate of Return (ARR)

- ☞ Total Cash Inflows (5 years):
 - ◆ Machine A: ₹8.50 lakhs
 - ◆ Machine B: ₹9.00 lakhs
- ☞ Less: Depreciation = ₹5.00 lakhs for both
- ☞ Total Earnings After Depreciation:
 - ◆ Machine A: ₹3.50 lakhs
 - ◆ Machine B: ₹4.00 lakhs
- ☞ Average Earnings per year:
 - ◆ Machine A: ₹0.70 lakh
 - ◆ Machine B: ₹0.80 lakh

Payback Period = 2 years 7.2 months

- ◆ ARR Calculation:

$$\text{ARR (A)} = \frac{0.70}{5.00} \times 100 = 14\%$$

$$\text{ARR (B)} = \frac{0.80}{5.00} \times 100 = 16\%$$

Ranking:

- ◆ Machine A: II
- ◆ Machine B: I

Conclusion

Based on the analysis:

- ◆ Machine A is better in terms of Payback Period, NPV, and Profitability Index.
- ◆ Machine B is better in terms of Average Rate of Return.

If the company prioritizes faster returns and overall value addition, Machine A is recommended. However, if consistent annual earnings are the key concern, Machine B may be preferred.

TEST YOURSELF

13. Harshit Ltd. is evaluating two **mutually exclusive projects**. The details are:

Particulars	Project A	Project B
Initial Investment (₹)	40,000	4
Life (Years)	4	7
Annual Net Cash Inflows (₹)	15,000	16,000
Scrap Value (₹)	5,000	3,000
Cost of Capital	15%	15%
PV Annuity Factor	2.8550 (4 yrs)	4.1604 (7 yrs)
PV Factor (Lumpsum)	0.5718 (4 yrs)	0.3759 (7 yrs)

Hint for Q13 (Harshit Ltd):

1. Find present value of running cost for both machines using given PV factors.
2. Add to the initial cost to get total PV of cost.
3. Divide by annuity factor to get equivalent annual cost.
4. Compare both – lower annual cost means better option.

Use this method if machines are to be replaced repeatedly.

14. Ridhima Ltd. is evaluating a project that has a life of 4 years, with yearly projected net profit of ₹10,000 after charging depreciation of ₹8,000 per annum. The total capital cost is 32,000, of which:

- ☛ ₹20,000 is payable **immediately (Year 0)**, and
- ☛ The remaining ₹ 12,000 is payable in **Year 1**.

In addition to the capital expenditure:

- ☛ Stock of ₹6,000 is to be invested in Year 0,
- ☛ Debtors worth ₹8,000 are to be invested in Year 1.

The **working capital (₹14,000 total)** will be recovered in **Year 5** (though the project ends in 4 years).

The machinery will have a **residual value of ₹2,000 at the end of Year 4**, which is subject to **tax @ 40%**.

- ☛ **Depreciation rate** = 25% (as per tax norms),
- ☛ **Income Tax** = 40%,
- ☛ **Tax payment delay** = Paid 9 months after the end of the year (i.e., discounted by 1 year),
- ☛ **Discount rate** = 10%,
- ☛ Discounting Factors:
 - ➡ Year 1: 0.9091
 - ➡ Year 2: 0.8264
 - ➡ Year 3: 0.7513
 - ➡ Year 4: 0.6830
 - ➡ Year 5: 0.6209

You are required to compute the **Net Present Value (NPV)** of the project and advise whether the project should be accepted.

Hint for Q14 (Ridhima Ltd):

1. Capital expenditure and working capital given in Year 0 and 1- treat them as outflows.
2. Net profit given is after depreciation, so add back depreciation to get cash flow.
3. Deduct tax from profit to get post-tax profit.
4. Add salvage value in Year 4 and working capital recovery in Year 5.
5. Apply 10% discount factor to each year's net cash flow.
6. Sum all PVs and subtract initial outflows to get NPV.
7. If NPV is positive, project is acceptable.

15. X Ltd. is evaluating a capital project, Project M, based on the following information:

- ☛ Annual cost saving: ₹ 40,000
- ☛ Useful life of the project: 4 years
- ☛ Internal Rate of Return (IRR): 15%
- ☛ Profitability Index (PI): 1.064
- ☛ Salvage value: Nil
- ☛ Missing values to be calculated: Net Present Value (NPV), Cost of Capital, Cost of Project, and Payback Period

Use the following discount factors:

Year	15%	14%	13%
1	0.870	0.877	0.885
2	0.756	0.769	0.783
3	0.658	0.675	0.693
4	0.572	0.592	0.613

Solution:

1. Calculation of Cost of Project (Initial Investment)

Since IRR is 15%, we calculate the present value of cost savings using PV factor of annuity @ 15% for 4 years.

Total PV factor @ 15% for 4 years = $0.87 + 0.756 + 0.658 + 0.572 = 2.856$

Cost of Project = ₹40,000 × 2.856 = ₹1,14,240 (rounded to ₹1,14,200 as given)

2. Calculation of Payback Period

Payback Period = Cost of Project ÷ Annual Cost Saving

= ₹1,14,200 / 740,000 = 2.855 years

= 2 years and 10.26 months ≈ 2 years 11 months

3. Calculation of Cost of Capital

Given:

☛ Profitability Index = 1.064

☛ Cost of Project = ₹1,14,200

Discounted Cash Inflows = Profitability Index × Cost of Project

= $1.064 \times ₹1,14,200 = ₹1,21,509$

Now, cumulative discount factor = ₹1,21,509 ÷ ₹40000 = 3.038

From the table, the cumulative factor for 4 years is closest to 3.038 at 12%

Hence, the Cost of Capital is 12%

4. Calculation of Net Present Value (NPV)

NPV Present value of cash inflows Initial investment

= ₹1,21,509 – ₹1,14,200 = ₹7,309

Conclusion:

Since the NPV is positive and the IRR is greater than the cost of capital, the project is financially viable and should be accepted.

16. Xpert Engineering Ltd.

Given:

☛ **Project A**

Initial Investment = 1,00,000 CFAT (Cash Flow After Tax) = 30,000 per year Life = 5 years

☛ **Project B**

Initial Investment = ₹1,25,000

CFAT = ₹27,000 per year

Life = 8 years

☛ Cost of Capital = 10%

Present Value Annuity Factors (PVAF) at 10%:

- ☛ For 5 years = 3.791
- ☛ For 8 years = 5.335

Solution:

Step 1: Calculate NPV

Project A:

PV of Inflows = ₹30,000 × 3.791 = ₹1,13,730
NPV = ₹1,13,730 – ₹1,00,000 = ₹13,730

Project B:

PV of Inflows = ₹27,000 × 5.335 = ₹1,44,045
NPV = ₹1,44,045 – ₹1,25,000 = ₹19,045

Step 2: Calculate Equivalent Annual NPV

This helps compare projects of unequal life if they are repeatable.

Project A:

$$\text{Equivalent Annual NPV} = ₹13,730 \div 3.791 = ₹3,622$$

Project B:

$$\text{Equivalent Annual NPV} = ₹19,045 \div 5.335 = ₹3,570$$

Conclusion:

- ◆ If it is a one-time project, choose **Project B**, as it gives higher NPV (19,045).
- ◆ If the project is to be repeated, choose **Project A**, as it gives a higher Equivalent Annual NPV (3,622).

17. Scenario:

Pranjul Ltd. XYZ Ltd., an infrastructure company, is evaluating a proposal to **Build, Operate and Transfer (BOT)** a 35 km road segment with a **total project cost of ₹200 crores**.

Project Financing:

- ☛ **Equity Share Capital:** ₹50 crores
- ☛ **Loan from financial institutions:** ₹150 crores @ 15% interest p.a.

Operational Details:

- ☛ Toll collection starts once the project is completed.
- ☛ **Annual Toll Revenue:** ₹50 crores
- ☛ **Annual operating & maintenance cost:** 5% of project cost = ₹10 crores
- ☛ **Annual Net Cash Inflow:** ₹50 crores – ₹10 crores = ₹40 crores
- ☛ **Project life:** 15 years (after which it is handed over at zero value)
- ☛ **Depreciation for accounting:** Straight Line over 15 years
- ☛ **Depreciation for tax (ignore for this case):** 10% WDV
- ☛ **Loan repayment:** 15 equal annual instalments (Principal + Interest)
- ☛ **Ignore corporate tax.**

Requirement:

1. Calculate **Project IRR**
2. Calculate **Equity IRR**
3. Explain the difference between Project IRR and Equity IRR

Solution:

1. Project IRR Calculation

- Project Cost (Initial Outlay): ₹200 crores
- Annual Net Cash Inflow: ₹40 crores
- Life of Project: 15 years
- Annuity Factor = Investment ÷ Annual Cash Inflow = ₹200 crores ÷ ₹40 crores = **5.000**

Locate the PV Annuity Factor for 15 years nearest to 5.000

Discount Rate	PVAF (15 yrs)	Present Value of Inflows	NPV
18%	5.092	$40 \times 5.092 = ₹203.68 \text{ Cr}$	₹3.68 Cr
19%	4.876	$40 \times 4.876 = ₹195.04 \text{ Cr}$	₹(4.96) Cr

Now interpolate between 18% and 19%:

$$\text{IRR} = 18\% + \left(\frac{3.68}{3.68 + 4.96} \right) \times 1\% = 18\% + \left(\frac{3.68}{8.64} \right) \times 1\% = 18\% + 0.426\% = 18.43\%$$

2. Calculation of Equity IRR:

Loan Details:

- Loan Amount = Rs. 150 crores
- Interest = 15%
- Tenure = 15 years
- Loan is repaid in equal instalments (principal + interest)

To compute annual loan repayment (EMI), we use the annuity factor for 15 years @15%:

Annuity Factor @ 15% for 15 years = 6.259

Annual loan repayment (principal + interest) = Rs. 150 crores/6.259 = Rs. 23.97 crores

Annual Net Inflow to Equity Shareholders =

= Net cash inflow - Loan repayment = Rs. 40 crores - Rs. 23.97 crores = Rs. 16.03 crores

Initial Equity Investment = Rs. 50 crores

We now find IRR where the PV of Rs. 16.03 crores for 15 years = Rs. 50 crores

Try different rates:

At 28%: PVAF @28% for 15 years ≈ 3.133

→ $16.03 \times 3.133 = 50.22$ (approx equal)

So, Equity IRR = 28%

3. Difference between Project IRR and Equity IRR:

- Project IRR** considers total project cash flows (before financing), i.e., it evaluates return on the total capital employed.
- Equity IRR** considers only cash flows available to equity shareholders **after servicing debt**, hence it reflects return **on equity capital only**.
- If debt is used, **Equity IRR > Project IRR**, due to leverage effect (only when the Project IRR > interest rate on loan).

18. X Ltd., an oil company, is planning to install a pipeline to transport crude oil from its wells to the refinery. The investment and operating savings depend on the pipeline diameter. The project has an estimated life of 10 years. The tax rate is 50% and depreciation is charged on a straight-line basis. There is no salvage value. The company requires a 15% post-tax return.

Details:

Pipeline Diameter (in inches)	Investment Required (Rs. lakhs)	Gross Annual Operating Savings (Rs. lakhs)
3	16	5
4	24	8
5	36	15
6	64	30
7	150	50

Required:

1. Calculate annual CFAT (Cash Flow After Tax) for each option.
2. Calculate Payback Period for each pipeline.
3. Calculate NPV for each diameter using PV factor at 15% for 10 years = 5.019.
4. Suggest the most suitable pipeline diameter.

Solution:**1. Calculation of CFAT (Rs. in lakhs):**

Diameter (inches)	Gross Savings	Depreciation	Tax on Earnings	Tax Shield on Depreciation	CFAT
3	5.0	1.6	2.5	0.8	3.3
4	8.5	2.4	4.0	1.2	5.2
5	15.0	3.6	7.5	1.8	9.3
6	30.0	6.4	15.0	3.2	18.2
7	50.0	15.0	25.0	7.5	32.5

2. Payback Period:

$$\text{Payback Period} = \frac{\text{Investment}}{\text{CFAT}}$$

Diameter (inches)	Investment	CFAT	Payback Period (Years)
3	16	3.3	4.85
4	24	5.2	4.62
5	36	9.3	3.87
6	64	18.2	3.52
7	150	32.5	4.62

Shortest Payback Period: 6 inches (3.52 years)

3. Net Present Value (NPV):

$$\text{NPV} = (\text{CFAT} \times \text{PV factor}) - \text{Investment}$$

PV factor for 10 years @15% = 5.019

Diameter (inches)	CFAT	PV of Cash Inflows	Investment	NPV
3	3.3	$3.3 \times 5.019 = 16.56$	16	0.56
4	5.2	$5.2 \times 5.019 = 26.10$	24	2.10
5	9.3	$9.3 \times 5.019 = 46.68$	36	10.68
6	18.2	$18.2 \times 5.019 = 91.35$	64	27.35
7	32.5	$32.5 \times 5.019 = 163.12$	150	13.12

Conclusion:

- ♦ The 6-inch pipeline has the shortest payback period and the highest NPV.
- ♦ Hence, X Ltd. should install the 6-inch pipeline as it offers the most beneficial return.

19. Indo Plastics Ltd. - Computerisation Project Evaluation

Background:

Indo Plastics Ltd. is considering computerising its ordering, inventory, and billing system. The relevant financial information is provided:

Estimated Annual Savings:

- ☞ Reduction in 4 clerical salaries (Rs. 50,000 each): Rs. 2,00,000
- ☞ Reduced production delays: Rs. 30,000
- ☞ Avoided lost sales: Rs. 25,000
- ☞ Timely billing gains: Rs. 18,000

Total savings = Rs. 2,73,000

Annual Operating Costs:

- ☞ 2 computer specialists (Rs. 80,000 each): Rs. 1,60,000
- ☞ Maintenance & operation: Rs. 22,000

Total expenses (excluding depreciation) = Rs. 1,82,000

Other Details:

- ☞ Purchase cost: Rs. 2,50,000
- ☞ Installation cost: Rs. 50,000
- ☞ Total cost = Rs. 3,00,000
- ☞ Depreciated over 5 years (straight-line): Rs. 60,000/year
- ☞ Tax rate: 40%
- ☞ Cost of capital: 12%
- ☞ Project life: 5 years
- ☞ Salvage value: Nil

Solution:

Step 1: Calculation of Annual Cash Flows

- ♦ **Operating Profit (before depreciation)** = Rs. 2,73,000 – Rs. 1,82,000 = Rs. 91,000
- ♦ **Less: Depreciation** = Rs. 60,000
- ♦ **Profit Before Tax** = Rs. 31,000
- ♦ **Less: Tax @ 40%** = Rs. 12,400
- ♦ **Profit After Tax** = Rs. 18,600
- ♦ **Cash Flow After Tax (CFAT)** = PAT + Depreciation = Rs. 18,600 + Rs. 60,000 = **Rs. 78,600 per year**

(i) Net Present Value (NPV) Method

- ☞ PV of annuity of Re. 1 for 5 years at 12% = 3.605
- ☞ Total PV of cash inflows = 78,600 × 3.605 = **Rs. 2,83,353**
- ☞ Initial Investment = Rs. 3,00,000
- ☞ **NPV** = 2,83,353 – 3,00,000 = **Rs. –16,647**

Conclusion: Since NPV is **negative**, the project is **not financially viable**.

(ii) Profitability Index (PI) Method

$$PI = \frac{PV \text{ of Inflows}}{\text{Initial Outlay}} = \frac{2,83,353}{3,00,000} = 0.945$$

Conclusion: Since PI is **less than 1**, the project should **not be accepted**.

(iii) Payback Period

Annual Cash Flow = Rs. 78,600

Initial Investment = Rs. 3,00,000

$$\text{Payback in full years} \left[\frac{3,00,000}{78,600} \right] = 3 \text{ years}$$

Cumulative after 3 years = Rs. 2,35,800

Remaining to recover = 3,00,000 – 2,35,800 = Rs. 64,200

Fraction of 4th year = 64,200 ÷ 78,600 = **0.82**

Total Payback Period = 3.82 years

Final Suggestion:

- ◆ The project has **negative NPV**,
- ◆ **PI < 1**, and
- ◆ A **payback period of 3.82 years**,
- ◆ **Thus, the project is not recommended based on financial criteria.**

20. Bijli Ltd. - Project Selection under Capital Constraint

Scenario:

Bijli Ltd. is evaluating 5 investment projects: M, N, O, P, and Q.

- ☛ Projects **N** and **Q** are **mutually exclusive**, i.e., **only one can be chosen**.
- ☛ The **total investment limit is Rs. 3,00,000**.
- ☛ The decision must be based on **Net Present Value (NPV)** and **Benefit-Cost Ratio (BCR)**
- ☛ The **cost of capital is 10%**.

Solution:

Project Data Table

Project	Investment (Rs.)	Cash Flow p.a.	No. of Years	PV Factor @10%	Present Value (PV)	NPV (Rs.)	BCR (PV÷ Investment)
M	50,000	18,000	10	6.145	1,10,610	60,610	2.212
N	1,00,000	50,000	4	3.170	1,58,500	58,500	1.585
O	1,20,000	30,000	8	5.335	1,60,050	40,050	1.334
P	1,50,000	40,000	16	7.824	3,12,960	1,62,960	2.086
Q	2,00,000	30,000	25	9.077	2,72,310	72,310	1.362

Feasible Combinations of Projects (Total ≤ Rs. 3,00,000)

	Combination	Investment (Rs.)	NPV (Rs.)	NPV Rank	BCR	BCR Rank
(i)	M + N + P	3,00,000	2,82,070	1	1.940	1
(ii)	M + N + O	2,70,000	1,59,160	4	1.589	4

(iii)	O + P	2,70,000	2,03,010	3	1.752	3
(iv)	M + Q	2,50,000	1,32,920	5	1.532	5
(v)	N + P	2,50,000	2,21,460	2	1.886	2
(vi)	N + Q	3,00,000	1,30,810	6	1.436	6

Conclusion:

The most optimal combination of projects is M, N, and P, as it gives the highest Net Present Value (Rs. 2,82,070) and the highest Benefit-Cost Ratio (1.940) within the capital limit of Rs. 3,00,000. Therefore, Bijli Ltd. should select Projects M, N, and P.

COST OF CAPITAL

A. CONCEPT OF COST OF CAPITAL

The term cost of capital refers to the maximum rate of return a firm must earn on its investment so that the market value of the company's equity shares does not fall. This aligns with the overall firm's objective of wealth maximization. This is possible only when the firm earns a return on the projects financed by equity shareholders' funds at a rate that is at least equal to the rate of return expected by them. If a firm fails to earn a return at the expected rate, the market value of the shares would fall and thus resulting in a reduction of the overall wealth of the shareholders.

Thus, a firm's cost of capital may be defined as **“the rate of return the firm requires from investment to increase the value of the firm in the marketplace”**.

There are three basic aspects of the concept of cost:

a. IT IS NOT A COST AS SUCH

A firm's cost of capital is the rate of return that it requires on the projects available. It is merely a 'hurdle rate'. of course, such a rate may be calculated based on the actual cost of different components of capital.

b. IT IS THE MINIMUM RATE OF RETURN

A firm's cost of capital represents the minimum rate of return that will result in at least maintaining (if not increasing) the value of its equity shares.

c. IT COMPRISES THREE COMPONENTS

A firm's cost of capital shall consist of three elements.

1. **Return at zero risk level:** This refers to the expected rate of return when a project involves no risk, whether business or financial.
2. **Premium for business risk:** The term business risk refers to the variability in operating profit (EBIT) due to a change in sales. In case a firm selects a project having more than the normal risk, the suppliers of funds for the project will expect a higher rate of return than the normal rate. The cost of capital will thus go up. The business risk is generally determined by the capital budgeting decisions.
3. **Premium for financial risk:** The term financial risk refers to the risk on account of the pattern of capital structure (or debt-equity mix). In general, it may be said that a firm having a higher debt content in its capital structure is more risky as compared to a firm that has a comparatively low debt content. This is because in the former case, the firm requires a higher operating profit to cover periodic interest payments and repayment of principal at the time of maturity as compared to the latter. Thus, the chances of cash insolvency are greater in the case of such firms. The suppliers of funds would therefore expect a higher rate of return from such firms as compensation for higher risk.

The above three components of cost of capital may be put in the form of the following equation:

$$K = r_0 + b + f$$

where

K = Cost of capital,

r_0 = return at zero risk level,

b = Premium for business risk;

f = Premium for financial risk.

B. IMPORTANCE OF COST OF CAPITAL

The determination of the firm's cost of capital is important from the point of view of both capital budgeting as well as capital structure planning decisions.

a. CAPITAL BUDGETING DECISIONS

In capital budgeting decisions, the cost of capital is often used as a discount rate based on which the firm's future cash flows are discounted to find out their present values. Thus, the cost of capital is the very basis for financial appraisal of new capital expenditure proposals. The decision of the finance manager will be irrational and wrong if case of cost of capital is not correctly determined. This is because the business must earn at least a rate that equals its cost of capital to make at least a break-even.

b. CAPITAL STRUCTURE DECISIONS

The cost of capital is also an important consideration in capital structure decisions. The finance manager must raise capital from different sources in a way that optimizes the risk and cost factors. The sources of funds that have less cost involved have high risk. Raising of loans may, therefore, be cheaper on account of income tax benefits, but it involves heavy risk because a slight fall in the earning capacity of the company may bring the firm near to cash insolvency. It is, therefore, necessary that the cost of each source of funds is carefully considered and compared with the risk involved with it.

C. CLASSIFICATION OF COST OF CAPITAL

Cost of capital can be classified as follows:

a. EXPLICIT COST AND IMPLICIT COST

The explicit cost of any source of finance may be defined as the discount rate that equates the present value of the funds received by the firm net of underwriting costs with the present value of the expected cash outflows. These outflows may be interest payments, repayment of principal, or dividends. This may be calculated by computing the value according to the following equation:

$$I_0 = \frac{C_1}{(1+K)} + \frac{C_2}{(1+K)^2} + \dots + \frac{C_n}{(1+K)^n}$$

where

I_0 = Net amount of funds received by the firm at time zero

C = Cash Outflow in the period concerned

n = Duration or number of years for which the funds are provided

K = Explicit cost of capital.

Thus, the explicit cost of capital may be taken as "the rate of return of the cash flows of financing opportunity". It is, in other words, the internal rate of return the firm pays for financing.

For example, if a company raised a sum of Rs. 1 lakh by way of debentures carrying interest at 9% and payable after 20 years, the cash inflow will be a sum of Rs. 1 lakh. However, annual cash outflow will be Rs. 9,000 for 20 years. The explicit cost will, therefore, be that rate of internal return which equates Rs. 1 lakh, the initial cash inflow, with Rs. 9,000 payable every year for 20 years and Rs. 1 lakh at the end of 20 years.

The implicit cost may be defined as “the rate of return associated with the best investment opportunity for the firm and its shareholders that will be forgone if the project presently under consideration by the firm were accepted”. When the earnings are retained by a company, the implicit cost is the income that the shareholders could have earned if such earnings had been distributed and invested by them. As a matter of fact, explicit costs arise when the funds are raised, while implicit costs arise whenever they are used. Viewed from this angle, funds raised from any source have implicit costs once they are invested.

b. FUTURE COST AND HISTORICAL COST

Future cost refers to the expected cost of funds to finance the project, while historical cost is the cost that has already been incurred for financing a particular project. In financial decision making, the relevant costs are future costs and not historical costs. However, historical costs are useful in projecting future costs and providing an appraisal of the past performance when compared with standard or predetermined costs.

c. SPECIFIC COST AND COMBINED COST

The cost of each component of capital (i.e., equity shares, preference shares, debentures, loans, etc.) is known as the specific cost of capital. In order to determine the average cost of capital of the firm, it becomes necessary first to consider the costs of specific methods of financing. This concept of cost is useful in those cases where the profitability of a project is judged on the basis of the cost of the specific sources from which the project will be financed. For example, if a company's estimated cost of equity share capital is 11%, a project that will be financed out of equity shareholders' funds would be accepted only when it gives a rate of return of at least 11%.

The composite or combined cost of capital is inclusive of all costs of capital from all sources, i.e., equity shares, preference shares, debentures, and other loans. In capital investment decisions, the composite cost of capital will be used as a basis for accepting or rejecting the proposal, even though the company may finance one proposal from one source of financing while financing another proposal from another source of financing. This is because it is an overall mix of financing over time, which is important in valuing the firm as an ongoing overall entity.

d. AVERAGE COST AND MARGINAL COST

The average cost of capital is the weighted average of the costs of each component of funds employed by the firm. The weights are in proportion to the share of each component of capital in the total capital structure.

The computation of average cost involves the following problems:

1. It requires measurement of the costs of each specific source of capital,
2. It requires assigning appropriate weights to each component of capital.
3. It raises a question whether the average cost of capital is at all affected by changes in the composition of the capital. The financing experts differ in their approaches. According to the traditional approach, the firm's cost of capital depends upon the method and level of financing, while according to the modern approach, as propounded by Modigliani and Miller, the firm's total cost of capital is independent of the method and level of financing.

Marginal cost of capital, on the other hand, is the weighted average cost of new funds raised by the firm. For capital budgeting and financing decisions, the marginal cost of capital is the most important factor to be considered.

D. APPROACHES

a. TRADITIONAL APPROACH

According to this approach, a firm's cost of capital depends upon the method and level of financing or its capital structure. A firm can, therefore, change its overall cost of capital by increasing or decreasing the debt-equity mix.

For example, if a company has 9% debentures (issued and payable at par), the cost of funds raised from this source comes to only 4.5% (assuming a 50% tax rate). Funds from other sources, such as equity shares and preference shares, also involve cost.

But the raising of funds through debentures is cheaper for the following reasons:

1. Interest rates are usually lower than dividend rates,
2. Interest is allowed as an expense resulting in a tax benefit, while dividend is not allowed as an expense while computing taxable profits of the company.

The traditional theorists, therefore, argue that the weighted average cost of capital will decrease with every increase in the debt content in the total capital employed. However, the debt content in the total capital employed should be maintained at a proper level because the cost of debt is a fixed burden on the profits of the company, and it may have adverse consequences in periods when a company has low profitability. Moreover, if the debt content is raised beyond a particular point, the investors will start considering the company too risky, and their expectations from equity shares will go up.

b. MODIGLIANI AND MILLER APPROACH

According to this approach, the corporation's total cost of capital is constant and it is independent of the method and level of financing. In other words, according to this approach, a change in the debt equity ratio does not affect the total cost of capital. According to the traditional approach, as explained above, the cost of capital is the weighted average cost of debt and the cost of equity. Each change in the debt-equity ratio automatically offsets a change in one with a change in the other on account of a change in the expectation of equity shareholders.

For example, the capital structure of a company is as follows:

9% Debenture	Rs. 1,00,000
Equity Share Capital	Rs. 1,00,000

The company has presented an even debt-equity ratio. It has been paying a dividend at the rate of 12% on equity shares. In case the debt equity ratio changes to say 60% debt and 40% equity, the following consequences will follow:

- ◆ Since the debt is cheaper, the overall cost of capital will come down.
- ◆ The expectation of the equity shareholders from the present dividend of 12% will go up because they will find the company now riskier.

Thus, the overall cost of capital of the company will not be affected by a change in the debt-equity ratio. Modigliani and Miller, therefore, argue that within the same risk class, a mere change of debt-equity ratio does not affect the cost of capital. Their following observations are in their article.

1. “Cost of Capital, Corporation Finance and Theory of Investment” needs careful consideration:

- (a) The total market value of the firm and its cost of capital are independent of its capital structure. The total market value of the firm can be computed by capitalizing the expected stream of operating earnings at a discount rate considered appropriate for its risk class.
- (b) The cut-off rate for investment purposes is completely independent of the way in which investment is financed.

2. Assumptions under the Modigliani-Miller Approach

The Modigliani-Miller Approach is subject to the following assumptions:

- (a) **Perfect capital market:** The securities are traded in a perfect capital market. This implies that:
 - (i) The investors are free to buy or sell securities.
 - (ii) The investors are completely knowledgeable and rational persons. All information and changes in conditions are known to them immediately.
 - (iii) The purchase and sale of securities involve no costs such as brokers' commissions, transfer fees, etc.
 - (iv) The investors can borrow against securities without restrictions on the same conditions as the firms can.
- (b) **Firms can be grouped in homogeneous risk classes:** Firms should be considered to belong to a homogeneous class if their expected earnings have identical risk characteristics. In other words, all firms can be categorised according to the return that they give, and a firm in each class has the same degree of business and financial risk.

- (c) **Same expectations:** All investors have the same expectation of the firm's net operating income (EBIT), which is used for the evaluation of a firm. There is a 100% dividend payout, i.e., firms distribute all of their net earnings to the shareholders.
- (d) **No corporate taxes:** In the original formulation Modigliani and Miller hypothesis assumes that there are no corporate taxes. This assumption was removed later.

3. Conclusion:

It may be said that in spite of the correctness of the basic reasoning of Modigliani and Miller, the traditional approach is more realistic on account of the following reasons:

- (a) The corporations are subject to income tax and, therefore, due to tax effect, the cost of debt is lower than the cost of equity capital.
- (b) The basic assumption of the Modigliani and Miller hypothesis that capital markets are perfect is seldom true.

On account of the above reasons, the Modigliani and Miller approach has come under severe criticism. Mr. Ezra Solomon has observed. "The thesis that the company's cost of capital is independent of its financial structure is not valid. As far as the leverage effect alone is concerned (and ignoring all other considerations that might affect the choice between debt and equity) there does exist a definable optimum position-namely, the point at which the marginal cost of more debt is equal to or greater than a company's average cost of more debt is equal to or greater than a company's average cost of capital".

However, Mr. E.W. Walker's remarks very aptly explain the utility of Modigliani and Miller's approach. According to him, "the criticisms lodged against Modigliani and Miller's thesis are valid thus limiting its use in actual situations. Nevertheless, the propositions as well as their criticisms should be carefully studied, since they will serve as an aid to understanding capital structure theory".

E. PROBLEMS DETERMINATION OF COST OF CAPITAL

It has already been stated that the cost of capital is one of the most crucial factors in most financial management decisions. However, the determination of the cost of capital of a firm is not an easy task. The finance manager is confronted with a large number of problems, both conceptual and practical, while determining the cost of capital of a firm. These **problems** can briefly be summarized as follows:

a. CONTROVERSY REGARDING THE DEPENDENCE OF THE COST OF CAPITAL UPON THE METHOD AND LEVEL OF FINANCING

There is a major controversy over whether or not the cost of capital depends upon the method and level of financing by the company. According to the traditional theorists, the cost of capital of a firm depends on the method and level of financing. In other words, according to them, a firm can change its overall cost of capital by changing its debt-equity mix. On the other hand, the modern theorists such as Modigliani and Miller argue that the firm's total cost of capital is independent of the method and level of financing. In other words, the change in the debt-equity ratio does not affect the total cost of capital.

An important assumption underlying the MM approach is that there is a perfect capital market. Since a perfect capital market does not exist in practice, the approach is not of much practical utility.

b. COMPUTATION OF THE COST OF EQUITY

The determination of the cost of equity capital is another problem. In theory, the cost of equity capital may be defined as the minimum rate of return that a company must earn on that portion of its capital employed, which is financed by equity capital, so that the market price of the shares of the company remains unchanged. In other words, it is the rate of return that the equity shareholders expect from the shares of the company, which will maintain the present market price of the equity shares of the company.

This means that the determination of the cost of equity capital will require quantification of the expectations of the equity shareholders. This is a difficult task because the equity shareholders value the equity shares of a company on the basis of a large number of factors, including financial as well as psychological factors. Different authorities have tried in different ways to quantify the expectations of the equity shareholders. Their methods and calculations differ.

c. COMPUTATION OF THE COST OF RETAINED EARNINGS AND DEPRECIATION FUNDS

The cost of capital raised through these sources will depend on the approach adopted for computing the cost of equity capital. Since there are different views, therefore, a finance manager has to face a difficult task in subscribing and selecting an appropriate approach.

d. FUTURE COSTS VERSUS HISTORICAL COSTS

It is argued that for decision-making purposes, the historical cost is not relevant. The future costs should be considered. It, therefore, creates another problem: whether to consider the marginal cost of capital, i.e., the cost of additional funds, or the average cost of capital, i.e., the cost of total funds.

e. PROBLEM OF WEIGHTS

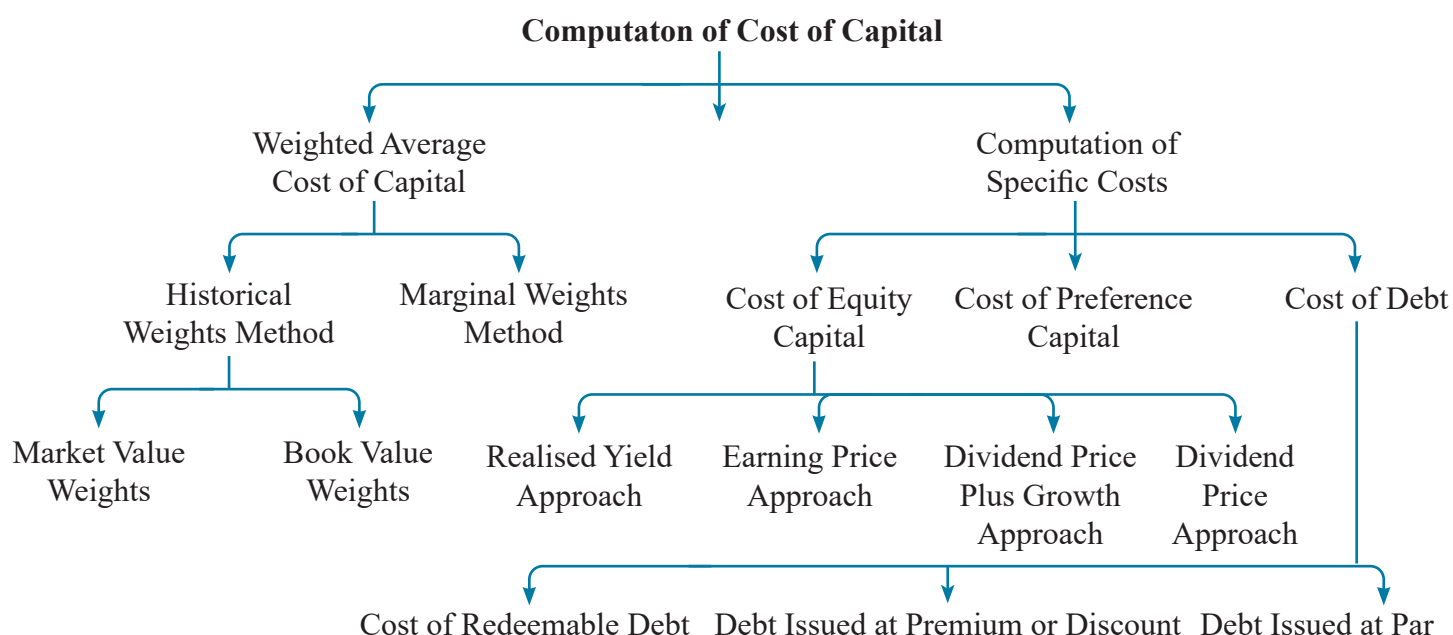
The assignment of weights to each type of fund is a complex issue. The finance manager has to make a choice between the risk value of each source of funds and the market value of each source of funds. The results would be different in each case.

It is clear from the above discussion that it is difficult to calculate the cost of capital with precision. It can never be a single given figure. At most, it can be estimated with a reasonable range of accuracy. Since the cost of capital is an important factor affecting managerial decisions, the finance manager must identify the range within which his cost of capital lies.

F. COMPUTATION OF COST OF CAPITAL

Computation of the cost of capital involves:

- ◆ Computation of the cost of each specific source of finance termed as computation of specific costs and
- ◆ The computation of composite cost is termed as weighted average cost.



a. COMPUTATION OF SPECIFIC COSTS

The cost of each specific source of finance, viz., debt, preference capital, and equity capital, can be determined as follows:

1. Cost of Debt

Debt may be issued at par, at a premium, or discount. It may be perpetual or redeemable. The technique of computation of cost in each case has been explained later.

- (a) **Debt issued at par:** The computation of the cost of debt issued at par is comparatively an easy task. It is the explicit interest rate adjusted further for the tax liability of the company. It may be computed according to the following formula:

$$K_d = (1 - T)R$$

where

K_d = Cost of debt;

T = Marginal tax rate;

R = Debenture interest rate.

For example, if a company has issued 9% debentures and the tax rate is 50%, the after-tax cost of debt will be 4.5%, calculated as given below:

$$K_d = (1 - T)R$$

$$= (1 - 0.5) 9 = 0.5 \times 9 = 4.5\%$$

The tax is deducted from the interest payable because interest is treated as an expense while computing the firm's income for tax purposes. However, the tax-adjusted rate of interest should be used only in those cases where the "earnings of the firm before interest and tax" (EBIT) is equal to or exceeds the interest. In case EBIT is negative, the cost of debt should be calculated before adjusting the interest rate for tax. For example, in the above cases, the cost of debt before adjusting for tax effect will be 9%.

- (b) **Debt issued at premium or discount:** In case the debentures are issued at a premium or discount, the cost of debt should be calculated on the basis of the net proceeds realized on account of the issue of such debentures or bonds. Such cost may further be adjusted, keeping in view the tax applicable to the company.
- (c) **Cost of redeemable debt:** In the preceding pages, while calculating the cost of debt, we have presumed that debentures/bonds are not redeemable during the lifetime of the company. However, if the debentures are redeemable after the expiry of a fixed period, the effective cost of debt before tax can be calculated by using the following formula:

$$C_d = \frac{i + \frac{MV - NP}{n}}{\frac{MV + NP}{2}} \times 100$$

where, C_d = cost of debt capital

i = annual interest payment

MV = Maturity Value

NP = Net Proceeds

n = number of years to maturity

In order to keep sufficient earnings available for equity shareholders for maintaining their present value, the company should ensure that it earns on the funds provided by raising loans at least equal to the effective interest rate payable on them. In case the firm earns less than the effective interest rate, earnings available for the equity shareholders will decrease. This would naturally affect adversely the market price of the company's equity shares.

It should be noted that while calculating the real cost of debt, not only the contractual interest rate but also certain other imputed costs or raising funds from debts should be considered. The more financing is of funds from debt, the higher is the expectation of the equity shareholders on their capital because of an increase in the risk factor. Moreover, with an increase in the amount of borrowed funds, the interest rate is also likely to rise. Thus, the imputed cost of raising funds through borrowings includes the increase in the expectation of the equity shareholders from their capital employed in the business. If it had not been so, the management would have at all times liked to finance their requirements only by means of raising long-term debts.

2. Cost of preference capital

The computation of the cost of preference capital, however, poses some conceptual problems. In case of borrowings, there is a legal obligation on the firm to pay interest at fixed rates, while in case of preference shares, there is no such legal obligation. Hence, some people argue that dividends payable on preference share capital do not constitute cost. However, this is not true. This is because, though it is not legally binding on the company to pay dividends on preference shares, it is generally paid whenever the company makes sufficient profits.

The failure to pay dividends may be of serious concern from the point of view of equity shareholders. They may even lose control of the company because of the preference shareholders getting the legal right to participate in the general meetings of the company with equity shareholders under certain conditions in the event of failure of the company to pay them their dividends. Moreover, the accumulation of arrears of preference dividends may adversely affect the right of equity shareholders to receive dividends. This is because no dividend can be paid to them unless the arrears of the preference dividend are cleared. On account of these reasons, the cost of preference capital is also computed on the same basis as that of debentures.

The method of its computation can be put in the form of the following equation:

$$C_p(\text{or } K_p) = \frac{PD}{NP} \times 100$$

Where, C_p = Cost of preference capital

PD = Preference dividend amount per share

NP = Net Proceeds per share

Cost of redeemable preference shares

In case of redeemable preference shares, the cost of capital is the discount rate that equals the net proceeds of sale of preference shares with the present value of future dividends and principal repayments. Such a cost can be calculated according to the same formula that has been given in the preceding pages for calculating the cost of redeemable debentures.

It should be noted that the cost of preference capital is not adjusted for taxes, since the dividend on preference capital is taken as an appropriation of profits and not as a charge against profits. Thus, the cost of preference capital is substantially greater than the cost of debt.

3. Cost of equity capital

The computation of the cost of equity capital is a difficult task. Some people argue, as observed in the case of preference shares, that the equity capital does not involve any cost. The argument put forward by them is that it is not legally binding on the company to pay dividends to the equity shareholders. This does not seem to be a correct approach because the equity shareholders invest money in shares with the expectation of getting dividends from the company. The company also does not issue equity shares without having any intention to pay them dividends. The market price of the equity shares, therefore, depends upon the return expected by the shareholders.

Conceptually cost of equity share capital may be defined as the minimum rate of return that a firm must earn on the equity-financed portion of an investment in a project in order to leave unchanged the market price of such shares.

For example, in case the required rate of return on equity shares is 16% and the cost of debt is 12%, and the company has the policy of financing with 75% equity and 25% debt, the required rate of return on the project could be estimated as follows:

$$16\% \times 0.75 = 12\%$$

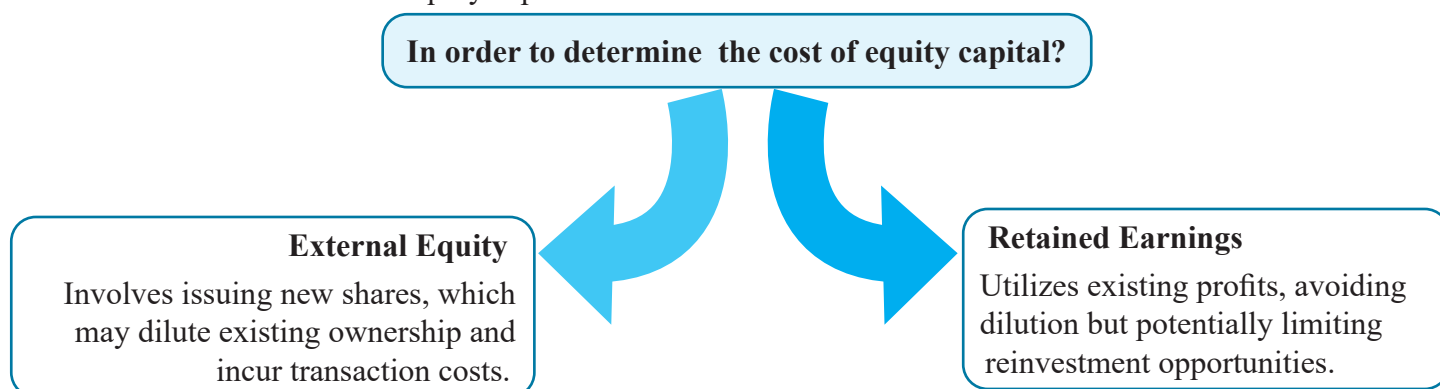
$$12\% \times 0.25 = \underline{3\%}$$

$$\underline{15\%}$$

This means that if the company accepts a project involving an investment of Rs.10,000, and giving an annual return of Rs. 1,500, the project would provide a return which is just sufficient to leave the market value unchanged of the company's equity shares. The rate of return on equity equity-financed portion can be calculated as follows:

Total Return	Rs. 1,500
Less: Interest on debentures $0.25 \times 12 \times 100$	<u>Rs. 300</u>
Amount Available for equity shareholders	<u>Rs. 1,200</u>
Rate of return on equity $= \frac{1,200 \times 100}{7,500}$	
= 16%	

Thus, the expected rate of return is 16%, which just equals the required rate of return on investment. If the project earns less than Rs. 1,500 a year, it would provide a return less than required by the investors. As a result, the market value of the company's shares would fall. Theoretically, this rate of return could be considered as the cost of equity capital.



The computation of the cost of each of these is explained below:

(a) The external equity or the new issue of equity shares

From the preceding discussion, it is implied that in order to find out the cost of equity capital, one must be in a position to determine what the shareholders as a class expect from their investment in equity shares. This is a difficult proposition because shareholders as a class are difficult to predict or quantify. Different authorities have conveyed different explanations and approaches. The following are some of the appropriate methods by which the cost of equity capital can be worked out:

(i) Dividend price (D/P) approach

According to this approach, the investor arrives at the market price of an equity share by capitalizing the set of expected dividend payments. Cost of equity capital has therefore been defined as “the discount rate that equates the present value of all expected future dividends per share with the net proceeds of the sale (or the current market price) of a share”.

In other words, the cost of equity capital will be the rate of expected dividends that will maintain the present market price of equity shares.

This approach rightly emphasizes the importance of dividends, but it ignores the fact that the retained earnings also have an impact on the market price of the equity shares. The approach, therefore, does not seem to be very logical.

(ii) Dividend price plus growth (D/P + g) approach

According to this approach, the cost of equity capital is determined on the basis of the expected dividend rate plus the rate of growth in dividends. The rate of growth in dividends is determined on the basis of the amount of dividends paid by the company for the last few years. The computation of the cost of capital according to this approach can be done by using the following formula:

$$K_e = D/NP + g$$

where

K_e = Cost of equity capital;

D = Expected dividend per share;

NP = Net proceeds per share;

g = Growth in expected dividend.

It may be noted that in the case of existing equity shares, the cost of equity capital can also be determined by using the above formula. However, the market price (MP) should be used in place of the net proceeds (NP) of the shares as given above.

(iii) Earning price (E/P) approach

According to this approach, it is the earnings per share that determine the market price of the shares. This is based on the assumption that the shareholders capitalize a stream of future earnings (as distinguished from dividends) in order to evaluate their shareholdings. Hence, the cost of capital should be related to that earnings percentage which could keep the market price of the equity shares constant. This approach, therefore, takes into account both dividends as well as retained earnings. However, the advocates of this approach differ regarding the use of both earnings and the market price figures. Some simply use of current earning rate and the current market price of the share of the company for determining the cost of capital. While others recommend the average rate of earnings (based on the earnings of the past few years) and the average market price (calculated on the basis of the market price for the last few years) of equity shares.

The formula for calculating the cost of capital according to the approach is as follows:-

$$K_e = \frac{E}{NP}$$

where

K_e = Cost of equity capital;

D = Earning per share;

NP = Net proceeds of an equity share,

However, in the case of existing equity shares, it will be appropriate to use the market price (MP) instead of the net proceeds (NP) for determining the cost of capital.

(iv) Realised yield approach

According to this approach, the cost of equity capital should be determined on the basis of the returns actually realized by the investors in a company on their equity shares. Thus, according to this approach, the past records in a given period regarding dividends and the actual capital appreciation in the value of the equity shares held by the shareholders should be taken to compute the cost of equity capital.

This approach gives fairly good results in the case of companies with stable dividends and growth records. In case of such companies, it can be assumed with a reasonable degree of certainty that the past behaviour will be repeated in the future also.

4. Cost of retained earnings

The companies do not generally distribute the entire profits earned by them by way of dividends among their shareholders. Some profits are retained by them for future expansion of the business. Many people feel that such retained earnings are absolutely cost-free. This is not the correct approach because the amount retained

by the company, if it had been distributed among the shareholders by way of a dividend, would have given them some earnings. The company has deprived the shareholders of these earnings by retaining a part of the profit for itself. Thus, the cost of retained earnings is the earnings forgone by the shareholders. In other words, the opportunity cost of retained earnings may be taken as the cost of the retained earnings. It is equal to the income that the shareholders could have otherwise earned by placing these funds in alternative investments.

For example, if the shareholders could have invested the funds in alternative channels, they could have gotten a return of 10%. This return of 10% has been forgone by them because the company is not distributing the full profits to them. The cost of retained earnings may, therefore, be taken at 10%.

The above analysis can also be understood in the following manner. Suppose the earnings not retained by the company are passed on to the shareholders, and are invested by the shareholders in the new equity shares of the same company, the expectation of the shareholders from the new equity shares would be taken as the opportunity cost of the retained earnings. In other words, if earnings were paid as dividends and simultaneously an offer for the right shares was made, the shareholders would have subscribed to the right shares on the expectation of a certain return. This expected return can be taken as the cost of retained earnings of the company.

Tax Adjusted Return

In the example given above, we have presumed that the shareholders will have with them the amount of retained earnings available when distributed by the company. In actual practice, it does not happen. The shareholders have to pay tax on the dividends received, incur brokerage costs for making investments, etc. The funds available to the shareholders are, therefore, less than what they would have been with the company, had they been retained by it. On account of this reason, the cost of retained earnings to the company would always be less than the cost of new equity shares issued by the company.

The following adjustments are made for ascertaining the cost of retained earnings:

- (a) **Income tax adjustment:** The dividends receivable by the shareholders are subject to income tax. Hence, the dividends actually received by them are not the amount of gross dividends but the amount of net dividends, i.e., gross dividends less income tax.
- (b) **Brokerage cost adjustment:** Usually, the shareholders have to incur some brokerage costs for investing the dividends received. Thus, the funds available to them for reinvesting will be reduced by this amount.

The opportunity cost of retained earnings to the shareholders is, therefore, the rate of return that they can obtain by investing the net dividends (i.e., after tax and brokerage) in an alternative opportunity of equal quality.

The cost of retained earnings after making adjustments for income tax and brokerage costs payable by the shareholders can be determined according to the following formula:

$$k_r = K_e(1 - T)(1 - B)$$

where

K_r = Required rate of return on retained earnings.

K_e = Shareholder's required rate of return.

T = Shareholders' marginal tax rate.

B = Brokerage cost.

The computation of the cost of retained earnings, after making adjustments for tax liabilities, is a difficult process because personal income tax rates will differ from shareholder to shareholder. Thus, it will be necessary to find out the personal income tax rates of the different shareholders of the company in case the cost of retained earnings is to be calculated according to the above approach. In the case of a widely held public company, there are a large number of shareholders of varying means and

incomes. It is, therefore, almost impossible to determine a single tax rate that would correctly reflect the opportunity cost of retained earnings to every shareholder. Even the computation of a weighted average tax would not give satisfactory results. Some authorities have, therefore, recommended the use of another approach termed by them as the external yield criterion. According to this approach, the opportunity cost of retained earnings is the rate of return that can be earned by investing the funds in another enterprise by the firm. Thus, according to this approach, the cost of retained earnings is simply the return on direct investment of funds by the firm and not what the shareholders are able to obtain on their investments. The approach represents an economically justifiable opportunity cost that can be applied consistently. Moreover, the need for determining the marginal tax rate for investors will not arise in case the approach is followed.

The “External Yield Criterion” suggested above has not yet been universally accepted. In the absence of a universally acceptable and practically feasible method for determining separately the cost of retained earnings, many accountants calculate the cost of retained earnings on the same pattern as that of equity shares. Moreover, when the cost of funds is raised by equity shares, the need for determining the separate cost for retained earnings does not at all arise. Some calculate it on the Dividend Payout Basis.

b. WEIGHTED AVERAGE COST OF CAPITAL

After calculating the cost of each component of capital, the average cost of capital is generally calculated on the basis of the weighted average method. This may also be termed as the overall cost of capital.

The computation of the weighted average cost of capital involves the following steps:

1. **Calculation of the cost of each specific source of funds:** This involves the determination of the cost of debt, equity capital, preference capital, etc., as explained before. This can be done either on a “before tax” basis or an “after tax” basis. However, it will be more appropriate to measure the cost of capital on an “after-tax basis”. This is because the return to the shareholders is an important figure in determining the cost of capital, and they can get dividends only after the taxes have been paid.
2. **Assigning weights to specific costs:** This involves the determination of the proportion of each source of funds in the total capital structure of the company. This may be done according to any of the following methods.
 - (a) **Marginal weights method:** In case of this method, weights are assigned to each source of funds, in proportion to the financial inputs the firm intends to employ. The method is based on the logic that our concern is with the new or incremental capital and not with capital raised in the past. In case the weights are applied in a ratio different than the ratio in which the new capital is to be raised, the weighted average cost of capital so calculated may be different from the actual cost of capital. This may lead to wrong capital investment decisions. However, the method of marginal weighting suffers from one major limitation. It does not consider the long-term implications of the firm’s current financing. A firm should give due attention to long-term implications while designing the firm’s financing strategy. **For example,** a firm may accept a project giving an after-tax return of 6% because it intends to raise the funds required by the issue of debentures having an after-tax cost of 5%. In case next year the firm intends to raise funds by issuing equity shares having a cost of 9%, it will have to reject a project that gives a return of only 8%. Thus, the marginal weighting method does not consider the fact that today’s financing affects tomorrow’s cost.
 - (b) **Historical weights method:** According to this method, the relative proportions of various sources to the existing capital structure are used to assign weights. Thus, in the case of this method, the basis of weights is the proportion of funds already employed by the firm. This is based on the assumption that the firm’s present capital structure is optimum and it should be maintained in the future also. Weights under the historical system may be either
 - (i) **Book value or**
 - (ii) **Market value weights.**

The weighted average cost of capital will be different, depending upon whether book value weights are used or market value weights are used.

The use of market value weights for calculating the cost of capital is theoretically more appealing on account of the following reasons:

1. The market values of the securities are closely approximated to the actual amount to be received from the sale of such securities.
2. The cost of each specific source of finance that constitutes the capital structure is calculated according to the prevailing market price.

However, the use of market value as weights is subject to the following practical difficulties:

1. The market values of the securities may fluctuate considerably.
2. Market values are not readily available as compared to the book values. The book values can be taken from the published records of the firm.
3. The analysis of the capital structure of the company, in terms of debt-equity ratio, is based on the book values and not on the market values.

Thus, market value weights are operationally inconvenient as compared to book value weights. However, market value weights are theoretically consistent and sound; hence, they are a better indicator of the firm's cost of capital.

3. Adding the weighted cost of all sources of funds to get on overall weighted average cost of capital.

TEST YOURSELF

1. **Sharda Ltd.** issues 10% **irredeemable debentures of ₹1,00,000**. The company falls under the **55% tax bracket**. You are required to **calculate the cost of debt (both before and after tax)** in the following three situations:

- (i) When the debentures are issued **at par**
- (ii) When the debentures are issued **at a 10% discount**
- (iii) When the debentures are issued **at a 10% premium**

Sol. Formula Used:

Cost of Debt (After Tax),

$$K_d = \frac{I(1 - T)}{NP}$$

Where:

- ◆ K_d = Cost of debt after tax
- ◆ I = Annual interest (10% of Face Value)
- ◆ T = Tax Rate (55%)
- ◆ NP = Net Proceeds (Amount actually received)

(i) When issued at Par:

- ◆ Face Value = ₹1,00,000
- ◆ Interest (I) 10% of ₹1,00,000 = ₹10,000
- ◆ Net Proceeds (NP) = 1,00,000

Before-tax Cost of Debt:

$$K_d(\text{before tax}) = \frac{₹10,000}{₹1,00,000} = 10\%$$

After-tax Cost of Debt:

$$K_d = \frac{₹10,000(1 - 0.55)}{₹1,00,000} = \frac{₹4,500}{₹1,00,000} = 4.5\%$$

(ii) When issued at 10% Discount:

◆ Net Proceeds (NP) = 1,00,000 - 10% = ₹90,000

◆ Interest (I) = ₹10,000

Before-tax Cost of Debt:

$$K_d(\text{before tax}) = \frac{₹10,000}{₹90,000} = 11.11\%$$

After-tax Cost of Debt:

$$K_d = \frac{₹10,000(1 - 0.55)}{₹90,000} = \frac{₹4,500}{₹90,000} = 5\%$$

(iii) When issued at 10% Premium:

◆ Net Proceeds (NP) = 1,00,000 + 10% = ₹1,10,000

◆ Interest (I) = ₹10,000

Before-tax Cost of Debt:

$$K_d(\text{before tax}) = \frac{₹10,000}{₹1,10,000} = 9.09\%$$

After-tax Cost of Debt:

$$K_d = \frac{₹10,000(1 - 0.55)}{₹1,10,000} = \frac{₹4,500}{₹1,10,000} = 4.1\%$$

Final Answer

Particulars	Before Tax Kd	After Tax Kd
Issued at Par	10.00%	4.5%
Issued at 10% Discount	11.11%	5%
Issued at 10% Premium	9.09%	4.1%

2. **Harshita Ltd.** issues 10% debentures of ₹1,00,000. The company realizes ₹98,000 after allowing a 2% commission to brokers. These debentures are due for redemption at the end of the 10th year. You are required to calculate the **effective cost of debt before tax and after tax**, assuming the tax rate is 55%.

Sol. Formula for cost of debt before tax:

$$K_d(\text{before tax}) = \frac{I + \left(\frac{P - NP}{n} \right)}{\left(\frac{P + NP}{2} \right)}$$

Where:

- ◆ $I = ₹10,000$ (10% of ₹1,00,000)
- ◆ $P = ₹1,00,000$ (Par Value)
- ◆ $NP = ₹98,000$ (Net Proceeds)
- ◆ $n = 10$ years

$$K_d = \frac{10,000 + \left(\frac{1,00,000 - 98,000}{10} \right)}{\left(\frac{1,00,000 + 98,000}{2} \right)} = \frac{10,000 + 200}{99,000} = \frac{10,200}{99,000} = 0.1030 \text{ or } 10.30\%$$

Now, to calculate cost of debt after tax:

$$K_d(\text{after tax}) = K_d(\text{before tax}) \times (1 - T) = 10.30\% \times (1 - 0.55) = 10.30\% \times 0.45 = 4.635\% \approx 4.64\%$$

Final Answer

- ◆ Cost of Debt (Before Tax): 10.30%
- ◆ Cost of Debt (After Tax): 4.64%

3. Debashish Ltd. raised preference share capital of 1,00,000 by issuing 10% preference shares of ₹10 each. You are required to calculate the cost of preference share capital in the following two cases:
- (i) When the preference shares are issued at a 10% premium
 - (ii) When the preference shares are issued at a 10% discount

Sol. Let:

- ◆ $D_p = 10,000$ (10% of ₹1,00,000)
- ◆ NP = Net Proceeds

- ◆ Formula: $K_p = \frac{D_p}{NP} \times 100$

(i) **When issued at 10% premium:**

$$NP = ₹1,00,000 + 10\% \text{ of } ₹1,00,000 = ₹1,10,000$$

$$K_p = \frac{10,000}{1,10,000} \times 100 = 9.09\%$$

(ii) **When issued at 10% discount:**

$$NP = ₹1,00,000 - 10\% \text{ of } ₹1,00,000 = ₹90,000$$

Final Answer

- ◆ Cost of Preference Capital (at 10% premium): **9.09%**
- ◆ Cost of Preference Capital (at 10% discount): **11.11%**

4. Tanuja Ltd. has issued **10% redeemable preference shares** amounting to ₹1,00,000, which are **redeemable at the end of the 10th year from the year** of their issue. The company incurred **underwriting costs of 2%**. You are required to calculate the **effective cost of preference share capital**.

Sol. Given:

$$D_p = ₹10,000 \text{ (10\% of ₹1,00,000)}$$

$$P = ₹1,00,000 \text{ (Par value)}$$

$$NP = ₹98,000 \text{ (Net Proceeds after deducting 2\% underwriting cost)}$$

$$n = 10 \text{ years}$$

Formula:

$$K_p = \frac{D_p + \left(\frac{P - NP}{n} \right)}{\left(\frac{P + NP}{2} \right)} \times 100$$

$$K_p = \frac{10,000 + \left(\frac{1,00,000 - 98,000}{10} \right)}{\left(\frac{1,00,000 + 98,000}{2} \right)} \times 100 = \frac{10,000 + 200}{99,000} \times 100 = \frac{10,200}{99,000} \times 100 = 10.30\%$$

Final Answer**Effective Cost of Preference Share Capital: 10.30%**

5. Ridhima Ltd. offers equity shares of 10 each to the public at a **premium of 10%**. The company pays **5% of the issue price as underwriting commission**. Equity shareholders are expected to receive a **dividend at the rate of 20%**.

You are required to calculate the cost of equity capital in the following two cases:

- When it is based on net proceeds from a new issue, and
- When it is based on the present market price, which is 15 per share.

Sol. (i) Cost of equity capital (based on new issue)

Formula: $K_e = \frac{D}{NP}$

Where:

- ◆ $D = 20\% \text{ of face value } ₹10 = ₹2$
- ◆ $\text{Issue Price} = ₹10 + 10\% \text{ premium} = ₹11$
- ◆ $\text{Underwriting commission} = 5\% \text{ of } ₹11 = ₹0.55$
- ◆ $\text{Net Proceeds } NP = ₹11 - ₹0.55 = ₹10.45$

$$K_e = \frac{2}{10.45} = 0.1914 \text{ or } 19.14\%$$

(ii) Cost of equity capital (based on market price)

Formula: $K_e = \frac{D}{MP}$

Where:

- ◆ $D = ₹2$
- ◆ $MP = ₹15$

$$K_e = \frac{2}{15} = 0.1333 \text{ or } 13.33\%$$

Final Answer

- ◆ **Cost of Equity (based on new issue): 19.14%**
- ◆ **Cost of Equity (based on market price): 13.33%**

6. The current market price of an equity share of **Praveen Ltd.** is ₹90. The current dividend per share is 4.50. If the dividends are expected to grow at the rate of **7% per annum**, calculate the **cost of equity capital**.

Sol. Formula: $K_e = \frac{D}{MP} + g$

Where:

- ◆ D = ₹4.50 Current dividend)
- ◆ MP = ₹90 (Market price)
- ◆ g = 7% = 0.07 Growth rate)

$$K_e = \frac{4.50}{90} + 0.07 = 0.05 + 0.07 = 0.12 \text{ or } 12\%$$

Final Answer

Cost of Equity Capital: 12%

7. From the following details of **Nakshatra Ltd.**, calculate the **cost of equity capital**:
- Each equity share is of ₹150
 - The **underwriting cost per share** is 2%
 - The following are the **dividends per share** paid by the company over the last five years:

Year	Dividend per Share (₹)
2021	10.50
2022	11.00
2023	12.50
2024	12.75
2025	13.40

- The company maintains a fixed dividend payout ratio
- The expected dividend on the new equity shares is 14.10 per share

Sol. To calculate the **cost of equity capital**, first calculate the **growth rate of dividends**:

$$\text{Growth Rate}(g) = \left(\frac{D_{2025}}{D_{2021}} \right)^{\frac{1}{4}} - 1 = \left(\frac{13.40}{10.50} \right)^{\frac{1}{4}} - 1 = 1.276^{0.25} = 6\%$$

Alternatively, based on compound interest tables, we find that ₹1 becomes ₹1.276 in 4 years at 6% interest.

So:

$$g = 6\%$$

Now calculate the **Net Proceeds (NP)**:

$$\text{Issue Price} = ₹150$$

$$\text{Underwriting Cost} = 2\% \text{ of } ₹150 = ₹3$$

$$\text{NP} = ₹150 - ₹3 = ₹147$$

Now apply the formula for **Cost of Equity Capital**:

$$K_e = \frac{D}{NP} + g = \frac{14.10}{147} + 6\% = 9.6\% + 6\% = 15.6\%$$

Final Answer

Cost of Equity Capital: 15.6%

8. The entire capital employed by **Praveen Ltd.** consists of **1,00,000 equity shares** of ₹100 each. The company's current earnings amount to 10 lakhs per annum. Praveen Ltd. now wants to raise **additional funds of ₹25 lakhs** by issuing **new equity shares**. The **floatation costs** are expected to be **10% of the face value** of the shares.

You are required to **calculate the cost of equity capital**, assuming that the **earnings are expected to remain stable** over the next few years.

Sol. Given:

- ◆ Total earnings = ₹10,00,000
- ◆ Number of existing shares = 1,00,000
- ◆ Earnings per share (E) = ₹10,00,000 / 1,00,000 = ₹10
- ◆ Face value of share = ₹100
- ◆ Floatation cost = 10% of ₹100 = ₹10
- ◆ Net proceeds (NP) = ₹100 – ₹10 = ₹90

Formula: $K_e = \frac{E}{NP}$

$$K_e = \frac{10}{90} = 0.1111 \text{ or } 11.11\%$$

Final Answer

Cost of Equity Capital: 11.11%

9. On **January 1, 2020**, an investor purchased **5 shares of Tanuja Ltd.** for a total of **₹240**. The investor held these shares for **6 years** and sold them in **January 2025** for **₹300**. The following dividends **were received each year** during the holding period:

Year	Dividend (₹)
2020	14.00
2021	14.00
2022	14.50
2023	14.50
2024	14.50

You are required to **calculate the cost of equity capital**.

Sol. To determine the **cost of equity capital**, we apply the **Realized Yield Approach**, which involves calculating the **Internal Rate of Return (IRR)** through the **trial and error method**.

Let's use a 10% discount rate and calculate the present value of all future cash inflows:

Year	Dividend/Sale (₹)	Discount Factor @10%	Present Value (₹)
2020	14.00	0.909	12.7
2021	14.00	0.826	11.6
2022	14.50	0.751	10.9
2023	14.50	0.683	9.9
2024	14.50	0.621	9.0
2025	300.00 (Sale)	0.621	186.3
	Total PV		240.4

Initial Investment: 240

Present Value of Inflows @10%: 240.4

Since the present value of expected cash inflows is approximately equal to the investment amount at a **10% discount rate**, the **cost of equity capital (IRR)** is: 10%

Note:

The **realized yield approach** assumes that:

1. The company's risk profile remains unchanged.
2. Investors' expectations for returns are stable.
3. Reinvestment opportunities earn the same rate as the realized yield.

- 10. Sharda Ltd.** is earning a **net profit of ₹50,000 per annum**. The **shareholders' required rate of return** is **10%**. If retained earnings are distributed as dividends, it is assumed that shareholders can invest the amount in similar securities offering a **10% return per annum**.

However, the shareholders:

- ◆ Will incur 2% brokerage cost on net dividends received for reinvestment
- ◆ Are in a 30% tax bracket

You are required to calculate the cost of retained earnings to the company.

- Sol.** To calculate the cost of retained earnings, we assess how much return shareholders effectively earn after taxes and brokerage if the profits are distributed instead of retained.

Step 1: Calculate Net Amount Available for Reinvestment

Particulars	Amount (₹)
Gross Dividends (Net Profit)	50,000
Less: Income Tax @ 30%	15,000
After-Tax Dividends	35,000
Less: Brokerage @ 2% on 35,000	700
Net Amount Available for Reinvestment	34,300

Step 2: Calculate Effective Return to Shareholders

Shareholders can invest 34,300 in similar securities at 10% return:

Earnings on reinvestment = 10% of ₹34,300 = 3,430

Step 3: Calculate Cost of Retained Earnings to the Company

If Sharda Ltd. retains the ₹50,000, it must generate 3,430 (same as shareholders would earn) to keep shareholders satisfied. So:

$$\text{Cost of Retained Earnings (K}_e\text{)} = \frac{3,430}{50,000} \times 100 = 6.86\%$$

Final Answer

Cost of Retained Earnings for Sharda Ltd. = 6.86%

- 11.** From the following capital structure of Tanuja Ltd., calculate the overall cost of capital (K_o) using:
- (a) Book Value Weights
 - (b) Market Value Weights

Capital Structure		
Source	Book Value (₹)	Market Value (₹)
Equity Share Capital (10/share)	45,000	90,000
Retained Earnings	15,000	–
Preference Share Capital	10,000	10,000
Debentures	30,000	30,000
Total	1,00,000	1,30,000

After-Tax Cost of Capital

- ◆ Equity Share Capital: 14%
- ◆ Preference Share Capital: 10%
- ◆ Retained Earnings: 13%
- ◆ Debentures: 5%

Sol.

(a) Weighted Average Cost of Capital - Book Value Weights

Source	Amount (₹)	Proportion	After-Tax Cost	Weighted Cost
Equity Share Capital	45,000	0.45	14%	6.30%
Retained Earnings	15,000	0.15	13%	1.95%
Preference Share Capital	10,000	0.10	10%	1.00%
Debentures	30,000	0.30	5%	1.50%
Total	1,00,000	1.00	–	10.75%

Conclusion: Weighted average cost of capital using **book value weights** is **10.75%**.

(b) Weighted Average Cost of Capital - Market Value Weights

Note: Retained earnings are included within equity in market value terms.

Source	Market Value (*)	Proportion	After-Tax Cost	Weighted Cost
Equity Share Capital	90,000	0.692	14%	9.688%
Preference Share Capital	10,000	0.077	10%	0.770%
Debentures	30,000	0.231	5%	1.155%
Total	1,30,000	1.000	–	11.613%

Conclusion: Weighted average cost of capital using **market value weights** is **11.613%**.

Final Answer

- ◆ Overall cost of capital (Book Value Weights): 10.75%
- ◆ Overall cost of capital (Market Value Weights): 11.613%

- 12. Chavi Ltd.** has total assets of 1,60,000, financed by ₹52,000 of debt, ₹90,000 of equity share capital, and 18,000 of general reserve. The firm's **net profit after interest and tax** for the year ended **31st March 1998** is ₹13,500. The company pays 8% interest on debentures, and its **₹100 equity shares** are currently trading at a **market price of ₹120 per share**.

You are required to **calculate the Weighted Average Cost of Capital (WACC)**.

Sol. Balance Sheet (Based on Given Data)

Liabilities	Amount (₹)	Assets	Amount (₹)
900 Equity Shares of ₹100 each	90,000	Total Assets	1,60,000
General Reserve	18,000		
8% Borrowed Funds (Debentures)	52,000		
Total Liabilities	1,60,000		1,60,000

Step 1: Earnings Per Share (EPS)

- ◆ Net profit after interest and tax = ₹13,500
- ◆ Number of equity shares = 900
- ◆ $EPS = 13,500 \div 900 = ₹15$

Step 2: Determination of Specific Costs

(i) Cost of Debt (K_d):

Assuming a tax rate of 50%,
 $K_d = 8\% \times (1 - 0.5) = 4\%$

(ii) Cost of Equity (K_e):

Using the Earnings Yield Approach:

$$K_e = \frac{EPS}{\text{Market Price}} \times 100 = \frac{15}{120} \times 100 = 12.5\%$$

Step 3: Weighted Average Cost of Capital (WACC)

Assuming **market value of debt equals book value**, and equity includes reserves:

Source	Market Value (₹)	Specific Cost (%)	Weighted Cost (₹)
Debt	52,000	4.0	2,080
Equity	1,08,000	12.5	13,500
Total	1,60,000		15,580

$$WACC(K_0) = \frac{15,580}{1,60,000} \times 100 = 9.74\%$$

Alternative Method: Using Book Value Weights

Since the market and book value of debt are assumed the same, and no separate cost is given for retained earnings, we can apply the same cost of equity (12.5%) to both equity capital and reserves.

- ◆ Equity + Reserves = ₹90,000 + 18,000 = ₹1,08,000
- ◆ Debt = 52,000
- ◆ Resulting WACC will also be **9.74%**, same as using market values.

Final Answer

Weighted Average Cost of Capital (WACC) = 9.74%

CAPITAL STRUCTURE

A. INTRODUCTION

Capital structure refers to the combination or mix of debt and equity that a company uses to finance its long-term operations.

The capital structure refers to a firm's financing of its overall operations and growth through various sources of funds.

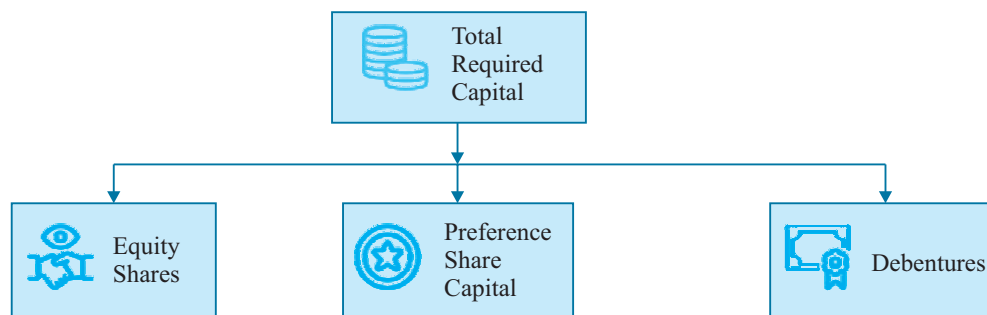
Raising of capital from different sources and its use in different assets by a company is made based on certain principles that provide a system of capital so that the maximum rate of return can be earned at a minimum cost.

This sort of system of capital is known as **capital structure**.

B. OPTIMAL CAPITAL STRUCTURE

- The optimal or the best capital structure implies the most economical and safe ratio between various types of securities.
- It is that mix of debt and equity that maximizes the value of the company and minimizes the cost of capital.
- Essentials of optimal capital structure:**
 - ☑ Minimum cost of capital,
 - ☑ Minimum risk,
 - ☑ Maximum return, and
 - ☑ Maximum control.

Sources of Required Capital



C. FACTORS DETERMINING THE CAPITAL STRUCTURES

a. GROWTH AND STABILITY OF SALES

The capital structure of a firm is highly influenced by the growth and stability of its sales. If the sales of a firm are expected to remain fairly stable, it can raise a higher level of debt. Stability of sales ensures that the firm will not face any difficulty in meeting its fixed commitments of interest payments and repayments of debt.

b. NATURE AND SIZE OF A FIRM

The nature and size of the firm also influence its capital structure. All public utility concerns have different capital structures as compared to other manufacturing concerns. Public utility concerns may employ more debt because of the stability and regularity of their earnings. On the other hand, a concern that cannot provide stable earnings due to the nature of its business will have to rely mainly on equity capital.

c. CONTROL

Whenever additional funds are required by a firm, the management of the firm wants to raise the funds without any loss or control over the firm. In case the funds are raised through the issue of equity shares, the control of the existing shareholders is diluted. Hence, they might raise the additional funds by way of fixed interest-bearing debt and preference share capital. Preference shareholders and debenture holders do not have voting rights. Hence, from the point of view of controlling debt financing is recommended.

d. FLEXIBILITY

Capital structure of the firm should be flexible, i.e, it should be such as to be capable of being adjusted according to the needs of the changing conditions. It should be possible to raise additional funds, wherever the need be, without much difficulty and delay. A firm should arrange its capital structure in such a way that it can substitute one form of financing for another.

e. CAPITAL MARKET CONDITIONS

Capital market conditions do not remain the same forever. Sometimes there may be depression, while at other times there may be a boom in the market. The choice of securities is also influenced by market conditions.

f. ASSET STRUCTURE

The liquidity and the composition of assets should also be kept in mind while selecting the capital structure. If fixed assets constitute a major portion of the total assets of the company, it may be possible for the company to raise more long-term total debts.

g. PURPOSE OF FINANCING

If funds are utilized for a productive purpose, debt financing is suitable, and the company should issue debentures, as interest can be paid out of the profits generated from the investment. If the funds are required for unproductive purposes or general development, we should prefer equity capital.

h. PERIOD OF FINANCE

The period for which the finances are required is also an important factor to be kept in mind while selecting an appropriate capital mix. If the finances are required for a limited period, debentures can be preferred to shares. If funds are needed permanently, equity share capital is more appropriate.

D. THEORIES OF CAPITAL STRUCTURE

Net Income Theory

Suggests that a firm's value increases with debt due to tax benefits.



Traditional Theory

Proposes an optimal capital structure balancing risk and return.



Net Operating Income Theory

Argues that a firm's value is independent of its capital structure.



Modigliani-Miller Theory

States that in perfect markets, capital structure doesn't affect firm value.



a. NET INCOME THEORY (NI APPROACH)

The **Net Income Approach** suggests that **debt financing** affects the **value of the firm**.

Propounded by **David Durand**, this approach is also known as the **Fixed Ke Theory**.

According to this theory, a firm can **increase its overall value** and **reduce the cost of capital** by increasing the **proportion of debt** in its capital structure.

Why is debt considered cheaper?

- ◆ **Interest rates** on debt are **lower than dividend rates** because of the lower risk.
- ◆ **Tax Benefit:** Interest is **tax-deductible**, reducing the firm's taxable income.

Example 01:

- ◆ Net operating Income = 7,000
- ◆ Debt = 30,000 at 8%
- ◆ Cost of equity (K_e) = 10%

Solution

$$\text{Interest} = 30,000 \times 8\% = 2,400$$

$$\text{Income to equity holders} = 7,000 - 2,400 = 4,600$$

$$\text{Market value of equity} = 4,600 / 10\% = 46,000$$

$$\text{Total value (V)} = 46,000 + 30,000 = 76,000$$

$$K_o = 7,000 / 76,000 = 9.21\%$$

1. Assumptions of the NI Approach:

- Cost of debt (K_d) is less than the cost of equity (K_e).
- Tax is ignored.
- K_d and K_e remain constant at all levels of capital structure.
- Overall cost of capital (K_o) decreases with more debt.

2. Relationship Among K_e , K_d , and K_o :

- Since K_d and K_e are assumed constant, their curves are horizontal and parallel.
- As debt increases, the overall cost of capital (K_o) falls.
- Hence, K_o is minimum when the proportion of debt is maximum.

3. Conclusion:

- According to the NI Approach, higher debt in the capital structure leads to:
 - Higher firm value**
 - Lower overall cost of capital**
- This theory supports using maximum debt to improve the firm's financial position.

b. NET OPERATING INCOME (NOI) APPROACH

According to the NOI approach, the value of a firm is independent of its capital structure. It states that the total market value of the firm remains constant, regardless of whether the firm is financed by debt or equity. The overall cost of capital (K_o) also remains unchanged.

Example 01:

Given:

- ◆ EBIT (Net Operating Income) = ₹ 1,00,000
- ◆ Debentures (Debt) = ₹ 5,00,000 at 6% interest
- ◆ Overall capitalization Rate (K_o) = 10%

Will the value of firm change if the debt increases to 7.5 lakh? What will be the impact on other factors.

(a) When Debt = ₹5,00,000

Step 1: Total value of Firm (V)

$$V = \frac{\text{EBIT}}{K_o} = \frac{1,00,000}{0.10} = ₹10,00,000$$

Step 2: Market value of Equity (E)

$$E = V - D = 10,00,000 - 5,00,000 = ₹ 5,00,000$$

Step 3: Interest on Debt (I)

$$I = 6\% \text{ of } ₹ 5,00,000 = ₹ 30,000$$

Step 4: Earnings Available to Equity Shareholders

$$\text{EBIT} - \text{Interest} = 1,00,000 - 30,000 = ₹ 70,000$$

Step 5: Equity Capitalization Rate (Ke)

$$K_e = \frac{\text{Earnings to Equity}}{\text{Market Value of Equity}} \times 100 = \frac{70,000}{5,00,000} \times 100 = 14\%$$

(b) When Debt = ₹7,50,000

Note: Under NOI approach, the total value of the firm remains unchanged at ₹10,00,000

Step 1: Market value of Equity (E)

$$E = 10,00,000 - 7,50,000 = ₹ 2,50,000$$

Step 2: Interest on Debt (I)

$$I = 6\% \text{ of } ₹ 7,50,000 = ₹ 45,000$$

Step 3: Earnings for Equity Shareholders

$$\text{EBIT} - \text{Interest} = 1,00,000 - 45,000 = ₹ 55,000$$

Step 4: Equity Capitalization Rate (Ke)

$$K_e = \frac{55,000}{2,50,000} \times 100 = 22\%$$

1. Assumptions of the NOI Approach:

- (a) The overall cost of capital (K_o) remains constant for all levels of capital structure.
- (b) The market capitalizes the firm as a whole; i.e., $V = \text{EBIT} / K_o$
- (c) There is no corporate taxation.
- (d) Debt does not affect the risk perception of investors; thus, K_o is constant.
- (e) The use of debt does not increase the cost of equity initially.
- (f) The capital markets are perfect, and investors have the same information as management.

2. Relationship among K_o , K_e , and K_d under the NOI Approach:

- (a) K_o (Overall Cost of Capital) remains constant
- (b) (Cost of Debt) is lower than K_e (usually fixed and known)
- (c) K_e (Cost of Equity) increases as more debt is used, to compensate equity holders for the higher financial risk
- (d) As debt increases, $\rightarrow$ risk to equity holders increases $\rightarrow K_e$ increases
- (e) But this rise in K_e is exactly offset by the cheaper cost of debt $\rightarrow$ so K_o remains unchanged

Formula:

$$K_o = \frac{E}{V} \times K_e + \frac{D}{V} \times K_d$$

Since K_o is constant, any increase in debt (D) increases, the required return on equity (K_e)

3. Conclusion:

- (a) Under the Net Operating Income Approach, the **value of the firm remains constant** at ₹10,00,000 regardless of the capital structure.
- (b) **Increasing debt** reduces the **market value of equity**, but the **overall value of the firm remains unchanged**.
- (c) As **financial risk increases** with higher debt, the **equity capitalization rate (K_e)** also increases.
- (d) **Capital structure decisions do not influence the total value of the firm or the overall cost of capital.**

c. THE TRADITIONAL APPROACH

According to the Traditional Approach, a firm can initially **increase its value** or **reduce its overall cost of capital (K_o)** by **using more debt** in the capital structure because **debt is a cheaper source of finance** than equity.

However, this benefit exists **only up to a certain point**. After this optimal level, any further increase in debt raises the **financial risk** of the firm, which leads to an **increase in the cost of equity (K_e)** and possibly the **cost of debt (K_d)** as well. As a result:

- ◆ **K_o initially decreases,**
- ◆ Then becomes **stable,**
- ◆ And finally **increases** after crossing the optimal capital structure level.

Thus, this theory suggests the existence of an **optimum capital structure** that **minimizes the cost of capital** and **maximizes the value of the firm**.

Example 01: Let's assume a firm has the following EBIT (Earnings Before Interest and Tax) and wishes to analyze how its overall cost of capital and firm value change with different debt levels:

Capital Structure	EBIT	Debt (₹)	Interest @10%	Equity (₹)	K_e	K_o	Value of Firm (V)
No debt (0%)	1,00,000	0	0	10,00,000	12%	12%	$1,00,000/0.12$ = ₹8,33,333
Moderate debt (30%)	1,00,000	3,00,000	30,000	7,00,000	14%	11%	$1,00,000/0.11$ = ₹9,09,090
High debt (60%)	1,00,000	6,00,000	60,000	4,00,000	18%	12.5%	$1,00,000/0.125$ = ₹8,00,000

Observation:

- ◆ As debt increases, K_o initially decreases (from 12% to 11%), and firm value increases.
- ◆ At debt moderate debt levels, K_o is minimized, and firm value is maximized.
- ◆ As debt increases, K_o initially decreases (from 12% to 11%), and firm value increases.

1. Assumptions of the Traditional Approach:

- (a) Debt is cheaper than equity.
- (b) There exists an optimal capital structure that maximizes the firm's value.
- (c) Initially, the cost of equity (K_e) remains constant or rises slowly.

- (d) Beyond a certain debt level, financial risk increases significantly.
- (e) After the optimum point, both K_e and K_d start to rise due to increased risk.
- (f) The overall cost of capital (K_o) does not remain constant like the NOI approach, but changes with capital structure.

2. Relationship among K_o , K_e , and K_d :

- (a) **At low debt levels:**
 - (i) $K_d < K_e$,
 - (ii) K_e remains fairly **stable**,
 - (iii) So, K_o decreases.
- (b) **At moderate debt levels:**
 - (i) K_e starts increasing slowly,
 - (ii) K_o remains relatively stable,
 - (iii) This is the **optimal capital structure**.
- (c) **At high debt levels:**
 - (i) K_e rises sharply due to high financial risk,
 - (ii) K_d may also increase if lenders feel insecure,
 - (iii) As a result, K_o **increases**.

Formula:

$$K_o = \left(\frac{E}{V} \times K_e \right) + \left(\frac{D}{V} \times K_d \right)$$

3. Conclusion:

The Traditional Approach presents a **realistic and practical view** of capital structure. It recognizes that:

- (a) Using some debt is **beneficial** as it lowers K_o and increases firm value.
- (b) There exists an **optimal mix of debt and equity** where the **firm value is maximized** and K_o is **minimized**.
- (c) Excessive debt **increases financial risk**, leading to higher K_e and K_d , and ultimately **increases K_o** , reducing the value of the firm.

Hence, the capital structure decision should be carefully balanced to achieve the optimal point.

d. MODIGLIANI AND MILLER (MM) APPROACH

The **Modigliani and Miller (MM) Approach** to capital structure is one of the most influential theories in financial management. It presents two contrasting views based on whether **corporate taxes are considered or not**.

1. In the Absence of Taxes – Theory of Irrelevance:

According to MM, in a **perfect capital market** without taxes, the **capital structure of a firm is irrelevant**. This means that whether a firm is financed through equity, debt, or a mix of both, the **total value of the firm remains unchanged**, and so does the **overall cost of capital (K_o)**. The logic is that although **debt is a cheaper source** of finance compared to equity, its increased usage raises the **financial risk for equity shareholders**, which leads to a rise in the **cost of equity (K_e)**. This **increase in K_e offsets the benefit of using low-cost debt**, resulting in no net advantage.

Furthermore, MM introduced the concept of **arbitrage** to support their argument. They stated that if two firms are identical in all aspects except for capital structure, they cannot have different market values. If they do, investors will exploit the difference by **buying shares of the undervalued firm** and **selling shares of the overvalued firm**, a process known as **arbitrage**. This buying and selling will continue until the **market values equalize**, proving that capital structure does not influence firm value.

2. In the Presence of Corporate Taxes – Theory of Relevance:

MM later revised their theory by including **corporate taxes**. They recognized that **interest on debt is tax-deductible**, meaning that using debt reduces a firm's tax liability. This leads to **higher after-tax profits** and thus **increases the value of the firm**. In this case, the **use of more debt lowers the overall cost of capital (Ko)** and increases the firm's market value. Therefore, **capital structure becomes relevant**, and the firm can achieve an **optimal capital structure** by **maximizing the proportion of debt** in its capital mix to take full advantage of the **tax shield on interest**.

3. Assumptions of the MM Approach

(a) Without Taxes (Irrelevance):

- (i) No corporate or personal taxes exist.
- (ii) Capital markets are perfect, with no transaction or bankruptcy costs.
- (iii) Firms and investors can borrow at the same interest rate.
- (iv) All investors have equal access to information.
- (v) Investors act rationally and seek to maximize wealth.
- (vi) Business risk is constant regardless of the capital structure.
- (vii) Arbitrage opportunities are available and effectively used.

(b) With Taxes (Relevance):

- (i) Corporate taxes exist, and interest on debt is tax-deductible.
- (ii) All other assumptions of the tax-free case remain unchanged.
- (iii) Debt financing provides a tax shield that increases firm value.
- (iv) There are no bankruptcy or agency costs in the basic model.

4. Relationship among Ko, Ke, and Kd:

(a) Without Taxes (Irrelevance):

- (i) **Kd < Ke**, but
- (ii) **Ke increases** with more debt due to financial risk.
- (iii) This **exactly offsets** the benefit of cheap debt.
- (iv) So, **Ko remains constant**.

$$K_o = \left(\frac{E}{V} \cdot K_e \right) + \left(\frac{D}{V} \cdot K_d \right)$$

(b) With Taxes (Relevance):

- (i) Interest provides **tax savings**, so
- (ii) The effective cost of debt is **lower after tax**, and
- (iii) **Ko decreases** as debt increases.

$$V = V_u + (\text{Tax Rate} \times \text{Debt})$$

5. Conclusion:

(a) Without Taxes (Irrelevance):

- (i) Capital structure has **no impact** on firm value or cost of capital.
- (ii) Firm value is based on **operating income**, not financing.

(b) With Taxes (Relevance):

- (i) Debt **adds value** to the firm due to **tax savings** on interest.
- (ii) More debt leads to **lower Ko** and **higher firm value**.
- (iii) Hence, **maximum debt** is desirable in capital structure.

E. LEVERAGES

Leverage refers to the strategic use of fixed costs to potentially enhance the profitability of a business. In broader terms, it reflects the influence or impact of one financial variable over another related financial variable. By incorporating leverage, a firm aims to magnify its earnings, but this also involves an associated risk.

There are two main types of leverage:

a. OPERATING LEVERAGE

Operating leverage arises from the use of fixed operating costs in a firm's cost structure. It is concerned with the operational side of the business and becomes significant when a company incurs costs that do not change with the level of production or sales.

1. Characteristics of Operating Leverage:

- (a) **Fixed Operating Costs:** These are costs that remain constant irrespective of changes in production volume. Examples include depreciation on buildings and machinery, insurance premiums, and salaries of permanent management staff.
- (b) **Variable Operating Costs:** These costs change directly in proportion to the level of output. Examples include the cost of raw materials, direct labor, sales commissions, and certain components of administrative expenses.
- (c) **Effect of Operating Leverage:** When fixed operating costs are high, a small increase in sales can lead to a disproportionately large increase in **operating profit** (Earnings Before Interest and Taxes - EBIT). Conversely, a small decline in sales can cause a significant drop in profits.
- (d) **Implication:** Operating leverage amplifies the effect of sales fluctuations on operating income. This means the firm is more sensitive to changes in sales volume.
- (e) **Risk Factor:** The presence of operating leverage adds to the business risk because it increases the variability of profits due to changes in sales, even though the fixed costs remain unchanged.

Operating leverage is defined as the firm's ability to use fixed operating costs to magnify the effects of changes in sales on its earnings before interest and taxes (EBIT).

2. Degree of Operating Leverage (DOL)

The **Degree of Operating Leverage (DOL)** is a **quantitative measure** that indicates how sensitive a firm's **operating profit (EBIT)** is to a change in **sales or output**. In other words, it shows how a percentage change in sales will impact the percentage change in EBIT due to the presence of fixed operating costs.

Formula for DOL:

At a given output level (Q units):

$$\text{DOL} = \frac{\% \text{ Change in Operating Profit (EBIT)}}{\% \text{ Change in Output Sales}}$$

Alternative formula:

◆ Based on output (Q units):

$$\text{DOL} = \frac{Q(P - V)}{Q(P - V) - FC}$$

where:

Q = Quantity of output/sales

P = Selling price per unit

V = Variable cost per unit

FC = Fixed operating costs

- ◆ Based on sales in rupees:

$$\text{DOL} = \frac{S - VC}{S - VC - FC} = \frac{\text{EBIT} + FC}{\text{EBIT}}$$

where:

S = Total Sales Revenue

VC = Total Variable Costs

FC = Fixed Costs

3. Interpreting Degree of Operating Leverage:

For example, if **DOL at 5,000 units = 5**, this means:

A **1% change in sales** from the 5,000 units level will result in a **5% change in EBIT**.

So, DOL acts as a multiplier that magnifies the effect of sales changes on operating profit.

Below is table showing the relationship between output, EBIT, and DOL:

Quantity (Q)	EBIT (₹)	DOL
0	– 1,00,000	0.00
1,000	– 75,000	– 0.33
2,000	– 50,000	– 1.00
3,000	– 25,000	– 3.00
4,000 (QBE)	0	Infinite
5,000	25,000	5.00
6,000	50,000	3.00
7,000	75,000	2.33
8,000	1,00,000	2.00

Insights:

- At the **Break-even point**, EBIT = 0, so DOL is infinite, as any small change in sales cause a drastic change in EBIT.
- As the firm **moves away from the break-even point**, EBIT increases, but **DOL decreases**.
- This means the further above break-even a firm operates, the less sensitive EBIT becomes to changes in sales.
- Firms with high fixed costs but high output levels (well above QBE) may still have a low DOL.

4. Importance of DOL for Financial Managers:

- Knowing the DOL helps financial managers **predict** the impact of changes in sales on operating profit.
- It aids in **decision-making**, especially when evaluating risks associated with expanding or contracting production levels.
- It helps in **strategic planning** for pricing, cost control, and capacity utilization.

5. DOL and Business Risk:

- Business risk refers to the **uncertainty** in a firm's operating income due to variability in **sales or production costs**.
- Operating leverage magnifies** the effect of these uncertainties.
- Therefore, **higher DOL implies higher business risk**, because even small fluctuations in sales or costs can lead to larger changes in EBIT.
- DOL should be seen as a measure of **potential risk**, which becomes critical only **when sales and cost variability are present**.

b. FINANCIAL LEVERAGE

Financial leverage refers to the **use of fixed cost financing**, such as **debt** or **preference shares**, to increase the potential return to equity shareholders. It focuses on how **changes in EBIT (Earnings Before Interest and Taxes)** affect the **Earnings Per Share (EPS)** available to equity shareholders.

As defined by Brigham:

“Financial leverage is the extent to which fixed-income securities (debt and preferred stock) are used in a firm’s capital structure.”

1. Characteristics of Operating Leverage:

- (a) Financial leverage arises from **fixed financial charges**, such as:
 - (i) Interest on debt
 - (ii) Dividends on preference shares
- (b) The firm must meet these financial obligations **regardless of its operating income**.
- (c) It measures how **sensitive EPS is to changes in EBIT**. A small change in EBIT can cause a much larger change in EPS if financial leverage is high.

2. Objective of Financial Leverage:

The goal is to use borrowed funds at a **fixed cost** to **earn a return higher than that cost**. This enhances the **returns available to equity shareholders**, assuming the firm performs well.

3. Types of Financial Leverage:

- (a) Favourable (Positive) Financial Leverage:
 - (i) Occurs when the **return on investment (ROI)** is **greater than** the **cost of borrowed funds**.
 - (ii) After paying fixed financing costs (like interest or preference dividend), the **residual profit belongs to equity shareholders**, increasing their earnings.
 - (iii) This is also referred to as **“trading on equity”**—using debt to enhance shareholder returns.
- (b) Unfavourable (Negative) Financial Leverage:
 - (i) Arises when the **return on investment is less than** the **cost of financing**.
 - (ii) The firm earns **less than it pays out** in interest or fixed charges.
 - (iii) This reduces or even eliminates returns available to equity shareholders.

4. Key points:

- (a) While financial leverage **can magnify profits**, it **also increases financial risk**.
- (b) If EBIT increases, EPS increases **more than proportionally**.
- (c) However, if EBIT decreases, EPS **declines sharply**, and the firm might face solvency issues.
- (d) Therefore, excessive reliance on financial leverage can be dangerous in volatile or declining markets.

5. Degree of Financial Leverage (DFL)

The **Degree of Financial Leverage (DFL)** is a **quantitative measure** that shows how sensitive a company’s **Earnings Per Share (EPS)** is to a change in its **Earnings Before Interest and Taxes (EBIT)**.

It indicates the **responsiveness of EPS** to changes in EBIT, due to the presence of **fixed financial charges** such as interest or preference dividends.

Formula for DFL:

$$\text{DFL} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}} = \frac{\Delta \text{EPS} / \text{EPS}}{\Delta \text{EBIT} / \text{EBIT}}$$

or, at a particular level of EBIT:

$$\text{DFL at EBIT} = \frac{\text{EBIT}}{\text{EBIT} - I - \frac{D_p}{1 - t}}$$

where:

- 🔑 EBIT = Earnings Before Interest and Tax
- 🔑 I = Interest on long-term debt
- 🔑 D_p = Preference dividend
- 🔑 t = Corporate income tax rate
- 🔑 N = Number of equity shares outstanding
- 🔑 EPS = Earnings per share

$$\text{EPS} = \frac{(\text{EBIT} - I)(1 - t) - D_p}{N}$$

Example: XYZ Ltd. currently has an equity capital base of ₹ 10,00,000, represented by 20,000 equity shares of ₹ 10 each. The company plans to raise an additional ₹ 5,00,000 to finance an expansion project. The additional funds can be raised through any of the following three financing alternatives:

- ◆ Issue of **equity shares**
- ◆ Issue of **debt at 12% interest**
- ◆ Issue of **preference shares with 11% dividend**

The company's current EBIT is ₹ 1,50,000 which is expected to increase to ₹ 2,70,000 post-expansion. The corporate tax rate is 40%.

You are required to calculate for each financing option:

- ◆ **Earnings per Share (EPS)** at EBIT levels of ₹ 1,50,000 and ₹ 2,70,000
- ◆ **Degree of Financial Leverage (DFL)** at EBIT = ₹ 2,70,000
- ◆ Provide a **comparative analysis** of all three financing alternatives.

Solution: Formula for Degree of Financial Leverage (DFL):

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBIT} - \text{Fixed Financial Charges (after - tax adjusted)}}$$

6. DFL Under Debt Financing Option:

- 🔑 Additional funds: ₹ 5,00,000
- 🔑 Interest rate: 12%
- 🔑 Interest: ₹ 5,00,000 × 12% = ₹ 60,000

DFL at EBIT = ₹ 2,70,000

Since interest is **tax-deductible**, no adjustment is required:

$$\text{DFL} = \frac{2,70,000}{2,70,000 - 60,000} = \frac{2,70,000}{2,10,000} = \boxed{1.29}$$

7. DFL Under Preference Share Financing Option:

- 🔑 Dividend on preference shares = 11% of ₹ 5,00,000 = ₹ 55,000

Since **preference dividends are not tax-deductible**, we adjust for after-tax impact using:

$$\text{Adjusted Fixed Financial Cost} = \frac{D_p}{(1 - t)} = \frac{55,000}{0.60} = ₹ 91,667$$

DFL at EBIT = ₹ 2,70,000

$$\text{DFL} = \frac{2,70,000}{2,70,000 - 91,667} = \frac{2,70,000}{1,78,333} \approx \boxed{1.51}$$

8. Interpretation and Insights:

- Even though the **absolute fixed cost** of preference dividends (₹ 55,000) is **less than** the interest cost (₹ 60,000), the **DFL is higher** under the preference share option.
- This is due to the **tax deductibility of interest**, which lowers the effective cost of debt financing.
- Higher DFL** means **greater volatility** in EPS with respect to changes in EBIT.
- So, although preference shares may carry less risk of cash insolvency compared to debt, they **create more EPS variability** due to their **non-tax deductible** nature.

9. Conclusion

- DFL is a useful risk-analysis tool for financial managers.
- It helps in assessing the financial risk and understanding how different financing options impact shareholder returns.
- High DFL indicates high financial risk, especially in volatile EBIT scenarios.
- Decision-makers should balance the trade-off between higher potential returns and increased earnings volatility when choosing between equity, debt, or preference share financing.

F. EARNING BEFORE INTEREST AND TAX – EARNING PER SHARE BREAK-EVEN OR INDIFFERENCE ANALYSIS

EBIT–EPS analysis is a financial tool used to evaluate the impact of different financing alternatives—such as equity, debt, or preference shares—on a company's earnings per share (EPS) at varying levels of earnings before interest and taxes (EBIT).

The key objective is to identify the **Indifference Point**, which is the level of EBIT at which two financing options result in **identical EPS**, thereby allowing informed financing decisions.

Example: XYZ Ltd. currently has ₹ 10,00,000 in equity capital (comprising 20,000 shares). The company proposes to raise an additional ₹ 5,00,000 for expansion, considering three alternative financing options:

- Equity shares:** Issue of 10,000 shares of ₹ 50 each.
- Debt Financing:** Borrowing ₹ 5,00,000 at 12% interest (₹ 60,000 per annum)
- Preference Shares:** Issue of preference shares carrying 11% dividend (₹ 55,000 per annum)

Assumptions:

- ◆ Expected EBIT after expansion: ₹ 2,70,000
- ◆ Corporate tax rate: 40%

EPS Computation at EBIT = ₹ 2,70,000

Particulars	Equity Financing	Debt Financing	Preference Share Financing
EBIT	2,70,000	2,70,000	2,70,000
Less: Interest	—	60,000	—

Earnings Before Tax (EBT)	2,70,000	2,10,000	2,70,000
Less: Tax @ 40%	1,08,000	84,000	1,08,000
Earnings After Tax (EAT)	1,62,000	1,26,000	1,62,000
Less: Preference Dividend	—	—	55,000
Earnings Available to Equity Holders	1,62,000	1,26,000	1,07,000
No. of Equity Shares	30,000	20,000	20,000
EPS (₹)	5.40	6.30	5.35

Interpretation

- ◆ **Debt financing** offers the highest EPS at the given EBIT due to the tax shield on interest.
- ◆ **Equity financing** yields the lowest EPS due to dilution (increased number of shares).
- ◆ **Preference share financing** offers moderate EPS but is less efficient than debt due to non-tax-deductibility of dividends.

Case 1:

$$\frac{(EBIT - 0)(1 - 0.4)}{30,000} = \frac{(EBIT - 60,000)(1 - 0.4)}{20,000}$$

$$\frac{EBIT \times 0.6}{30,000} = \frac{(EBIT - 60,000) \times 0.6}{20,000}$$

Multiply both sides by LCM (60,000):

$$EBIT \times 2 = (EBIT - 60,000) \times 3$$

$$2EBIT = 3EBIT - 1,80,000$$

$$EBIT = 1,80,000$$

Case 2:

$$\frac{(EBIT - 0)(1 - 0.4)}{30,000} = \frac{(EBIT - 55,000)(1 - 0.4)}{20,000}$$

$$\frac{EBIT \times 0.6}{30,000} = \frac{(EBIT - 55,000) \times 0.6}{20,000}$$

Multiply both sides by LCM (60,000):

$$EBIT \times 2 = (EBIT - 55,000) \times 3$$

$$2EBIT = 3EBIT - 1,65,000$$

$$EBIT = 2,75,000$$

a. INDIFFERENCE POINT ANALYSIS

The indifference point is the level of EBIT at which EPS remains the same under two financing alternatives. It helps in selecting the most appropriate financing option based on expected profitability.

1. Given Data

- ◆ Debt vs Equity: Indifference point = ₹ 1,80,000 EBIT
- ◆ Preference vs Equity: Indifference point = ₹ 2,75,000 EBIT

2. Decision Guidelines

- ◆ EBIT below ₹ 1,80,000: Equity is preferred over debt
- ◆ EBIT above ₹ 1,80,000: Debt becomes more beneficial

- ♦ EBIT below ₹ 2,75,000: Equity is better than preference shares
- ♦ EBIT above ₹ 2,75,000: Preference shares yield better EPS than equity

Formula

To determine the indifference point algebraically between two financing options:

$$\frac{(\text{EBIT} - I_1)(1 - t) - Dp_1}{N_1} = \frac{(\text{EBIT} - I_2)(1 - t) - Dp_2}{N_2}$$

Where:

EBIT = Earnings Before Interest and Tax

I = Interest expense

Dp = Preference dividend

t = Corporate income tax rate

N = Number of equity shares

3. Cost Structures



4. Marginal Costing

- ♦ **Contribution Margin** = Sales – Variable Costs
- ♦ **Profit** = Contribution – Fixed costs

Marginal Cost Statement (Single Product Example)

Particulars	Amount (₹)
Sales Revenue	XXXXX
Less: Variable Costs	XXXXX
Contribution	XXXXX
Less: Fixed Costs	XXXXX
Net Profit	XXXXX

5. Break-even Analysis

Break-even analysis is a fundamental financial and managerial accounting tool used to determine the level of sales at which a business neither earns a profit nor incurs a loss. It helps management understand the relationship between **costs, sales volume, and profits**, and supports decision-making regarding pricing, production levels, and cost control.

(a) Objective of Break-even Analysis

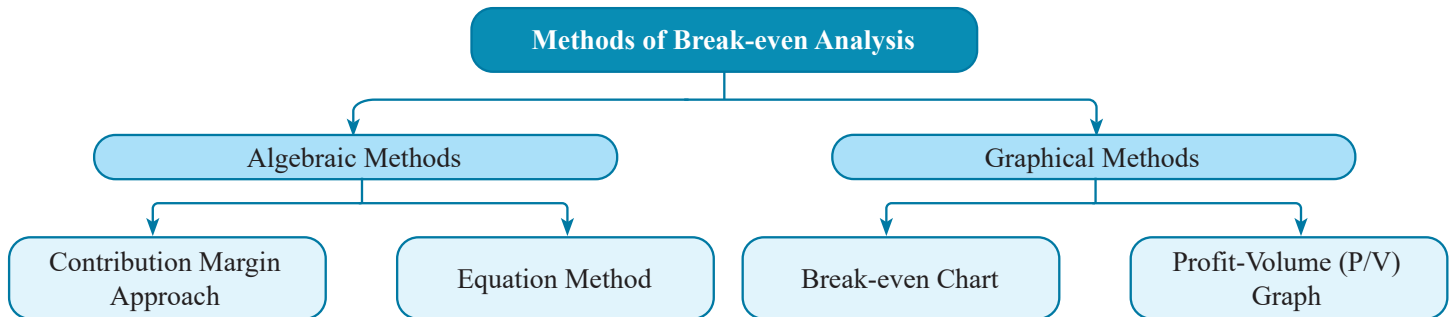
The primary objective is to determine the **Break-even Point (BEP)** — the sales volume at which **total revenues are exactly equal to total costs**, resulting in **zero net income**.

At this point:

- ◆ Total Contribution = Total Fixed Costs
- ◆ Profit = 0

(b) Methods for Determining the Break-even Point

The break-even point can be determined using two main approaches:



(i) Algebraic Methods

These are mathematical methods to compute BEP using formulas.

(1) Contribution Margin Approach

This approach uses the concept of contribution, which is the difference between sales and variable costs.

$$\text{Break-even Sales (Units)} = \frac{\text{Fixed Costs}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$

$$\text{Break-even Sales (₹)} = \frac{\text{Fixed Costs}}{\text{Contribution Margin Ratio}}$$

Where:

- ◆ Contribution per unit = Selling Price per unit – Variable Cost per unit
- ◆ Contribution Margin Ratio = Contribution per unit ÷ Selling Price per unit

(2) Equation Method

This method sets up an equation where:

$$\text{Sales Revenue} = \text{Fixed Costs} + \text{Variable Costs}$$

$$\text{or, (Selling Price} \times \text{Quantity)} = \text{Fixed Costs} + (\text{Variable Cost per unit} \times \text{Quantity})$$

Solving this equation provides the break-even quantity.

(ii) Graphical Methods

A graphical presentation visually demonstrates the break-even point and the relationship among cost, volume, and profit.

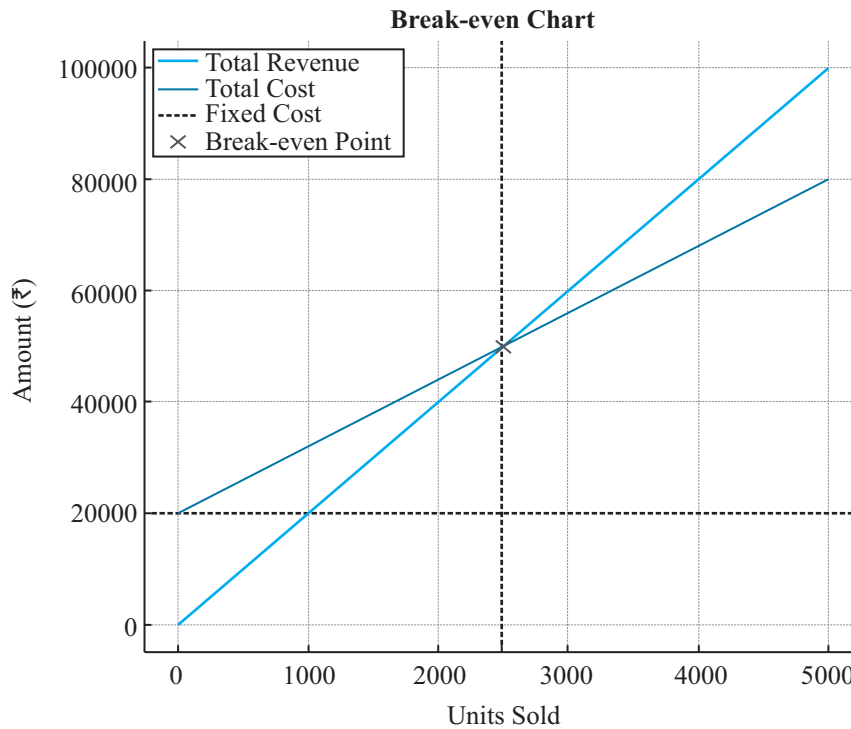
(1) Break-even Chart

A **break-even chart** is a graphical representation that illustrates:

- Total Revenue Line
- Total Cost Line (Fixed + Variable)
- Fixed Cost Line
- Break-even Point (intersection of total revenue and total cost lines)

Key Features:

- The **horizontal axis (X-axis)** shows the level of activity (units sold or produced).
- The **vertical axis (Y-axis)** shows the amount in Rupees (sales, costs, profits).
- The point where the **total revenue line intersects the total cost line** is the **break-even point**.



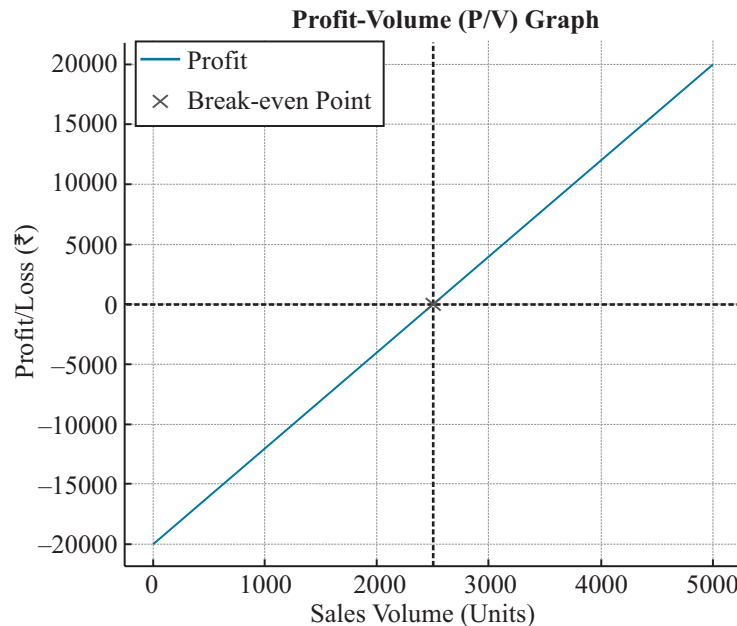
- ◆—: Total Revenue
- ◆—: Total Cost (Fixed + Variable)
- ◆---: Fixed Cost
- ◆ Intersection Point: Break-even Point (Revenue = Total Cost)

(2) Profit-Volume (P/V) Graph

The **P/V graph** directly shows the relationship between profit (or loss) and sales volume.

Key Features:

- The X-axis represents the level of output or sales.
- The Y-axis represents profit or loss.
- The graph starts from the negative profit (loss) zone and moves upwards.
- The point where the profit line crosses the X-axis is the **break-even point**.
- The slope of the line indicates the **contribution margin**.



Profit-Volume (P/V) Graph:

- 🔵 **Blue:** Profit as sales volume increases
- 🔵 **X-axis intersection:** Break-even Point where Profit = 0
- 🔵 Shows movement from **loss zone to profit zone**

TEST YOURSELF

1. **Harshita Ltd.** has an EBIT of ₹ 4,00,000 and belongs to a risk class of 10%, i.e., its overall cost of capital is 10%. What is the value of equity capital if it employs debt at 5% to the extent of 30%, 40%, or 50% of the total capital of ₹ 20,00,000?

Assume that the Net Operating Income (NOI) approach applies.

Solution.

Particulars	30% Debt	40% Debt	50% Debt
EBIT (A)	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000
Overall Cost of Capital (Ko)	10%	10%	10%
Value of the Firm ($V = \text{EBIT}/K_o$)	₹ 40,00,000	₹ 40,00,000	₹ 40,00,000
Value of Debt (D) (30%, 40%, 50% of ₹ 20,00,000)	₹ 6,00,000	₹ 8,00,000	₹ 10,00,000
Value of Equity ($E = V - D$)	₹ 34,00,000	₹ 32,00,000	₹ 30,00,000
Interest on Debt @5% (B)	₹ 30,000	₹ 40,000	₹ 50,000
Net Profit for Equity Shareholders ($A - B$)	₹ 3,70,000	₹ 3,60,000	₹ 3,50,000
Cost of Equity ($K_e = \text{Net Profit}/\text{Equity}$)	10.88%	11.25%	11.67%

Conclusion

The cost of equity capital (K_e) increases with an increase in the proportion of debt capital. This is due to the increased financial risk borne by equity shareholders as debt increases in the capital structure.

2. **Nakshatra Ltd.** has reported a Net Operating Income (EBIT) of ₹ 5,00,000. The company has currently employed ₹ 15,00,000 of debt at an interest rate of 10%. The overall cost of capital (K_o) for the firm is 15%. You are required to calculate:
- Total value of the Firm
 - Cost of Equity

Solution.

(i) Statement Showing Value of the Firm

Particulars	Amount (₹)
Net Operating Income (EBIT)	5,00,000
Less: Interest on Debt (10% of ₹15,00,000)	1,50,000
Earnings Available to Equity Shareholders	3,50,000
Total Cost of Capital (K_o)	15%
Value of the Firm ($V = \text{EBIT}/K_o$)	33,33,333

(ii) Calculation of Cost of Equity (K_e)

Particulars	Amount (₹)
Market Value of Debt (D)	15,00,000
Market Value of Equity ($S = V - D$)	18,33,333

Earnings for Equity Shareholders	3,50,000
$K_e = \frac{\text{Earnings for Equity Shareholders}}{\text{Market Value of Equity}} \times 100$ $K_e = \frac{3,50,000}{18,33,333} \times 100 = 19.09\%$	

Alternatively, you can use the formula:

$$K_e = K_0 + (K_0 - K_d) \times \frac{D}{S}$$

Where

☛ $K_0 = 15\%$

☛ $K_d = 10\%$

☛ $D = ₹ 15,00,000$

☛ $S = ₹ 18,33,333$

$$K_e = 15\% + (15\% - 10\%) \times \frac{15,00,000}{18,33,333} = 15\% + 5\% \times 0.8181 = 15\% + 4.09\% = 19.09\%$$

3. Debashish Ltd. and Suraj Ltd. are identical companies in all respects except for their capital structures.

☛ Debashish Ltd. has 50% debt and 50% equity,

☛ Suraj Ltd. has 20% debt and 80% equity,

(All percentages are based on market values.)

The borrowing rate for both companies is 8% and we assume a no-tax world with perfect capital markets.

(a) (i) If you own 2% of the shares of Debashish Ltd., determine your return if the company has Net Operating Income (NOI) of ₹ 3,60,000 and the overall capitalization rate (K_0) is 18%.

(ii) Calculate the implied rate of return on equity of Debashish Ltd.

(b) Suraj Ltd. has the same Net Operating Income (NOI) as Debashish Ltd.

(i) Determine the implied required equity return of Suraj Ltd.

(ii) Explain why it differs from Debashish Ltd.

Solution. (a) (i) Value of Debashish Ltd.

$$\text{Value of the firm (V)} = \frac{\text{NOI}}{K_0} = \frac{3,60,000}{0.18} = ₹ 20,00,000$$

☛ Market value of debt (50%) = ₹ 10,00,000

☛ Market value of equity (50%) = ₹ 10,00,000

Earnings available to shareholders:

$$\text{Interest on debt} = 8\% \times 10,00,000$$

$$= ₹ 80,000 \text{ Earnings to equity} = ₹ 3,60,000 - ₹ 80,000 = ₹ 2,80,000$$

Return on 2% shareholding

$$2\% \times ₹ 2,80,000 = ₹ 5,600$$

(ii) Implied rate of return on equity (K_e) for Debashish Ltd.

$$K_e = \frac{2,80,000}{10,00,000} \times 100 = \boxed{28\%}$$

(b) (i) **For Suraj Ltd.:**

- ◆ Total firm value = ₹ 20,00,000
- ◆ Market value of debt (20%) = ₹ 4,00,000
- ◆ Market value of equity (80%) = ₹ 16,00,000

Interest on debt = $8\% \times ₹ 4,00,000 = ₹ 32,000$ Earnings to equity = ₹ 3,60,000 – ₹ 32,000 = ₹ 3,28,000

$$K_e = \frac{3,28,000}{16,00,000} \times 100 = \boxed{20.5\%}$$

(ii) **Reason for Difference:**

The required return on equity for **Suraj Ltd.** is **lower** than that of **Debashish Ltd.** because it uses **less debt** in its capital structure.

According to the **Net Operating Income (NOI) approach**, the cost of equity increases as the proportion of debt increases (to compensate for higher financial risk). Since Debashish Ltd. uses more debt (50%) compared to Suraj Ltd. (20%), its equity shareholders demand a higher return (28% vs. 20.5%)

This supports the idea that equity capitalization rate is a linear function of debt-equity ratio, and under a no-tax, perfect match scenario, overall firm value remains constant, regardless of the capital structure.

4. Let us consider the case of two firms – **Debashish Ltd. and Tanuja Ltd.** – which are identical in all respects except for their capital structures.

- ◆ **Debashish Ltd.** is fully financed by **equity only**
- ◆ **Tanuja Ltd.** is financed by a **mix of equity and debt**.

Below are their financial details:

Financial Particulars of Debashish Ltd. and Tanuja Ltd.

(Amount in ₹)

Particulars	Debashish Ltd.	Tanuja Ltd.
Total Capital Employed	10,00,000	10,00,000
Equity Capital	10,00,000	6,00,000
Debt	Nil	4,00,000
Net Operating Income	1,00,000	1,00,000
Debt Interest	0	20,000
Market Value of Debt	0	4,00,000
Equity Earnings	1,00,000	80,000
Equity Capitalization Rate	10%	12%
Market Value of Equity	10,00,000	
Total Market Value of Firm	10,00,000	10,66,667
Average Cost of Capital	10%	9.37%
Debt-Equity Ratio	0	0.6

From the above, it appears that Tanuja Ltd., which is leveraged, has a higher total market value than **Debashish Ltd.** According to the **Modigliani-Miller (MM) approach**, this kind of situation cannot persist in a perfect capital market.

Solution:

To demonstrate the MM arbitrage argument:

- ◆ Suppose an investor holds 1% equity in Tanuja Ltd.
- ◆ Market value = 1% of ₹ 6,66,667 = ₹ 6,667
- ◆ He can sell this investment and do the following:
 - ☞ Borrow ₹ 4,000 at 5% interest on a personal basis
 - ☞ Invest ₹ 10,667 (₹ 6,667 + ₹ 4,000) to buy 1.0667% equity in Debashish Ltd.

This will yield:

Particulars	Amount in ₹
Income from investment in Debashish Ltd. (1.0667% of ₹ 1,00,000)	1,066.70
Less: Interest on personal debt (5% of ₹ 4,000)	200.00
Net Income	866.70

This net income of ₹866.70 is higher than the earlier ₹ 800 (1% of ₹ 80,000 from Tanuja Ltd.).

As a result, the market value of equity of Tanuja Ltd. will fall, and that of Debashish Ltd. will rise due to investor action - until they become equal, eliminating the arbitrage opportunity.

Now consider the **reverse situation**, where **Debashish Ltd. has a higher total market value than Tanuja Ltd.**, as shown below:

Particulars	Debashish Ltd.	Tanuja Ltd.
Debt Interest	0	20,000
Market Value of Debt	0	4,00,000
Equity Earnings	1,00,000	80,000
Equity Capitalization Rate	8%	12%
Market Value of Equity	12,50,000	6,66,667
Total Market Value	12,50,000	10,66,667

In this case:

- ◆ An investor owning **1% equity in Debashish Ltd.** (₹12,500) can:
 - ☞ Sell it for ₹12,500
 - ☞ Reinvest by buying 1.01% of **Tanuja Ltd's equity and debt**, using ₹ 10,773

This would yield a net income **increase of ₹ 1,727, without increasing risk.**

Again, investor actions will push the **market value of Debashish Ltd. down** and **Tanuja Ltd. up**, until the total **values equalize**, confirming MM's argument that **capital structure is irrelevant in perfect markets.**

5. From the following selected operating data, determine the **degree of operating leverage (DOL)**. Which company has the greater amount of **business risk**? Explain with reason.

(Amount in ₹)

Particulars	Bijli Ltd.	Reo Ltd.
Sales	25,00,000	30,00,000
Fixed costs	7,50,000	15,00,000
Variable costs	50% of sales	25% of sales

Solution. Statement of Profit:

Particulars	Bijli Ltd.	Reo Ltd.
Sales	25,00,000	30,00,000
Less: Variable costs	12,50,000 (50%)	7,50,000 (25%)
Contribution	12,50,000	22,50,000
Less: Fixed costs	7,50,000	15,00,000
Operating Profit (EBIT)	5,00,000	7,50,000

Degree of Operating Leverage (DOL):

$$DOL = \frac{\text{Contribution}}{\text{Operating Profit (EBIT)}}$$

◆ For Bijli Ltd.

$$DOL = \frac{12,50,000}{5,00,000} = \boxed{2.5}$$

◆ For Reo Ltd.

$$DOL = \frac{22,50,000}{7,50,000} = \boxed{3.0}$$

Conclusion:

Since Reo Ltd. has a higher operating leverage (3.0) compared to Bijli Ltd. (2.5), it means:

- ◆ Reo Ltd. is exposed to higher business risk.
- ◆ Small changes in sales will cause greater proportional changes in operating profit (EBIT) for Reo Ltd. due to its higher fixed costs.
- ◆ Therefore, Reo Ltd.'s earnings are more volatile and sensitive to sales fluctuations.

6. Tanuja Ltd. has the following capital structure:

Particulars	Amount (₹)
Equity Share Capital	1,00,000
10% Preference Share Capital	1,00,000
8% Debentures	1,25,000

The company's current **Earnings Before Interest and Tax (EBIT)** is ₹ 50,000.

Assume that the company falls under a **50% tax bracket**.

You are required to calculate the **financial leverage** of Tanuja Ltd.

Solution.

Particulars	Amount (₹)
Earnings Before Interest & Tax (EBT)	50,000
Less: Interest on Debentures (8% of ₹ 1,25,000)	10,000
Earnings Before Tax (EBT)	40,000
Less: Income Tax @ 50% of EBT	20,000
Profit After Tax	20,000

Financial Leverage Calculation:

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{50,000}{40,000} = \boxed{1.25}$$

Interpretation:

A financial leverage of 1.25 means that for every 1% change in EBIT, the EBT will change by 1.25%, indicating the presence of debt and its effect on the firm's earnings volatility due to fixed interest obligations.

7. **Ridhima Ltd.** is evaluating two alternative financial plans to raise a total investment of ₹ 50,000. The details of the two plans are as follows:

Particulars	Plan A	Plan B
Debentures (Interest @ 10%)	₹ 40,000	₹ 10,000
Equity Share Capital (₹ 10 each)	₹ 10,000	₹ 40,000
Total Investment Required	₹ 50,000	₹ 50,000
Number of Equity Shares	1,000 shares	4,000 shares

Assume that **Earnings Before Interest and Tax (EBIT)** are ₹ 5000 and ₹ 12,500 respectively under both plans. The applicable tax rate is 50%.

You are required to calculate the **Earnings Per Share (EPS)** under both plans for each EBIT level.

Solution. When EBIT is ₹5,000

Particulars	Plan A	Plan B
EBIT	5,000	5,000
Less: Interest (10% of debt)	4,000	1,000
Earnings Before Tax (EBT)	1,000	4,000
Less: Tax @ 50%	500	2,000
Earnings for Equity Shareholders	500	2,000
No. of Equity Shares	1,000	4,000
EPS (Earnings ÷ No. of shares)	₹ 0.50	₹ 0.50

When EBIT is ₹12,500

Particulars	Plan A	Plan B
EBIT	12,500	12,500
Less: Interest	4,000	1,000
Earnings Before Tax (EBT)	8,500	11,500
Less: Tax @ 50%	4,250	5,750
Earnings for Equity Shareholders	4,250	5,750
No. of Equity Shares	1,000	4,000
EPS (Earnings ÷ No. of shares)	₹ 4.25	₹ 1.44

Conclusion:

- ◆ At lower EBIT (₹ 5,000), both plans result in the same EPS of ₹ 0.50.
- ◆ At higher EBIT (₹ 12,500), Plan A yields a significantly higher EPS (₹ 4.25) than Plan B (₹ 1.44), due to the benefit of financial leverage.
- ◆ Hence, Plan A is more favorable when the company expects higher EBIT, but it carries higher financial risk due to greater debt.

8. Shefali Ltd. has the following capital structure and wants to determine its **Financial Break-Even Point (FBEP)**:

Particulars	Amount (₹)
Equity Shares (Face Value ₹ 100)	₹ 5,00,000
12% Preference Shares (Face Value ₹ 100)	₹ 5,00,000
10% Debentures (Face Value ₹ 100)	₹ 10,00,000
Tax Rate	40%

You are required to compute the Financial Break-Even Point of Shefali Ltd.

Solution.

Step 1: Calculate Fixed Financial Obligations

- ◆ **Interest on Debentures** (10% of ₹ 10,00,000) = ₹ 1,00,000
- ◆ **Preference Dividend** (12% of ₹ 5,00,000) = ₹ 60,000

Since preference dividend is paid from **post-tax income**, we gross it up:

$$\text{Grossed-up Preference Dividend} = \frac{\text{₹ 60,000}}{1 - 0.40} = \text{₹ 1,00,000}$$

Financial Break-Even Point (FBEP)

$$\text{FBEP} = \text{Interest} + \text{Grossed-up Preference Dividend} = \text{₹ 1,00,000} + \text{₹ 1,00,000} = \text{₹ 11,00,000}$$

Interpretation:

Shefali Ltd.'s EPS will be zero when EBIT = ₹ 11,00,000, as the entire operating profit goes toward servicing debt and preference dividends.

Verification Table:

Particulars	Amount (₹)
EBIT	11,00,000
Less: Interest on Debentures	1,00,000
EBT	10,00,000
Less: Tax @ 40%	4,00,000
EAT (Earnings After Tax)	6,00,000
Less: Preference Dividend	6,00,000
Earnings for Equity Holders	Nil
No. of Equity Shares (FV ₹100)	5,000
EPS	₹ 0.00

9. Prashant Ltd. is planning to launch a new project requiring a total capital outlay of **₹400 lakhs**. The company is considering **two financing options**:

- ◆ **Option A:** Raise the entire amount through **equity shares** of ₹ 100 each .
- ◆ **Option B:** Raise **₹ 200 lakhs through equity shares** and **₹ 200 lakhs through a loan** at 15% interest.

Assuming the applicable **tax rate is 40%**, calculate the **EBIT level** at which the **equity shareholders would** be indifferent between the two financing options.

Solution.

Particulars	Option A (Full Equity)	Option B (Equity + Debt)
Equity Capital (₹ 100 FV)	₹ 400 Lakhs	₹ 200 Lakhs
15% Loan	Nil	₹ 200 Lakhs
Total Capital	₹ 400 Lakhs	₹ 400 Lakhs
No. of Equity Shares	4,00,000	2,00,000

Let the indifference level of EBIT = X

Set up EPS equality equation:

$$\frac{(X - 0)(1 - 0.4)}{4,00,000} = \frac{(X - 30,000)(1 - 0.4)}{2,00,000} = \frac{0.6 X}{4,00,000} = \frac{0.6(X - 30,000)}{2,00,000}$$

Multiply both sides by 4,00,000:

$$0.6 X = 1.2 X - 36,00,000$$

$$36,00,000 = 1.2 X - 0.6 X = 0.6 X \Rightarrow X = ₹ 60,00,000$$

Verification:

Particulars	Option A (₹)	Option B (₹)
EBIT	60,00,000	60,00,000
Less: Interest	Nil	30,00,000
EBT	60,00,000	30,00,000
Less: Tax @40%	24,00,000	12,00,000
Earnings for Equity Holders	36,00,000	18,00,000
No. of Equity Shares	4,00,000	2,00,000
EPS	₹ 9	₹ 9

Conclusion:

The EBIT level at which both financing options result in the same EPS is ₹ 60, 00,000.

At this level, Prashant Ltd.'s equity shareholders would be indifferent between full equity financing and the combination of equity and debt.

10. Ritik Ltd. has reported the following financial details:

- ☛ Sales: ₹ 25,00,000
- ☛ Variable Cost: ₹ 15,00,000
- ☛ Fixed Cost: ₹ 5,00,000
- ☛ Debt: ₹ 12,50,000 at an interest rate of 8%

You are required to calculate the **Combined leverage** of Ritik Ltd.

Solution.**Step 1: Prepare Statement of Profit**

Particulars	Amount (₹)
Sales	25,00,000
Less: Variable Cost	15,00,000
Contribution	10,00,000
Less: Fixed Cost	5,00,000
Operating Profit (EBIT)	5,00,000

Step 2: Calculate Operating Leverage

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{10,00,000}{5,00,000} = \boxed{2}$$

Step 3: Calculate Financial Leverage

Particulars	Amount (₹)
EBIT	5,00,000
Less: Interest (8% of ₹ 12,50,000)	1,00,000
Earnings Before Tax	4,00,000

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{5,00,000}{4,00,000} = \boxed{1.25}$$

Step 4: Calculate Combined Leverage

$$\text{Combined Leverage} = \text{Operating Leverage} \times \text{Financial Leverage} = 2 \times 1.25 = 2.5$$

OR directly:

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{10,00,000}{4,00,000} = \boxed{2.5}$$

Conclusion:

The Combined Leverage of Ritik Ltd. is 2.5, which indicates that a 1% change in sales will cause a 2.5% change in earnings before tax (EST), highlighting the impact of both operating and financial leverage.

11. The following financial data is available for two companies: **Sameer Ltd.** and **Sana Ltd.**

You are required to compare the **sensitivity of earnings** of the two companies to a **30% change in the level** of their current assets, using the concept of **Working Capital leverage**.

Data:

Particulars	Sameer Ltd. (₹)	Sana Ltd. (₹)
Fixed Assets	4,00,000	1,00,000
Current Assets	10,00,000	4,00,000
Total Assets	14,00,000	14,00,000
EBIT (Earnings Before Interest & Tax)	1,50,000	1,50,000

Solution. Formula for Working Capital Leverage (WCL):

For Sameer Ltd.:

- ◆ 30% decrease in Current Assets = 30% of ₹ 10,00,000 = ₹ 3,00,000
- ◆ New Current Assets = ₹ 10,00,000 – ₹ 3,00,000 = ₹ 7,00,000
- ◆ New Total Assets = ₹ 4,00,000 + ₹ 7,00,000 = ₹ 11,00,000

$$\text{WCL (Sameer Ltd.)} = \frac{10,00,000}{11,00,000} = \boxed{0.91}$$

For Sana Ltd.:

- ◆ 30% decrease in Current Assets = 30% of ₹ 4,00,000 = ₹ 1,20,000
- ◆ New Current Assets = ₹ 4,00,000 – ₹ 1,20,000 = ₹ 2,80,000
- ◆ New Total Assets = ₹ 1,00,000 + ₹ 2,80,000 = ₹ 3,80,000

$$\text{WCL (Sana Ltd.)} = \frac{4,00,000}{12,80,000} = \boxed{0.3125}$$

Conclusion:

- ◆ Sameer Ltd. has a higher Working Capital leverage (0.91) than Sana Ltd. (0.3125).
- ◆ This implies that earnings of Sameer Ltd. are more sensitive to changes in current assets, meaning it faces higher working capital risk.
- ◆ A 30% change in working capital affects Sameer Ltd.'s performance more strongly than Sana Ltd.

12. Calculate the **Operating leverage**, **Financial leverage**, and **Combined Leverage** under **Situation 1** and **Situation 2** for **Financial Plans X and Y** respectively, based on the following details related to the operating and capital structure of a company. Also, determine **which combination gives the highest** and **which gives the least** combined leverage.

Details:

- ◆ Installed capacity = 5,000 units
- ◆ Actual production & sales = 60% of installed capacity = 3,000 units
- ◆ Selling Price per unit = ₹ 25
- ◆ Variable Cost per unit = ₹ 15
- ◆ Fixed Cost:
 - ☛ Situation 1 = ₹ 10,000
 - ☛ Situation 2 = ₹ 12,000

Financial Plans:

Particulars	Plan X (₹)	Plan Y (₹)
Equity	25,000	50,000
Debt@10%	50,000	25,000
Total	75,000	75,000

Step 1: Contribution Calculation

- ◆ Contribution per unit = Selling Price – Variable Cost = ₹ 25 – ₹ 15 = ₹ 10
- ◆ Total Contribution = 3,000 units × ₹ 10 = ₹ 30,000

Step 2: Computation of Leverages**Plan X**

Particulars	Situation 1 (₹)	Situation 2 (₹)
Contribution	30,000	30,000
Fixed cost	10,000	12,000
EBIT	20,000	18,000
Interest (10% of 50,000)	5,000	5,000
EBT	15,000	13,000
Operating Leverage (30,000 ÷ EBIT)	1.5	1.67
Financial Leverage (EBIT ÷ EBT)	1.33	1.38
Combined Leverage (OL × FL)	2.00	2.31

Plan Y

Particulars	Situation 1 (₹)	Situation 2 (₹)
Contribution	30,000	30,000
Fixed cost	10,000	12,000
EBIT	20,000	18,000
Interest (10% of 25,000)	2,500	2,500
EBT	17,500	15,500

Operating Leverage (30,000 ÷ EBIT)	1.5	1.67
Financial Leverage (EBIT ÷ EBT)	1.14	1.16
Combined Leverage (OL × FL)	1.71	1.94

Conclusion:

- ◆ **Highest Combined leverage = 2.31, under Situation 2 - Plan X**
- ◆ **Lowest Combined Leverage = 1.71, under Situation 1 - Plan Y**

13. Lucy Ltd. is evaluating three alternative financial plans. You are required to calculate the **Financial Leverage (FL)** under each plan.

Particulars	Plan I	Plan II	Plan III
Equity Capital	₹ 2,000	₹ 1,000	₹ 3,000
Debt	₹ 2,000	₹ 3,000	₹ 1,000
EBIT	₹ 400	₹ 400	₹ 400
Interest rate	10% p.a. on all debt		

Solution.

Step 1: Calculate Interest on Debt

- ◆ Plan I: 10% of ₹2,000 = ₹ 200
- ◆ Plan II: 10% of ₹3,000 = ₹ 300
- ◆ Plan III: 10% of ₹1,000 = ₹ 00

Step 2: Prepare Financial leverage Table

Particulars	Plan I	Plan II	Plan III
EBIT	₹ 400	₹ 400	₹ 400
Less: Interest	₹ 200	₹ 300	₹ 100
EBT (Earnings Before Tax)	₹ 200	₹ 100	₹ 300
Financial Leverage (EBIT ÷ EBT)	2.00	4.00	1.33

Conclusion:

- ◆ **Highest financial leverage = 4.00 under Plan II (highest debt, highest risk)**
- ◆ **Lowest financial leverage = 1.33 under Plan III (lowest debt, lowest risk)**

Financial leverage indicates the sensitivity of earnings before tax (EST) to changes in EBIT. Higher the leverage, higher the financial risk.

14. Calculate the **Operating Leverage** and **Financial Leverage** under **Situations A, B, and C**, and **Financial Plans 1, 2, and 3** respectively from the following information. Also, determine **which combination gives the highest and the least** leverage values.

Given Information:

- ☛ **Installed Capacity** = 1,200 units
- ☛ **Actual Production & Sales** = 800 units
- ☛ **Selling Price per Unit** = ₹ 15
- ☛ **Variable Cost per Unit** = ₹ 10

Fixed Costs under Situations:

- ☛ **Situation A** = ₹ 1,000
- ☛ **Situation B** = ₹ 2,000
- ☛ **Situation C** = ₹ 3,000

Financial Plans:

Sources of Fund	Plan 1	Plan 2	Plan 3
Equity	₹ 5,000	₹ 7,500	₹ 2,500
Debt	₹ 5,000	₹ 2,500	₹ 7,500
Cost of Debt	12% per annum for all plans		₹ 300

Solution:**Step 1: Operating Leverage Calculation**

Total Sales (S) = 800 units × ₹ 15 = ₹ 12,000

Total Variable Cost (VC) = 800 units × ₹ 10 = ₹ 8,000

Contribution (S – VC) = ₹ 4,000

Situation	Fixed Cost	EBIT (Contribution – Fixed Cost)	Operating leverage (Contribution ÷ EBIT)
A	₹ 1,000	₹ 3,000	1.33
B	₹ 2,000	₹ 2,000	2.00
C	₹ 3,000	₹ 1,000	4.00

Step 2: Financial Leverage Calculation**Situation A (EBIT = ₹ 3,000)**

Plan	Interest (12% on Debt)	EBT	Financial Leverage (EBIT ÷ EBT)
1	₹ 600 (12% of ₹ 5,000)	₹ 2,400	1.25
2	₹ 300 (12% of ₹ 2,500)	₹ 2,700	1.11
3	₹ 900 (12% of ₹ 7,500)	₹ 2,100	1.43

Situation B (EBIT = ₹ 2,000)

Plan	Interest	EBT	Financial Leverage
1	₹ 600	₹ 1,400	1.43
2	₹ 300	₹ 1,700	1.18
3	₹ 900	₹ 1,100	1.82

Situation C (EBIT = ₹ 1,000)

Plan	Interest	EBT	Financial Leverage
1	₹ 600	₹ 400	2.50
2	₹ 300	₹ 700	1.43
3	₹ 900	₹ 100	10.00

Conclusion:

- ◆ Highest Operating Leverage = 4.00 in Situation C
- ◆ Least Operating Leverage = 1.33 in Situation A
- ◆ Highest Financial Leverage = 10.00 under Plan 3 – Situation C
- ◆ Least Financial Leverage = 1.11 under Plan 2 – Situation A

15. A company has the following financial characteristics:

- ☛ Debt-to-Equity Ratio (D/E) = 0.70 : 1.00
- ☛ Tax Rate = 34%
- ☛ Unlevered Beta (β) = 0.75

You are required to calculate the Hamada Coefficient (Leveraged Beta, β).

Solution. We use the Hamada formula:

$$\beta_L = \beta_U \left[1 + (1 - T) \left(\frac{D}{E} \right) \right]$$

Substitute the values:

- ◆ $\beta = 0.75$
 - ◆ $T = 34\% = 0.34$
 - ◆ $D/E = 0.70$
- $$\begin{aligned}\beta_L &= 0.75[1 + (1 - 0.34) \times 0.70] \\ &= 0.75[1 + 0.66 \times 0.70] \\ &= 0.75[1 + 0.462] \\ &= 0.75 \times 1.462 \\ &= 1.0965 = 1.09\end{aligned}$$

Conclusion:

- ◆ The leveraged beta (β) is 1.09
- ◆ The financial leverage increases the company's risk by: $1.09 - 0.75 = 0.34$ or 34%
- ◆ This increase reflects the additional risk taken by equity shareholders due to the presence of debt in the capital structure.

DIVIDEND DECISIONS

A. INTRODUCTION: MEANING OF DIVIDEND

A dividend is the portion of a company's profit distributed to its shareholders as a return on their investment. It is a reward for holding shares in the company.

As per ICAI, "A dividend is a distribution to shareholders out of profit or reserves available for this purpose."

B. DIVIDEND POLICY

Dividend policy refers to the company's approach to deciding what portion of profits will be distributed as dividends and what portion will be retained for future growth. It plays a key role in balancing shareholder expectations with the company's need for internal financing.

As per **Weston** and **Brigham**, dividend policy determines how earnings are split between dividend payouts and retained earnings.

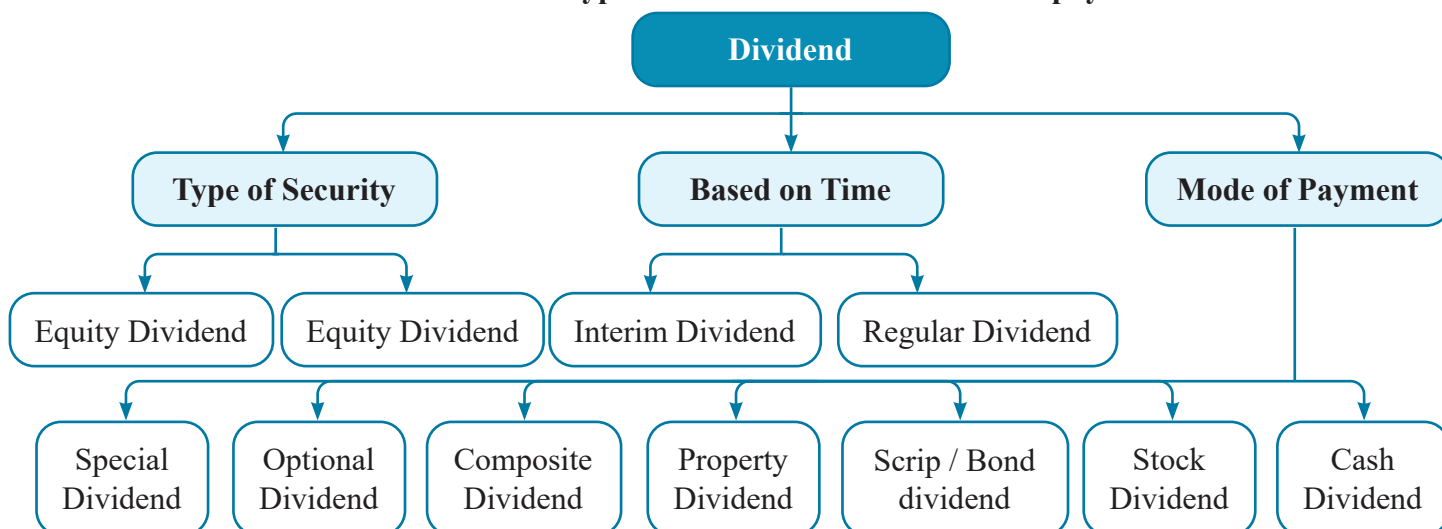
Formulating a dividend policy is challenging because:

- Dividends provide an immediate return to shareholders (cash flow).
- Retained earnings are a major source of financing expansion.
- Both dividend and growth are important, but often conflict.
- High dividends reduce funds for growth; low dividends may dissatisfy investors.
- Companies aim to retain profits if returns exceed the cost of capital, but must also consider shareholder preferences.

Thus, dividend policy is a strategic decision that affects both shareholder satisfaction and the company's long-term financial planning.

C. KINDS (FORMS) OF DIVIDEND

Dividends can be classified into different types based on source and mode of payment:



a. BASED ON THE SOURCE

1. **Profit Dividend** – Paid out of profits during regular business.
2. **Liquidation Dividend** – Paid from capital during winding up.

b. BASED ON THE MODE OF PAYMENT

1. **Equity Dividend** – Paid on equity shares; rate recommended by the Board and approved by shareholders.
2. **Preference Dividend** – Paid on preference shares at a fixed rate, before equity dividend.
3. **Interim Dividend** – Declared during the year before final accounts are closed, usually due to high earnings.
4. **Regular Dividend** – Paid at the usual rate; preferred by investors needing steady income.
5. **Cash Dividend** – Paid in cash; reduces the company's liquidity but is preferred by shareholders.
6. **Stock Dividend** – Also called bonus shares; given when cash is limited, capitalizes earnings without cash outflow.
7. **Scrip/Bond Dividend** – A promise to pay a dividend in the future via bonds/notes, carries interest; used when a company lacks immediate funds.
8. **Property Dividend** – Paid in the form of physical assets instead of cash; rare and used in special situations.
9. **Composite Dividend** – Partly in cash and partly in property.
10. **Optional Dividend** – Shareholders are given a choice between a cash dividend and a property dividend.
11. **Extra/Special Dividend** – One-time dividend declared in case of unusually high profits or reserves, without changing the regular dividend pattern.

D. STOCK SPLIT

A **stock split** is a decision by a company's board to increase the number of outstanding shares by issuing more shares to existing shareholders, **without affecting the overall value of their holdings**.

a. PURPOSE

Companies generally split stocks when the share price becomes too high, making it less attractive to small investors. A stock split reduces the per-share price and increases liquidity.

b. EFFECT ON SHAREHOLDERS

Although the number of shares held increases, the **total investment value remains the same**.

Example:

◆ Before Split:

Shares = 2,000

Price per share = 20

Total Value = ₹40,000

◆ After 2:1 Split:

Shares 4,000

Price per share = 10

Total Value = 40,000

The split makes shares more affordable but does not change the total value of the investment.

c. REASONS BEHIND STOCK SPLIT

Companies split their stock for the following main reasons:

1. **Affordability for Investors:**

High share prices may become unaffordable. A stock split reduces the price per share, making it easier for more investors to buy.

2. **Improved Liquidity:**

More shares in the market increase trading volume, making it easier to buy and sell without affecting the share price much. It also helps companies repurchase shares at lower costs.

3. **Higher Profit Potential for Investors:**

After a split, investors hold more shares. If the share price increases in the future, they benefit more from the rise.

4. **Stock Exchange Listing Requirement:**

Sometimes companies perform a reverse stock split to meet minimum share price requirements and stay listed on the stock exchange.

E. SHARE REPURCHASE

Share repurchases or stock buybacks occur **when a company buys back its own shares from the market using its cash reserves**. This reduces the total number of outstanding shares and increases the ownership percentage of remaining shareholders.

a. REASONS FOR SHARE REPURCHASE

1. **Undervalued Stock:**

The company believes its shares are undervalued and sees buyback as a good investment.

2. **Confidence Signal:**

Investors take a buyback as a sign of management's confidence in future performance.

3. **Avoid Dividend Commitment:**

If extra cash is temporary, the company may prefer a buyback over a regular dividend increase to avoid future payout obligations.

4. **Tax Advantage:**

Buybacks may offer a tax advantage since the tax is paid by the company and not by the shareholders directly, unlike dividends, which are taxed according to the investor's slab.

5. **Offset Stock Option Dilution:**

To balance the effect of issuing new shares under employee stock option plans.

b. IMPACT OF SHARE REPURCHASE:

1. **Increased EPS/CFPS:**

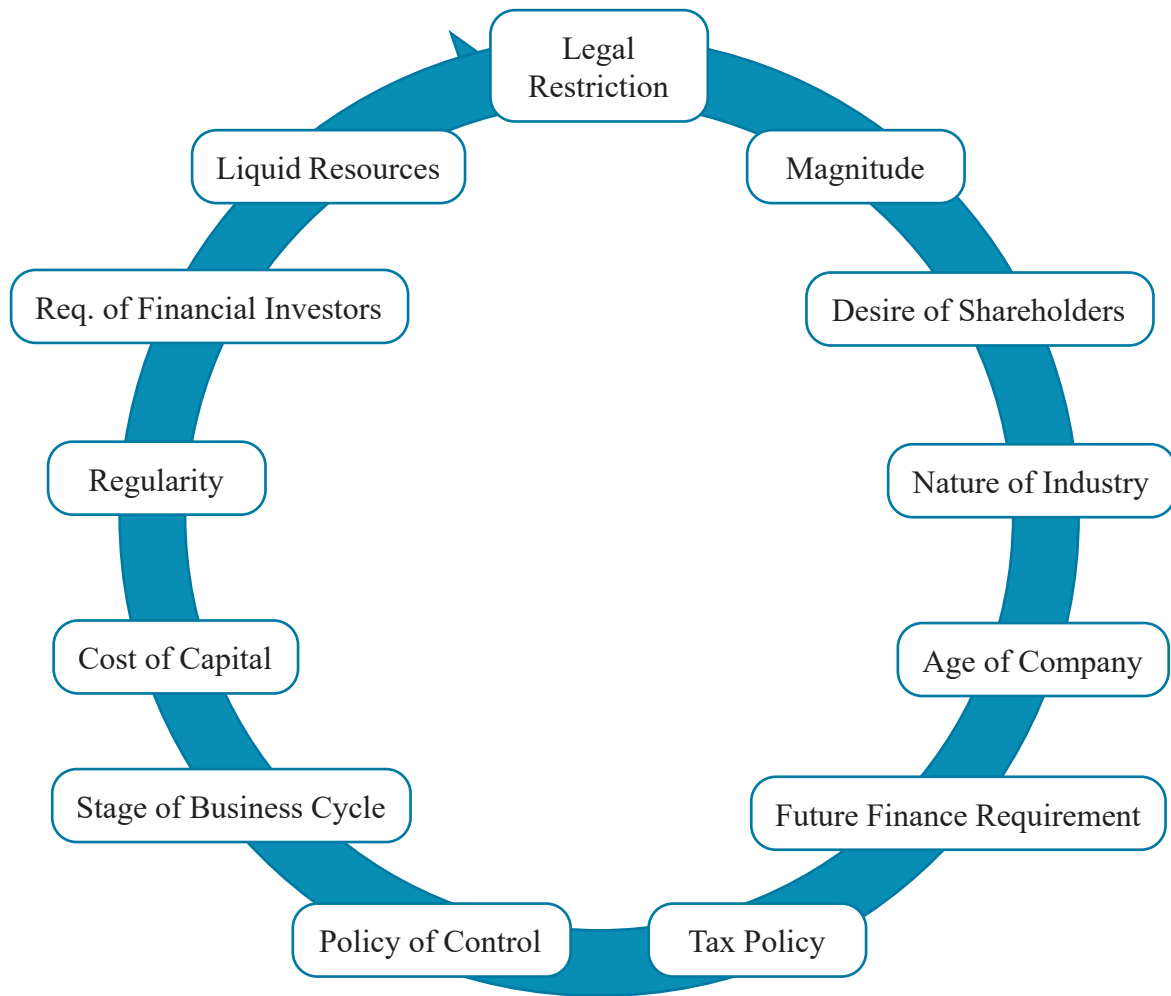
Since the number of shares decreases, EPS and CFPS improve, but this can be **artificial**, not due to real business growth.

2. **Possible Price Rise:**

A lower supply of shares (with constant demand) can lead to an increase in share price. However, this rise may not always reflect actual financial improvement.

F. DETERMINANTS OF DIVIDEND POLICY

Companies consider several factors while formulating their dividend policy. These include legal, financial, operational, and market-based factors:



a. LEGAL RESTRICTIONS

Dividends must be paid from profits, not capital. Some loan agreements may also restrict dividend payments to protect creditors.

b. MAGNITUDE & TREND OF EARNINGS

Higher and stable profits increase the ability to pay dividends. Past and present earnings trends are key factors.

c. DESIRE & TYPE OF SHAREHOLDERS

Different shareholders (e.g., retired vs. wealthy investors) have different dividend preferences. Companies often try to balance these.

d. NATURE OF INDUSTRY

Industries with stable demand (like FMCG) often pay regular dividends, unlike cyclical industries.

e. AGE OF THE COMPANY

New companies prefer to retain earnings for growth, while mature companies may pay higher dividends.

f. FUTURE FINANCIAL REQUIREMENTS

Firms may retain earnings if they expect future investment or expansion needs.

g. TAXATION POLICY

High personal tax on dividends may discourage investors, indirectly affecting a company's dividend policy.

h. POLICY OF CONTROL

To avoid diluting ownership, management may prefer retained earnings over issuing new shares for funding.

i. BUSINESS CYCLE STAGE

During boom phases, dividends may rise; during recessions, firms may conserve cash and cut dividends.

j. COST OF CAPITAL

If external finance is expensive, firms prefer to retain earnings, leading to conservative dividend policies.

k. REGULARITY OF DIVIDENDS

Some companies prioritize consistent dividend payments to build investor trust, even if current profits are low.

l. INSTITUTIONAL INVESTOR REQUIREMENTS

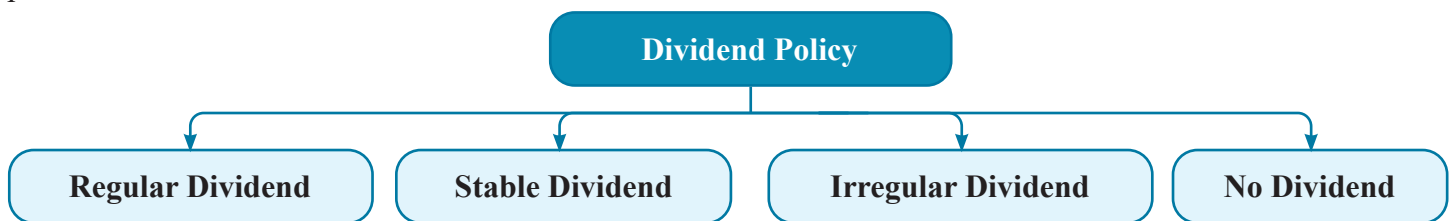
Big investors (like banks and FIs) often demand regular dividend policies.

m. LIQUIDITY POSITION

Even if profits exist, a lack of liquid cash may restrict dividend payments.

G. TYPES OF DIVIDEND POLICY

Companies adopt various dividend policies depending on their earnings stability, liquidity position, and future plans.



a. REGULAR DIVIDEND POLICY

A fixed dividend is paid every year, regardless of fluctuations in profits.

1. Suitable for

Companies with stable earnings and well-established business operations.

2. Advantages

- (a) Builds shareholders' confidence
- (b) Helps in long-term financing
- (c) Maintains stable market value of shares
- (d) Provides regular income to dependent investors (like retired persons)
- (e) Avoids high tax on accumulated profits

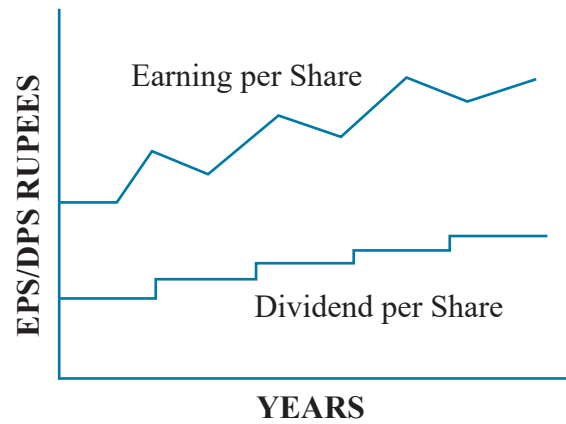
b. STABLE DIVIDEND POLICY

Emphasizes consistency in dividend payments, even if the dividend is small.

1. Forms of Stable Dividend Policy:

(a) Constant Dividend Per Share

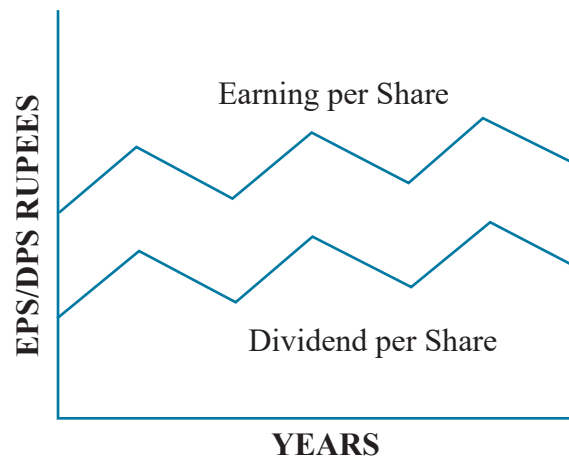
A fixed amount is paid per share every year, even if profits fluctuate. Reserves may be used in low-profit years.



Constant Dividend per share

(b) **Constant Payout Ratio**

A fixed percentage of net profits is paid as dividends. The dividend amount varies according to annual earnings.



Constant Dividend Pay-out Ratio

(c) **Stable Dividend Plus Extra Dividend**

A regular low dividend is paid every year, with an additional dividend declared in years of high profits.

2. Advantages of Stable Dividend Policy:

- (a) Indicates consistent business operations
- (b) Stabilizes the market price of shares
- (c) Builds investor trust
- (d) Provides regular income to shareholders
- (e) Attracts institutional investors
- (f) Improves the company's credit standing
- (g) Supports national income stability

c. IRREGULAR DIVIDEND POLICY

Dividend is declared inconsistently or only when profits are high.

Reasons

- 1. Uncertain or irregular earnings
- 2. Weak financial performance
- 3. Lack of liquid resources
- 4. Fear of weakening financial standing

d. NO DIVIDEND POLICY

No dividend is paid. Profits are retained for future business expansion or due to a poor cash/liquidity position.

Reasons for No Dividend

1. Business is in the early growth phase
2. The company needs funds for reinvestment
3. Weak working capital position

H. ESSENTIALS OF A SOUND DIVIDEND POLICY

A well-planned dividend policy reflects the financial discipline and future vision of a company. The key essentials are:

a. STABILITY IN DIVIDENDS

1. Consistent and regular dividend payments are more appreciated than fluctuating high and low payouts.
2. Regular moderate dividends build shareholder trust and reduce speculation in the stock market.

b. GRADUALLY RISING DIVIDENDS

1. As the company's earnings grow, the dividend rate should gradually increase.
2. In years of high profits, the company may declare a **special or extra dividend** in addition to the regular one.

c. DISTRIBUTION OF CASH DIVIDEND

1. Preferably, dividends should be paid in **cash**, ensuring liquidity to shareholders.
2. **Stock dividends** (bonus shares) may be declared if cash reserves are high, but should be limited to avoid over-capitalization.

d. MODERATE START

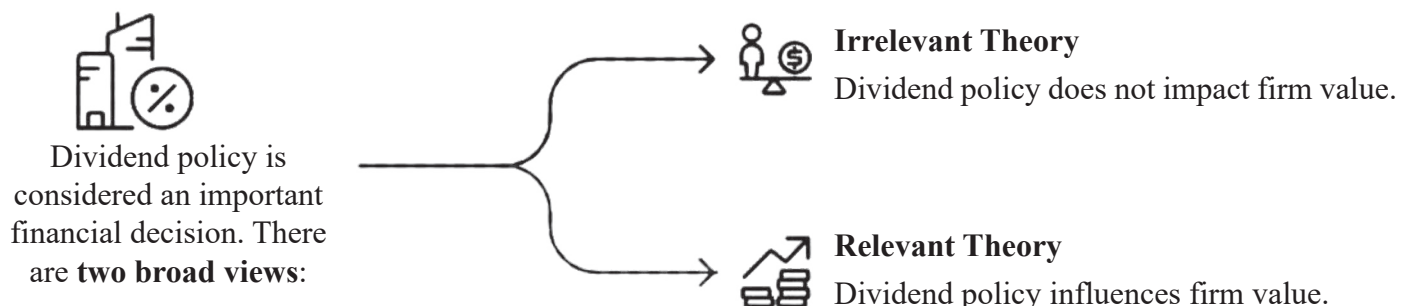
1. In the initial years, companies should begin with **lower dividend rates** to strengthen their financial base.
2. As the company grows, dividends can be gradually increased.

e. OTHER FACTORS

1. Dividends should be declared **only from current or retained profits**.
2. If past losses exist, they must be set off before declaring any dividend.
3. **Interim dividends** (paid during the year) may also be declared to maintain investor confidence.

I. DIVIDEND THEORIES/DIVIDEND MODELS

Relationship Between Dividend Policy and Value of Firm



a. IRRELEVANT CONCEPT OF DIVIDEND

1. Modigliani and Miller Approach (M-M Model)

M-M state that **dividend policy has no impact** on the **value of the firm** or **shareholders' wealth**.

The firm's value depends **only on its earning capacity and investment decisions**, not on how earnings are split between dividends and retained earnings.

(a) Assumptions of the M-M Model:

(i) Perfect Capital Market:

- (1) All investors have equal access to information.
- (2) No transaction or flotation costs.
- (3) No investor can influence market prices individually.

(ii) Rational Investors: Rational investors make decisions based on risk-return analysis, aiming to maximize their wealth logically.

(iii) No Taxes or equal tax treatment for dividends and capital gains.

(iv) Fixed Investment Policy: The company's investment decisions remain unchanged and are not influenced by its dividend policy.

(v) No Risk or Uncertainty: Investors can forecast future prices and dividends accurately.

(b) Explanation of the Theory:

- (i) If a firm pays dividends, it may need to raise **external funds** to maintain its investment plan.
- (ii) Raising funds externally leads to:
 - (1) Increase in the number of shares, and
 - (2) Reduction in EPS (Earnings Per Share).
- (iii) Any **gain from dividends** is **offset by a fall in share price**, due to the dilution of earnings from issuing new shares.

M-M Valuation Formula

$$P_0 = \frac{D_1 + P_1}{1 + CR}$$

or

$$CR = \frac{D_1 + (P_1 - P_0)}{P_0}$$

Where:

- ◆ P_0 = Market price of the share at the beginning of the period
- ◆ P_1 = Market price at the end of the period
- ◆ D_1 = Dividend at the end of the period
- ◆ CR or K_e : Cost of equity capital

Under the assumptions of the M-M model, **dividends are irrelevant**. Shareholders are indifferent between dividends and capital gains, as both provide the same return.

Example 01:

MM Approach – Payment or Non-payment of Dividend Does Not Affect Firm Value

Harshit Ltd. is a company that falls under a risk class for which the appropriate capitalization rate is 10%. The company has 30,000 equity shares outstanding, and the current market price per share is ₹80. The company is considering whether or not to declare a dividend of ₹4 per share at the end of the current financial year. It expects to earn a net income of ₹3,00,000 during the year. Additionally, Harshit Ltd. is planning to undertake new investment projects requiring a total capital of ₹6,00,000.

Using the Modigliani-Miller (MM) approach, demonstrate that the payment or non-payment of dividend does not affect the value of the firm. Show your calculations and prove that the market value of the company remains the same under both situations-(a) when the dividend is paid, and (b) when no dividend is paid.

Solution:

Given:

- ◆ Capitalization rate (K_e or CR) = 10%
- ◆ Number of existing shares = 30,000
- ◆ Market Price per share (P_0) = ₹80
- ◆ Expected Dividend (D_1) = ₹4 per share
- ◆ Net Income (E) = ₹3,00,000
- ◆ New Investment Required (I) = ₹6,00,000

(A) When Dividend is Paid

Step 1: Find P_1 (Price at end of year)

$$P_0 = \frac{D_1 + P_1}{1 + K_e} \Rightarrow 80 = \frac{4 + P_1}{1.1} \Rightarrow 80 \times 1.1 = 4 + P_1 \Rightarrow 88 = 4 + P_1 \Rightarrow P_1 = 84$$

Step 2: Calculate Funds Required Through New Issue

- ◆ Total Dividend Paid = ₹30,000 × ₹4 = ₹1,20,000
- ◆ Retained Earnings = ₹3,00,000 – ₹1,20,000 = ₹1,80,000
- ◆ Funds needed = ₹6,00,000 – ₹1,80,000 = ₹4,20,000

Step 3: New Shares Issued

- ◆ Issue Price = P_1 = ₹84
- ◆ New shares = ₹4,20,000 ÷ ₹84 = **5,000 shares**

Step 4: Total Shares After Issue

- ◆ Existing = 30,000
- ◆ New = 5,000
- ◆ **Total = 35,000 shares**

Step 5: Value of Firm (MM Formula)

$$\text{Value of firm} = \frac{\text{Value of total shares} - (I - E)}{1 + K_e}$$

$$= \frac{35,000 \times 84 - (6,00,000 - 3,00,000)}{1.1} = \frac{29,40,000 - 3,00,000}{1.1} = \frac{26,40,000}{1.1} = ₹24,00,000$$

(B) When Dividend is Not Paid

Step 1: P_1 (Price at end of year)

$$P_0 = \frac{0 + P_1}{1.1} \Rightarrow P_1 = 80 \times 1.1 = ₹88$$

Step 2: Funds Needed Through New Issue

- ◆ Retained Earnings = Full ₹3,00,000
- ◆ Remaining Investment = ₹6,00,000 – ₹3,00,000 = ₹3,00,000
- ◆ Issue Price = ₹88
- ◆ New Shares = ₹3,00,000 ÷ ₹88 ≈ 3,409 shares

Step 3: Total Shares After Issue

- ◆ Existing = 30,000
- ◆ New = 3,409
- ◆ Total = 33,409 shares

Step 4: Value of Firm

$$\begin{aligned}\text{Value of firm} &= \frac{33,409 \times 88 - (6,00,000 - 3,00,000)}{1.1} = \frac{29,39,992 - 3,00,000}{1.1} \\ &= \frac{26,39,992}{1.1} \approx 23,99,993 \approx ₹24,00,000\end{aligned}$$

Final Conclusion:

In both situations

(A) Dividend Paid and

(B) Dividend Not Paid,

Firm value = ₹24,00,000

Hence, proved as per MM Theory that:

“Dividend policy is irrelevant to the value of the firm.”

Example 02:

Tanuja Ltd. belongs to a risk class for which the appropriate capitalization rate is 12%. It currently has 30,000 equity shares outstanding, each selling at ₹100 in the market. The company is planning to declare a dividend of ₹6 per share at the end of the current financial year. It expects a net income of ₹3,00,000 for the year and has an investment proposal requiring ₹6,00,000.

Show with the help of the Modigliani-Miller approach that the payment or non-payment of dividend does not affect the value of the firm. Also calculate:

1. The number of new shares to be issued
2. Market value of the firm at the end of the year in both cases.

Solution:

Given:

- ◆ $P_0 = ₹100$
- ◆ $D_1 = ₹6$ (if dividend is paid), $D_1 = ₹0$ (if not paid)
- ◆ $Ke = 12\%$
- ◆ E (Net Income) = ₹3,00,000
- ◆ I (Investment Required) = ₹6,00,000
- ◆ Existing shares = ₹30,000

Step 1: Price at Year-End (P_1) using M-M Formula

M – M Formula:

$$P_0 = (D_1 + P_1)/(1 + Ke)$$

(A) When Dividend is Paid:

$$100 = (6 + P_1)/1.12$$

$$\Rightarrow 100 \times 1.12 = 6 + P_1$$

$$\Rightarrow 112 = 6 + P_1$$

$$\Rightarrow P_1 = ₹106$$

(B) When Dividend is Not Paid:

$$100 = (0 + P_1)/1.12$$

$$\Rightarrow P_1 = ₹112$$

Step 2: Comparative Table - Value of Firm in Both Cases

Particulars	Dividend is Paid	Dividend is NOT Paid
Net Income (E)	₹3,00,000	₹3,00,000
Total Dividend (₹6 × 30,000)	₹1,80,000	Nil
Retained Earnings	₹1,20,000	₹3,00,000
Investment Requirement (1)	₹6,00,000	₹6,00,000
External Funds Needed	₹4,80,000	₹3,00,000
Market Price per Share (P_1)	₹106	₹112
No. of New Shares Issued	$4,80,000 \div 106 = 4,528$	$3,00,000 \div 112 = 2,679$
Existing Shares	30,000	30,000
Total Shares after Issue	34,528	32,679
Market Value of Firm	$34,528 \times 106 = ₹36,60,000$	$32,679 \times 112 = ₹36,60,000$

Conclusion

Whether Tanuja Ltd. declares a dividend or not, the total market value of the firm remains 36,60,000 in both cases.

According to the Modigliani-Miller (M-M) theory, dividend policy has no impact on the value of the firm under the assumptions of perfect capital markets, rational investors, and no taxes.

(C) Criticism of M-M Hypothesis

Although the Modigliani-Miller (M-M) hypothesis suggests that dividend policy is irrelevant to the value of a firm, this theory is based on certain unrealistic assumptions. Its major criticisms are:

(i) Perfect Capital Market:

In reality, perfect capital markets do not exist. Not all investors have equal access to information, and there may be manipulation and delays in information flow.

(ii) Tax Differential:

Tax rates differ for capital gains and dividends. Since capital gains are often taxed at a lower rate than dividends, shareholders may prefer earnings to be retained rather than distributed.

(iii) Floatation Cost:

When companies raise funds externally, they incur costs like underwriting, legal, and brokerage charges. These costs make external financing more expensive than using retained earnings.

(iv) Transaction Costs:

Investors incur brokerage and other charges while selling their shares to realize capital gains. This makes receiving dividends more convenient and preferable for many investors.

(v) Uncertainty:

Future capital gains are uncertain due to market volatility. Therefore, shareholders often prefer a certain dividend in the present rather than uncertain gains in the future.

(vi) Rigid Investment Policy:

Companies may not always stick to a fixed investment policy as assumed by the M-M theory. Investment opportunities may change depending on market and business conditions.

b. RELEVANT CONCEPT OF DIVIDEND

1. Walter's Approach

Prof. James E. Walter proposed that dividend policy is relevant and affects the value of the firm, depending on the relationship between:

◆ r = Return on Investment

◆ k = Cost of Capital

(a) Types of Firms:

(i) Growth Firms ($r > k$):

Should retain earnings. Zero dividend maximizes share value.

(ii) Declining Firms ($r < k$):

Should distribute all earnings as dividends. 100% payout is best.

(iii) Normal Firms ($r = k$):

Dividend policy is irrelevant. Share value remains unchanged.

Formula:

$$P = \frac{D + (E - D) \times \frac{r}{k_e}}{K_e}$$

Where:

◆ P = Market price per share

◆ D = Dividend per share

◆ E = Earnings per share

◆ r = Internal rate of return

◆ k = Cost of equity capital

(b) Assumptions:

(i) Financing only through retained earnings

(ii) r and K remain constant

(iii) All earnings are either fully retained or distributed

(iv) EPS and DPS remain constant

(v) The Firm has infinite life

(c) Criticism of Walter's Approach :

(i) Firms also use external financing

(ii) r and K are not constant in reality

(iii) The model ignores practical factors like tax, risk, and legal obligations

(iv) Capitalization rate (k) is difficult to determine

Example 01: Ms. Ridhima Ltd. has earnings per share of ₹16. The company's cost of equity capital is 12.5%. The return that the company can earn on its retained earnings is 10%. The management is considering three dividend payout ratios: 25%, 50%, and 75%.

You are required to:

Determine which payout ratio would maximize the market price of the equity share as per **Walter's Model**.

Solution:**Given:**

- ◆ E (EPS) = ₹16
- ◆ K_e (Cost of equity capital) = 12.5% = 0.125
- ◆ r (Rate of return on retained earnings) = 10% = 0.10
- ◆ **Walter's Formula:**

$$P = \frac{D + (E - D) \times \frac{r}{K_e}}{K_e}$$

Case 1: When Payout = 25%

- ◆ Dividend (D) = 25% of ₹16 = ₹4

$$P = \frac{4 + (16 - 4) \times \frac{0.10}{0.125}}{0.125} = \frac{4 + 12 \times 0.8}{0.125} = \frac{13.6}{0.125} = ₹108.80$$

Case 2: When Payout = 50%

- ◆ Dividend (D) = 50% of ₹16 = ₹8

$$P = \frac{8 + (16 - 8) \times \frac{0.10}{0.125}}{0.125} = \frac{8 + 8 \times 0.8}{0.125} = \frac{14.4}{0.125} = ₹115.20$$

Case 3: When Payout = 75%

- ◆ Dividend (D) = 75% of ₹16 = ₹12

$$P = \frac{12 + (16 - 12) \times \frac{0.10}{0.125}}{0.125} = \frac{12 + 4 \times 0.8}{0.125} = \frac{15.2}{0.125} = ₹121.60$$

Conclusion

Since the market price per share is highest at ₹121.60 when the payout ratio is 75%, Ms. Ridhima Ltd. should adopt a 75% dividend payout policy to maximize shareholders' wealth as per Walter's Model.

Example 02: Tanuja Ltd. has earnings per share (EPS) of ₹8, and the applicable capitalization rate is 10%. The company is considering adopting a dividend payout ratio of 25%, 50%, or 75%.

Using **Walter's formula**, compute the market price per share under three different scenarios:

- ◆ When the return on retained earnings is 15%
- ◆ When the return on retained earnings is 10%
- ◆ When the return on retained earnings is 5%

Also, based on your findings, state the inference regarding dividend policy and share value.

Solution:**Given:**

- ◆ E (Earnings per Share) = ₹8
- ◆ K_e (Capitalization Rate) = 10% = 0.1
- ◆ **Walter's Formula:**

$$P = \frac{(D + (E - D) \times \frac{r}{K_e})}{K_e}$$

Case A: Return on Retained Earnings (r) = 15%

Payout Ratio	Dividend (D)	Market Price (P) Calculation	P (₹)
25%	₹2	$(2 + (82) \times 0.15/0.10)/0.10 = (2 + 9)/0.10$	₹110
50%	₹4	$(4 + (8 - 4) \times 0.15/0.10)/0.10 = (4 + 6)/0.10$	₹100
75%	₹6	$(6 + (8 - 6) \times 0.15/0.10)/0.10 = (6 + 3)/0.10$	₹90

Case B: Return on Retained Earnings (r) = 10%

Payout Ratio	Dividend (D)	Market Price (P) Calculation	P (₹)
25%	₹2	$(2 + (82) \times 0.10/0.10)/0.10 = (2 + 6)/0.10$	₹80
50%	₹4	$(4 + (8 - 4) \times 0.10/0.10)/0.10 = (4 + 4)/0.10$	₹80
75%	₹6	$(6 + (8 - 6) \times 0.10/0.10)/0.10 = (6 + 2)/0.10$	₹80

Case C: Return on Retained Earnings (r) = 5%

Payout Ratio	Dividend (D)	Market Price (P) Calculation	P (₹)
25%	₹2	$(2 + (82) \times 0.05/0.10)/0.10 = (2 + 3)/0.10$	₹50
50%	₹4	$(4 + (8 - 4) \times 0.05/0.10)/0.10 = (4 + 2)/0.10$	₹60
75%	₹6	$(6 + (8 - 6) \times 0.05/0.10)/0.10 = (6 + 1)/0.10$	₹70

Conclusion

When $r > K_e$ (15%), the lower the payout, the higher the market price (retention preferred).

When $r = K_e$ (10%), payout has no impact on the market price.

When $r < K_e$ (5%), the higher the payout, the higher the market price (distribution preferred).

Thus, Walter's model suggests that a firm's dividend policy is relevant, and the relationship between return on investment and cost of capital plays a crucial role in maximizing shareholder wealth.

Example 03:

Tanuja Ltd. earns 10 per share, is capitalized at the rate of 20%, and the return on its investment is 25%.

You are required to determine the **market price per share** using **Walter's Model**.

Solution:**Given:**

- ◆ Earnings per share (E) = 10
- ◆ Capitalization rate (K_e) = 20% = 0.20
- ◆ Return on investment (r) = 25% = 0.25
- ◆ Let dividend per share (D) = ₹0 (Assuming retention of full earnings)

Walter's Formula:

$$P = \frac{(D + (E - D) \times \frac{r}{K_e})}{K_e}$$

Substitute the values:

$$P = \frac{0 + (10 - 0) \times \frac{0.25}{0.20}}{0.20}$$

$$P = \frac{10 \times 1.25}{0.20} = \frac{12.50}{0.20}$$

$$P = ₹62.50$$

Conclusion

The market price per share of Tanuja Ltd., based on Walter's model, is ₹62.50, assuming full retention of earnings and given rates of return and capitalization.

2. Gordon's Dividend Relevance Model – Summary

Myron Gordon proposed a dividend relevance theory, suggesting that dividend policy has a direct effect on the value of a firm. The model emphasizes the relationship between dividend payout, retained earnings, and the market value of equity.

(a) Key Assumptions:

- (i) All-equity firm (no debt financing).
- (ii) No external financing – growth is financed solely through retained earnings.
- (iii) Constant internal rate of return (r).
- (iv) Cost of capital (K_e) is constant and greater than the growth rate, i.e., $K_e > g$.
- (v) Constant retention ratio (b).
- (vi) Perpetual earnings and firm life – i.e., growth is constant and infinite.
- (vii) No corporate taxes.

Gordon's Formula:

$$P = \frac{E(1-b)}{K_e - br}$$

Or,

$$P = \frac{D}{K_e - g}$$

Where:

- ◆ P or V_e = Market price of equity share
- ◆ E = Earnings per share
- ◆ $D = E(1-b)$ = Dividend per share
- ◆ b = Retention ratio
- ◆ K_e = Cost of equity capital
- ◆ r = Internal rate of return
- ◆ $g = br$ = Growth rate of earnings and dividends

(b) Implications of the Model:

(i) Growth Firm ($r > K_e$):

Retaining earnings increases the value of the firm. Optimal dividend payout is low or zero.

(ii) Normal Firm ($r = K_e$):

Dividend policy does not affect market value. It is irrelevant.

(iii) Declining Firm ($r < K_e$):

Distributing all earnings as dividends increases the value of the firm. Optimal payout is 100%.

Example 01:

The following information is available for Debasish Ltd. regarding its rate of return on investment, cost of capital, and earnings per share:

- ◆ Rate of Return on Investment (r):
 - (i) 15%
 - (ii) 12%
 - (iii) 10%

- ◆ Cost of Capital (K_e): 12%
- ◆ Earnings per Share (E): ₹10

Using Gordon's Dividend Valuation Model, compute the market value per share under the following dividend payout ratios:

Dividend Payout Ratio (D/P)	Retention Ratio (b)
(a) 100%	0
(b) 80%	20% (0.20)
(c) 40%	60% (0.60)

Solution:

Gordon's Model Formula:

$$P = \frac{E(1-b)}{K_e - br}$$

Where:

- ◆ P = Price of share
- ◆ E = Earnings per share = 10
- ◆ b = Retention ratio
- ◆ K_e Cost of capital = 12%
- ◆ r = Rate of return on investment

Case (i): $r = 15\%$ (Growth firm: $r > K_e$)

(a) $b = 0$ (100% payout)

$$P = \frac{10(1-0)}{0.12-0} = \frac{10}{0.12} = ₹83.33$$

(b) $b = 0.20$ (80% payout)

$$P = \frac{10(1-0.20)}{0.12-(0.20 \times 0.15)} = \frac{8}{0.12-0.03} = \frac{8}{0.09} = ₹88.89$$

(c) $b = 0.60$ (40% payout)

$$P = \frac{10(1-0.60)}{0.12-(0.60 \times 0.15)} = \frac{4}{0.12-0.09} = \frac{4}{0.03} = ₹133.33$$

Case (ii): $r = 12\%$ (Normal firm: $r = K_e$)

(a) $b = 0$

$$P = \frac{10}{0.12} = ₹83.33$$

(b) $b = 0.2$

$$P = \frac{8}{0.12-(0.20 \times 0.12)} = \frac{8}{0.096} = ₹83.33$$

(c) $b = 0.6$

$$P = \frac{4}{0.12-(0.60 \times 0.12)} = \frac{4}{0.048} = ₹83.33$$

Case (iii): $r = 10\%$ (Declining firm: $r < K_e$)

(a) $b = 0$

$$P = \frac{10}{0.12} = ₹83.33$$

(b) $b = 0.2$

$$P = \frac{8}{0.12 - (0.20 \times 0.10)} = \frac{8}{0.12 - 0.02} = \frac{8}{0.10} = ₹80.00$$

(c) $b = 0.6$

$$P = \frac{4}{0.12 - (0.60 \times 0.10)} = \frac{4}{0.12 - 0.06} = \frac{4}{0.06} = ₹66.67$$

Conclusion:

- ◆ When $r > K_e$: Lower payout (more retention) increases the value of the share.
- ◆ When $r = K_e$: Dividend policy is irrelevant; value remains constant.
- ◆ When $r < K_e$: Higher payout increases the value; retention decreases value.

3. Gordon's REVISED MODEL

Gordon extended his original dividend valuation model to **incorporate risk and uncertainty** in the valuation of shares.

(a) Key Assumptions of the Revised Model:

- (i) Investors are risk-averse.
- (ii) Investors prefer certainty – they value **near dividends more** than **uncertain future capital gains**.
(This is called the “**Bird-in-the-Hand**” theory.)

(b) Implications of Risk and Uncertainty:

- (i) When a company **retains earnings**, it delays dividend payment.
- (ii) Due to **uncertainty**, investors **discount distant dividends** (future returns) at a **higher rate** than near dividends.
- (iii) This results in an **increase in the cost of capital** (K_e) as the **retention ratio increases**.
- (iv) A **higher dividend payout** is preferred to maintain or increase the **market value of shares**.

Formulas Related to Gordon's Revised Model:

1. Future Dividend:

$$Dt = D_0(1 + g)^t$$

Where:

- ◆ Dt = Dividend at time t
- ◆ D_0 = Current dividend
- ◆ g = Growth rate of dividends
- ◆ t = Time period

2. Market Price of Share:

$$P_0 = \frac{D_1}{K_e - g} = \frac{D_0(1 + g)}{K_e - g}$$

Where:

- ◆ P_0 = Present market price of share
- ◆ D_1 = Expected dividend at year 1
- ◆ K_e = Cost of equity capital
- ◆ g = Constant growth rate of dividends

3. Cost of Equity Capital (Ke):

$$K_e = \frac{D_1}{P_0} + g = \frac{D_0(1+g)}{P_0} + g$$

(c) Criticism of Gordon's Revised Model:

- (i) Not all scholars agree with Gordon's argument.
- (ii) Some believe capital gains and dividends are treated equally by rational investors.
- (iii) Still, the **Gordon Revised Model** helps justify investor behavior under **risk and uncertainty**.

Example 01: The dividends of **A&G Ltd.** are expected to grow at a rate of **25% for 2 years**, after which the growth rate is expected to fall to **5%** permanently. The dividend paid last year was **Rs.2**. If the investor requires a **12% return**, determine the **intrinsic value of the stock**.

Present Value Factors at 12%:

Year	PV Factor
1	0.893
2	0.797
3	0.712

Solution:

Given:

- ◆ $D_0 = \text{Rs.}0.2$ (last year's dividend)
- ◆ Growth rate for Year 1 and 2 = 25%
- ◆ Growth rate from Year 3 onwards = 5%
- ◆ Required rate of return (K_e) = 12%

Step 1: Calculate expected dividends

- ◆ $D_1 = D_0(1+g) = 2 \times 1.25 = \text{Rs.}0.2 \times 1.25$
- ◆ $D_2 = D_1(1+g) = 2.5 \times 1.25 = \text{Rs.}0.3 \times 1.25$
- ◆ $D_3 = D_2 \times (1+g_{\text{terminal}}) = 3.125 \times 1.05 = \text{Rs.}0.3 \times 1.05$

Step 2: Calculate terminal price at the end of Year 2

$$P_2 = \frac{D_3}{K_e - g} = \frac{3.281}{0.12 - 0.05} = \frac{3.281}{0.07} = \text{Rs.}46.86$$

Step 3: Calculate present value of all future cash flows

Year	Amount (Rs.)	PV Factor @12%	Present Value (Rs.)
1	2.50	0.893	2.23
2	3.125	0.797	2.49
2	46.86	0.797	37.34

Step 4: Total present value of the stock

Intrinsic value of stock = $2.23 + 2.49 + 37.34 = \text{Rs.}42.06$

Final Answer: Present Value is 42.06

TEST YOURSELF

1. Nakshatra Ltd. belongs to a risk class for which the capitalization rate is 20%. The company has 1,00,000 existing equity shares, each currently selling at Rs. 100. The company is considering declaring a dividend of Rs. 5 per share at the end of the current year.

Using the Modigliani and Miller Dividend Irrelevance Model (assuming no taxes), calculate the price of an equity share at the end of the year under the following two situations:

- (i) When the dividend is declared
- (ii) When the dividend is not declared

Also, explain how the wealth of shareholders remains the same whether the dividend is paid or not, according to the MM model.

Solution:

Given:

$$P_0 = \text{Rs.} 100$$

$$K_e = 20\% \text{ or } 0.20$$

$$D_1 = \text{Rs.} 5 \text{ (when dividend is declared)}$$

No taxes

Using MM model formula:

$$P_0 = (D_1 + P_1) / (1 + K_e)$$

- (i) When dividend is declared ($D_1 = 5$)

$$100 = (5 + P_1) / 1.2$$

$$\Rightarrow 100 \times 1.2 = 5 + P_1$$

$$\Rightarrow 120 = 5 + P_1$$

$$\Rightarrow P_1 = 115$$

So, the price at the end of the year = Rs.115

Total value to shareholder = Dividend + Price = 5 + 115 = Rs.120

- (ii) When dividend is not declared ($D_1 = 0$)

$$100 = (0 + P_1) / 1.2$$

$$P_1 = 100 \times 1.2$$

$$\Rightarrow P_1 = 120$$

So, the price at the end of the year = Rs.120

Total value to shareholder = 0 + 120 = Rs.120

Conclusion

Whether the company pays a dividend or not, the total wealth of the shareholder remains the same (Rs.120). This proves the Modigliani and Miller Dividend Irrelevance Theory that dividend policy has no effect on shareholder wealth.

2. Ms. Ridhima Ltd. has 25,000 equity shares outstanding and its shareholders' expected rate of return is 10%. The current market price of a share is Rs.100. It is expected that the firm would pay a dividend of Rs.5 per share in the next year. The firm has a project requiring new investment of Rs.5,00,000. The expected net income of the firm is Rs.2,50,000.

Calculate the market value of the firm under both the conditions:

- (i) When dividends are paid
- (ii) When dividends are not paid

Also calculate the number of new equity shares to be issued in each case to meet the investment requirement, using the Modigliani and Miller approach (assume no taxes).

Solution:**Given:**

Number of existing shares = 25,000

K_e (cost of equity) = 10%

P_0 = Rs.0.1

D_1 = Rs.0.5 (if dividend is paid)

Investment needed = Rs.5,00,000

Net income = Rs.2,50,000

Using MM model:

Case (i): When dividend is paid ($D_1 = 5$)

$$P_0 = (D_1 + P_1)/(1 + K_e)$$

$$100 = (5 + P_1)/1.1$$

$$100 \times 1.1 = 5 + P_1$$

$$110 = 5 + P_1$$

$$P_1 = 105$$

To find the amount to be raised externally:

Total investment required = Rs.5,00,000

Net income = Rs.2,50,000

Dividend = $25000 \times 5 = \text{Rs.}0.1, 25000$

Retained earnings = $2,50000 - 1,25000 = \text{Rs.}0.1, 25000$

Funds to be raised by issuing new shares = $5,00,000 - 1,25,000 = \text{Rs.}3,75,000$

Number of new shares to be issued = $3,75,000/105 = 3571.42$

Market value of the firm:

Total shares after issue = $25,000 + 3571.42 = 28,571.42$

Market price per share = Rs.105

Market value of firm = $28,571.42 \times 105 = \text{Rs.}30,00,000$

Case (ii): When dividend is not paid ($D_1 = 0$)

$$100 = (0 + P_1)/1.1$$

$$P_1 = 100 \times 1.1 = \text{Rs.} 110$$

All income is retained:

Amount to be raised = $5,00,000 - 2,50,000 = \text{Rs.}2,50,000$

Number of shares to be issued = $2,50,000/110 = 2272.72$

Market value of firm = $(25,000 + 2272.72) \times 110 = 27,272.72 \times 110 = \text{Rs.}30,00,000$

Conclusion

Whether dividend is paid or not, the total market value of the firm remains the same i.e. Rs.30,00,000, proving the Modigliani and Miller dividend irrelevance theory.

3. A Ltd. had 50,000 equity shares of Rs.10 each outstanding on January 1, 1999. The shares are currently quoted at par in the market. The company intends to pay a dividend of Rs.2 per share for the current year. It belongs to a risk class for which the capitalization rate is 15%.

Using the Modigliani and Miller (MM) Model and assuming no taxes, calculate the price of the company's share at the end of the year under the following situations:

- (i) When dividend is declared
- (ii) When no dividend is declared

Also, determine the number of new equity shares the company must issue to meet its investment requirement of Rs.2,00,000, assuming net income of Rs.1,10,000.

Show that the payment or non-payment of dividend does not affect the value of the firm as per MM approach.

Solution:

Given:

Number of shares = 50,000

Face value = Rs.10 Po Rs.10 (quoted at par)

D_1 = Rs.2 per share (if dividend declared)

K_e = 15% Investment required = Rs.2,00,000

Net income = Rs.1,10,000

Tax = Nil

(i) When dividend is declared ($D_1 = 2$) ;

Using MM formula:

$$P_0 = (D_1 + P_1)/(1 + K_e)$$

$$10 = (2 + P_1)/1.15$$

$$10 \times 1.15 = 2 + P_1$$

$$11.5 = 2 + P_1$$

$$P_1 = 9.5$$

Funds required = Rs.2,00,000

Net income = Rs.1,10,000

Dividend paid = 50000×2 = Rs.1,00,000

Retained earnings = $1,10,000 - 1,00,000$ = Rs.10,000

External funds required = $2,00,000 - 10,000$ = Rs.1,90,000

Number of shares to be issued = $1,90,000/9.50$ = 20,000 shares

Total shares = 50,000 (existing) + 20,000 (new) = 70,000 shares

Market price per share at year-end = Rs.9.50

Total value of firm = $70,000 \times 9.50$ = Rs.6,65,000

But this includes value after new investment, so to find value of the firm before financing:

Value of firm = 50000×10 = Rs.5,00,000

(ii) When dividend is not declared ($D_1 = 0$);

Using MM formula:

$$10 = (0 + P_1)/1.15$$

$$P_1 = 10 \times 1.15 = Rs.11.50$$

Retained earnings = Net income = Rs.1,10,000

External funds required = $2,00,000 - 1,10,000$ = Rs.90,000

Number of shares to be issued = $90,000/11.50$ = 7,826.08 shares

Total shares = $50,000 + 7,826.08$ = 57,826.08 shares

Market price per share at year-end = Rs.11.50

Total value of firm = $57,826.08 \times 11.50$ = Rs.6,65,000

Again, to find original value of the firm before investment:

Value of firm = 50000×10 = Rs.5,00,000

Conclusion

Whether dividend is paid or not, the value of the firm remains the same, i.e., Rs.5,00,000. This supports the Modigliani and Miller dividend irrelevance theory.

4. Harshit Ltd. has earnings per share (EPS) of Rs.20. The capitalization rate applicable to the company is 15%, and any retained earnings can be reinvested to earn a return of 18%. The company is considering three different dividend payout ratios: 25%, 50%, and 75%.

Using Walter's Model, determine which payout ratio would maximize the wealth of shareholders. Calculate the market price per share under each option and suggest the best payout ratio.

Solution:

When payout is 25% → Market price = Rs.153.33

When payout is 50% → Market price = Rs.146.67

When payout is 75% → Market price = Rs.140

Hence, 25% payout ratio maximizes shareholder wealth.

5. The par value of equity shares of Sharda Ltd. is Rs.100 per share. The company's earnings per share (EPS) is Rs.15. The capitalization rate in the market is 15%. The management is considering the following dividend payout options:

- | | |
|-------------------------|-------------------------|
| (a) Payout ratio of 0% | (b) Payout ratio of 20% |
| (c) Payout ratio of 40% | (d) Payout ratio of 60% |

Which of these alternatives would be considered the best under the following scenarios regarding the productivity (return) on retained earnings:

- | | | |
|---------|---------|---------|
| (a) 20% | (b) 15% | (c) 10% |
|---------|---------|---------|

Use Walter's Model to determine the market price per share in each case and identify the best option under each scenario.

Solution:

When return on retained earnings is 20%:

Market Prices: 133.33, 126.67, 120, 113.33

→ Best option: **Payout ratio 0%**

When return on retained earnings is 15% (equal to market capitalization rate):

Market Prices: 100, 100, 100, 100

→ All options are **equally preferred**

When return on retained earnings is 10%:

Market Prices: 66.67, 73.33, 80, 86.67

Best option: Payout ratio 60%

6. Praveen Ltd. earns Rs.5 per share. The rate of capitalization (K_e) is 10%, and the company earns a return of 18% on its investments. Using **Walter's Model**, calculate the **market price per share** if the company follows a **25% dividend payout ratio**.

Also, determine whether this payout ratio is optimum as per Walter's theory.

7. Calculate the market price of a share of **Debasish Ltd.** under:

- | |
|----------------------------|
| (i) Walter's Formula |
| (ii) Dividend Growth Model |

Solution:

Given:

Earnings per share = Rs.5

Dividend per share = Rs.3

Cost of Capital = 60%

Internal rate of return on investment = 20%

Retention ratio = 50%

Answer (Hint):**(i) As per Walter's Model Price = Rs.34.37****(ii) As per Dividend Growth Model Price Rs.41.67****Case Study****Case Study 1: Dividend Decision - Tanuja Ltd.**

Tanuja Ltd. has 10,00,000 outstanding equity shares at the beginning of the financial year. The current market price per share is Rs.150. At the end of the year, the Board of Directors is considering declaring a dividend of Rs.8 per share. The capitalization rate for the company is 12%. The company has earned a net income of Rs.2 crore for the year and has an investment requirement of Rs.4 crore.

You are required to:

- (a) Calculate the market price per share at the end of the year using the Modigliani-Miller Approach:
 - (i) When the dividend of Rs.8 per share is declared
 - (ii) When no dividend is declared
- (b) Calculate the number of new equity shares the company needs to issue in both cases:
 - (i) When the dividend is paid
 - (ii) When the dividend is not paid
- (c) Prove that the total market value of the company remains unchanged regardless of whether the dividend is paid or not. Also, calculate the firm's current market value in both cases.

Solution: Given:

- ◆ Existing shares = 10,00,000
- ◆ Current price = Rs.150
- ◆ Dividend = Rs.8
- ◆ $K_e = 12\%$
- ◆ Net income = Rs.2,00,00,000
- ◆ Investment required = Rs.4,00,00,000

(a) Share Price using MM model**(i) When dividend is declared:**

$$P_0 = \frac{D_1 + P_1}{1 + K_e} \Rightarrow 150 = \frac{8 + P_1}{1.12} \Rightarrow 150 \times 1.12 = 8 + P_1 \Rightarrow 168 = 8 + P_1 \Rightarrow P_1 = 160$$

(ii) When dividend is not declared:

$$P_0 = \frac{0 + P_1}{1.12} \Rightarrow P_1 = 150 \times 1.12 = 168$$

(b) Number of New Shares to be Issued**(i) When dividend is declared:**

- ◆ Dividend paid = $10,00,000 \times 8 = \text{Rs.}80,00,000$
- ◆ Retained earnings = $2,00,00,000 - 80,00,000 = \text{Rs.}1,20,00,000$
- ◆ External funds needed = $4,00,00,000 - 1,20,00,000 = \text{Rs.}2,80,00,000$
- ◆ Price per share = Rs.160

$$\text{New shares} = \frac{2,80,00,000}{160} = 1,75,000 \text{ shares}$$

(ii) When dividend is not declared:

- Full income retained = Rs.2,00,00,000
- External funds = 4,00,00,000 – 2,00,00,000 = Rs.2,00,00,000
- Price per share = Rs.168

$$\text{New shares} = \frac{2,00,00,000}{168} \approx 1,19,050 \text{ shares}$$

(c) Total Market Value of the Firm

We calculate total shares after issue x market price:

(i) When dividend is declared:

- Total shares = 10,00,000 + 1,75,000 = 11,75,000
- Price = Rs.160

$$\text{Value of firm} = 11,75,000 \times 160 = \text{Rs.18,80,00,000}$$

But this includes value after Rs.4 crore investment. To find original value, use:

$$\text{Value of firm at beginning} = 10,00,000 \times 150 = \text{Rs.15,00,00,000}$$

(ii) When dividend is not declared:

- Total shares = 10,00,000 + 1,19,050 = 11,19,050
- Price = Rs.168

$$\text{Value of firm} = 11,19,050 \times 168 = \text{Rs.18,80,00,400}$$

Again, **original firm** value remains:

$$= 10,00,000 \times 150 = \text{Rs.15,00,00,000}$$

Conclusion

According to the Modigliani-Miller dividend irrelevance theory, the value of Tanuja Ltd. remains the same whether the company distributes dividends or not. The value of the firm is Rs.15,00,000 (or Rs.1.5 crore) in both cases, indicating that dividend policy does not affect shareholder wealth.

Case Study 2: Dividend Policy Impact Using Walter's Model

Three companies - Tanuja Ltd., Sharda Ltd., and Harshit Ltd. are from the steel sector and have the same Earnings Per Share (EPS) of Rs.10. The market capitalization rate (K_e) is 10% for all three companies. However, their internal rate of return (r) differs.

Company	Internal Rate of Return (r)	Capitalization Rate (K_e)	EPS
Tanuja Ltd.	15%	10%	10
Sharda Ltd.	10%	10%	10
Harshit Ltd.	8%	10%	10

You are required to calculate the market value per share of each company under the Walter's Model in the following situations:

- (a) When no dividend is paid
- (b) When dividend is Rs.8 per share
- (c) When dividend is Rs.10 per share

Answer (Hint):

Dividend per share	Tanuja Ltd.	Sharda Ltd.	Harshit Ltd.
(a) Rs.0	150	100	80
(b) Rs.8	230	180	160
(c) Rs.10	250	200	180

Interpretation: Use Walter's Model to show the effect of dividend payout on share value based on whether $r > K_e$, $r = K_e$, or $r < K_e$.

WORKING CAPITAL MANAGEMENT

A. MEANING OF WORKING CAPITAL

Working capital means that which is issued to carry out the **day-to-day** operations of a business.

Capital required for a business can be classified under two main categories.

- ◆ Fixed capital
- ◆ Working capital

Every business needs funds for two purposes: for its establishment and to carry on its day-to-day operations. Long-term funds are required to create production facilities through the purchase of fixed assets such as plant and machinery, land, buildings, furniture, etc. Investment in these assets represents that part of a firm's capital, which is blocked on a permanent or fixed basis, called fixed capital.

Funds are also needed for short-term purposes, i.e., for the purchase of raw materials, payment of wages, and other day-to-day operations of the business. These funds are known as working capital. In other words, working capital refers to a firm's capital, which is required for short-term assets or current assets. Funds thus invested in current assets keep revolving fast and being constantly converted into cash, and this cash flow is again converted into other current assets. Hence, it is known as *circulating or short-term capital*.

B. CONCEPT OF WORKING CAPITAL

a. GROSS WORKING CAPITAL

It is simply called working capital refers to the firm's investment in current assets, so the total current assets of the firm are known as gross working capital.

b. NET WORKING CAPITAL

It represents the difference between current assets and current liabilities. Net working capital may be positive or negative. Positive net working capital occurs when current assets are more than current liabilities. But when current liabilities become more than current assets, then it is negative working capital.

In brief, we can say that working capital is necessary for the smooth functioning and proper utilization of fixed assets.

C. TYPES OF WORKING CAPITAL

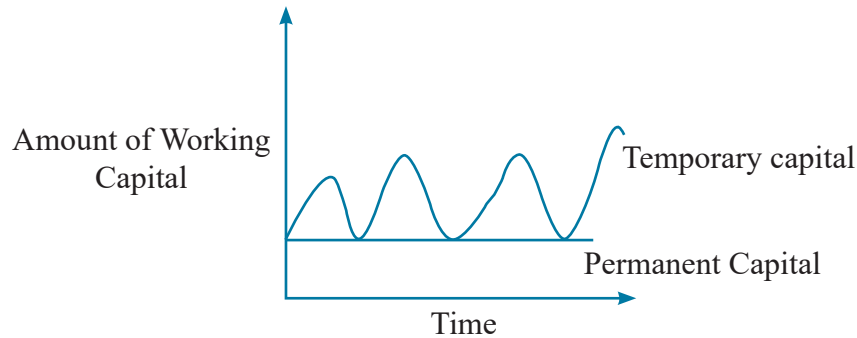
a. PERMANENT WORKING CAPITAL

As the operating cycle is a continuous process, the need for working capital also arises continuously. But the magnitude of current assets needed is not always the same; it increases and decreases over time. However, there is always a minimum level of current assets. This level is known as permanent or fixed working capital.

b. TEMPORARY WORKING CAPITAL

The extra working capital needed to support the changing production and sales activities is called variable or functioning, or temporary working capital.

This can be shown in the following Diagram:



D. NEED FOR WORKING CAPITAL

The need for working capital cannot be overemphasized. The need for working capital arises due to the time gap between production and the realization of cash from sales. So, the working capital or investment in current assets becomes a necessary need for working capital. It arises due to the following reasons:-

a. OPERATING CYCLE

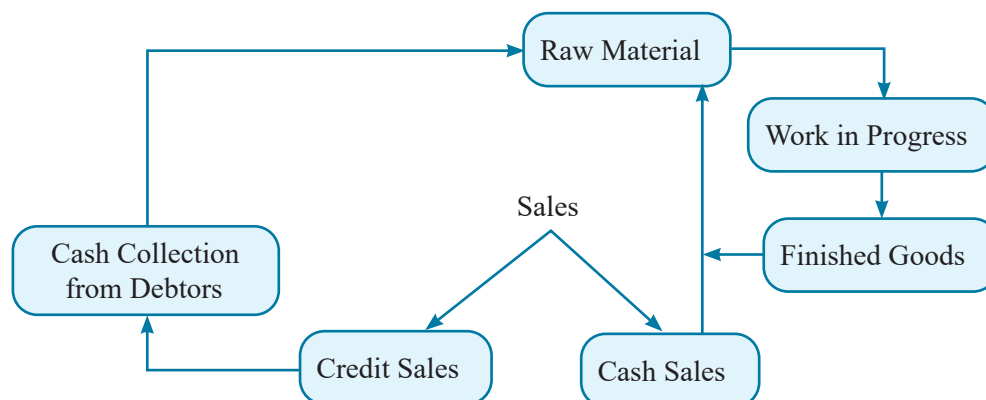
“Operating cycle is the time duration required for converting sales into cash after the conversion of resources into inventories.”

First of all, a firm purchases raw material, then after some processing, it is converted into work-in-progress, and after this further processing is done to convert work-in-progress into finished goods. After the raw material is converted into finished goods, sales are made. Sales are not always full cash sales; there are also credit sales. These credit sales after some period are converted into cash. So the whole process takes time. This time taken is known as the length of the operating cycle.

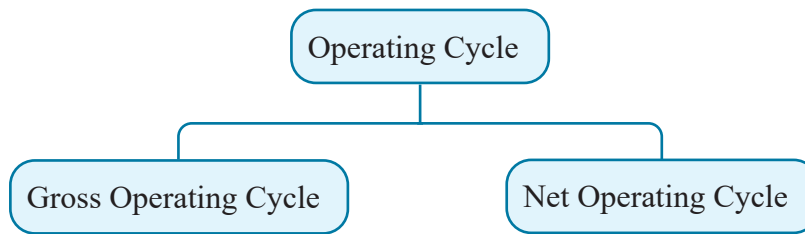
So operating cycles include:

- ◆ Raw Material Conversion Period (RMCP)
- ◆ Work-in – progress conversion period (WIPCP)
- ◆ Finished goods conversion period (FCP)
- ◆ Debtors Conversion period (DCP)

So operating cycle can be known as following:-



Operating Cycle Types



1. Gross Operating cycle

Gross Operating cycle is the total time period from the conversion of Raw Material into finished goods and finished goods into sales and then sales into cash.

$$\text{GOC} = \text{RMCP} + \text{WIPCP} + \text{FCP} + \text{DCP}$$

2. Net Operating Cycle

As we provide period to debtors for the payments, our creditors also provide period to us for payment to them. So this reduces our requirement of working capital. This also affects the operating cycle. Operating cycle's length reduces with so many days as provided by the creditors to us. The difference between gross operating cycle and period allowed by the creditors for payment is known as net operating cycle.

$$\text{NOC} = \text{GOC} - \text{CPP}$$

E. WORKING CAPITAL REQUIREMENT FOR THE ANTICIPATED NEEDS FOR FUTURE

These needs may be for Raw materials or Finished Goods. Sometimes, because of the unavailability of Raw Material or due to seasonal availability of Raw Material, some advance stock of Raw Material becomes necessary for the company. In a similar way, due to the sudden rise in demand for finished goods, in the future, more finished goods will be kept in stock. For both reasons, more working capital is required because funds will be involved in these safety stocks.

F. DETERMINANTS OF WORKING CAPITAL

The following are the main determinants of working capital.

a. NATURE AND SIZE OF BUSINESS

The working capital of a firm depends upon the nature of its business, e.g., Public utility undertakings like electricity; water supply need very little working capital because they offer only cash sales, whereas trading & financial firms have a very small investment in fixed assets but require a large sum of money invested in working capital. The size of the business also determines the working capital requirement, and it may be measured in terms of the scale of operations. The greater the size of the operation, the larger will be the requirement for working capital.

b. MANUFACTURING CYCLE

The manufacturing cycle also creates the need for working capital. The manufacturing cycle starts with the purchase and use of Raw materials and completes with the production of finished goods. If the manufacturing cycle is longer, more working capital will be required, or vice versa.

c. SEASONAL VARIATION

In certain industries, like VTM, raw materials are not available throughout the year. They have to buy raw material in bulk during the season to ensure an uninterrupted flow and process it during the year. Generally, during the busy season, a firm requires more working capital than in the slack season.

d. PRODUCTION POLICY

Production policy also determines the working capital level of a firm. If the firm has a steady production policy, it may require need of continuous working capital. But if the firms adopt a fluctuating production policy, meaning to produce more during the lead demand season, then more working capital may be required at that time, but not in other periods during a financial year. So the different production policies give rise to different types of needs for working capital.

e. FIRM'S CREDIT POLICY

The firm's credit policy directly affects the working capital requirement. If the firm has a liberal credit policy, then the longer credit period will be provided to the debtors so which will lead to more working capital requirements. With the liberal credit policy operating cycle length increases and vice versa.

f. SALES GROWTH

Working capital requirement is directly related to sales growth. If the sales are growing, more working capital will be needed due to the arising need for more Raw materials, finished goods, and credit sales.

g. BUSINESS CYCLE

Business cycle refers to the alternate expansion and contraction of the general business. In a period of boom, a larger amount of working capital is required where whereas in a period of depression lesser amount of working capital is required.

h. EARNING CAPACITY & DIVIDEND POLICY

If the firm has enough earnings and it is not paying dividends, then it will not be in need of external borrowings. If a firm wants to increase its earning power, then more working capital will be required, also to pay more dividends, more profits are needed, which gives rise to more working capital. The company is paying a 42% dividend to its shareholders.

i. PRICE LEVEL CHANGES

Changes in the price level also affect the working capital requirements. Generally, the rising prices will require the firm to maintain a larger amount of working capital, as more funds will be required to maintain the same current assets.

j. CONDITION OF SUPPLY

The inventory of raw material, spares, and stores depends on the condition of supply. If the supply is prompt, the firm can manage with a small inventory. However, if the supply is unpredictable, then the firm to ensure continuity of production should acquiring stocks as and when they are available and have to carry larger inventory on average.

k. OTHER FACTORS

Certain other factors such as operating efficiency, management ability, irregularities of supply, import policy, asset structure, importance of labour, banking facilities, and time lag. etc., also influence the requirement of working capital.

So these are the main determinants of working capital. The importance of the influence of these determinants on working capital may differ from firm to firm.

G. MAJOR DECISIONS IN WORKING CAPITAL MANAGEMENT

There are two major decisions management relating to working capital management:-

- a. What should be the ratio of current assets to sales?
- b. What should be the appropriate mix of short-term financing and long-term financing for financing these current assets?

a. CURRENT ASSETS ABOUT SALES

If the firm can forecast accurately the factors that affect the working capital, the investment in current assets can be designed uniquely. When uncertainty characterizes the above factors, as it usually does, the investment in current assets cannot be specified uniquely. In case of uncertainty, the outlay on current assets should consist of a base component meant to meet normal requirements and a safety component meant to cope with unusual requirements. The safety component depends upon low conservatism or aggressiveness in the current assets policy of a firm. If the firm purchases a very conservative current asset policy, it would carry a high level of current assets in relation to sales. If a firm adopts a moderate current assets policy, it would carry a moderate level of current assets about sales. Finally, if a firm follows a highly aggressive current assets policy, it would carry a low level of current assets relative to sales. VTM is following the current assets policy, showing a moderate level of current assets about sales, as is evident from the ratio analysis.

b. DETERMINING A SHORT TERM AND LONG TERM FINANCING MIX FOR FINANCING OF CURRENT ASSETS

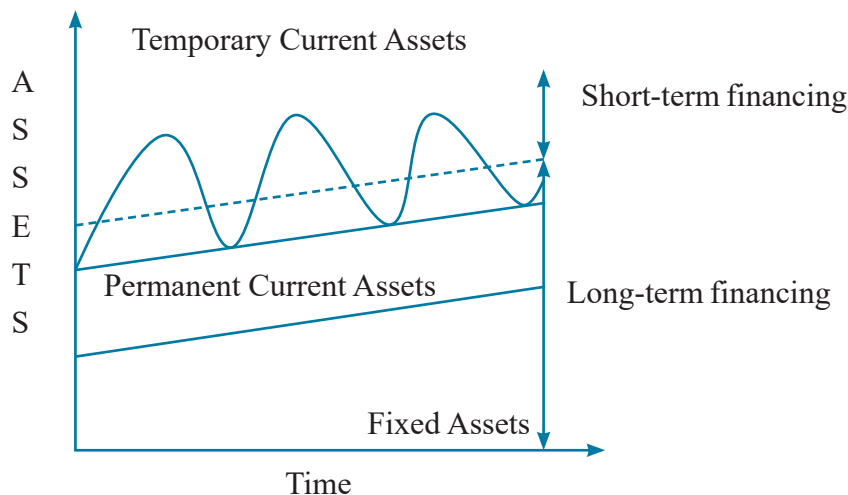
There are three approaches in this regard, which are discussed below:

1. HEDGING APPROACH

This approach is also called the matching approach. In this approach, there is a proper matching of the expected life of the asset with the duration of the fund. Usually, according to this approach, long-term sources are used for financing permanent current assets and fixed assets & short-term sources are used for financing temporary current assets:

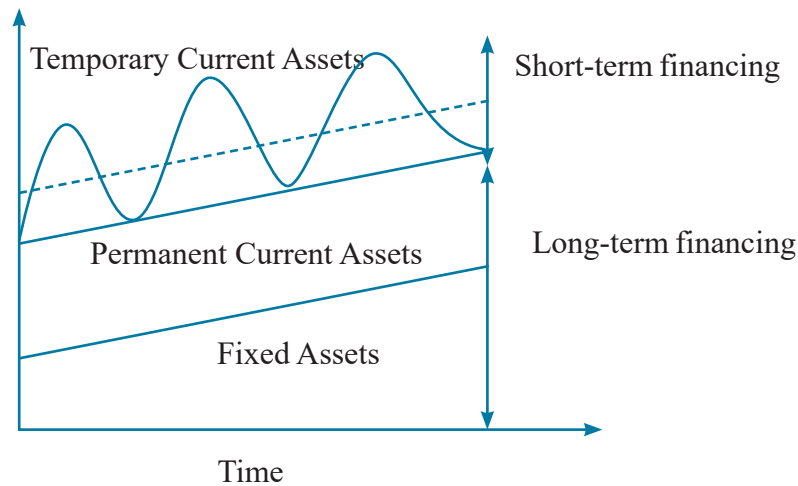
2. CONSERVATIVE APPROACH

In this approach, there is more reliance on long-term financing in comparison to short-term financing. Even some part of the temporary current comparison to finance from long-term sources because long-term sources are less risky in comparison to short-term sources.



3. AGGRESSIVE APPROACH

In this approach, there is more reliance on short-term financing, and even a part of permanent current assets is financed from short-term financing.



In VTM, the current assets are financed from short-term sources as well as long-term sources, so they follow a conservative approach.

H. CASH MANAGEMENT

Cash management refers to the management of cash balance and the bank balance, and also includes short-term deposits. Cash is an important current asset for the operation of the business. Cash is the most liquid and can be used to make immediate payments. Insufficiency of cash at any stage may prevent a firm from discharging its liabilities or force it to sell its other assets immediately. On the other hand, extreme liquidity may lead to uneconomic investments. This underlines the significance of cash management. The term cash includes coins, currency, and cheques held by the firm and balances in its bank accounts. Sometimes, near-cash items such as marketable securities or bank item deposits are included in cash.

A financial manager is required to manage the cash flows (both inflows and outflows) arising out of the operations of the firm. For this, he will have to forecast the cash inflows from sales and outflows for costs, etc. This will enable the financial manager to identify the timings as well as the amount of future cash flows. Cash management does not end here, and the financial manager may also be required to identify the sources from which cash may be produced on a short-term basis or the outlets where excess cash may be invested for a short term.

Cash is the basic input needed to keep the business running continuously. It is also the ultimate output expected to be realized by selling the product manufactured by the firm. Cash shortages will simply disturb the firm's manufacturing operations, where excessive cash will simply remain idle. Thus, a firm should keep sufficient cash, neither more nor less. Hence, a major function of the financial manager is to maintain a sound cash position. Cash management is one of the key areas of working capital management. Apart from the fact that it is the most liquid current asset, cash is the common denominator to which all the current assets can be reduced because the major liquid assets, i.e., receivables and inventory, are eventually converted into cash.

Cash management is concerned with the management of:

- ◆ Cash inflows and outflows of the unit
- ◆ Cash flows within the unit
- ◆ Cash balance held by the unit at a point in time, by financing deficit or investing surplus cash

I. MOTIVES FOR HOLDING CASH

a. TRANSACTION MOTIVE

It means a firm holds cash for the conduct of business. The daily requirements of cash come under this motive. So, enough cash for smooth business is required for transactional purposes.

b. PRECAUTIONARY MOTIVE

Under this motive, cash is held for most contingencies in the future. There may be so many reasons why the need for cash arises.

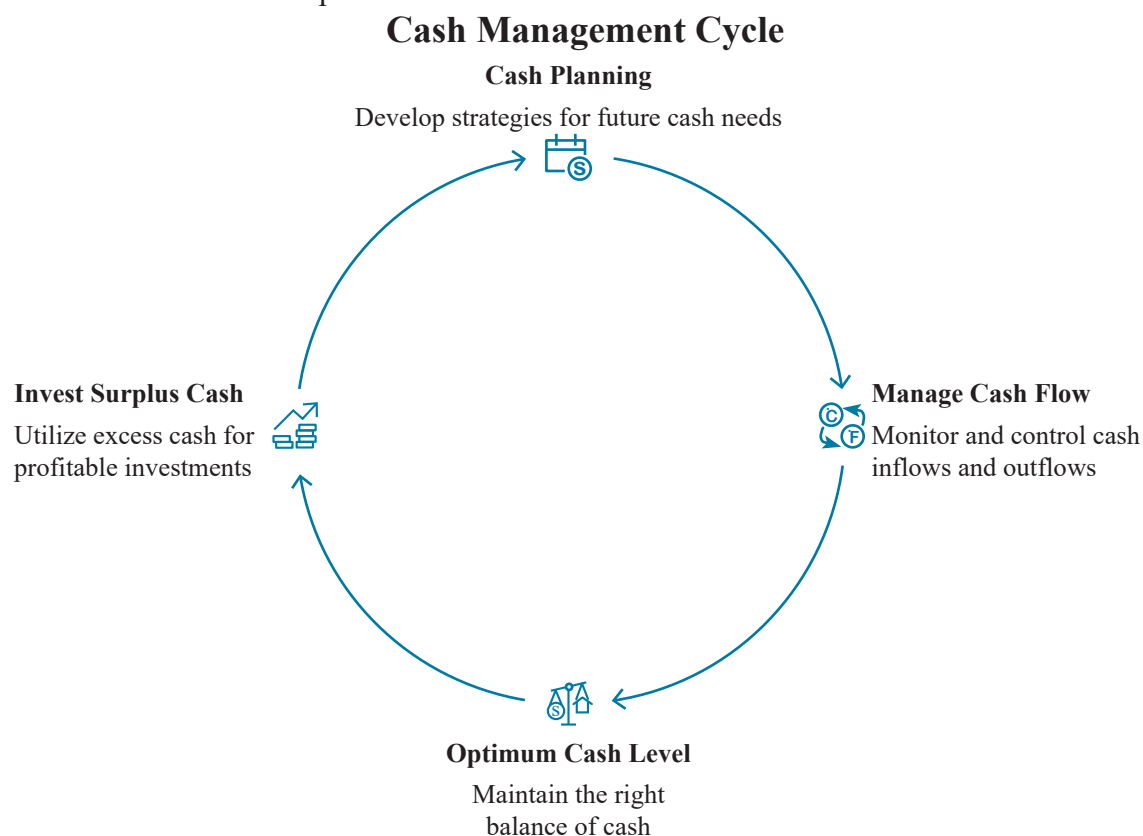
These reasons can be:

1. A sudden rise in the demand, which leads to more production
2. Due to inflation
3. Due to any mishap in the future, like loss by fire, theft, etc. The cash is held for precautionary motives in advance. This cash may be held as marketable securities, which are highly liquid and low-risk.

c. SPECULATIVE MOTIVE

The speculative motive relates to holding cash for investing in a profit-making opportunity as when arises. The opportunity may arise in securities, in material purchasing, or any other type. By holding cash for speculative motives, the firm can choose the most profitable opportunity, yet, by and large, business firms do not engage in speculation because the primary motives to hold cash are transaction and precautionary motives.

In VTM, it holds cash only for transactional purposes. Speculation and precautionary motives, cash is held by the corporate office. If the VTM requires some more cash to meet any future contingency, then it informs about it to corporate office, and the corporate office sends cash to the VTM as per the requirement. But the VTM has to give the reasons for extra cash to the corporate office.



1. CASH PLANNING

Cash planning is a technique to plan to control the use of cash. Cash planning can help to anticipate future cash flows and the need for the firm and reduce the possibility of idle cash. Cash planning may be done on a daily, weekly, and monthly basis. Cash budget is the most significant device to plan for and control cash receipts and payments.

The unit under study makes cash planning through the following tools:

- ◆ Cash Budget
- ◆ Rolling cash flow statement
- ◆ Daily cash flow statement

The cash budgets are prepared by the firm on a monthly and yearly basis. Through the cash budget, the unit can make estimates of cash receipts and disbursements during a future period of time. Their estimates show the requirement for cash in the unit.

Another device used for cash planning is a six-monthly rolling cash flow statement prepared every month. This statement shows the projections of inflows and outflows of cash during the next six months.

This statement can help in making various decisions. If the unit wants to make any capital payment, these statements can tell when there is a surplus of cash, and payments can be made during the month.

A daily cash flow statement is prepared to see the daily cash position. Their estimations are sent to the corporation office at Ludhiana so that the needed cash is obtained at night.

2. MANAGING CASH FLOWS

A significant part of cash management is the management of cash flow, both inflows and outflows, without any loss to the unit and without impairing its goodwill in the market. These are made at the head office in Ludhiana, so the main source of cash inflows to VTM is the cash credit limit, which is as follows:

The main limits are controlled by H.O. The sub-limits have been allocated to the unit for fulfilling the day-to-day requirements of working capital. The daily bank position of sublimits is fixed to H.O. for monitoring the daily bank position. In case of drawn sub-limits, the funds get transferred from the main limits. The interest rate paid for this is near about 11.25%. The cash credit limits are sanctioned by the bank against the hypothecation stocks and fluctuating assets as security.

The unit can withdraw from these limits as and when needed. The amount received from the sale of yarn is debited at the head office in the main account. To increase the efficiency of cash management, the surplus funds are transferred to other units if those units need cash, thus increasing the overall profitability.

The main outflow of the unit is on raw material cost. Different types of raw materials are purchased from different states. Normally, cotton is purchased during peak season when good quality cotton is available. Generally, payment for cotton is made when cotton is received in the mill, and the credit period depends on the states from which cotton is purchased.

Cash outflows also arise on account of the purchase of stores, spares, and all other materials. Normal credit for these products is mainly 30 to 60 days, and the full credit period is used.

3. DETERMINING OPTIMUM BALANCE

An efficient finance manager always aims at preparing the cash and bank balance at the optimum level, i.e., neither is it too high that it remains idle and the firm loses interest on it, nor is it too low that the firm is always in a cash-tight position. *For example:* The unit always keeps Rs. 1.5 lacs for the routine expenses, around the days of wages, the amount is approx. Rs. 4 lacs per day kept in hand, thus the unit maintains the appropriate amount of cash balance and meets the firm's obligations as and when they are due.

4. INVESTING IDLE CASH

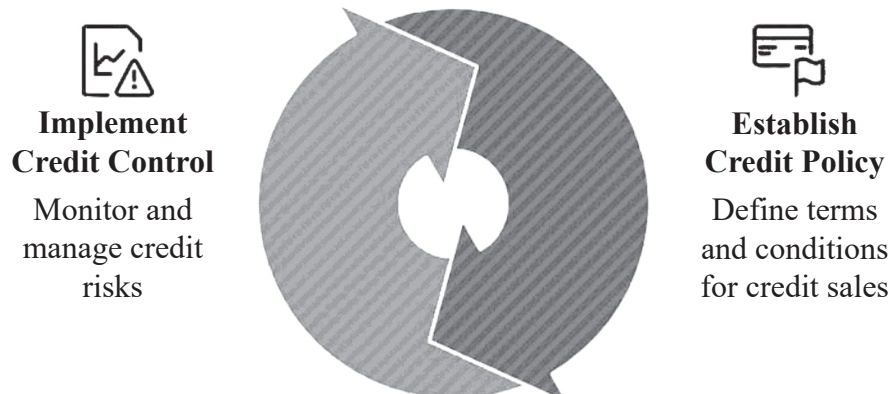
Since the main input of the company is seasonal. Therefore, the company has to maintain a high level of assets during the cotton season, which falls between October to March. During April to September, the company gets its cash credit limits reduced by the respective banks. The company has a very good system of managing its current assets. The current assets of the unit are managed at the corporate level, and the unit seeks funds according to its requirements, calculated on a day-to-day basis. Hence, there are no idle funds at the unit level. As the funds are monitored/controlled at the corporate level, therefore, it becomes the prime responsibility of H.O. to have a good policy of investing idle funds in an appropriate security, keeping in view the requirement of funds in the future and liquidity of the security in which the investment is being made.

J. RECEIVABLES MANAGEMENT

Accounts receivable are simply an extension of credit to the firm's customers, allowing them a reasonable period in which to pay for the goods. Most firms treat accounts receivable as a marketing tool to promote sales and profits. They represent an extension of credit and investment of funds and must be carefully managed. Every firm must develop a credit policy that includes setting credit standards, defining credit terms, and employing methods for the timely collection of receivables. The receivables (including the debtors and the bills) constitute a significant portion of working capital and are an important element of it. The receivables emerge whenever goods are sold on credit, and customer receivables are created when a firm sells goods or services to its customers and accepts, instead of the immediate cash payment, the promise to pay within a specified period. Thus, receivables are a type of loan extended by the seller to the buyer to facilitate the purchase process. As against the ordinary type of loan, the trade credit in the form of receivables is not a profit-making service but an inducement or facility to the buyer-customer of the firm.

Receivables are a direct result of credit sales. Credit sale is resorted to by a firm to push up sales, which ultimately results in pushing up the profits earned by the firm. At some time, selling goods on credit results in the blocking of funds in accounts receivable. Additional funds are required for operating needs of the business, which involve extra costs in terms of interest. Moreover, an increase in receivables also increases the chances of bad debts. The creation of accounts receivable is beneficial as well as dangerous. The finance manager has to follow a policy that uses cash funds as economically as possible by extending receivables without adversely affecting the chance of increasing sales and making more profits. Receivables Management generally means what type of credit policy a firm should adopt so that sales and profits can be promoted on the one hand and funds can be economically utilized on the other hand.

Cycle of Receivables Management



a. THE CREDIT POLICY

It may be defined as the set of parameters and principles that govern the extension of credit to customers. This requires the determination of:

1. The Credit Standard

When a firm sells on credit, it takes about the paying capacity of the customers. Therefore, to be on the safer side, it must set a credit standard that should be applied in selecting customers for credit sales. The credit standards of a firm represent the basic criteria for the extension of credit to customers.

So the credit standard is the combination of three C's. These are:

- (a) Character of a person
- (b) Capacity of a person
- (c) Condition of a person

2. Credit Terms

The credit terms refer to the set of stipulations under which the credit is extended to the customers.

The credit terms may relate to the following:

(a) Credit Period

The credit period is an important aspect of the credit policy. It refers to the length of time customers are allowed to pay for their purchases. It may differ from one market to another. The credit period generally varies from 15 days to 60 days.

In some cases, the credit period may be zero, and only cash sales are made. It refers to the time duration in terms of net date, e.g., if a firm's credit terms are "net 30," it means the customers are expected to pay within 30 days from the date of sales. The shorter the credit period, the better it is for the firm. But the firm has to lengthen its credit period to increase sales. But one must compare the cost of extended credit with the incremental profits. If this cost is less than it will be beneficial for the company to increase the credit period.

(b) Discount Terms

It is a reduction in the payment offer to customers to induce them to repay their credit obligations within a specified period.

In practice, the credit terms would include:

- (i) The rate of cash discount
- (ii) The cash discount period
- (iii) The net credit period

b. CREDIT CONTROL

The next important step in the management of receivables is the control of these receivables.

The following are the directions for controlling the receivables:

1. The Collection Procedure

The overall collection procedure of the firm should neither be too lenient nor too strict. A strict collection policy can affect the goodwill and damage the growth prospects of the sales. If a firm has a lenient credit policy, the customer with a natural tendency towards slow payments may become slower to settle his accounts. One possible way of ensuring early payments from customers may be to charge interest on overdue balances.

2. Monitoring of Receivables

To control the level of receivables, the firm should apply regular checks, and there should be a continuous monitoring system.

For this, several measures are available as follows:

- (a) A common method to monitor receivables is the collection period or the number of days' outstanding receivables.
- (b) Another technique available for monitoring the receivables is known as the ageing schedule. Ageing schedule down book debts according to the length of time they have been outstanding. The ageing schedule provides more information about the collection experience. It helps to sort out the slow-paying debtors.

K. MANAGEMENT OF INVENTORY

Inventory is a very important part of current assets. Approximately 60% of the current assets are inventories. So, the proper management of inventory is required for successful working capital management. As a larger amount of funds is involved in the inventories, they must be carried with care for proper utilization of funds.

L. NATURE OF INVENTORY

In inventories, we include:

- a. **Raw Material:** There are those basic inputs that are converted into work-in-progress after the manufacturing process. Raw materials are purchased for production and storage purposes.
- b. **Work-in-Progress:** These inventories are semi-manufactured products. These products are those that are ready for sale.
- c. **Finished Goods:** These are completely manufactured products. These products are those that are ready for sale.

Here is another type of inventory also, which is not directly related to production but facilitates in production process. These inventories are known as supplies. Cleaning material, oil, fuel, electric tube, etc, are the supplies.

M. OBJECTIVES OF INVENTORY MANAGEMENT

There are so many objectives of inventory management. These objectives may differ from firm to firm.

The main objectives of inventory management are:

- a. To make adequate investments in inventories so that funds can be best utilized.
- b. Smooth production in the present and future.
- c. Time availability of inventories.
- d. Smooth and uninterrupted sales processes.
- e. Minimize the cost related to inventories.
- f. To meet the future price change.
- g. To get an adequate return on investment.

N. INVENTORY MANAGEMENT TECHNIQUES

For inventory management, the two questions must be answered. First, how much should be ordered in one order so that excess or insufficient inventories can be avoided? Secondly, the order should be placed so that the timely availability of inventory is there. To get answers to these two questions, we use *two techniques*, which are as follows:

a. ECONOMIC ORDER QUANTITY (EOQ)

Economic order quantity provides the answer to our first question. By this, we come to know how much we must order at a single time. So that all the costs related to inventory are minimal.

The EOQ is the inventory level that minimizes the total of ordering and carrying costs.

Determining an optimum inventory level involves two types of costs:

1. Ordering Costs

All these costs are incurred in placing one order. It includes: requisition, transportation, receiving, inspecting, clerical, and staff services. Ordering costs are fixed per order. Therefore, they decline as the order size increases.

2. Carrying Costs

Costs incurred for maintaining a given level of inventory are called carrying costs. It includes storage, insurance, handling, and taxes. Carrying costs vary with inventory.

To calculate economic order quantity there is formula:

$$EOQ = 2AO/C$$

Where,

A = Annual requirement

O = Ordering cost per order

C = Carrying cost of inventory

b. REORDER POINT

The reorder point is the inventory level at which an order should be placed to replenish the inventory.

To calculate the reorder point, we should know

1. Lead Time

2. Average Usage

3. EOQ

Lead Time is the time normally taken to replenish inventory after the order has been placed.

Average Usage is the inventory used on average daily basis or average weekly basis.

So, Reorder Point = Lead Time × Average Usage

If the firm also maintains safety stock then the reorder point will be:

Reorder Point = (Lead Time × Average Usage) + Safety Stock

So when the inventory of reorder point will be in store then the order will be placed for purchase of inventory.

In this way, the production process will not stop because the inventory will be available for that period.

TEST YOURSELF

1. The cost sheet of PQR Ltd. provides the following data:

1	Raw material	50
2	Direct Labour	20
3	Overheads (including depreciation of ₹10)	40
4	Total cost	110
5	Profits	20
6	Selling price	130

- A. Average raw material in stock is for one month.
- B. Average material in work-in-progress is for half month.
- C. Credit allowed by suppliers: one month; credit allowed to debtors: one month.
- D. Average time lag in payment of wages: 10 days;
- E. Average time lag in payment of overheads 30 days.
- F. 25% of the sales are on cash basis.
- G. Cash balance to be ₹1,00,000.
- H. Finished goods lie in the warehouse for one month.

You are required to prepare a statement of the working capital needed to finance a level of the activity of 34,000 units of output. Production is carried on evenly throughout the year and wages and overheads accrue similarly. State your assumptions, if any, clearly.

Sol. Statement Showing Working Capital Requirements

Particulars	Amount
Current Assets	
Cash	1,00,000
Raw Material ($4,500 \times 1 \times 50$)	2,25,000
Work in Progress (W.N. 2)	1,68,750
Finished Goods ($4,500 \times 1 \times 100$)	4,50,000
Debtors ($4,500 \times 1 \times 100 \times 75\%$)	3,37,500
A. Total Current Assets	12,81,250
Current Liabilities	
Creditors ($4,500 \times 1 \times 50$)	2,25,000
Wages ($4,500 \times 10/30 \times 20$)	30,000
Overheads ($4,500 \times 1 \times 30$)	1,35,000
B. Total Current Liabilities	3,90,000
C. Net Working Capital (A – B)	8,91,250

Working Notes:

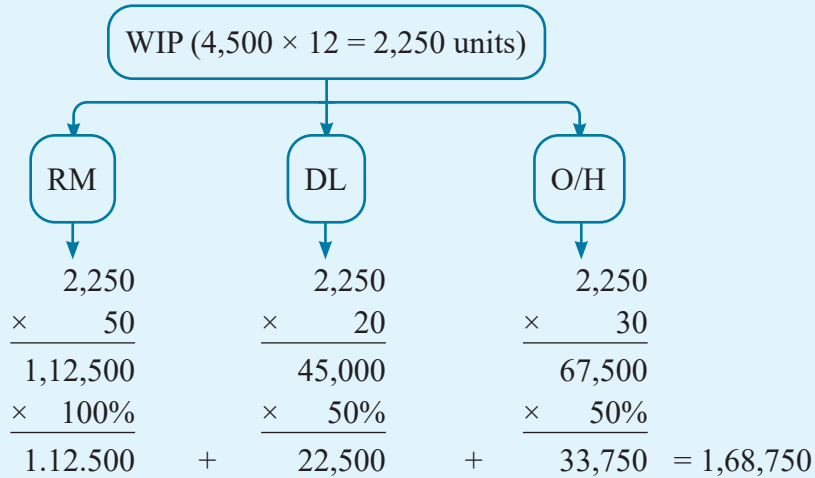
1. Computation of Monthly Production:

Annual Production = 54,000 units

$$\text{Monthly Production} = \frac{\text{Annual Production}}{12} = \frac{54,000}{12} = 4,500 \text{ units per month.}$$

2. Computation of amount of WIP:

Degree of completion of WIP is not given; hence we assume that WIP constitutes Raw Material 100%, Labour 50% & Overheads 50%.



3. As Depreciation is a non cash item, it is to be excluded for the computation of Working Capital.

4. As profit is not our investment, it is to be excluded for the computation of Working Capital

2. The following information has been extracted from the records of a Company:

Product cost sheet

Raw materials	45
Direct Labour	20
Overheads	40
Total	105
Profit	15
Selling price	120

1. Raw materials are in stock on an average for two months.
2. The materials are in process on an average for one month. The degree of completion is 50% in respect of all elements of cost.
3. Finished goods stock on an average is for one month.
4. Time lag in payment of wages and overheads is 12 weeks.
5. Time lag in receipt of proceeds from debtors is 2 months.
6. Credit allowed by suppliers in one month.
7. 20% of the output is sold against cash.
8. The company expects to keep a Cash balance ₹ 1,00,000.
9. The company is poised for a manufacture of 1,44,000 units in the next year. You are required to prepare a statement showing the Working Capital requirements of the company.

Sol. Statement Showing Working Capital Requirements

Particulars	Amount
Current Assets	
Cash	1,00,000
Raw Material ($12,000 \times 2 \times 45$)	10,80,000
Work in Progress (W.N. 2)	6,30,050
Finished Goods ($12,000 \times 1 \times 105$)	12,60,000
Debtors ($12,000 \times 2 \times 105 \times 80\%$)	20,16,000

A. Total Current Assets	50,86,000
Current Liabilities	
Creditors (12,000 × 1 × 45)	5,40,000
Wages (12,000 × 1.5/4 × 20)	90,000
Overheads (12,000 × 1.5/4 × 40)	1,80,000
B. Total Current Liabilities	8,10,000
C. Net Working Capital (A – B)	42,76,000

Working Notes:

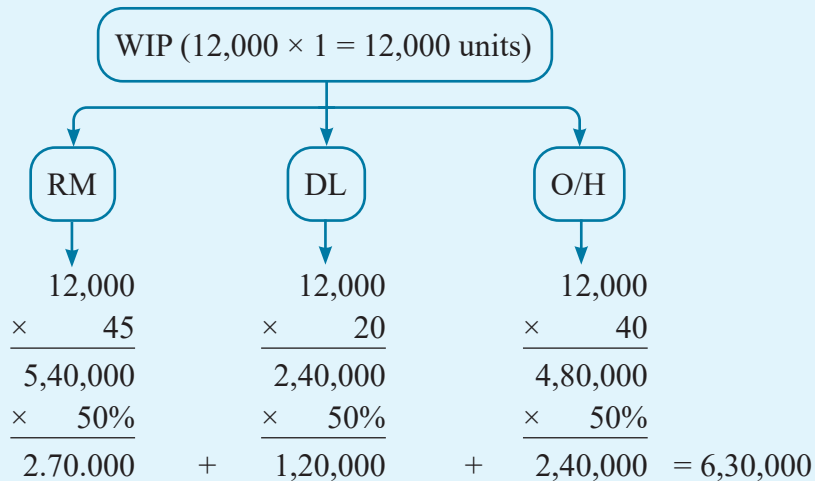
1. Computation of Monthly Production:

Annual Production = 1,44,000 units

Monthly Production $\frac{\text{Annual Production}}{12} = \frac{1,44,000}{12} = 12,000$ units per month.

2. Computation of amount of WIP

Degree of completion of WIP is 50% in respect of all elements of cost; hence WIP constitutes Raw Material 50%, Labour 50% & Overheads 50%



3. As profit is not our investment, it is to be excluded for the computation of Working Capital

3. Grow More Ltd. is presently operating at 60% leve, producing 36,000 units Per annum. In view of favorable market conditions, it has been decided that from 1st January 2004, the Company would operate at 90% capacity. The following information is available:

Existing cost-price structure per unit is given below:

Raw material	4.00
Wages	2.00
Overheads (Variable)	2.00
Overheads (Fixed)	1.00
Profits	1.00

1. It is expected that the cost of raw material, wages rate, expenses and sales per unit will remain unchanged in 2004.
2. Raw materials remain in stores for 2 months before these are issued to production. These units remain in production process for 1 month.
3. Finished goods remain in godown for 2 months.

- Credit allowed to debtors is 2 months. Credit allowed by creditors is 3 months.
- Lag in wages and overhead payments is 1 month. It may be assumed that wages and overhead accrue evenly throughout the production cycle.

You are required to:

- Prepare profit statement at 90% capacity level; and
- Calculate the working requirements on an estimated basis to sustain the increased production level.

Assumption made if any, should be clearly indicated.

Sol. Statement Showing Working Capital Requirement

Particulars	Amount
Current Assets	
Cash	—
Raw Material (4,500 × 2 × 4)	36,000
Work in Progress (W.N. 4)	28,500
Finished Goods (4,500 × 2 × 8) + (3000 × 2)	78,000
Debtors (4,500 × 2 × 8) + (3000 × 2)	78,000
A. Total Current Assets	2,20,500
Current Liabilities	
Creditors (4,500 × 3 × 4)	54,000
Wages (4,500 × 1 × 2)	9,000
Variable Overheads (4,500 × 1 × 2)	9,000
Fixed Overheads (3000 × 1)	3,000
B. Total Current Liabilities	75,000
C. Net Working Capital (A – B)	1,45,500

Working Notes:

1. Computation of Annual Production at 90% Capacity level:

60% 36,000 units

90% X units

$$X = \frac{36,000 \times 90}{60} = 54,000 \text{ units}$$

2. Computation of Monthly Production at 90% capacity level:

Annual Production = 54,000 units

$$\text{Monthly Production} = \frac{\text{Annual Production}}{12} = \frac{54,000}{12} = 4,500 \text{ units per month.}$$

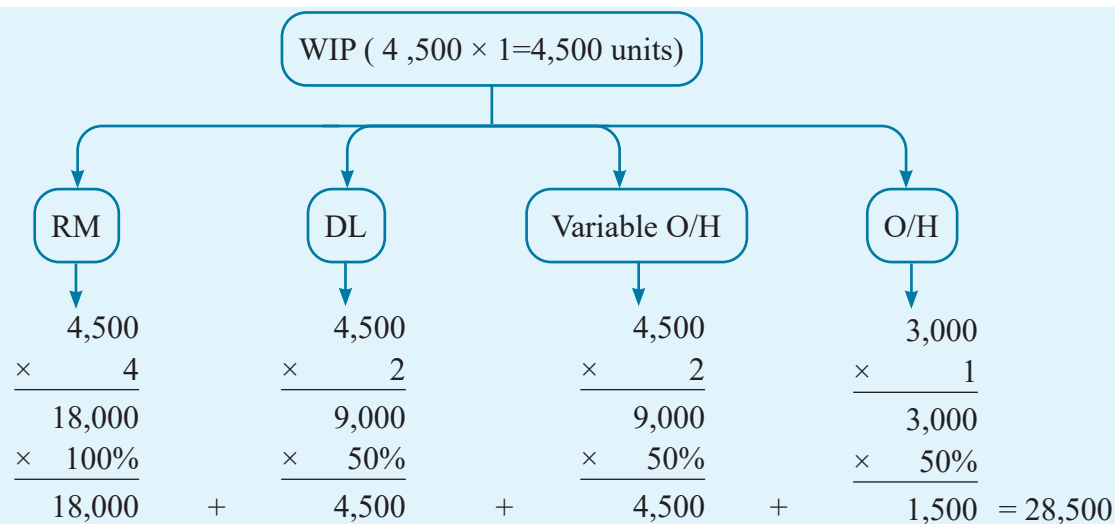
3. Computation of Monthly Fixed Overheads:

Annual Fixed Overheads = Current Annual Production × Fixed Overhead per unit = 36,000 × 1 = 36,000

$$\text{Monthly Fixed Overheads} = \frac{\text{Annual Fixed Overheads}}{12} = \frac{36,000}{12}$$

4. Computation of amount of WIP:

Degree of completion of WIP is not given; hence we assume that WIP constitutes Raw Material 100%, Labour 50% & Overheads 50%.



5. As profit is not our investment, it is to be excluded for the computation of Working Capital.

Profit Statement at 90% Capacity Level

Particulars	₹
Sales (54000 × 4)	5,40,000
(–) Raw Material Consumed (54000 × 4)	2,16,000
(–) Direct Labour (54000 × 2)	1,08,000
(–) Variable Overheads (54000 × 2)	1,08,000
(–) Fixed Overheads (3000 × 12)	36,000
Profit	72,000

4. Hi-tech Ltd. Plans to sell 30,000 units next year. The expected cost of goods Sold is as follows:

Particulars	₹
Raw material	100
Manufacturing expenses	30
Selling administration and financial expenses	20
Selling price	200

The duration at various stages of the operating cycle is expected to be as follows:

Raw material stage	2 months
Work-in-progress stage	1 months
Finished stage	1½ months
Debtors stage	1 month

Assuming the monthly sales level of 2,500 units, estimate the gross working

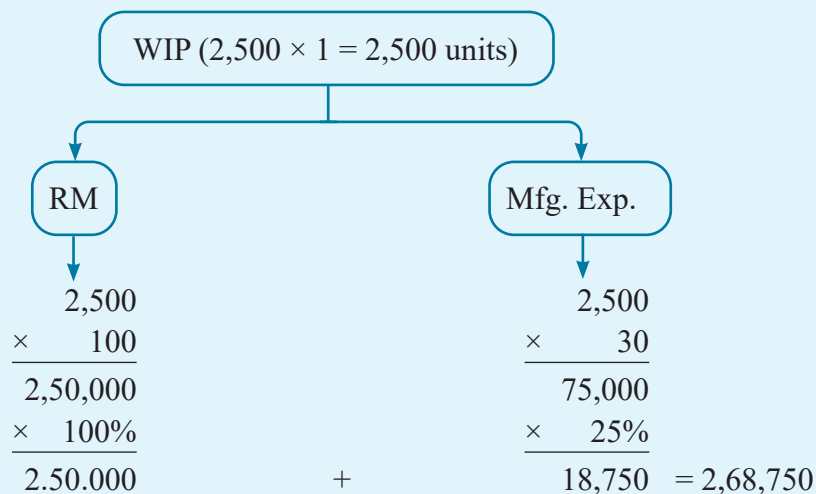
Capital requirement if the desired cash balance is 5% of the gross working capital requirement, and work-in-progress is 25% complete with respect to manufacturing expenses.

Sol. Statement Showing Working Capital Requirements

Particulars	Amount
Current Assets	
Cash	68,750
Raw Material ($2,500 \times 2 \times 100$)	5,00,000
Work in Progress (W.N. 1)	2,68,750
Finished Goods ($2,500 \times 12 \times 130$)	1,62,500
Debtors ($2,500 \times 1 \times 150$)	3,75,000
Gross Working Capital	13,75,000

Working Notes:**1. Computation of amount of WIP:**

Degree of completion of WIP is 25% with respect to manufacturing expenses; hence WIP constitutes Raw Material 100%, Manufacturing Expenses 25%.

**2. Computation of amount of cash:**

Cash = 5% of Gross Working Capital

Current Assets other than cash = 95% of Gross Working Capital

95% $\times$ 13,06,250

5% $\times$ X

$$X = \frac{13,06,250 \times 5}{95} = 68,750 \quad \text{Cash} = 68,750$$

3. As profit is not our investment, it is to be excluded for the computation of Working Capital.**5.** Calculate the amount of working capital requirement for SRCC Ltd. from the following information:

Particulars	₹
Raw material	160
Direct labor	60
Overheads	120
Total cost	340
Profit	60
Selling price	400

Raw materials are held in stock on an average for one month. Materials are in process on an average for half-a-month. Finished goods are in stock on an average for one month.

Credit allowed by suppliers is one month and credit allowed to debtors is two months. Time lag in payment of wages is 1 1/2 weeks. Time lag in payment of overhead expenses is one month. One fourth of the sales are made on cash basis.

Cash in hand and at the bank is expected to 50,000: and expected level of production amounts to 1,04,000 units for a year of 52 weeks. You may assume that production is carried on evenly throughout the year and a time period of four weeks is equivalent to a month.

Sol. Statement Showing Working Capital Requirements

Particulars	Amount
Current Assets	
Cash	50,000
Raw Material ($8,000 \times 1 \times 160$)	12,80,000
Work in Progress (W.N. 2)	10,00,000
Finished Goods ($8,000 \times 1 \times 340$)	27,20,000
Debtors ($8,000 \times 2 \times 340 \times 75\%$)	40,80,000
A. Total Current Assets	91,30,000
Current Liabilities	
Creditors ($8,000 \times 1 \times 160$)	12,80,000
Wages ($8,000 \times 1.5/4 \times 60$)	1,80,000
Overheads ($8,000 \times 1 \times 120$)	9,60,000
B. Total Current Liabilities	24,20,000
C. Net Working Capital (A – B)	67,10,000

Working Notes:

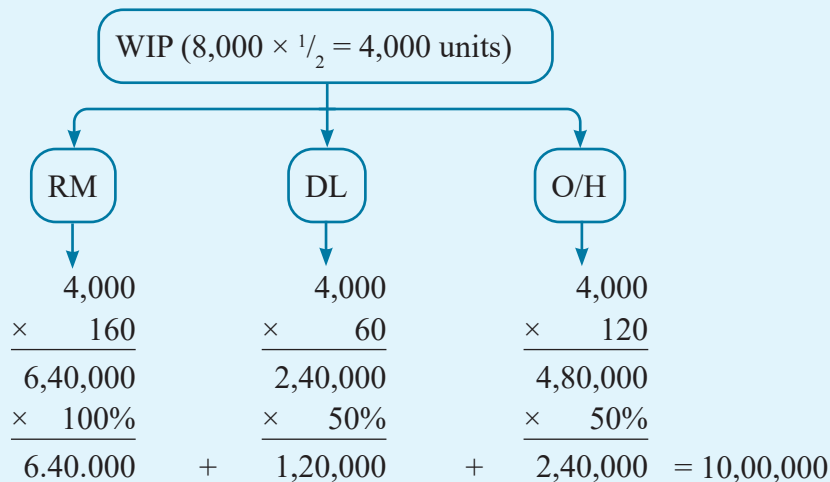
1. Computation of Monthly Production:

Annual Production = 1,04,000 units

Monthly Production = $\frac{\text{Annual Production}}{52} \times 4 = \frac{1,04,000}{52} \times 4 = 8,000$ units per month.

2. Computation of amount of WIP:

Degree of completion of WIP is not given; hence we assume that WIP constitutes Raw Material 100%, Labour 50% & Overheads 50%.



3. As profit is not our investment, it is to be excluded for the computation of Working Capital

SECURITY ANALYSIS

A. INTRODUCTION TO INVESTMENT AND SECURITIES

“Investment is the act of committing money or resources now, with the hope of future gain.”

– Zvi Bodie, 2016

Investment is essentially the trade-off of today’s consumption for tomorrow’s wealth. It can take various forms—from real estate and commodities to jewellery—but this chapter focuses on **financial assets** like **shares, bonds, and debentures**.

Unlike real assets (e.g., land, machinery), which are used to produce goods and services, **financial assets** represent **claims on income or assets** of real entities. They’re instruments like equity or debt securities issued by firms to raise capital and offer investors a more liquid and tradable option in vibrant markets.

Every investor faces key questions:

- a. **Where to invest?** (Shares, bonds, gold?)
- b. **How long to invest?**
- c. **What level of risk is acceptable?**

Balancing return expectations with risk appetite is essential in smart investing.

B. WHAT ARE SECURITIES?

Securities are financial instruments that allow the transfer of funds from investors (fund providers) to entities (fund seekers). They act as **evidence of ownership or debt** and entitle holders to returns and repayment at maturity or redemption.

a. TYPES OF SECURITIES

Broadly categorized into:

1. **Debt Securities** (like bonds, debentures)
2. **Equity Securities** (like shares, stocks)

Under **Section 2(h)** of the **Securities Contracts (Regulation) Act, 1956**, securities include:

- (a) Shares, scrips, stocks, and debentures of companies
- (b) Derivatives
- (c) Units of collective investment schemes or mutual funds
- (d) Government securities
- (e) Security receipts (under the SARFAESI Act)
- (f) Any instruments notified by the Central Government
- (g) Rights or interests in the above

In essence

Securities are the legal and financial bridge between capital seekers and capital providers. Understanding them is key to navigating the Indian securities market efficiently.

b. INVESTMENT: A SMART FINANCIAL COMMITMENT

Investment is the strategic placement of funds into assets to earn income or grow capital over time. It's a decision that involves *foregoing present consumption* in pursuit of *future financial gain*, but not without **time** and **risk**.

1. Core Attributes of Investment:

- (a) **Time** – Returns unfold over a future horizon.
- (b) **Risk** – Returns are uncertain, while the sacrifice (investment) is real.

2. The Triad of Investment Objectives:

Every smart investment should strike a balance between:

(a) Security

- (i) Refers to the **safety of the principal**.
- (ii) Protecting capital is a priority; returns are secondary.

Example: Treasury Bills (T-Bills) are safer than **corporate bonds** due to government backing.

(b) Liquidity

- (i) The ability to **convert investment into cash quickly**, without loss.
- (ii) Highly liquid assets: **Stocks**, some **bonds**.

Example: Fixed-term deposits, non-tradable bonds.

(c) Yield

- (i) The **net return** on investment.
- (ii) Even a secure, liquid asset may be unattractive if the yield is too low.
- (iii) A good investment gives **adequate yield** while respecting the investor's risk appetite.

c. INVESTMENT VS. SPECULATION

Benjamin Graham's Golden Rule

"An investment operation is one which, upon thorough analysis, promises safety of principal and an adequate return."

Key Differences

Aspect	Investment	Speculation
Basis	Careful analysis	Rumors or hunches
Objective	Capital protection + adequate return	High, quick profit
Risk Level	Measured and calculated	High and often unmeasured
Market	Primary and secondary markets	Mostly secondary market
Time Horizon	Long-term	Short-term

d. INVESTMENT VS. GAMBLING: A WORLD APART

While both **investment** and **gambling** involve money and an element of uncertainty, they differ **fundamentally in intent, process, and outcome**.

Gambling and betting are games of **pure chance**, with returns **dependent on unpredictable events**. There is **no research or analysis** involved—just luck, risk, and adrenaline.

Key Differences Between Investment and Gambling

Basis of Comparison	Investment	Gambling
Planning Horizon	Long-term (wealth-building over time)	Short-term (instant or quick outcomes)
Decision Foundation	Based on analysis of intrinsic value, trends, and financial fundamentals	Based on tips, hunches, and rumors
Nature	Planned and strategic	Unplanned and impulsive
Type of Risk	Commercial risk (related to market/business dynamics)	Artificial risk (created purely by uncertainty and chance)
Returns	Aim for a reasonable, risk-adjusted return	Known payoff, but often negative in long run
Primary Motive	Capital safety and consistent growth	Entertainment with the hope of sudden gain

C. SECURITY ANALYSIS: THE FIRST STEP IN INVESTMENT DECISIONS

Security Analysis is the **art and science of evaluating individual securities** (like stocks or bonds) to determine their **true or intrinsic value**. The goal? **To identify if a security is overvalued or undervalued** compared to its current market price.

a. TWO CORE APPROACHES TO SECURITY ANALYSIS

1. Fundamental Analysis

This is a **deep-dive, 3-tier evaluation** based on internal and external factors:

(a) Economic Analysis (Macro View)

The company's performance is linked to the nation's economic health.

Key indicators:

- (i) **GDP Growth** – Signals expansion or contraction of the economy.
- (ii) **Savings & Investment** – Fuel capital markets.
- (iii) **Inflation** – Mild inflation is healthy; high inflation is harmful.
- (iv) **Interest Rates** – Lower rates = cheaper finance = higher profitability.
- (v) **Budget & Tax Structure** – Deficits, incentives, or surpluses impact cost structures and investor sentiment.
- (vi) **Other Factors** – Monsoon, agriculture, infrastructure, demographic trends.

(b) Industry Analysis (Sectoral View)

Focuses on the performance and structure of the industry the company belongs to.

Key considerations:

(i) Life Cycle Stage:

- (1) *Pioneering*: High innovation, high profit, attracts new firms.
- (2) *Expansion*: Stable growth, increased competition, better capital access.
- (3) *Stagnation*: No innovation, limited growth, difficult capital access.

(ii) Barriers to Entry/Exit, Tech Disruption, Substitutes, Demand Shifts.

(c) Company Analysis (Micro View)

(i) Look inside the company using:

- (1) **Internal Sources**: Financial statements – income statement, balance sheet, cash flow.
- (2) **External Sources**: Analyst reports, industry news, regulatory filings.

(ii) Traditional Tools:

- (1) Dividend forecast
- (2) Price-Earnings (P/E) ratios

- (3) Dividend Discount Models (DDM)

(iii) Modern Tools:

- (1) Regression & Correlation Analysis
- (2) Trend Analysis
- (3) Decision Trees & Simulations

b. RATIO ANALYSIS: UNLOCKING THE NUMBERS BEHIND FINANCIAL STATEMENTS

A **ratio** is a mathematical relationship between two related financial figures. In the context of financial analysis, it acts as a diagnostic tool to measure a company's performance, financial health, and efficiency.

Ratios are calculated primarily from two key financial statements:

◆ **Balance Sheet**

◆ **Profit & Loss Account (Income Statement)**

Financial ratios are broadly classified into **six categories**:

1. Liquidity Ratios

Purpose: Measure the firm's ability to meet short-term obligations.

Ratio	Formula	Insight
Current Ratio	Current Assets/Current Liabilities	Can the company pay its bills in the short run?
Quick Ratio (Acid Test)	(Current Assets - Inventory)/ Current Liabilities	Can the company meet obligations without selling inventory?

2. Turnover (Activity) Ratios

Purpose: Evaluate how efficiently assets are utilized.

Ratio	Formula	Insight
Inventory Turnover	Net Sales/Inventory or COGS/ Avg. Inventory	How fast inventory is sold?
Receivables Turnover	Net Credit Sales/Avg. Accounts Receivable	How efficiently receivables are collected?
Capital Employed Turnover	Net Sales/Avg. Capital Employed	How well capital is used to generate revenue?
Working Capital Turnover	Net Sales/Working Capital	Revenue generated per unit of working capital.
Asset Turnover	Net Sales/Avg. Total Assets	Asset efficiency in generating sales.

3. Leverage Ratios

Purpose: Assess financial structure and risk from debt.

Ratio	Formula	Insight
Debt-to-Assets	Total Debt/Total Assets	% of assets financed by debt.
Debt-to-Equity	Total Debt/Equity	Leverage used by the company.
Debt-to-Capitalization	Total Debt/(Debt + Equity + Others)	Debt weight in capital mix.
Net Debt-to-Capital	(Debt-Cash)/(Capital - Cash)	Adjusted leverage position.
Operating Leverage	(Sales-VC)/Profit	Sensitivity of profit to sales changes.
Financial Leverage (DFL)	EBIT/EBT	Effect of interest expense on earnings.

4. Profitability Ratios

Purpose: Measure the company's ability to generate profit.

Ratio	Formula	Insight
Gross Profit Margin	Gross Profit/Net Sales $\times 100$	Profit after COGS.
Operating Profit Margin	Operating Profit/Net Sales $\times 100$	Profit after operating expenses.
Net Profit Margin	Net Profit/Net Sales $\times 100$	Bottom-line profitability.
Return on Equity (ROE)	Net Profit/Shareholder's Equity $\times 100$	Efficiency of shareholder capital.
Return on Assets (ROA)	Net Profit/Total Assets $\times 100$	Asset efficiency in generating income.
ROCE	EBIT/Capital Employed	Overall return on all funds used.

5. Coverage Ratios

Purpose: Evaluate the firm's ability to meet debt and interest obligations.

Ratio	Formula	Insight
Interest Coverage	Operating Income/Interest Expense	Can interest be paid comfortably?
Debt Service Coverage	Operating Income/Total Debt Service	Overall debt repayment ability.
Cash Coverage	Cash/Interest Expense	Interest paid from available cash.
Asset Coverage	(Assets – Intangibles – Net Liabilities)/Interest Expense	Debt backed by assets.

6. Valuation (Market Value) Ratios

Purpose: Help investors assess stock value and investment potential.

Ratio	Formula	Insight
P/E Ratio	Share Price/EPS	What investors pay per rupee of earnings.
P/B Ratio	Market Price/Book Value per Share	Market value vs. book value.
P/S Ratio	Mkt Cap/Sales or MVS/SPS	Valuation per unit of sales.
P/CF Ratio	Share Price/Operating CF per Share	Valuation against cash flow.
PEG Ratio	P/E/EPS Growth	Growth-adjusted valuation; $PEG \leq 1$ is ideal.

Where,

P/E	Price to Earnings Ratio
EPS	Earnings Per Share
P/B	Price to Book Ratio
P/S	Price to Sales Ratio
Mkt Cap	Market Capitalization
MVS	Market Value of Sales
SPS	Sales Per Share
P/CF	Price to Cash Flow Ratio
CF	Cash Flow
PEG	Price/Earnings to Growth Ratio

7. Why Ratio Analysis Matters

- (a) **Compare** performance across time and companies.
- (b) **Identify** trends, risks, and strengths.
- (c) **Support** decisions for investment, lending, or strategic planning.

c. COMPARATIVE FINANCIAL STATEMENTS

Comparative Financial Statements present the **financial position of a company over two or more periods**, allowing for easy identification of trends, growth, or decline.

They Provide

- ◆ **Time-based insights** into performance.
- ◆ Both **absolute changes (in ₹)** and **percentage changes (%)** between years.

1. Benefits:

- Simplifies year-over-year analysis.
- Highlights strengths, weaknesses, and trends.
- Aids in strategic and investment decision-making.

2. Example: Comparative Balance Sheet of Tanuja Ltd.

Given Balance Sheet Data:

Particulars	2021 (₹)	2022 (₹)
I. Equity and Liabilities		
Share Capital	3,00,000	3,50,000
Long-term Borrowings	2,00,000	1,00,000
Trade Payables	1,00,000	1,50,000
Total	6,00,000	6,00,000
II. Assets		
Fixed Assets (Tangible)	3,00,000	4,00,000
Trade Receivables	3,00,000	2,00,000
Total	6,00,000	6,00,000

Solution:

Comparative Balance Sheet

Particulars	2021 (₹)	2022 (₹)	Change (₹)	% Change
I. Equity and Liabilities				
Share Capital	3,00,000	3,50,000	+50,000	+16.67%
Long-term Borrowings	2,00,000	1,00,000	-1,00,000	-50.00%
Trade Payables	1,00,000	1,50,000	+50,000	+50.00%
Total	6,00,000	6,00,000	0	0.00%
II. Assets				
Fixed Assets (Tangible)	3,00,000	4,00,000	+1,00,000	+33.33%
Trade Receivables	3,00,000	2,00,000	-1,00,000	-33.33%
Total	6,00,000	6,00,000	0	0.00%

d. TREND ANALYSIS

Trend analysis examines financial data over multiple periods using a base year as a reference. It shows whether key metrics like sales, profits, or costs are increasing or decreasing over time.

1. Key Benefits

- Detects growth trends, declines, or stagnation.
- Highlights management efficiency or inefficiency.
- Uncovers red flags, e.g., rising sales but stagnant profits or bloated inventories.

2. Example

- (a) If sales rise consistently but inventory also increases, it might signal overstocking or falling market demand.
- (b) If sales grow but profits don't, it could suggest cost control issues or inefficient operations.

e. COMMON SIZE STATEMENTS

Common-size financial statements present each line item as a percentage of a chosen base figure, simplifying comparison across:

- ◆ Different years (trend)
- ◆ Different companies (benchmarking)

1. Where It's Used

- (a) **Balance Sheet:** Each item as % of Total Assets.
- (b) **Income Statement:** Each item as % of Net Sales (top-line revenue).

2. Example

Common Size Example: Apple Inc.

Period: 2023-2025 (All figures in ₹ Million)

Item	2023	2024	2025	% of Sales (2023)	% (2024)	% (2025)
Total Revenue	215,639	229,234	265,595	100.0%	100.0%	100.0%
Cost of Revenue	131,376	141,048	163,756	60.9%	61.5%	61.7%
Gross Profit	84,263	88,186	101,839	39.1%	38.5%	38.3%
R&D Expenses	10,045	11,581	14,236	4.7%	5.1%	5.4%
SG&A	14,194	15,261	16,705	6.6%	6.7%	6.3%
Net Income	45,687	48,351	59,531	21.2%	21.1%	22.4%

3. Advantages

- (a) Simplifies complex data for quick interpretation.
- (b) Facilitates trend analysis over the years.
- (c) Ideal for cross-company comparisons, especially when firms vary in size.

4. Disadvantages

- (a) No universal benchmarks for line-item percentages.
- (b) Comparisons may be misleading if statements aren't prepared consistently.

Key Takeaways

Tool	Core Use	Best For
Trend Analysis	Tracks financial performance over time	Spotting trends, growth or decline
Common Size Statement	Shows proportions for better comparability	Peer comparison, margin analysis

f. FUND FLOW ANALYSIS

Fund Flow Analysis is a financial tool that tracks the movement of funds (i.e., working capital) in and out of a business over a period, unlike the balance sheet, which gives a static view at a point in time.

It answers important financial questions like:

- ◆ How were profits used?
- ◆ How were dividends and debts financed?

- ◆ What was the source of funds for buying fixed assets?
- ◆ Where did the proceeds from asset sales go?

1. Key Component: Funds Flow Statement

It consists of:

- Sources of Funds** – Where the money came from (e.g., net profit, issuance of shares, sale of assets).
- Application of Funds** – How the money was used (e.g., purchase of fixed assets, repayment of loans, dividend payments).

2. Example:

- Tanuja's Limited presents the following information, and you are required to calculate funds from operations.

Profit and Loss Account

Particulars	Amount (Rs. in Lakhs)	Particulars	Amount (Rs. in Lakhs)
To Operation Expenses	1,00,000	By Gross Profit	2,00,000
To Depreciation	40,000	By Gain on Sale of Plant	20,000
To Loss on Sale of Building	10,000		
To Advertisement Suspense Account	5,000		
To Discount Allowed	500		
To Discount on Issue of Shares written off	500		
To Goodwill written off	12,000		
To Net Profit	52,000		2,20,000

Solution:

Calculation of Funds from Operations

Particulars	Amount (Rs. in Lakhs)	Amount (Rs. in Lakhs)
Net Profit (given)		52,000
Add: Non-fund or non-operating items which have been debited to Profit & Loss A/c:		
Depreciation	40,000	
Loss on Sale of Building	10,000	
Advertisement written off	5,000	
Discount written off	500	
Goodwill written off	12,000	67,500
		1,19,500
Less: Non-fund or non-operating items which have been credited to Profit & Loss A/c:		
Gain on Sale of Plant	20,000	20,000
Funds from Operations		99,500

Alternatively,

Adjusted Profit & Loss Account

Particulars	Amount (Rs. in Lakhs)	Particulars	Amount (Rs. in Lakhs)
To Depreciation	40,000	By Opening balance	—
To Loss on Sale of Building	10,000	By Gain on Sale of Plant	20,000
To Advertisement Suspense A/c	5,000	By Funds from Operations (Balancing Figure)	99,500
To Discount written off	500		
To Goodwill written off	12,000		
To Closing balance	52,000		
	1,19,500		1,19,500

(b) The Balance Sheets of Tanuja's Corporation as on 31st December, 2020 and 31st December, 2021 are as follows:

Liabilities	2020 (Rs.)	2021 (Rs.)	Assets	2020 (Rs.)	2021 (Rs.)
Share Capital	5,00,000	7,00,000	Land Buildings and	80,000	1,20,000
Profit & Loss	1,00,000	1,60,000	Plant Machinery and	5,00,000	8,00,000
General Reserve	50,000	70,000	Stock	1,00,000	75,000
Sundry Creditors	1,53,000	1,90,000	Debtors	1,50,000	1,60,000
Bills Payable	40,000	50,000	Cash	20,000	20,000
Outstanding Expenses	7,000	5,000			
	8,50,000	11,75,000		8,50,000	11,75,000

Additional Information:

- Rs 50,000 depreciation has been charged on plant and machinery during 2021.
- A piece of machinery was sold for Rs. 8,000 during 2021. It had cost Rs 12,000, depreciation of Rs. 7,000 had been provided on it.

Prepare a Schedule of Changes in Working Capital and a Statement showing the Sources and Applications of Funds for 2021.

Solution:

Schedule of Changes in Working Capital

Items	2020	2021	Changes in Working Capital	
			Increase	Decrease
Current Assets:				
Stock	1,00,000	75,000	—	25,000
Debtors	1,50,000	1,60,000	10,000	—
Cash	20,000	20,000	—	—
	2,70,000	2,55,000		

Current Liabilities:				
Sundry Creditors	1,53,000	1,90,000	—	37,000
Bills Payable	40,000	50,000	—	10,000
Outstanding Expenses	7,000	5,000	2,000	—
	2,00,000	2,45,000		
Working Capital (Current Assets - Current Liabilities)	70,000	10,000		
Net Decrease in Working Capital		60,000	60,000	
	70,000	70,000	72,000	72,000

Statement and Sources of Application of Funds
(For the year ended December 31, 2021)

Sources	Rs.	Application	Rs.
Funds from Operations (1)	1,27,000	Purchase of Land and Buildings	40,000
Issue of Shares	2,00,000	Purchase of Plant and Machinery (2)	3,55,000
Sale proceeds of machinery	80,000		
Decrease in working capital	60,000		
	3,95,000		3,95,000

Workings

(1) Adjusted Profit & Loss Account

Particulars	Rs.	Particulars	Rs.
To Plant & Machinery A/c (Depreciation)	50,000	By Balance b/d (Opening Balance)	1,00,000
To General Reserve (Transferred during 2021)	20,000	By Plant & Machinery (Profit on sale)	3000
To Balance c/d	1,60,000	By Funds from Operations (Balancing figure)	1,27,000
	2,30,000		2,30,000

(2) Plant & Machinery

Particulars	Rs.	Particulars	Rs.
To Balance b/d	5,00,000	By Bank (Sale of machinery)	8000
To Profit & Loss A/c (profit on sale)	3,000	By Profit & Loss A/c (Depreciation)	50,000
To Bank A/c (Purchase of Machinery & Plant)	3,55,000	By Balance c/d	8,00,000
(Balancing figure)	8,58,000		8,58,000

g. CASH FLOW STATEMENT

A Cash Flow Statement shows inflows and outflows of cash and cash equivalents during a financial period.

It is prepared using data from:

- ◆ The Balance Sheet
- ◆ The Profit & Loss Account
- ◆ Additional transaction details

1. It helps assess:

- (a) Liquidity and solvency
- (b) Cash-based profitability
- (c) How profit differs from cash position

2. Classification of Cash Flows

- (a) **Operating Activities** – Core business operations (e.g., cash from sales, payments to suppliers)
- (b) **Investing Activities** – Buying/selling of long-term assets (e.g., land, machinery)
- (c) **Financing Activities** – Issuing shares, borrowing, repaying loans, paying dividends

3. Example:

- (a) From the following Profit and Loss Account of Tanuja's Ltd., calculate Net Cash Flows from operating activities.

Particulars	₹	Particulars	₹
To depreciation	40,800	By gross profit	3,94,400
To rent	72,000	By profit on sale of building	53,600
To administrative expenses	48,000	By profit on sale of furniture	50,400
To salaries	64,000	By income tax refund	18,400
To loss on sale of plant	12,800		
To provision for bad debts	64,000		
To goodwill written off	34,400		
To loss on sale of machinery	19,600		
To provision for tax	24,000		
To proposed dividend	48,000		
To net profit	89,200		
Total	5,16,800	Total	5,16,800

Solution:

Calculation of Net Cash Flow from Operating Activities of Tanuja's Limited

Net Cash Flow from Operating Activities (in ₹)

Particulars	Amount (₹)
Profit for the Year	89,200
Add: Non-Cash & Non-Operating Expenses	
Provision for Bad Debts	64,000
Depreciation	40,800
Goodwill Written Off	34,400
Loss on Sale of Plant	12,800

Loss on Sale of Machinery	19,600
Provision for Tax	24,000
Proposed Dividend	48,000
Total Additions	2,43,600
Subtotal (Before Deductions)	3,32,800
Less: Non-Operating Incomes	
Profit on Sale of Building	53,600
Profit on Sale of Furniture	50,400
Income Tax Refund	18,400
Total Deductions	1,22,400
Net Cash Flow from Operating Activities	2,10,400

(b) From the following information extracted from the book of Tanuja's Ltd. for the year 2019-20, calculate the net cash flow from investing activities.

Net Cash Flow from Investing Activities of Tanuja's Ltd.

(All figures in ₹)

Given Comparative Balances

Particulars	2018-19 (₹)	2019-20 (₹)
Furniture	₹1,00,000	₹1,20,000
Machinery	₹15,00,000	₹18,00,000
Building	₹20,00,000	₹19,80,000
Land (at cost)	₹18,00,000	₹16,00,000
Investment (long-term)	₹90,000	₹2,10,000

Additional information is given as follows:

- ◆ Depreciation charged on furniture during the year was ₹10,000.
- ◆ Depreciation on machinery charged during the year was ₹25,000.
- ◆ The machinery, the book value of which was ₹80,000, sold for ₹75,000.
- ◆ The land was sold at a profit of ₹90,000.

Solution:

Cash Flow from Investing Activities

Particulars	₹
Cash Receipts	
Sale of Land	₹2,90,000
Sale of Machinery	₹75,000
Total Inflows	₹3,65,000
Cash Payments	
Purchase of Furniture	₹30,000
Purchase of Machinery	₹4,05,000
Purchase of Investment	₹1,20,000
Total Outflows	₹5,55,000
Net Cash Flow from Investing	(₹1,90,000)

Note: A negative cash flow indicates that outflows exceed inflows.

Working Notes (₹)

Furniture Account

Particulars	₹	Particulars	₹
To Balance b/d	₹1,00,000	By Depreciation	₹10,000
To Bank - Purchase	₹30,000	By Balance c/d	₹1,20,000
Total	₹1,30,000	Total	₹1,30,000

Machinery Account (All figures in ₹)

Particulars	Amount (₹)	Particulars	Amount (₹)
To Balance b/d	₹15,00,000	By Depreciation	₹25,000
To Bank - Purchase of Machinery	₹4,05,000	By Bank - Sale of Machinery	₹75,000
		By Loss on Sale of Machinery (₹80,000 ₹75,000)	₹5,000
		By Balance c/d	₹18,00,000
Total	₹19,05,000	Total	₹19,05,000

Building Account

Particulars	₹	Particulars	₹
To Balance b/d	₹20,00,000	By Depreciation (Bal. fig.)	₹20,000
		By Balance c/d	₹19,80,000
Total	₹20,00,000	Total	₹20,00,000

Land Account

Particulars	₹	Particulars	₹
To Balance b/d	₹18,00,000	By Bank - Sale of Land	₹2,90,000
To Profit & Loss (Gain)	₹90,000	By Balance c/d	₹16,00,000
Total	₹18,90,000	Total	₹18,90,000

D. TECHNICAL ANALYSIS

Technical Analysis is a method of predicting **future price movements** of stocks based on the **study of past market data**, primarily **price and volume**.

Unlike **fundamental analysis**, which relies on company financials (profits, dividends, assets), technical analysis focuses purely on **market behaviour**.

KEY ASSUMPTIONS OF TECHNICAL ANALYSIS

1. **Price reflects everything:** All known and unknown information (including future expectations) is already reflected in the stock price.
2. **Prices move in trends:** Stock prices typically follow identifiable upward or downward movements.
3. **History repeats itself:** Market patterns (like head & shoulders, double tops, etc.) repeat over time due to recurring investor behaviour.

4. **Demand & Supply rule:** Prices are determined by the equilibrium between buyers and sellers.

5. **Charts can detect trends:** Shifts in demand and supply create identifiable patterns on charts.

Technical analysts ignore qualitative factors like **sentiment, ethics, or management behaviour**—they believe **the chart tells all**.

E. DOW JONES THEORY: THE FOUNDATION OF TECHNICAL ANALYSIS

Formulated by **Charles H. Dow** (founder of Dow Jones & Co. and editor of *The Wall Street Journal*), this theory identifies long-term price patterns in markets and classifies them into **three trends**:

a. PRIMARY TREND (1–3 YEARS)

1. **Direction:** The **major movement** of the market.

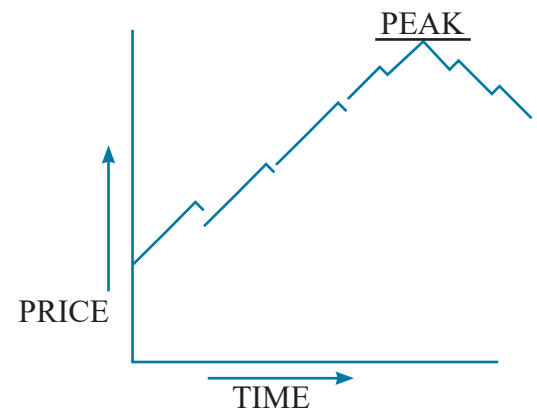
2. **Types:**

(a) **Bullish Phase** – Overall rising market with higher highs and higher lows.

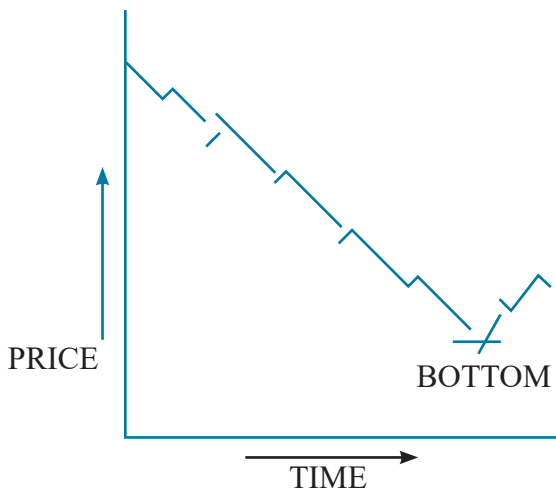
(b) **Bearish Phase** – Overall falling market with lower lows and lower highs.

Graph of Bullish Phase

In a bullish phase, after each peak, there is a fall but the subsequent rise is higher than the previous one. The prices reach higher level with each rise. After the peak has been reached, the primary trend now turns to a bearish phase.



Graph of a Bearish Phase



In a bearish phase, the overall trend is one of decline in share values. After each fall, there is a slight rise, but the subsequent fall is even sharper.

b. SECONDARY TREND (3 WEEKS TO 3 MONTHS)

1. **Temporary reversal** of the **primary trend**.

2. **In a bull market**, temporary price **decline** (profit-taking, corrections).

3. **In a bear market**, temporary price **rises** (rallies).

Usually retraces **1/3 to 2/3** of the primary trend's movement.

c. MINOR TREND (UP TO 3 WEEKS)

1. Short-term daily fluctuations.
2. Represent **noise or temporary reactions** to news or speculation.
3. Not reliable for long-term decisions, but helpful for **timing entries and exits**.

F. TOOLS OF TECHNICAL ANALYSIS

Technical analysis relies on two key variables:

Price movements

Trading volumes

a. TECHNICAL CHARTS

Charts are graphical representations of price and volume data over time. The aim is to:

- ◆ Detect **market trends** and **reversal signals**
- ◆ Forecast future **price direction**.
- ◆ Interpret **volume spikes** with price action (e.g., high price + high volume = strength)

A drop in price **with volume increase** = **bearish sign**

A price rise **with volume increase** = **bullish sign**

1. Line Chart

- (a) Connects the **closing prices** of each day.
- (b) Ignores intraday highs/lows and opening price.
- (c) Good for a **simple overview** of trend direction over time.

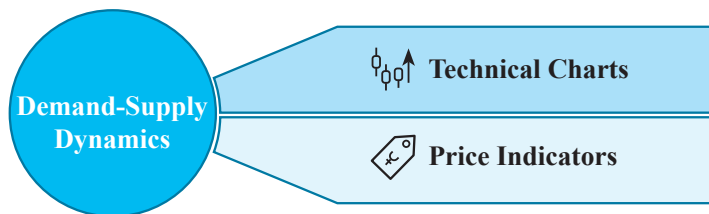
Example Use: Tracking long-term growth of a stock.



2. Bar Chart

- (a) Shows **OHLC**: Open (left tick), High-Low (vertical line), Close (right tick).
- (b) Often color-coded: **Black for gains, Red for losses**.
- (c) Reveals **intraday price range** and daily sentiment.

Visualizing Demand-Supply Dynamics



Example Use: Analyzing daily volatility or breakout potential.



3. Candlestick Chart

- (a) Visual and intuitive: Shows **open, high, low, close** with a “**body**” and **wicks**.
- (b) **Red/black body** = price fell.
 White/green body = price rose.
- (c) Common patterns: **Doji, Hammer, Shooting Star**

Example Use: Detecting trend reversals and entry/exit signals.



4. Point & Figure Chart

- (a) Focuses **only on price changes**—ignores time and volume.
- (b) Uses **X** for rising prices, **O** for falling prices.
- (c) Columns change when the trend reverses.

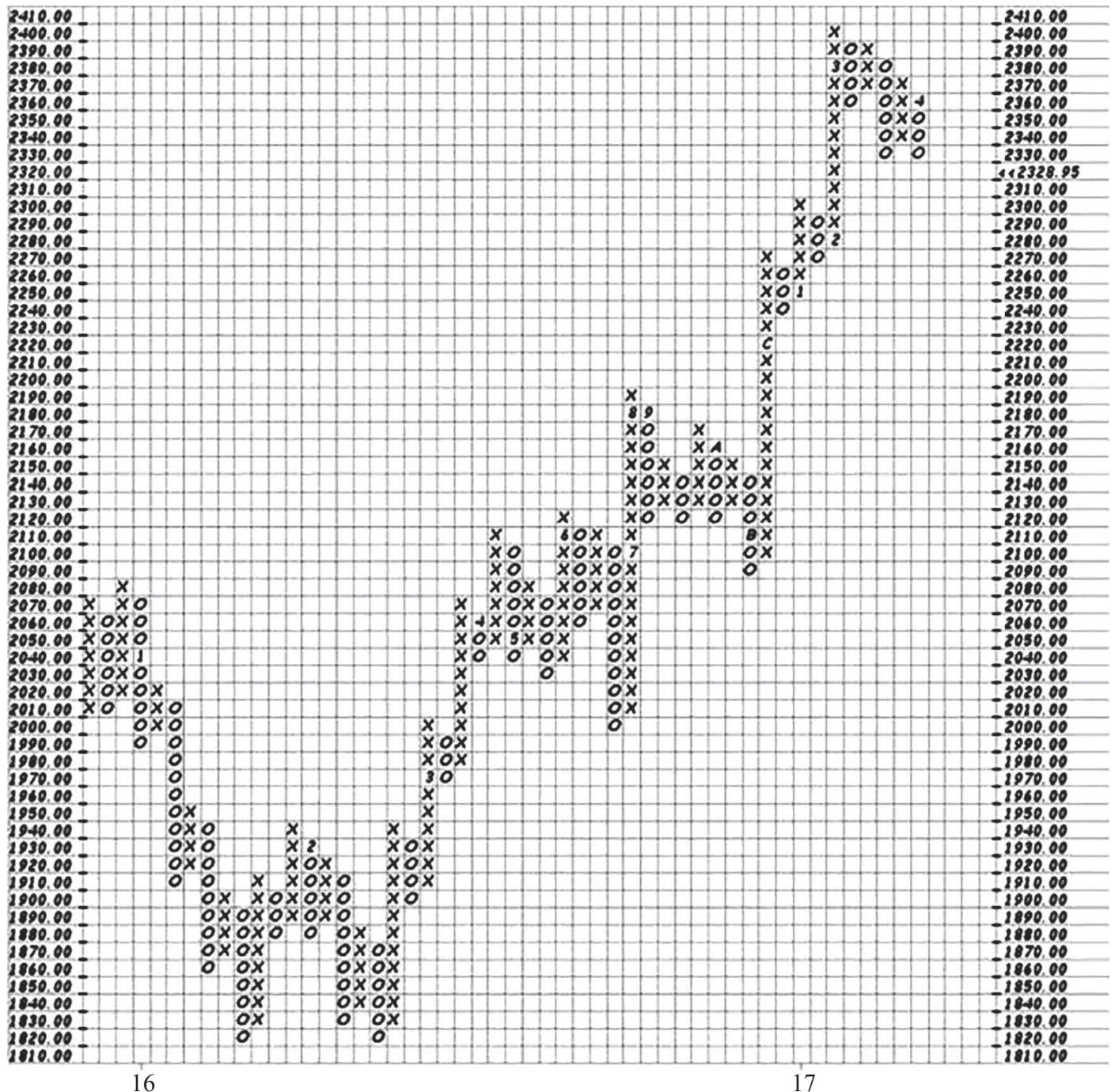
Example Use: Identifying breakouts and support/resistance levels.

\$\$SPX S&P 500 Large Cap Index INDX

13-Apr-2017, 16:00 ET, daily, O: 2,341.98, H: 2,348.26, L: 2,328.95, C: 2,328.95, V: 1765447936,
Chg: -15.98 (-0.68%)

No recent chart pattern found

Scaling Traditional [Reversal 3]



b. CHART PATTERNS: INTERPRETING MARKET SIGNALS

Once technical charts (line, bar, candlestick, etc.) are plotted, analysts look for **repetitive price patterns** that suggest **future market behaviour**. These patterns reflect **investor psychology**, **market sentiment**, and **support-resistance dynamics**.

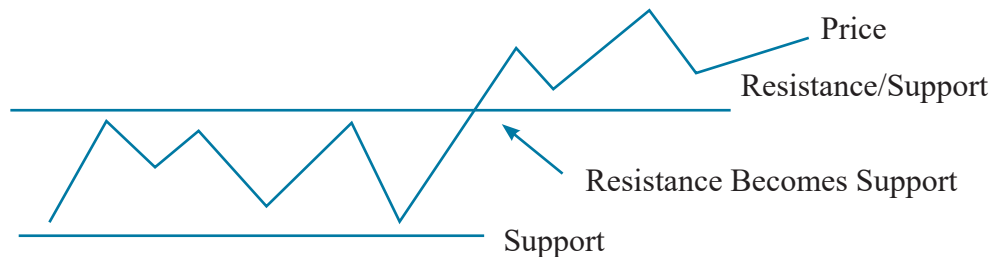
1. Support and Resistance Levels

Term	Meaning
Support Level	A price level where the stock tends to stop falling and bounces back up . Indicates buying interest .
Resistance Level	A price level where the stock tends to stop rising and starts falling . Indicates selling pressure .

Breaching Support = Start of bearish phase

Breaking Resistance = Start of bullish phase

Support and Resistance



2. Head and Shoulders Pattern

This classic pattern signals a trend reversal.

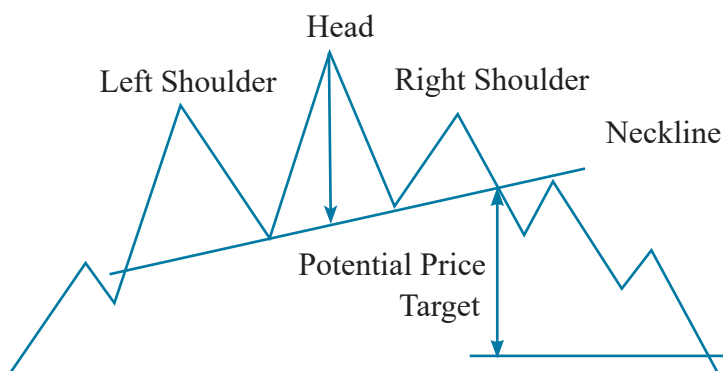
(a) Head and Shoulders Top (HST) – Bearish reversal

- (i) Left Shoulder: Price rise + fall
- (ii) Head: Higher rise + fall
- (iii) Right Shoulder: Lower rise + fall
- (iv) Neckline: Support level, once broken → signals decline
- (v) Used to predict a downtrend after an upward movement.

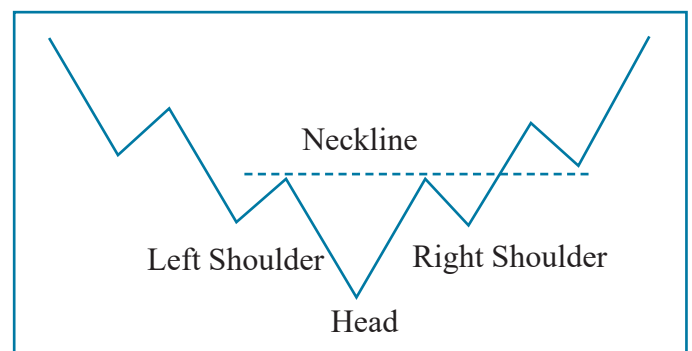
(b) Inverse Head and Shoulders (IHST) – Bullish reversal

- (i) Same as HST but inverted
- (ii) Predicts upward breakout after a downtrend

Head and Shoulders Top (HST) Pattern

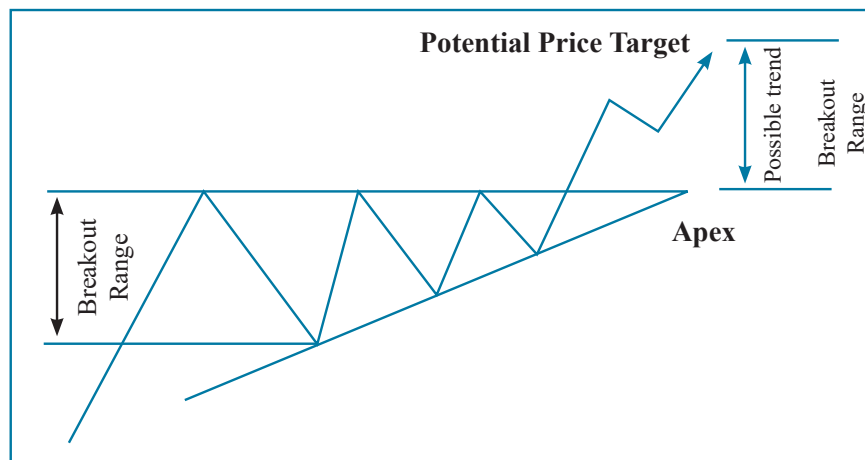


Inverse Head and Shoulder Top (IHST) Pattern



3. Triangle or coil formation

This pattern represents a pattern of uncertainty. Hence it is difficult to predict which way the price will break out.

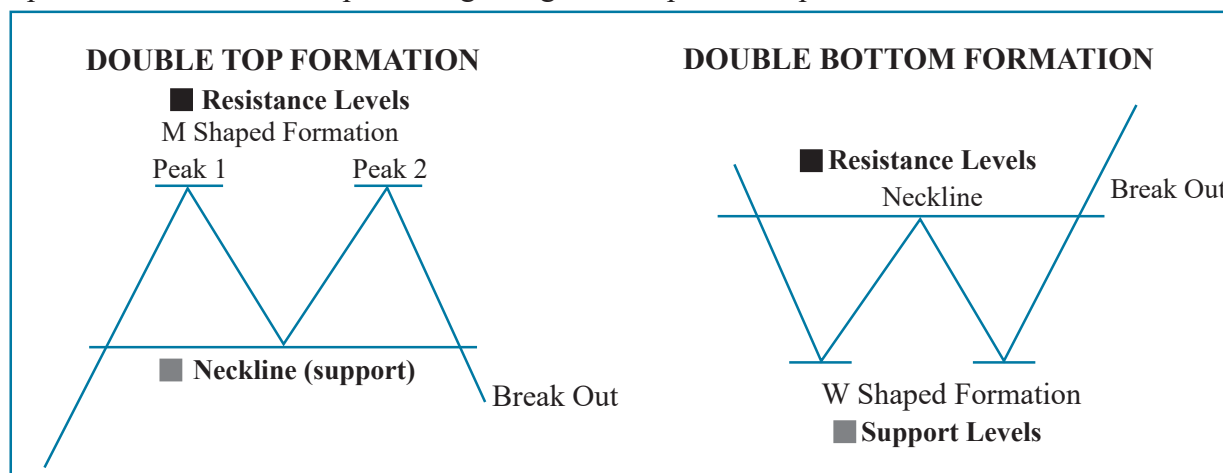


4. Double Top Formation

It represents a bearish development, signaling that the price is expected to fall.

5. Double bottom formation

It represents a bullish development, signaling that the price is expected to rise.



6. Limitations of Charts:

Despite their popularity, **charts** are not foolproof. They come with several **limitations**:

Limitation	Explanation
Subjective Interpretation	Analysts may draw different conclusions from the same chart pattern.
Frequent Short-Term Changes	Chart signals can shift rapidly, leading to conflicting buy/sell recommendations .
Charts Ignore Fundamental Factors	Decisions are often made purely on chart patterns , ignoring earnings, news, or macro trends.
False Signals	Patterns may fail to complete or lead to premature entries/exits.

c. TECHNICAL INDICATORS

1. Advance-Dcline Ratio (A/D Ratio)

Formula:

$A/D \text{ Ratio} = \text{Number of Advancing Stocks} / \text{Number of Declining Stocks}$

(a) > 1 = Bullish Trend

(b) < 1 = Bearish Trend

Used to confirm whether a market rise/fall is broad-based or limited to a few stocks.

2. Market Breadth Index

Formula:

$$\text{Market Breadth} = 2 \times (\text{Advances} - \text{Declines}) / \text{Total Traded Stocks}$$

Example:

If out of 1,000 stocks, 400 rose, 300 fell:

$$\text{Market Breadth} = 2 \times (400 - 300) / 300 = \mathbf{0.67}$$

(a) **Rising Breadth + Rising Index = Strong Bullish Signal**

(b) **Falling Breadth + Rising Index = Market may reverse**

3. Moving Averages

(a) Shows the average price over a fixed number of days (e.g., 50-day, 100-day, 200-day).

(b) Smoothness price data and indicates trend direction.

Price vs. MA	Signal
Price above MA	Bullish trend
Price below MA	Bearish trend

Longer MA (e.g., 200-day) → Fewer false signals, ideal for long-term investors.

4. Relative Strength Index (RSI)

A momentum oscillator that measures speed and magnitude of recent price movements.

RSI ranges from 0 to 100

(a) **> 70 → Overbought (Sell Zone)**

(b) **< 30 → Oversold (Buy Zone)**

Formula:

$$\text{RSI} = 100 - \left(\frac{100}{1 + \text{RS}} \right) \text{ where } \text{Rs} = \frac{\text{Average Gain}}{\text{Average Loss}}$$

RSI Calculation Example – Tanuja's Ltd.

Date	Price (*)	Gain	Loss
Oct 1	300	—	—
Oct 6	304	4	—
Oct 7	319	15	—
Oct 8	317	—	2
Oct 11	319	2	—
Oct 12	333	14	—
Oct 13	331	—	2
Oct 14	332	1	—
Oct 18	348	16	—
Oct 19	346	—	2

◆ **Total Gain** = 52 over 6 days → Avg Gain = 8.67

◆ **Total Loss** = 6 over 3 days → Avg Loss = 2

◆ **RS** = 8.67/2 = 4.335

◆ **RSI** = 100 (100/(1 + 4.335)) = **81.26**

Since RSI > 70 ⇒ Overbought zone ⇒ Signal to **Sell**

Conclusion: Role of Technical Indicators

Indicator	Purpose
Advance-Decline Ratio	Trend confirmation
Market Breadth Index	Market-wide strength analysis
Moving Averages	Trend direction & support/resistance levels
RSI	Momentum and overbought/oversold conditions

These indicators **complement charts** and improve decision-making by reducing false signals and providing **mathematical backing**.

5. Aroon Indicator

The **Aroon Indicator** is a **trend strength and reversal indicator**, developed by **Tushar Chande** in 1995. It helps identify:

- ◆ If a stock is trending or consolidating
- ◆ Whether the trend is upward or downward
- ◆ The strength of that trend

Components

(a) **Aroon Up**: Measures how recently a **high** occurred.

(b) **Aroon Down**: Measures how recently a **low** occurred.

Both are calculated over a **25-period range** (default).

Formulas:

$$\text{Aroon UP} = \left(\frac{25 - \text{Periods Since High}}{25} \right) \times 100$$

$$\text{Aroon Down} = \left(\frac{25 - \text{Periods Since Low}}{25} \right) \times 100$$

Interpretation

Aroon Up	Aroon Down	Signal
> 70	< 30	Strong uptrend (Buy)
< 30	> 70	Strong downtrend (Sell)
Both < 50	Both low	Consolidation (No trend)

Crossovers	
Aroon Up crosses above Aroon Down	Bullish signal
Aroon Down crosses above Aroon Up	Bearish signal

(c) Price Rate of Change (ROC)

ROC (Rate of Change) is a **momentum oscillator** that measures the **percentage change in price** over a specific number of past periods.

Formula:

$$\text{ROC} = \left(\frac{\text{Closing Price}_p - \text{Closing Price}_{p-n}}{\text{Closing Price}_{p-n}} \right) \times 100$$

- ◆ p = Most recent closing price
- ◆ n = Number of periods ago

Interpretation

ROC Value	Meaning
> 0	Positive momentum, uptrend
< 0	Negative momentum, downtrend
≈ 0	Consolidation, no strong trend

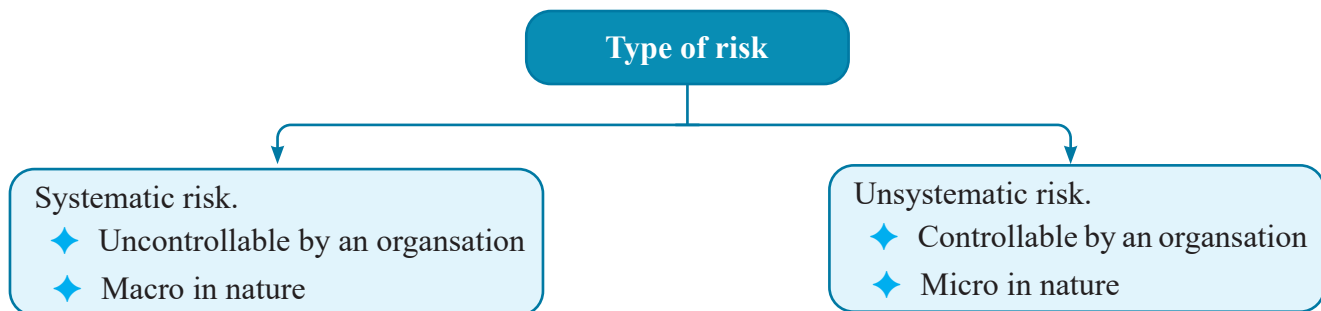
(a) **Rising ROC above zero** → Confirms bullish trend

(b) **Falling ROC below zero** → Confirms bearish trend

G. UNDERSTANDING RISK IN SECURITY ANALYSIS

Risk refers to the possibility that the **actual return** from an investment may be **less than the expected return**.

In finance, **risk** is broadly categorized into two types:



a. SYSTEMATIC RISK (MARKET RISK)

Systematic risk arises from **external, uncontrollable macro-level factors** that affect **all companies and industries** to varying degrees.

This type of risk **cannot be diversified** away.

1. Sources of Systematic Risk

- (a) Economic changes (e.g., recession, interest rate hikes)
- (b) Political instability (e.g., elections, war)
- (c) Global events (e.g., pandemics, oil price shocks)

2. Types of Systematic Risk

Type Explanation

(a) Interest Rate Risk	Fluctuations in market interest rates affect returns on fixed-income securities like bonds.
(b) Market Risk	Arises due to price volatility in the stock market. Affects all listed securities .
(c) Purchasing Power (Inflationary) Risk	Inflation erodes real returns , reducing the purchasing power of future cash flows.

b. UNSYSTEMATIC RISK (COMPANY-SPECIFIC RISK)

Unsystematic risk is linked to **internal, controllable factors** specific to a **particular company or industry**.

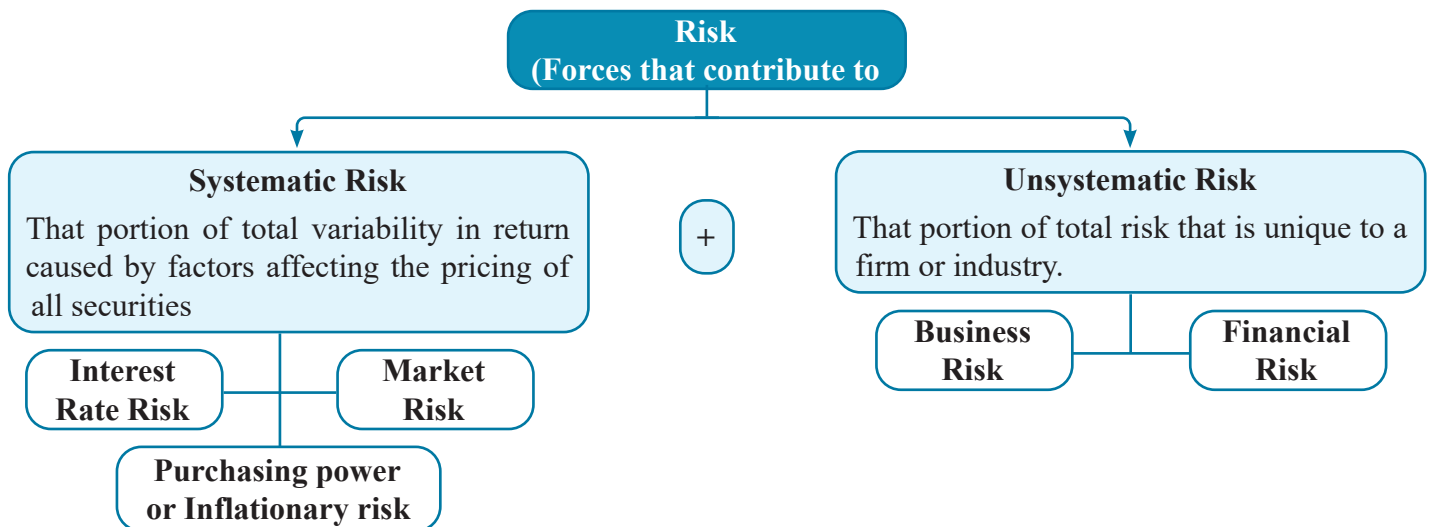
This type of risk **can be reduced** through **diversification**.

1. Sources of Unsystematic Risk

- (a) Poor management decisions
- (b) Labour strikes
- (c) Operational inefficiencies
- (d) Technological disruptions in one sector
- (e) Controllable by an organisation
- (f) Micro in nature

2. Types of Unsystematic Risk

Type	Explanation
(a) Business or Liquidity Risk	Arises from operational issues , such as poor sales, product obsolescence, or business cycles. Also known as liquidity risk .
(b) Financial or Credit Risk	Related to capital structure and borrowing . High debt levels increase the risk of default or bankruptcy.



H. RETURN OF THE SECURITY

Return is the **gain or loss** from an investment over a period. It is the **primary motivation** for investors and represents the **reward for bearing risk**.

a. COMPONENTS OF RETURN

Component	Description
Current Return	Regular income from investment (e.g., dividends or interest)
Capital Return	Change in price/value of the security (gain/loss)

Total Return = Current Return + Capital Return

- ◆ Current Return is always ≥ 0
- ◆ Capital Return can be positive, zero, or negative

b. MEASURING RETURN

1. Holding Period Return (HPR)

Formula:

$$\text{HPR} = \frac{\text{Income} + (\text{End Value} - \text{Initial Value})}{\text{Initial Value}}$$

Used to calculate total return **over any holding period**, regardless of its length.

2. Annualized Holding Period Return

To compare returns over **different timeframes**, use:

$$\text{Annualized HPR} = \left(\frac{\text{End Value} + \text{Income}}{\text{Initial Value}} \right)^{\frac{1}{n}} - 1$$

🔵 n = Number of years

Example 1: Mr. A's Investment in ₹

◆ Initial Investment = ₹10,000

◆ Final Value after 10 years = ₹23,800

$$\text{Total Return} = \frac{₹23,800 - ₹10,000}{₹10,000} = \frac{₹13,800}{₹10,000} = 1.38 = 138\%$$

Example 2: Portfolio With Tax & Inflation Adjustments (₹)

◆ Beginning Value = ₹1,00,000

◆ Dividends = ₹7,500

◆ Capital Appreciation = 12,500

◆ Ending Value = 1,20,000

◆ Tax Rate = 30%

◆ Inflation Rate = 10%

Nominal Return

$$\text{Total Return} = \frac{₹1,20,000 - ₹1,00,000}{₹1,00,000} = 20\%$$

After-Tax Return

◆ Dividend After Tax = ₹7,500 × 0.70 = ₹5,250

◆ Capital Gain After Tax = ₹12,500 × 0.70 = ₹8,750

$$\text{After-Tax Return} = \frac{₹1,00,000 + ₹5,250 + ₹8,750}{₹1,00,000} - 1 = \frac{₹1,14,000}{₹1,00,000} - 1 = 14\%$$

After-Tax Real Return

$$\text{Real Return} = \frac{₹1,14,000}{₹1,10,000} - 1 = 3.6364\%$$

Summary:

◆ Nominal Return = 20%

◆ After-Tax Return = 14%

◆ After-Tax Real Return = 3.6364%

Example 3: 3-Year Investment With Dividend (₹)

◆ Initial Investment = ₹10,000

◆ Dividend Each Year = ₹100

◆ Total Dividends = ₹300

◆ Selling Price after 3 Years = ₹12,000

$$\text{HPR} = \frac{₹300 + (₹12,000 - ₹10,000)}{₹10,000} = \frac{₹2,300}{₹10,000} = 23\%$$

Example 4: Single-Year Investment (₹)

- ◆ Buy Price = ₹50
- ◆ Dividend = ₹5
- ◆ Current Price = ₹60

$$\text{HPR} = \frac{\text{₹5} + (\text{₹60} - \text{₹50})}{\text{₹50}} = \frac{\text{₹15}}{\text{₹50}} = 30\%$$

Example 5: Comparing Two Mutual Funds (₹)

Fund	Start Value	End Value	Distributions	Years Held
X	₹100	₹150	₹5	3
Y	₹200	₹320	₹10	4

Solution:

HPR Calculations

$$\text{HPR}_X = \frac{\text{₹5} + (\text{₹150} - \text{₹100})}{\text{₹100}} = \frac{\text{₹55}}{\text{₹100}} = 55\% \quad \text{HPR}_Y = \frac{\text{₹10} + (\text{₹320} - \text{₹200})}{\text{₹200}} = \frac{\text{₹130}}{\text{₹200}} = 65\%$$

Annualized Return:

$$\text{Annualized}_X = (1 + 0.55)^{1/3} - 1 = 15.73\% \quad \text{Annualized}_Y = (1 + 0.65)^{1/4} - 1 = 13.34\%$$

Conclusion: Even though Fund Y had higher total HPR (65%), Fund X is the better investment based on annualized return.

I. APPROACHES TO SECURITY VALUATION

Before investing in a security, an investor must **assess its value**. There are **three key approaches** used by analysts:

a. FUNDAMENTAL ANALYSIS APPROACH

Every security has an **intrinsic value** based on **future earning potential**.

1. Key Factors Considered:

- Earnings per share (EPS)
- Business environment
- Management quality
- Industry position
- Capital structure
- Dividend policy

2. Decision Rule:

- If **Intrinsic Value** > **Market Price** → **BUY** (undervalued)
 - If **Intrinsic Value** < **Market Price** → **SELL** (overvalued)
- Used By:** Long-term investors and value investors (like Warren Buffett)

b. TECHNICAL ANALYSIS APPROACH

Stock price reflects **market psychology** (demand-supply), not fundamentals.

1. Tools Used:

- Chart Patterns** (Head & Shoulders, Triangles)
- Dow Theory**
- Indicators** (RSI, MACD, Moving Averages)

2. Assumptions:

- (a) History repeats itself
- (b) Price moves in trends
- (c) Charts can predict future movement

Used By: Traders, speculators, short-term investors

c. EFFICIENT MARKET HYPOTHESIS (EMH)

Core Idea:

Market prices reflect **all available information** at any given time.

It is **impossible to consistently outperform the market**.

1. EMH Types:

- (a) **Weak Form:** Prices reflect all past price data
- (b) **Semi-Strong Form:** Prices reflect all public info
- (c) **Strong Form:** Prices reflect all public & private info

2. Criticism:

- (a) Assumes no transaction cost
- (b) Assumes all investors interpret info the same way
- (c) Real-world anomalies and inefficiencies exist

Used By: Index fund investors and passive investors

d. COMPARISON SUMMARY

Approach	Focus	Belief	Tool
Fundamental Analysis	Business fundamentals	Stock has intrinsic value	Financial statements, DCF
Technical Analysis	Price trends, volume	Market trends repeat; psychology matters	Charts, indicators
Efficient Market Theory	Market efficiency	Stock always priced fairly; can't beat the market	Random walk model

e. CLARIFICATION OF VALUATION TERMS

Term	Meaning
Face Value	Also called nominal value ; the original issue value (e.g., ₹10/share)
Book Value	Net worth per share = $\frac{\text{Share Capital} + \text{Reserves} - \text{Liabilities}}{\text{No. of shares}}$

|Market Value | Current trading price of the share in the stock market |

|Intrinsic Value| The **true value** of the share based on future income or DCF |

Example:

If a share has:

◆ **Face Value** = ₹10

◆ **Issue Price** = ₹30

◆ **Premium** = ₹20

Why? Because the company has:

- 1. Strong profitability history
- 2. Market leadership
- 3. High reserves
- 4. Growth potential

Final Thought

1. **Fundamentalists** look at “what the stock is worth”
2. **Technicians** look at “what the price is doing”
3. **EMH theorists** say “the price is already correct”

CASE STUDY

Let us clarify the concept of book value a little further. Assuming that a company has been incorporated with an authorized capital of 2 crore shares of ₹10/- each and the company operates profitably for three years, the broad financial position of the company shall be as under:

(₹In lacs)

Item	Year 1	Year 2	Year 3
Income	600	1,200	2,400
Expenditure	800	1,000	1600
Profit/Loss	(200)	200	800
Equity Capital	2,000	2,000	2,000
Free Reserves	–200	0	800
Face Value/share	10	10	10
Book Value/share (Share capital + free reserves)	9	10	14

Book value of the share of the company became less than face value at the end of the first year due to the loss incurred by it. The book value was equal to the face value at the end of the second year due to recoupment of the loss. At the end of the third year the book value become ₹14/- due to building up of reserves. If, after the end of the third year the issuer wishes to come up with an offering of additional shares, the offer price will not be less than ₹14.

In actual market conditions does the book value track the market value? We may observe the trend of few company

Sl. No.	Name of the Company	Face ValuePer Share	Book Value	Market Value(As on 30 September)
(1)	(2)	(3)	(4)	(5)
1	HB Ltd.	1	13.8	168.70
2	FI Ltd.	10	136.3	250.50
3	IT Ltd.	5	314.3	3411.30
4	ITB Ltd.	10	175.8	349.80
5	BIS	10	299.3	229.70
6	RP Ltd.	10	20.4	22.60
7	RL Ltd.	10	138.2	562.60
8	IDD Ltd.	10	101.9	144.85
9	MTGL Ltd.	10	151.2	109.50
10	SC Ltd.	2	61.4	210.70
11	RLD Ltd.	5	190.5	821.15
12	HCT Ltd.	2	60.5	205.90
13	HPC Ltd.	10	173.8	172.45
14	CIP Ltd.	10	152.8	946.35
15	NES Ltd.	10	27.5	565.85

16	HH Ltd.	2	34.3	248.75
17	TISC Ltd.	10	93.6	116.50
18	LCET Ltd.	10	77.1	129.15
19	T&L Ltd.	10	133.0	167.00
20	BA Ltd.	10	283.2	380.05
21	BHL Ltd.	10	182.6	160.15
22	HIND Ltd.	10	621.5	522.10
23	ZTE Ltd.	1	99.2	57.50
24	BSS Ltd.	10	194.3	216.20
25	GRA Ltd.	10	295.3	309.00
26	GSIM Ltd.	10	75.3	367.15
27	GLX Ltd.	10	155.14	163.00
28	ASC Ltd.	10	171.17	138.15
29	CAS Ltd.	10	32.1	189.80
30	CIG Ltd.	10	18.2	131.20

We note that the market value is not equal to the book value for shares of any of the leading companies of the country. In fact, there is wide divergence between these two. The divergence is mostly on the upper side except in some cases. We can conclude, therefore, that book value is not a perfect indicator of the intrinsic value of a security. At best it can be an indicator of the floor value or base value below which the market value in normal circumstances should not slide. Book value is a historic indicator. It depicts what the company has earned and saved in the past. It does not reflect the future earning potential of the company.

Having considered that the book value is not an appropriate measure for ascertaining the real or intrinsic value of a security, let us take up a more rigorous process of evaluating securities called fundamental analysis.

J. FUNDAMENTAL APPROACH TO VALUATION

To determine the **intrinsic value** of a security based on **future income streams**, adjusted for **risk and time value of money**.

A security is **undervalued** if its **Intrinsic value** > **current market price**, and **overvalued** if the reverse.

a. KEY CONCEPTS IN FUNDAMENTAL VALUATION

1. Time Value of Money (TVM):

A rupee today is worth more than a rupee tomorrow — hence, **future income must be discounted**.

2. Tools Used:

- | | |
|-----------------|------------------------------|
| (a) Discounting | (c) Dividend discount models |
| (b) Compounding | (d) Earnings-based valuation |

Equity Valuation – Single-Year Holding

Let's begin with a simple example:

Given:

- ◆ Purchase Price $(P_0) = ₹500$
- ◆ Dividend $(D_1) = ₹20$
- ◆ Selling Price at year-end $(P_1) = ₹530$

Returns:

$$\text{Dividend Yield} = \frac{D_1}{P_0} = \frac{200}{500} = 0.04 = 4\%$$

$$\text{Capital Gain Yield} = \frac{P_1 - P_0}{P_0} = \frac{530 - 500}{500} = 0.06 = 6\%$$

$$\text{Total Return (r)} = 4\% + 6\% = 10\%$$

Present Value View

$$P_0 = \frac{D_1 + P_1}{(1 + r)}$$

Let's test this with $r = 10\%$

$$P_0 = \frac{20 + 530}{1.10} = \frac{550}{1.10} = ₹500$$

At 10%, the intrinsic value = market price.

What if required return is 15%?

$$P_0 = \frac{20 + 530}{1.15} = \frac{550}{1.15} = ₹478.26$$

So, paying ₹500 would be **overvalued** for a 15% required return.

Reverse Case: Target Price Calculation

If:

- ◆ Required return $r = 15\%$
- ◆ Purchase price = ₹500
- ◆ Dividend = ₹20

Find the target P_1 (selling price):

$$500 = \frac{20}{1.15} + \frac{P_1}{1.15} \Rightarrow 554.72$$

You must sell at **₹554.72** to earn 15%.

Multiple-Year Holding: Dividend Discount Model (DDM)

When dividends grow forever at constant rate (g), we use the Gordon Growth Model:

$$P_0 = \frac{D_1}{r - g}$$

Where:

- ◆ D_1 = Dividend expected next year
- ◆ r = Required rate of return
- ◆ g = Expected dividend growth rate

Example:

- ◆ $D_1 = ₹25$
- ◆ $r = 12\%$
- ◆ $g = 5\%$

$$P_0 = \frac{25}{0.12 - 0.05} = \frac{25}{0.07} \Rightarrow ₹357.14$$

So, the intrinsic value is 357.14. If the stock is available at 320, it is **undervalued**.

Rearranged Gordon Model for Return Estimation

$$r = \frac{D_1}{P_0} + g$$

Example:

- ◆ $D_1 = 30$
- ◆ $P_0 = 400$
- ◆ $g = 8\%$

$$r = \frac{30}{400} + 0.08 = 0.075 + 0.08 = 15.5\%$$

Thus, the **investor's expected return is 15.5%.**

Conclusion & Strategy

Approach	Use When
Single-Period Model	Clear target price and dividend known
Gordon Growth Model	Dividend growth expected to be constant indefinitely
Reverse DDM ($r = D/P + g$)	To evaluate investor's expected return

K. ALTERNATIVE APPROACHES TO VALUATION

Beyond the traditional **Fundamental** and **Technical Analysis**, two important theories provide a different lens for understanding stock price behaviour:

a. RANDOM WALK THEORY

The **Random Walk Theory** asserts that **stock price movements are random and unpredictable**, making it **impossible to forecast future prices** based on past movements.

1. Key Features:

- (a) Price changes are **independent of past movements**.
- (b) No reliable trends or patterns exist.
- (c) Believes **market manipulation, news shocks, and irrational investor behaviour** drive prices.
- (d) **Charts and trends are ineffective** for forecasting.

2. Implications:

- (a) **Technical analysis is irrelevant** as past data is not useful.
- (b) **Fundamental analysis is not negated**, as long-term trends can still emerge.
- (c) Suitable for **passive investment strategies** like index investing.

3. Advantages:

- (a) Simplifies investment decisions — **no need for stock selection or timing**.
- (b) Reduces overreliance on news and speculation.
- (c) Encourages diversification.

b. EFFICIENT MARKET HYPOTHESIS (EMH)

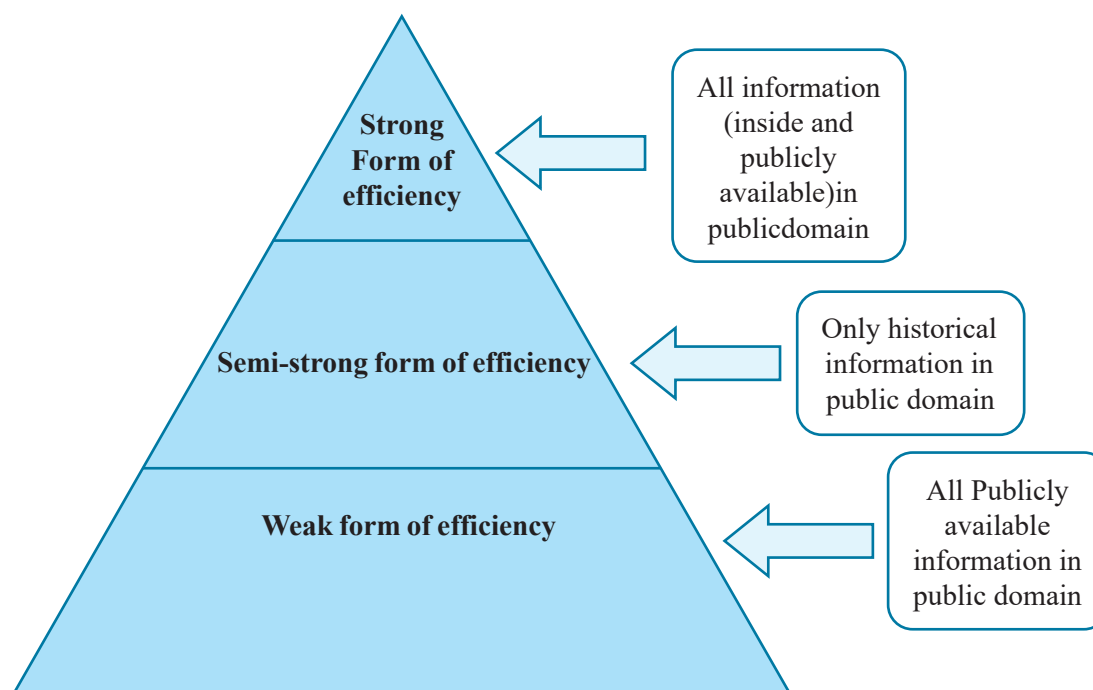
EMH states that **stock prices always reflect all available information**, making it **impossible to consistently outperform the market**.

Introduced by Eugene Fama (1970)

1. Key Assumptions:

- (a) All investors have **free and equal access** to information.
- (b) **No transaction costs** or taxes.

- (c) Investors behave **rationally** and aim to maximize returns.
- (d) **All known information is already priced in.**



2. Types of EMH (Based on Information Efficiency):

Form	Definition	Implication
Weak Form	Prices reflect past trading data like prices and volumes.	Technical analysis is ineffective.
Semi-Strong Form	Prices reflect all publicly available info (e.g., news, earnings).	Both fundamental and technical analysis are ineffective.
Strong Form	Prices reflect all information , including inside info .	Even insiders cannot outperform the market.

3. Test Techniques Used:

- (a) **Runs Test:** Checks for randomness in price movements.
- (b) **Correlation Analysis:** Measures the relation between past and future prices.
- (c) **Filter Test:** Tests the profitability of trend-following strategies.

4. EMH vs Other Approaches

Aspect	Fundamental Analysis	Technical Analysis	EMH
Based On	Earnings, management, ratios	Price trends, charts, volume	Public/private info and randomness
Assumes	Markets can be inefficient	History repeats itself	Markets are efficient
Predictive Power	Medium to High (long term)	Short-term trends	Very low-cannot beat market

5. Key Takeaways

- (a) **Random Walk Theory:** Day-to-day price movements are **random**, hence prediction is unreliable.
- (b) **EMH:** Prices reflect all available information; **only new (unexpected) info causes price changes**.
- (c) In **strong EMH**, **no investor** (not even insiders) can consistently earn excess returns.

When Can an Investor Outperform?

Only if:

- ◆ “They have access to inside (non-public) information, OR”
- ◆ “They have superior analytical skills to interpret public data innovatively.”

c. CAPITAL ASSET PRICING MODEL (CAPM)

CAPM is used to **estimate the expected return** on a security based on:

1. Its **systematic risk (β)**,
2. The **market return**, and
3. The **risk-free return**.

CAPM Formula

$$E(R) = R_f + \beta (R_m - R_f)$$

Where:

$E(R)$ = Expected return on the security

R_x = Risk-free rate

R_m = Expected return on the market

β = Beta of the security (measure of systematic risk)

Example 1: Calculate Expected Return (in ₹ terms)

$R_x = 2.5\%$ (10-year Govt. bond)

$R_m - R_x = 7.5\%$ (Market risk premium)

$\beta = 1.25$

$E(R) = 2.5\% + 1.25 \times 7.5\% = 2.5\% + 9.375\% = 11.875\%$

Expected Return = 11.88%

Example 2: Find Growth Rate (g)

$P_0 = 24.625$

$D_1 = 1.32$

$\beta = 1.35$

$R_m = 13.5\%$, $R_x = 8.2\%$

Step 1: Use CAPM to find required return R:

$E(R) = 8.2\% + 1.35 \times (13.5\% - 8.2\%) = 8.2\% + 7.095\% = 15.295\%$

Step 2: Use Dividend Discount Model (Gordon):

$$P_0 = \frac{D_1}{R - g} \Rightarrow g = R - \frac{D_1}{P_0}$$

$$g = 0.15295 - \frac{1.32}{24.625} = 0.15295 - 0.0536 = 0.0993 = 9.93\%$$

Expected Growth Rate = 9.93%

Example 3: Fair Price of Stock

$\beta = 1.15$

$D_1 = 1$

$$g = 6\%$$

$$R_m = 18\%$$

$$R_x = 12\%$$

Step 1: Find Expected Return:

$$E(R) = 12\% + 1.15 \times (18\% - 12\%) = 12\% + 6.9\% = 18.9\%$$

Step 2: Use Gordon Model:

$$P_0 = \frac{D_1}{R - g} = \frac{1}{0.189 - 0.06} = \frac{1}{0.129} = ₹7.75$$

Fair Price = ₹7.75

Example 4: Calculate Beta Before & After Announcement

Before:

$$D_1 = ₹6, g = 5\%$$

$$P_0 = ₹120$$

$$R_m = 12\% \quad R_x = 6\%$$

$$R = \frac{D_1}{P_0} + g = \frac{6}{120} + 0.05 = 0.05 + 0.05 = 10\%$$

$$\beta = \frac{E(R) - R_x}{R_m - R_x} = \frac{10\% - 6\%}{12\% - 6\%} = \frac{4}{6} = 0.667$$

After (New Price = ₹125):

$$R = \frac{6}{125} + 0.05 = 0.048 + 0.05 = 9.8\%$$

$$\beta = \frac{9.8\% - 6\%}{12\% - 6\%} = \frac{3.8}{6} = 0.633$$

Beta Before = 0.667, After = 0.633

d. KEY TAKEAWAYS FROM CAPM

Component	Meaning
Risk-Free Rate	Return with zero risk (Govt bonds).
Beta (β)	Sensitivity of stock to market movement.
Market Premium	$(R_m - R_x)$: Extra return expected over risk-free rate.
CAPM Output	Tells you the expected return for taking that risk.

TEST YOURSELF

1. What is security analysis? Why do we need to carry it out?
2. What are the various kinds of risks?
3. What are the various techniques of security analysis?
4. Describe some techniques of Technical analysis.
5. Write a short note on common-size financial statements.

6. Tanuja Ltd. A Ltd. has declared a dividend of ₹10 per share, which is 100% of the face value. The Earnings Per Share (EPS) of the company is ₹40, and the Return on Equity (ROE) is 20%. If an investor requires a 20% rate of return, calculate the intrinsic value or price per share of the company.

(Hint: Use the Gordon Growth Model where applicable)

Sol. Given:

Dividend = ₹10

EPS = ₹40

ROE = 20%

Required return (K_e) = 20%

Step 1: Calculate Dividend Payout Ratio

Payout Ratio = Dividend/EPS = $10/40 = 0.25$ (25%)

Retention Ratio (b) = $1 - 0.25 = 0.75$ (75%)

Step 2: Calculate Growth Rate

$g = ROE \times b = 20\% \times 0.75 = 15\%$

Step 3: Use Gordon Growth Model

Price = $D/(k_e - g) = 10/(0.2 - 0.15) = 10/0.05 = \text{mp}200$

Step 4: Adjust for value of growth

Price without growth = $EPS/K_e = 40/0.2 = ₹200$

But actual price = ₹230

So, Value of Growth = $230 - 200 = ₹30$

Final Answer: ₹230 per share

7. The analysts believe that the equity shares of **Ridhima Ltd** will give a return of **20%** if the economy grows faster. If the economy continues to grow at the current rate, the return is expected to be **10%**, and if the economy slows down, the return may fall to **5%**. The probability of fast growth, normal growth, and slowdown (recession) is estimated to be **50%**, **30%**, and **20%** respectively.

Calculate the **average (expected) return** on Ridhima Ltd's equity shares.

Sol. This is a case of **expected return using probability-weighted average**.

Step 1: Write down returns and probabilities:

Fast growth: Return = 20%, Probability = 50% or 0.50

Normal growth: Return = 10%, Probability = 30% or 0.30

Recession: Return = 5%, Probability = 20% or 0.20

Step 2: Use expected return formula:

Expected Return = $(\text{Return}_1 \times \text{Probability}_1) + (\text{Return}_2 \times \text{Probability}_2) + (\text{Return}_3 \times \text{Probability}_3)$

$= (20 \times 0.50) + (10 \times 0.30) + (5 \times 0.20)$

$= 10 + 3 + 1 = 14$

Final Answer: 14% average return

8. Sun Star Ltd. is a listed company whose share prices for last 14 trading session are given below. Calculate Relative Strength Index: June 2025

Trading session number	Share price (₹)
1	334
2	344
3	335
4	312
5	334
6	348
7	314
8	309
9	316
10	339
11	347
12	317
13	333
14	340

Solution:

Formula:

$$RSI\ 100\ [100 / (1 + RS)]$$

Where,

RS (Relative Strength) = Average Gain / Average Loss

We are using a 14-period RSI, so we will calculate average gain and average loss over the last 14 days.

Step 1: Calculate Daily Changes, Gains, and Losses

Day	Price	Change	Gain	Loss
1	334	—	—	—
2	344	+10	10	0
3	335	–9	0	9
4	312	–23	0	23
5	334	+22	22	0
6	348	+14	14	0
7	314	–34	0	34
8	309	–5	0	5
9	316	+7	7	0
10	339	+23	23	0
11	347	+8	8	0
12	317	–30	0	30
13	333	+16	16	0
14	340	+7	7	0

Step 2: Total Gain and Loss

$$\text{Total Gain} = 10 + 22 + 14 + 7 + 23 + 8 + 16 + 7 = 107$$

$$\text{Total Loss} = 9 + 23 + 34 + 5 + 30 = 101$$

Step 3: Average Gain and Loss (14 days)

$$\text{Average Gain} = 107/14 = 7.64$$

$$\text{Average Loss} = 101/14 = 7.21$$

Step 4: Calculate RS

$$\text{RS} = 7.64/7.21 = 1.06$$

Step 5: Calculate RSI

$$\text{RSI} = 100 - [100/(1 + 1.06)]$$

$$= 100 - (100/2.06)$$

$$= 100 - 48.54$$

$$= 51.46$$

Final Answer: RSI = 51.46

OPERATIONAL APPROACH TO FINANCIAL DECISIONS

A. INTRODUCTION TO COST LEADERSHIP AND COSTING CONCEPTS

In the fierce battlefield of business, where every company races to win consumer attention and global relevance, standing out is no longer optional—it's survival.

Michael Porter's strategy map offers three routes to victory: product differentiation, niche focus, and the powerhouse.

Cost Leadership: This strategy isn't just about being cheap; it's about being *efficiently excellent*—delivering quality at the lowest cost possible. And the real engine behind this power move? **Cost and Management Accounting (CMA)**.

A company that masters its costs, controls its waste, and maximizes its resources earns the golden ticket to cost leadership. CMA becomes the compass guiding decisions, identifying profit leaks, and fueling smart pricing strategies that outshine competitors.

a. COSTING: THE LANGUAGE OF VALUE

At its core, **costing** is more than just numbers—it's the financial storytelling of a business's resource journey. It assigns value to every product, process, and person involved in delivering goods or services. Whether it's manufacturing a car or running a consultancy, costing helps leaders understand where money flows, and more importantly, where it sticks.

Think of cost as the financial shadow of every decision. It's what you pay, whether in cash, time, or opportunity, for every input. Not all costs become expenses immediately; some wait their turn as assets until they're finally used up.

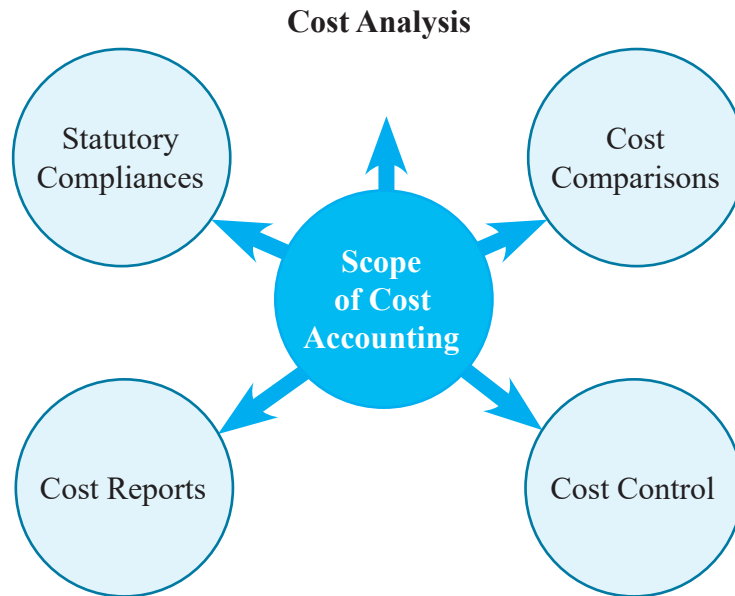
b. COST ACCOUNTING: WHERE SCIENCE MEETS STRATEGY

Defined by ICWAI and CIMA, **Cost accounting** is where logic meets leadership. It's a **science** because it follows a structured framework of methods and rules. But it's also an **art**, because interpreting numbers and applying insights to real-world problems takes judgment, experience, and strategic thinking.

A skilled cost accountant doesn't just crunch numbers; they shape decisions. From pricing a product right, to finding cost-saving opportunities, to supporting expansion plans, cost accounting becomes a silent strategist behind every smart move.

B. NATURE AND SCOPE OF COSTING: THE STRATEGIC MIND BEHIND THE NUMBERS

In the dynamic world of business, **costing** is more than just an accounting tool — it's a strategic compass. According to CIMA (London), costing is the art and science of estimating the value behind goods and services using structured methods and principles. But it's not just about numbers—it's about **narratives hidden in numbers**.



a. NATURE OF COSTING: FOUR DIMENSIONS OF INTELLIGENCE

1. **A Body of Knowledge:** Cost accounting isn't merely an extension of financial accounting—it's a knowledge system of its own, governed by unique principles and tailored industry practices. It blends strategy with structure.
2. **A Science with a Pulse:** While not as exact as physics, cost accounting has scientific depth. It draws from multiple disciplines like production, law, data processing, and materials management, forming a multidisciplinary framework.
3. **An Art of Application:** Like a painter choosing the right brushstroke, a cost accountant must choose the right method, tweak assumptions, and interpret cost behavior smartly to solve real business problems—pricing, planning, or control.
4. **A Profession with Power:** With the rise of global institutions like ICWAI (India) and CIMA (UK), cost accounting has become a highly respected profession. It empowers individuals to lead with financial clarity, making it a key pillar of strategic management.

b. SCOPE OF COSTING: FROM ESTIMATION TO EXECUTION

Costing stretches far beyond cost calculation. It includes:

1. **Cost Collection & Classification:** Sorting and organizing data for direct and indirect cost components.
2. **Cost Allocation & Apportionment:** Assigning untraceable costs fairly across processes and units.
3. **Cost Estimation:** Predicting future costs through data analysis, production measurement, and rational estimation.
4. **Cost Systems:** Including **Historical**, **Estimated**, and **Standard Costing**.
5. **Cost Techniques:** Like **Marginal**, **Absorption**, and **Uniform Costing**—each serving specific managerial purposes.

c. OBJECTIVES OF COSTING: WHY IT MATTERS

1. **Cost Ascertainment:** The first rule: know your cost before you price. Costing ensures every rupee spent is tracked on materials, labor, and overheads. This helps in accurate product pricing and drives profit planning.

2. **Cost Control:** Estimating is good, but **controlling is critical**. Through budgeting and variance analysis, costing helps organizations compare planned vs. actual performance. Deviations signal action, leading to smarter spending and higher margins.
3. **Guidance for Management:** Costing is the manager's silent advisor. Should we launch a new product? Can we expand production? Is outsourcing viable? These questions find answers through cost data. It fuels strategy, flags inefficiencies, and reveals new opportunities.

C. TYPES OF COSTING: UNDERSTANDING THE COST PUZZLE

In the world of managerial decisions, *costs aren't all created equal*. Each type tells a different story, and knowing the difference can shape smarter strategies.

a. BASIC CLASSIFICATIONS

1. **Fixed Cost:** These are the silent constants—costs that *don't change* with output. **Example:** rent, salaried staff.
2. **Variable Cost:** The flexible ones—*costs that rise or fall* as production does. **Example:** raw materials, fuel.
3. **Direct Cost:** Directly linked to a product or service—*traceable and tangible*. **Example:** fabric for garments.
4. **Indirect Cost:** Shared resources not directly assigned to a unit. **Example:** lighting, admin salaries.

b. DECISION-ORIENTED COSTS

1. **Relevant Cost:** The costs that matter when *choosing between options*. If it changes your decision, it's relevant.
2. **Irrelevant Cost:** Stays the same no matter what you decide. *No impact on the outcome*.
3. **Sunk Cost:** Already spent. It can't be recovered. Like a non-refundable movie ticket—you *can't un-spend* it.
4. **Shutdown Cost:** Costs that *don't go away* even if production halts. **Example:** rent, depreciation, insurance.
5. **Imputed/Notional Cost:** Costs with *no cash outflow* but with real value. Like using your own building rent-free.
6. **Out-of-Pocket Cost:** Actual expenses that require cash outflow. Direct impact on cash flow and budgeting.

E. WHY COSTING MATTERS: MAJOR ADVANTAGES

A good costing system does more than track money—it gives your business a brain. Here's how:

a. MEASURES AND IMPROVES EFFICIENCY

Costing helps identify wastage and inefficiencies. If the cost of making shoes goes up, is it material price or mismanagement? Costing helps you find out.

b. IDENTIFIES PROFITABLE AND LOSS-MAKING AREAS

Know which products are earning profits and which are silently bleeding resources.

c. ASSISTS IN PRICING

You can't price right without knowing your cost. Overpricing? You lose customers. Underpricing? You lose profits. Costing ensures you hit the sweet spot.

d. SUPPORTS PRICE REDUCTION DECISIONS

In tough times, cutting prices is a survival tactic. But how low can you go without sinking the ship? Costing shows you the floor.

e. AIDS IN RESOURCE PLANNING

Proper data on labor, machines, and material capacity prevents overwork or idle resources.

f. CONTROLS MATERIAL USAGE AND PREVENTS LOSSES

Constant tracking of stock levels avoids overstocking, theft, or shortages, supporting smart inventory management.

g. ENABLES MACHINE VS. LABOUR DECISIONS

Should you buy a machine or hire more people? Cost accounting helps compare long-term expenses and benefits.

h. GUIDES EXPANSION DECISIONS

Want to scale production? Cost data tells you whether it's profitable or risky.

i. DIAGNOSE LOSSES

Not all losses come from high costs or low prices. Sometimes, it's underutilized capacity. Costing uncovers hidden causes.

j. SHARPENS DECISION-MAKING

Costing encourages calculated decisions, not blind guesses—reducing errors, like quoting prices without considering material cost.

k. VERIFIES FINANCIAL ACCOUNTS

Cost and financial accounts are reconciled to cross-check the accuracy of results, ensuring trust and transparency.

F. COSTING ISN'T PERFECT: LIMITATIONS TO WATCH OUT FOR

Every tool has its edges—costing is no different.

a. EXPENSIVE TO MAINTAIN

Skilled professionals and detailed systems cost money. ROI must justify the expense.

b. COMPLEX AND TIME-CONSUMING

Numerous forms, classifications, and reconciliations make the process lengthy and resource-intensive.

c. NOT UNIVERSALLY APPLICABLE

No one-size-fits-all. Each industry requires customized methods—using the wrong techniques can lead to misdirection.

d. UNSUITABLE FOR SMALL BUSINESSES

High cost and complexity make it impractical for micro or small firms.

e. LACK OF STANDARDIZATION

Different accountants may interpret the same data differently, making outcomes subjective and inconsistent.

f. APPROXIMATIONS REDUCE ACCURACY

Assumptions in costing (like depreciation, standard hours) may dilute exact accuracy.



G. BASIC PRINCIPLES OF COSTING: NAVIGATING THE COSTING LANDSCAPE

Costing isn't just about collecting numbers—it's about following principles that ensure **accuracy, relevance, and fairness** in cost data. These foundational ideas help organizations make meaningful managerial decisions.

a. CAUSE-AND-EFFECT RELATIONSHIP: THE GOLDEN RULE

Every cost must be traceable to its **origin**—its cause. This ensures fairness in cost distribution. Only departments or processes that use a resource should bear its cost. For example, if electricity is used in one production line, it shouldn't be charged to another.

Why it matters: Prevents arbitrary allocations and keeps costs fair and logical.

b. DON'T DRAG THE PAST INTO THE FUTURE

Costs that were incurred and **not recovered in the past** should be left in the past. Including them in future costs distorts profitability and misguides future pricing or investment decisions.

Why it matters: Keeps cost sheets clean and current, focused on today's reality.

c. COST ONLY WHAT'S ACTUALLY INCURRED

Only record **real, current expenses**—not future guesses or irrelevant charges. For example, don't add selling expenses to the cost of an item that's still being manufactured.

Why it matters: Maintains the integrity of unit costs, avoiding misleading valuations.

d. SAY NO TO ABNORMAL COSTS

Losses from theft, accidents, or managerial negligence are **abnormal** and should be excluded from regular cost calculations. These don't reflect standard performance and can lead to bad decisions if considered normal.

Why it matters: Keeps costs realistic and helps identify inefficiencies.

e. FOLLOW DOUBLE ENTRY WHERE POSSIBLE

Using **double-entry** in cost accounts enhances transparency and accuracy. It creates a built-in check system, making reconciliation with financial accounts easier.

Why it matters: Reduces errors and builds confidence in cost reports.

H. INTERCONNECTION OF COSTING, ACCOUNTING & FINANCE: A UNIFIED FINANCIAL ECOSYSTEM

While each branch has its purpose, in practice, they're like four players on the same team—different roles, one mission: **financial clarity and performance**.

a. COST ACCOUNTING

Focuses on **internal efficiency**—measuring, recording, and analyzing the cost of production and services. It helps set prices, control wastage, and improve productivity.

b. MANAGEMENT ACCOUNTING

Turns cost and financial data into **actionable insights**. It supports decision-making, planning, and performance evaluation—like a GPS for managerial strategy.

c. FINANCIAL ACCOUNTING

Looks outward. It prepares **external financial statements** like P&L, balance sheets, and cash flows, used by investors, regulators, and stakeholders.

d. FINANCIAL MANAGEMENT

Focuses on the **future**—capital structure, funding decisions, investment planning, and maximizing shareholder value.

e. HOW THEY INTERACT

1. **Cost data** feeds **management accounting** for budgeting and performance checks.
2. **Financial accounting** provides statutory reports, which are cross-verified with **costing data**.
3. **Financial management** uses both cost and financial records to make investment and financing decisions.

f. THINK OF IT LIKE THIS:

1. **Costing** tells you *how much you spend*.
2. **Management Accounting** tells you *what to do about it*.
3. **Financial Accounting** tells the world *how you performed*.
4. **Financial Management** decides *where you're going next*.



I. CLASSIFICATION OF COSTS

Cost classification helps organize expenses based on different criteria for better analysis and control.

a. BY NATURE (ELEMENTS)

1. **Material Cost:** Raw materials used in production (*excludes indirect items like cleaning supplies*).
2. **Labour Cost:** Wages and benefits (direct for production; indirect for support staff).
3. **Other Expenses:** Utilities, depreciation, services, and notional costs (**Example:** own building rent).

b. BY FUNCTION

Production, Admin, Selling & Distribution costs.

c. BY BEHAVIOUR

Fixed (unchanged), **Variable** (changes with output), **Semi-variable** (mixed).

d. BY TRACEABILITY

1. **Direct:** Traced to product.
2. **Indirect:** Shared across activities.

e. BY REGULARITY

1. **Normal:** Usual costs.
2. **Abnormal:** Unexpected (e.g., theft, fire).

f. FOR DECISION MAKING

Relevant, Irrelevant, Sunk, Imputed, Out-of-Pocket, Shutdown costs.

J. MARGINAL COSTING (IN A NUTSHELL)

It's the **cost of producing one extra unit**, which mainly includes variable costs.

Example:

Variable Cost = ₹25/unit

Fixed Cost = ₹1,00,000

Marginal Cost of unit = ₹25

Use: Helpful for pricing, cost control, and production decisions.

a. NEED FOR MARGINAL COSTING

Marginal costing focuses only on **variable costs**, treating **fixed costs** as period costs. It is essential in modern management for clearer decisions, especially in pricing, cost control, and profit planning.

Why It's Needed:

1. **Profit Planning** becomes easy as cost-volume-profit relationships are directly visible.
2. **Inventory changes** don't affect profit figures—only sales affect them.
3. **Reports align with management logic**, unlike complex absorption costing.
4. **The fixed costs' impact** is clearly shown, helping to better control and provide visibility.
5. Helps **compare products, departments, or customers** without distortion from fixed cost allocation.
6. Complements **standard costing and flexible budgets** for cost control.
7. Useful in **price setting during low demand** or tenders.
8. Required for **break-even analysis** and profitability forecasts.

b. KEY FEATURES OF MARGINAL COSTING

1. **Clear separation** of fixed and variable costs, including semi-variable elements.
2. **Simplifies profit fluctuations**, focusing only on cost-volume behavior.
3. **Supports faster decisions**, especially about product viability and pricing.
4. Acts as a **cost control tool**, offering detailed, level-based reporting.
5. Aids in **future profit planning**, using contribution and margin ratios.
6. Helps evaluate **product or unit performance** clearly.

7. Valuable for **forecasting and budgeting**.
8. Determines **optimum product and sales mix** for maximum profit.
9. **Stock valuation** is done at variable cost only.
10. **Fixed costs are not included** in inventory or unit cost—they're written off immediately.
11. Pricing is based on **marginal cost and contribution**.
12. Merges **cost recording with reporting** for better insights.

In a nutshell: Marginal costing is not just a cost method—it's a **strategic decision-making tool** that gives sharper insights, faster reactions, and stronger financial control.

c. ASCERTAINMENT OF PROFIT UNDER MARGINAL COST

Let's understand with the help of an example

Ascertainment of Profit under Marginal Cost

The term **"contribution"** refers to a sum of money equal to the selling price of a good less the marginal cost. One way to characterize contribution is as follows:

Contribution = Selling Price – Marginal Cost

Contribution = Fixed Expenses + Profit

Contribution – Fixed Expenses = Profit

Income Statement under Marginal Costing

Example:

Income Statement For the year ended 31-03-2021

Particulars	Amount	Total
Sales		25,00,000
Less: Variable cost		
Cost of goods manufactured	12,00,000	
Variable Selling Expenses	3,00,000	
Variable Administration Exp	50000	
		15,50,000
Contribution		9,50,000
Less: fixed cost		
Fixed Administration Exp	70,000	
Fixed selling Expenses	1,30,000	200,000
		7,50,000

d. ADVANTAGES OF MARGINAL COSTING

Marginal costing is more than a cost technique—it's a **decision-making powerhouse**. Here's how it helps management:

1. **Effective Cost Control:** By separating fixed and variable costs, it sharpens control over marginal (variable) costs.

2. **Simplifies Overhead Treatment:** Fixed costs are kept out of product costing, reducing over/under-absorption issues.
3. **Realistic Valuation:** Stock and WIP are valued only on variable cost, giving a more practical and uniform valuation.
4. **Supports Managerial Decisions:** Helps in pricing, tendering, expanding product lines, and “make or buy” choices.
5. **Aids Production Planning:** Cost-volume-profit (CVP) tools and break-even charts guide optimal output levels.
6. **Better Results with Standard Costing:** When paired with standard costing, it enhances accuracy in cost control.
7. **Price Fixation Clarity:** Fixed vs. variable cost split helps in pricing strategy, especially in competitive markets.
8. **Helps Budgeting:** Makes flexible budgeting easier with a clear classification of expenses.
9. **Tendering Support:** Knowing variable cost helps set a profitable floor price when bidding.
10. **Make or Buy Analysis:** Reveals if outsourcing is more cost-effective than in-house production.
11. **Clearer Presentations:** Graphs and statements under marginal costing are easy to interpret—ideal for non-financial managers.

K. BREAK-EVEN ANALYSIS: THE NO-PROFIT-NO-LOSS POINT

The break-even point is where **Total Revenue = Total Cost**—no profit, no loss. It’s the minimum sales level a business must reach to cover all costs.

Formula: Break-Even Point (Units) = Fixed Costs / (Selling Price – Variable Cost per Unit)

a. IMPORTANCE OF BREAK-EVEN ANALYSIS:

1. **Forecasting Profits:** Links cost, volume, and profit for future planning.
2. **Flexible Budgets:** Helps design budgets for different activity levels.
3. **Performance Review:** Assesses how changes in volume affect profits and costs.
4. **Pricing Strategy:** Projects the effect of different prices on profits.
5. **Overhead Allocation:** Helps assess how fixed costs behave at different volumes.

Graphical Tool: Break-even charts visually show profits/losses across sales volumes using cost and revenue lines.

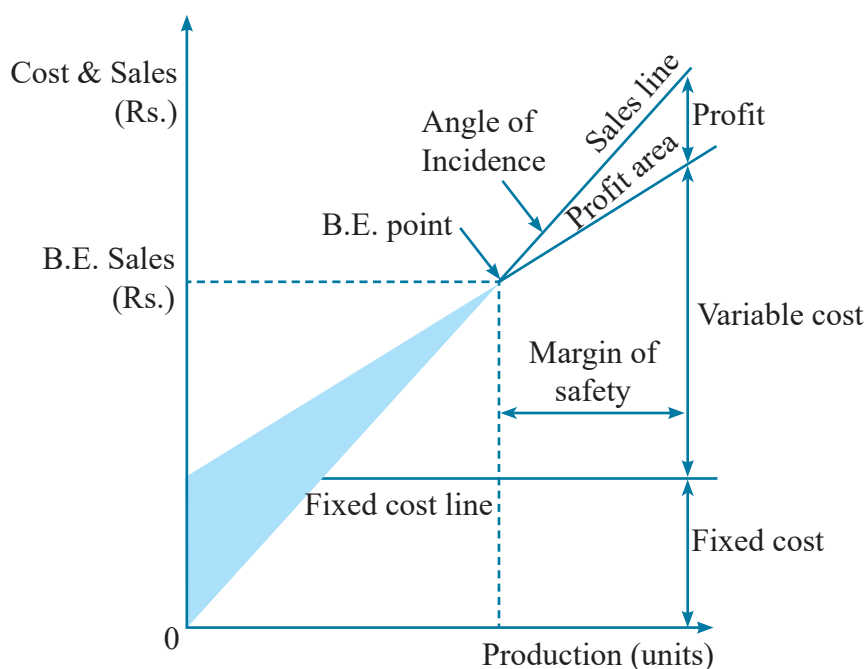
b. STEPS TO CONSTRUCT A BREAK-EVEN CHART

Building a break-even chart is a visual way to understand when a business starts making profits. Here’s how to create it:

Step-by-Step Guide:

1. **X-Axis (Horizontal):** Choose a scale for **sales units**.
2. **Y-Axis (Vertical):** Choose a scale for **costs and revenues**.
3. **Fixed Cost Line:** Draw a **horizontal line** from the fixed cost level on the Y-axis.
4. **Total Cost Line:** Start from the fixed cost point and draw a line **adding variable costs**.

5. **Sales Line:** Draw a line from the origin to the maximum sales point.
6. **Break-Even Point (BEP):** It's where the sales line intersects the total cost line.
7. **Mark BEP:** Project lines from the intersection to the X and Y axes to find sales units and revenue at BEP.
8. **Loss & Profit Zones:**
 - ☛ Left of BEP = Loss
 - ☛ Right of BEP = Profit
9. **Margin of Safety:** = Actual Sales – Break-Even Sales
10. **Angle of Incidence:** Shows rate of profit after BEP—steeper angle = higher profit rate.



c. BREAK-EVEN ANALYSIS CONCEPTS & FORMULAE

Core Assumption: Selling price/unit, variable cost/unit, and total fixed costs remain constant.

Key Formulae:

1. **Break-Even Point (Units):** = Fixed Cost ÷ Contribution per Unit
2. **Break-Even Sales (in ₹):** = Fixed Cost ÷ P/V Ratio or = Break-even units × Selling price per unit
3. **P/V Ratio:** = (Contribution ÷ Sales) × 100
4. **Desired Sales (to earn target profit):** = (Fixed Cost + Desired Profit) ÷ P/V Ratio
5. **At Break-Even Point:** Contribution = Fixed Costs → Profit = 0

d. PURPOSE OF BREAK-EVEN ANALYSIS

1. Determine **viability**
2. Aid in **pricing, budgeting, and cost control**
3. Predict profit or loss based on **sales volume changes**.

e. BREAK-EVEN ANALYSIS: ASSUMPTIONS

Break-even analysis works on certain ideal conditions, which may differ in real-world scenarios:

1. **Cost Classification:** Assume a clear division of costs into **fixed** and **variable**. Semi-variable costs are difficult to classify precisely.
2. **Fixed Costs Remain Constant:** Assumes fixed costs do **not change** across all levels of activity—unrealistic in practice.
3. **Constant Variable Cost per Unit:** Assumes variable cost **varies proportionately** with output. But efficiency, discounts, or waste may alter it.

4. **No Inventory Change:** Assumes production = sales (i.e., **no opening or closing stock**), which is rare in actual business operations.
5. **Constant Selling Price & Sales Mix:** Assumes **no changes in selling price** or product mix, though market conditions often require adjustments.
6. **Static Operations:** Presumes **no changes in productivity, methods, or efficiency**, which is impractical over time.
7. **Single Product Focus:** Only one product can be analyzed per chart. For multi-product firms, separate break-even charts are needed.
8. **Capital Ignored:** Break-even analysis **does not consider capital employed**, which affects return and profitability.
9. **Straight-Line Assumption:** Assumes **total cost and revenue increase linearly**. In reality, cost/revenue curves are often **non-linear**.

L. PROFIT-VOLUME RATIO (P/V RATIO)

Shows the **relationship between contribution and sales**, indicating profitability potential.

Formula: $P/V \text{ Ratio} = (\text{Contribution}/\text{Sales}) \times 100$

a. IMPROVING P/V RATIO

1. **Increase Selling Price**
2. **Reduce Variable Costs** (via efficiency)
3. **Focus on High P/V Ratio Products**

b. LIMITATIONS OF P/V RATIO

1. **Oversimplified Profit Focus:** Depends heavily on the **excess of revenue over variable cost**—ignores other profitability factors.
2. **Capital Ignored:** Doesn't consider additional **capital/fixed costs** required for higher output.
3. **Only a Guideline:** Useful for **relative comparison**, but not for **final decision-making**.
4. **Cost Classification Required:** Inaccurate classification of costs **can distort the P/V ratio**.
5. **Conditional Comparison:** A high P/V ratio is meaningful **only when other factors (like capital and fixed cost) remain constant**.

Quick Tip for Exams:

Break-even = No Profit, No Loss, and P/V Ratio = How much of the sales contributes to profit. Both tools are **decision aids**, not absolute indicators.

M. MARGIN OF SAFETY (MOS)

The **Margin of Safety** represents the amount by which **actual or expected sales exceed the break-even sales**. It indicates how much sales can drop before the business starts incurring losses.

In simple terms: $MOS = \text{“Profit Cushion” over Break-Even Sales}$

a. SIGNIFICANCE

1. A **higher MOS** implies strong profitability and financial stability.
2. A **lower MOS** indicates **high fixed costs** and **vulnerability to losses** during a sales decline.
3. Crucial during **economic downturns** or sales dips to assess how long a business can stay afloat.

b. FORMULAS:

1. **Absolute Margin of Safety (in ₹):** $MOS = \text{Actual Sales} - \text{Break-Even Sales}$
2. **Using Profit & P/V Ratio:** $MOS = \text{Profit} / \text{P/V Ratio}$
3. **Using Unit Economics:** $MOS = \text{Profit} \times (\text{Selling Price per Unit}) / (\text{Selling Price per Unit} - \text{Variable Cost per Unit})$
4. **Margin of Safety (%):** $MOS\% = (\text{Margin of Safety} / \text{Total Sales}) \times 100$

Key Takeaway: The **greater the margin of safety**, the **stronger the business is** in absorbing shocks and sustaining profits.

c. HOW TO IMPROVE MARGIN OF SAFETY (MOS)

A **higher Margin of Safety** means **greater financial strength** and a greater **buffer against losses**. Businesses can improve MOS through the following strategic actions:

1. **Maximize Actual Sales:** Focus on increasing real sales while keeping break-even sales low to widen the safety cushion.
2. **Boost Sales Volume:** More units sold = higher total contribution, improving profitability and MOS.
3. **Increase Selling Price:** Raising prices (without affecting demand) increases unit contribution and reduces the break-even point.
4. **Change Product Mix:** Promote or shift towards products with **higher contribution margins** to raise total profit.
5. **Reduce Fixed Costs:** Cutting overheads like rent or salaries lowers the break-even point, thereby widening the margin.
6. **Reduce Variable Costs:** Cost savings in materials, labor, or overheads improve per-unit contribution.
7. **Drop Unprofitable Products:** Removing low or negative margin products from the sales mix increases overall profitability and strengthens MOS.

Quick Tip: $\text{Margin of Safety} = \text{Actual Sales} - \text{Break-Even Sales}$

So, either **increase actual sales** or **decrease break-even sales** to improve it.

N. ANGLE OF INCIDENCE: A MEASURE OF PROFITABILITY

The **Angle of Incidence** is the angle formed at the **intersection of the Total Cost Line and Sales Line** in a break-even chart.

Wider Angle = Higher Profit Margin
Narrow Angle = Lower Profit Margin

a. WHAT IT INDICATES

1. **Large Angle** = Business earns **high profits** quickly after crossing BEP.
2. **Small Angle** = Business earns **low profits** after BEP, indicating slower growth in profitability.
3. It's a **visual indicator** of how steep the profit curve is post-break-even.

b. RELATIONSHIP: BEP, MARGIN OF SAFETY & ANGLE OF INCIDENCE

Relationship: BEP, Margin of Safety & Angle of Incidence

Concept	Meaning	Indicator
Break-Even Point (BEP)	Sales level where Total Cost = Total Revenue	No profit, no loss
Margin of Safety (MOS)	Actual Sales - BEP Sales	Profit buffer
Angle of Incidence	Angle between Sales Line & Total Cost Line	Profit growth rate

Best Case Scenario for a Business

- ◆ Low BEP
- ◆ High Margin of Safety
- ◆ Large Angle of Incidence

Together, these reflect strong financial health and high profitability after break-even.

Test Yourself

1. Cost-Volume-Profit Analysis

Problem Statement

From the following particulars, calculate:

- (a) The Profit/Volume (P/V) Ratio
- (b) The Fixed Cost

Given:

- Year I
Sales ₹1,95,000
Profit ₹9,000
- Year II
Sales = ₹2,25,000
Profit ₹15,000

Solution:

Given:

Step 1: Calculate P/V Ratio

$$\begin{aligned}\text{P/V Ratio} &= \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 \\ &= \frac{15,000 - 9,000}{2,25,000 - 1,95,000} \times 100 = \frac{6,000}{30,000} \times 100 \times 100 = 20\%\end{aligned}$$

Step 2: Calculate Variable Cost (using Year II figures)

$$\begin{aligned}\text{Variable Cost Sales (1P/V Ratio)} \\ &= 2,25,000 \times (10.20) = 2,25,000 \times 0.80 = ₹1,80,000\end{aligned}$$

Step 3: Calculate Fixed Cost

Fixed Cost = Sales Variable – Cost Profit

$$= 2,25,000 - 1,80,000 - 15,000 = ₹30,000$$

Final Answers:

◆ P/V Ratio = 20%

◆ Fixed Cost = ₹30,000

2. Profit Determination Using Marginal Costing - Tanuja Ltd.

Problem Statement: From the following particulars of Tanuja Ltd., calculate the amount of profit earned during the year using the **Marginal Costing Technique**:

Particulars	Product A	Product B	Product C
Output (Units)	10,000	20,000	60,000
Selling Price (per unit)	₹10	₹10.	₹5
Variable Cost (per unit)	₹6	₹7.50	₹4.50

☞ Total Fixed Costs = ₹80,000

Solution:**Statement of Cost and Profit (Marginal Costing Approach)**

Particulars	Product A (₹)	Product B (₹)	Product C (₹)	Total (₹)
Sales Revenue	$10000 \times 10 = 100000$	$20000 \times 10 = 200000$	$60000 \times 5 = 300000$	60,000
Marginal Costs	$10000 \times 6 = 60000$	$20000 \times 7.5 = 150000$	$60000 \times 4.5 = 270,000$	480,000
Contribution	40,000	50,000	30,000	120,000
Less: Fixed Costs				80,000
Net Profit				40,000

Conclusion:

Using the marginal costing technique, Tanuja Ltd. earned a profit of ₹40,000 during the year.

This method highlights how the contribution (i.e., the difference between sales and marginal cost) helps first cover fixed costs, and the remaining balance is profit. The concept of contribution is thus a powerful tool in managerial decision-making, especially under cost-volume-profit analysis.

- 3. Problem Statement:** Two companies, Tanuja Ltd. and Debashish Ltd., sell the same type of product. Their Income Statements are as follows:

Particulars	Tanuja Ltd. (₹)	Debashish Ltd. (₹)
Sales	2,40,000	2,40,000
Less: Variable Cost	96,000	1,20,000
Fixed Costs	64,000	40,000
Profit	80,000	80,000

Required:

State which company is likely to earn **greater profit** under the following conditions:

1. When there is **heavy demand** for the product.
2. When there is **poor demand** or decline in sales.

Solution:

Let's compare the contribution and P/V Ratio for both companies using marginal costing:

Particulars	Tanuja Ltd. (₹)	Debashish Ltd. (₹)
Sales	2,40,000	2,40,000
Less: Variable Cost	96,000	1,20,000
Contribution	1,44,000	1,20,000
P/V Ratio (Contribution Sales)	0.60 (60%)	0.50 (50%)

Analysis:

◆ **In case of heavy demand:**

- ☞ Tanuja Ltd. has a higher P/V Ratio (60%), meaning every extra ₹100 in sales contributes ₹60 toward covering fixed costs and generating profit.
- ☞ Therefore, Tanuja Ltd.'s profit will rise faster than that of Debashish Ltd. when sales increase.

◆ **In case of poor demand:**

- ☞ A fall in sales by ₹100 reduces contribution by ₹60 for Tanuja Ltd., and ₹50 for Debashish Ltd.
- ☞ Thus, Debashish Ltd. will experience a smaller decline in contribution and profit, making it safer during sales downturns.

Marginal Costing Recap:

Sales Variable Cost + Fixed Cost Profit

Sales Variable Cost Contribution Contribution Fixed Cost Profit (or Loss)

- ◆ **To earn profit:** Contribution > Fixed Cost
- ◆ **At break-even point:** Contribution = Fixed Cost
- ◆ **To maximize profit:** Maximize Contribution

Conclusion:

- ◆ **Tanuja Ltd. is expected to earn higher profit in case of heavy demand due to its higher P/V ratio.**
 - ◆ **Debashish Ltd. is better protected during a sales decline, due to lower fixed costs and a more stable contribution pattern.**
4. Tanuja Ltd. made total sales of ₹1,00,000 during a certain period and reported a net profit of ₹10,000. The fixed overheads for the same period amounted to ₹15,000.

You are required to calculate:

1. P/V Ratio
2. Sales needed to generate a profit of ₹15,000
3. Net profit if sales are ₹1,50,000
4. Break-even sales (BEP)

Solution:

1. Calculation of P/V Ratio

Formula:

$$\text{P/V Ratio} = \left(\frac{F + P}{S} \right) \times 100$$

Where:

☛ F = Fixed Overheads = ₹15,000

☛ P = Profit ₹10,000

☛ S = Sales ₹1,00,000

$$P/V \text{ Ratio} = \left(\frac{15,000 + 10,000}{1,00,000} \right) \times 100 = \left(\frac{25,000}{1,00,000} \right) \times 100 = 25\%$$

2. Sales required to earn a profit of ₹15,000

Using the formula again:

$$P/V \text{ Ratio} = \left(\frac{F + P}{S} \right) \times 100$$

$$25 = \left(\frac{15,000 + 15,000}{S} \right) \times 100 \Rightarrow S = \left(\frac{30,000}{25} \right) \times 100 = ₹1,20,000$$

3. Net Profit from Sales of ₹1,50,000

Again using:

$$P/V \text{ Ratio} = \left(\frac{F + P}{S} \right) \times 100$$

$$25 = \left(\frac{15,000 + P}{1,50,000} \right) \times 100 \Rightarrow 15,000 + P = 37,500 \Rightarrow P = 37,500 - 15,000 = ₹22,500$$

4. Break-even Sales (BEP)

At Break-even, Profit = 0, so:

$$P/V \text{ Ratio} = \left(\frac{F}{BEP \text{ Sales}} \right) \times 100$$

$$25 = \left(\frac{15,000}{BEP} \right) \times 100 \Rightarrow BEP = \left(\frac{15,000}{25} \right) \times 100 = ₹60,000$$

Final Answers Summary:

	Requirement	Answer
(i)	P/V Ratio	25%
(ii)	Sales for profit of ₹15,000	₹1,20,000
(iii)	Net profit at sales of ₹1,50,000	₹22,500
(iv)	Break-even Sales	₹60,000

5. The following data pertains to Tanuja Ltd., a manufacturing company:

☛ **Plant Capacity:** 4,00,000 units per annum

☛ **Current Utilization:** 40%

☛ **Actual Figures for the Year:**

◆ **Selling Price:** ₹50 per unit

◆ **Material Cost:** ₹20 per unit

◆ **Variable Manufacturing Cost:** ₹15 per unit

◆ **Fixed Costs:** ₹27,00,000

To improve capacity utilization, the company is considering the following proposals:

1. Reduce the selling price by 10%
2. Spend an additional ₹3,00,000 on sales promotion

Required:

Determine the number of units that must be produced and sold annually to achieve a net profit of ₹5,00,000.

Solution:

Step 1: Calculate Revised Selling Price

- ◆ Original Selling Price = ₹50 per unit
 - ◆ Reduction = 10% of ₹50
 - ◆ $10\% \text{ of } 50 = \frac{10}{100} \times 50 = ₹5$
- Revised Selling Price $50 - 5 = ₹45$ per unit

Step 2: Calculate Total Variable Cost per Unit

Component	Cost per Unit (₹)
Material Cost	20
Variable Manufacturing Cost	15
Total Variable Cost	35

Step 3: Contribution per Unit

Contribution per unit Revised Selling Price Variable Cost = $45 - 35 = ₹10$

Step 4: Total Contribution Required

Particular	Amount (₹)
Fixed Costs	27,00,000
Additional Promotion Expenses	3,00,000
Desired Profit	5,00,000
Total Contribution Needed	35,00,000

Step 5: Units to be Sold to Earn Desired Contribution

$$\text{Required Units} = \frac{\text{Total Contribution Required}}{\text{Contribution per Unit}} = \frac{35,00,000}{10} = 3,50,000 \text{ units}$$

Final Answer: Tanuja Ltd. must produce and sell 3,50,000 units per annum to earn a profit of ₹5,00,000, considering the price reduction and increased promotional spend.

6. Marginal Costing Analysis and Product Recommendation - Tanuja Ltd.

Statement of Marginal Cost (per unit)

Particulars	Product A (₹)	Product B (₹)
Sales Price	100	100
Direct Materials	24	16
Direct Wages @ ₹2/hr	6	10
Variable Overheads	4	6
Marginal Cost	34	32
Contribution	66	68

Based on this data, we are to determine:

1. Contribution per kg of material
2. Contribution per hour of labour
3. Contribution per hour of machine time
4. Recommend the best product to produce under each limiting factor scenario.

Solution:

Step 1: Material Consumption (kg)

◆ Cost per kg = ₹4 ◆ **Product A:** $\frac{24}{4} = 6 \text{ kg}$ ◆ **Product B:** $\frac{16}{4} = 4 \text{ kg}$

Step 2: Contribution per kg of Material

◆ **Product A:** $\frac{66}{6} = ₹11$ ◆ **Product B:** $\frac{68}{4} = ₹17$

Step 3: Labour Hours Used

◆ Wages @ Rs. 2 per hour ◆ **Product A:** $\frac{6}{2} = 3 \text{ hours}$

◆ **Product B:** $\frac{10}{2} = 5 \text{ hours}$

Step 4: Contribution per Hour of Labour

◆ **Product A:** $\frac{66}{3} = ₹22$ ◆ **Product B:** $\frac{68}{5} = ₹13.60$

Step 5: Machine Hours Used

◆ **Product A:** 4 hours ◆ **Product B:** 3 hours

Step 6: Contribution per Hour of Machine Time

◆ **Product A:** $\frac{66}{4} = ₹16.50$ ◆ **Product B:** $\frac{68}{3} = ₹22.67$

Conclusion & Recommendations:

Based on marginal costing analysis:

Limiting Factor	Recommendation	Justification
Raw Material (kg)	Product B	Higher contribution per kg (₹17 > ₹11)
Labour Hours	Product A	Higher contribution per labour hour (₹22 > ₹13.60)
Machine Hours	Product B	Higher contribution per machine hour (₹22.67 > ₹16.50)

Thus, Tanuja Ltd. should prioritize product selection based on the most critical limiting resource:

- ◆ If material is scarce → produce Product B
- ◆ If labour is scarce → produce Product A
- ◆ If machine time is limited → produce Product B

7. (Decision-Making using CVP Analysis)

Sharda Ltd. has recorded sales of ₹30,00,000 during the financial year. The variable cost amounts to 60% of the sales, and the company has earned a profit of ₹8,00,000.

You are required to compute the following:

1. P/V Ratio
2. Fixed Costs

3. Break-even Sales
4. Sales level that would still be profitable if:
 - ◆ The selling price is reduced by 10%, and
 - ◆ The fixed costs increase by ₹1,00,000

Solution:

Given:

- ◆ Sales = ₹30,00,000
- ◆ Variable Cost = 60% of Sales = ₹18,00,000
- ◆ Profit = ₹8,00,000

Step-by-Step Calculations

- (i) **Contribution:** Contribution = Sales – Variable Cost = ₹30,00,000 – ₹18,00,000 = ₹12,00,000
- (ii) **Fixed Cost:** Profit = Contribution – Fixed Cost $\Rightarrow$ ₹8,00,000 = ₹12,00,000 – FC $\Rightarrow$ Fixed Cost = ₹4,00,000
- (i) **P/V Ratio:** $\left(\frac{\text{Contribution}}{\text{Sales}} \right) \times 100 = \left(\frac{12,00,000}{30,00,000} \right) \times 100 = 40\%$
- (iii) **Break-even Sales:** $\text{BEP (₹)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{₹4,00,000}{40\%} = ₹10,00,000$
- (iv) **Revised Scenario:**
 - ◆ New Selling Price = 90% of original $\rightarrow ₹30,00,000 \times 90\% = ₹27,00,000$
 - ◆ Fixed Cost increases = ₹4,00,000 + ₹1,00,000 = ₹5,00,000
 - ◆ Variable Cost remains the same = ₹18,00,000
 - ◆ New Contribution = ₹27,00,000 – ₹18,00,000 = ₹9,00,000
 - New P/V Ratio: $\left(\frac{₹9,00,000}{₹27,00,000} \right) \times 100 = 33.33\%$

Revised Break-even Sales (to cover increased fixed cost):

$$\text{Required Sales} = \frac{₹5,00,000}{33.33\%} = ₹15,00,000$$

Answer Summary:

Part	Result
(i) P/V Ratio 40%	
(ii) Fixed Cost = 4,00,000	
(iii) Break-even Sales = ₹10,00,000	
(iv) Revised sales (after price cut and cost increase) = ₹15,00,000	

8. (Break-even Analysis & Activity Level Calculation)

Harshit Ltd. manufactures a product and is currently operating at **80% of its production capacity**, generating a turnover of ₹8,00,000 by selling the product at ₹25 per unit. The cost details are as follows:

- ◆ **Material Cost:** ₹7.50 per unit
- ◆ **Labour Cost:** ₹6.25 per unit
- ◆ **Semi-variable Cost:** ₹1,80,000
 - ◆ (Includes a variable component of ₹3.75 per unit)
- ◆ **Fixed Cost:** ₹90,000 (up to 80% capacity); an additional ₹20,000 is incurred beyond this level.

You are required to calculate:

Activity level at the Break-even Point (BEP)

Solution:

Step 1: Calculate Number of Units Sold

$$\text{Units Sold} = \frac{\text{Total Sales}}{\text{Selling Price per Unit}} = \frac{₹8,00,000}{₹25} = 32,000 \text{ units}$$

Step 2: Determine Fixed Component of Semi-variable Cost

$$\text{Variable Part of Semi-variable Cost} = ₹3.75 \times 32,000 = ₹1,20,000$$

$$\text{Fixed Part} = ₹1,80,000 - ₹1,20,000 = ₹60,000$$

Step 3: Total Fixed Costs

$$\text{Fixed Costs} = ₹90,000 + ₹60,000 = ₹1,50,000$$

Step 4: Calculate Variable Cost per Unit

$$\text{Variable Cost per Unit} = ₹7.50 + ₹6.25 + ₹3.75 = ₹17.50$$

Step 5: Contribution per Unit

$$\text{Contribution per Unit} = ₹25.00 - ₹17.50 = ₹7.50$$

Step 6: Break-even Sales in Units

$$\text{BEP (units)} = \frac{\text{Total Fixed Cost}}{\text{Contribution per Unit}} = \frac{₹1,50,000}{₹7.50} = 20,000 \text{ units}$$

Step 7: Calculate Activity Level at BEP

$$\text{Activity Level at BEP} = \left(\frac{20,000}{32,000} \right) \times 80\% = 50\%$$

Final Answer: Activity Level at BEP = 50%

9. (Break-even Analysis with Cash Flow Consideration)

Sharda Ltd. sold 2,75,000 units of its product at a price of ₹37.50 per unit. The variable costs include:

☛ **Manufacturing Cost:** ₹14.00 per unit

☛ **Selling Cost:** ₹3.50 per unit

The total fixed costs incurred uniformly throughout the year are ₹35,00,000, which includes ₹15,00,000 as depreciation. There are no beginning or ending inventories.

You are required to compute:

1. Break-even Sales Quantity
2. Cash Break-even Sales Quantity

Solution:

Step 1: Variable Cost per Unit

$$\text{Total Variable Cost per unit} = ₹14.00 + ₹3.50 = ₹17.50$$

Step 2: Contribution per Unit

$$\text{Contribution per unit} = ₹37.50 - ₹17.50 = ₹20.00$$

Step 3: Break-even Sales Quantity

$$\text{Break-even Units} = \frac{\text{Fixed Costs}}{\text{Contribution per Unit}} = \frac{₹35,00,000}{₹20} = 1,75,000 \text{ units}$$

Step 4: Cash Fixed Costs

$$\text{Cash Fixed Costs} = ₹35,00,000 - ₹15,00,000 (\text{Depreciation}) = ₹20,00,000$$

Step 5: Cash Break-even Sales Quantity

$$\text{Cash Break-even Units} = \frac{₹20,00,000}{₹20} = 1,00,000 \text{ units}$$

Final Answer for Question 9:

- ◆ Break-even Sales Quantity = 1,75,000 units
- ◆ Cash Break-even Sales Quantity = 1,00,000 units

10. (Composite Break-even Point for Multi-product Firm)

Sharda Ltd. sells two products - J and K. The sales mix is 4 units of Product J for every 3 units of Product K (i.e. ratio = 4:3).

☞ Contribution margin per unit for:

◆ Product J = ₹40

◆ Product K = ₹20

☞ Fixed Costs per month = ₹6,16,000

You are required to compute:

1. Composite Break-even Sales (in units)
2. Break-even Sales (in units) for Product J and Product K separately

Solution:

Step 1: Total Mix = 4 (J) + 3 (K) = 7 units

Step 2: Weighted Average (Composite) Contribution per Unit

$$\left(\frac{4}{7} \times ₹40 \right) + \left(\frac{3}{7} \times ₹20 \right) = ₹22.86 + ₹8.57 = ₹31.43$$

Step 3: Composite Break-even Point (Units)

$$\frac{₹6,16,000}{₹31.43} \approx 19,600 \text{ units}$$

Step 4: Break-even Units for Product J and K

$$\text{Product J: } 19,600 \times \frac{4}{7} = 11,200 \text{ units}$$

$$\text{Product K: } 19,600 \times \frac{3}{7} = 8,400 \text{ units}$$

Final Answer for Question 10:

- ◆ Composite Break-even Sales = 19,600 units
- ◆ Break-even Units of Product J = 11,200 units
- ◆ Break-even Units of Product K = 8,400 units

11. (Margin of Safety and Net Profit Calculation)

Harshit Ltd. has a Profit-Volume (P/V) Ratio of 50% and a Margin of Safety ratio of 40%. The total sales volume for the period is ₹1,00,000.

You are required to calculate the Net Profit earned by the company.

Solution:**Step 1:** Calculate Margin of Safety (₹)

$$\text{Margin of Safety Ratio} = \left(\frac{\text{Margin of Safety}}{\text{Actual Sales}} \right) \times 100 \Rightarrow 40 = \frac{\text{Margin of Safety}}{₹1,00,000} \times 100 = ₹40,000$$

Step 2: Use P/V Ratio to Calculate Profit

$$\text{Margin of Safety} \times \frac{\text{Profit}}{\text{P/V Ratio}} \Rightarrow ₹40,000 = \frac{\text{Profit}}{0.50} \Rightarrow \text{Profit} = ₹40,000 \times 0.50 = ₹20,000$$

Final Answer: Net Profit = ₹20,000**12. (Break-even Revenue and P/V Ratio Calculation)**

Harshit Ltd. has provided the following details:

Sales Volume: 20,000 units**Selling Price per Unit:** ₹5**Variable Cost per Unit:** ₹3**Fixed Cost per Year:** ₹8,000**You are required to determine:**

1. Profit-Volume (P/V) Ratio
2. Break-even Revenue (₹)

Solution:**Step 1:** Calculate P/V Ratio

$$\text{Contribution per unit } ₹5 - ₹3 - ₹2 \Rightarrow \text{P/V Ratio} = \left(\frac{₹2}{₹5} \right) \times 100 = 40\% \text{ or } 0.40$$

Step 2: Calculate Break-even Revenue

$$\text{Break-even Revenue} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{₹8,000}{0.40} = ₹20,000$$

Final Answer:◆ **P/V Ratio = 40%**◆ **Break-even Revenue = ₹20,000****13. (Comprehensive CVP Analysis)****Tanuja Ltd.** has provided the following information:**Year-wise Financial Details**

Particulars	2005-2006	2006-2007
Sales (at 10/unit)	₹2,00,000	₹2,50,000
Net Profit	₹30,000	₹50,000

You are required to calculate:

1. Profit/Volume (P/V) Ratio
2. Break-even Sales Revenue
3. Total Variable Cost for both years
4. Sales Revenue needed to earn a profit of ₹60,000
5. Profit or Loss when sales are ₹1,00,000

Solution:**(a) P/V Ratio**

$$\text{P/V Ratio} = \left(\frac{\Delta \text{Profit}}{\Delta \text{Sales}} \right) \times 100 = \left(\frac{50,000 - 30,000}{2,50,000 - 2,00,000} \right) \times 100 = \frac{20,000}{50,000} \times 100 = 40\%$$

(b) Break-even Sales

Using data for 2006-2007: $\text{P/V Ratio} \left(\frac{F + P}{S} \right) \times 100 \Rightarrow 40 = \left(\frac{F + 50,000}{2,50,000} \right) \times 100$ $F + 50,000 = ₹1,00,000$
 $\Rightarrow \text{Fixed Cost} = ₹50,000$

$$\text{Break-even Sales} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} \times 100 = \frac{50,000}{40} \times 100 = ₹1,25,000$$

(c) Total Variable Cost

For 2005-2006: $\text{P/V Ratio} = \left(\frac{S - V}{S} \right) \times 100 \Rightarrow 40 = \left(\frac{2,00,000 - V}{2,00,000} \right) \times 100 \Rightarrow 80,000 = 2,00,000 - V \Rightarrow V = ₹1,20,000$

For 2006-2007: $\text{P/V Ratio} = \left(\frac{S - V}{S} \right) \times 100 \Rightarrow 40 = \left(\frac{2,50,000 - V}{2,50,000} \right) \times 100 \Rightarrow 1,00,000 = 2,50,000 - V \Rightarrow V = ₹1,50,000$

(d) Required Sales to Earn Profit of ₹60,000

$$\text{Required Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} \times 100 = \frac{50,000 + 60,000}{40} \times 100 = \frac{1,10,000}{40} \times 100 = ₹2,75,000$$

(e) Profit or Loss when Sales = 1,00,000

$$\text{P/V Ratio} = \left(\frac{F + P}{S} \right) \times 100 \Rightarrow 40 = \left(\frac{50,000 + P}{1,00,000} \right) \times 100 \Rightarrow 40,000 = 50,000 + P \Rightarrow P = -₹10,000$$

Therefore, there is a Loss of ₹10,000

Final Answers for Question 13. Tanuja Ltd.:

	Requirement	Result
(a)	P/V Ratio	40%
(b)	Break-even Sales	₹1,25,000
(c)	Variable Cost 2005-06	₹1,20,000
(c)	Variable Cost 2006-07	₹1,50,000
(d)	Sales for 60,000 Profit	₹2,75,000
(e)	Loss at 1,00,000 Sales	₹10,000

14. (P/V Ratio and Margin of Safety Calculation)

Ms. Ridhima Ltd. provides the following financial information:

- 👉 **Total Sales:** ₹10,00,000
- 👉 **Total Variable Cost:** ₹4,00,000
- 👉 **Net Profit:** ₹3,00,000

You are required to compute the following:

1. Profit/Volume (P/V) Ratio

2. Break-even Sales (Value)

3. Margin of Safety (₹)

Solution:

Step 1: Contribution

$$\text{Contribution} = \text{Sales} - \text{Variable Cost} = ₹10,00,000 - ₹4,00,000 = ₹6,00,000$$

Step 2: Fixed Cost

$$\text{Fixed Cost} = \text{Contribution} - \text{Profit} = ₹6,00,000 - ₹3,00,000 = ₹3,00,000$$

Step 3: P/V Ratio

$$\text{P/V Ratio} = \left(\frac{\text{Contribution}}{\text{Sales}} \right) \times 100 = \left(\frac{₹6,00,000}{₹10,00,000} \right) \times 100 = 60\%$$

Step 4: Break-even Sales (Value)

$$\text{BEP} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{₹3,00,000}{0.60} = ₹5,00,000$$

Step 5: Margin of Safety

$$\text{Margin of Safety} = \text{Actual Sales} - \text{Break-even Sales} = ₹10,00,000 - ₹5,00,000 = ₹5,00,000$$

Final Answers:

1. **P/V Ratio = 60%**

2. **Break-even Sales (Value) = 5,00,000**

3. **Margin of Safety = ₹5,00,000**

15. (Fixed Cost and Break-even Sales from Margin of Safety)

Ms. Ridhima Ltd. reports the following financial data:

☛ **Total Sales (Turnover): ₹10,00,000**

☛ **Margin of Safety Ratio: 30%**

☛ **Variable Cost: 60% of Sales**

You are required to determine:

1. Fixed Cost

2. Break-even Sales (₹)

Solution:

Step 1: Variable Cost

$$\text{Variable Cost} = 60\% \text{ of Sales} = 0.60 \times ₹10,00,000 = ₹6,00,000$$

Step 2: Contribution

$$\text{Contribution} = \text{Sales} - \text{Variable Cost} = ₹10,00,000 - ₹6,00,000 = ₹4,00,000$$

Step 3: P/V Ratio

$$\text{P/V Ratio} = \left(\frac{₹4,00,000}{₹10,00,000} \right) \times 100 = 40\%$$

Step 4: Margin of Safety (₹)

$$\text{Margin of Safety} = 30\% \text{ of } ₹10,00,000 = ₹3,00,000$$

Step 5: Profit

$$\text{Profit} = \text{Margin of Safety} \times \text{P/V Ratio} = ₹3,00,000 \times 0.40 = ₹1,20,000$$

Step 6: Fixed Cost

$$\text{Fixed Cost} = \text{Contribution} - \text{Profit} = ₹4,00,000 - ₹1,20,000 = ₹2,80,000$$

Step 7: Break-even Sales

$$\text{BEP (₹) Total Sales} = \text{Margin of Safety} = ₹10,00,000 - ₹3,00,000 = ₹7,00,000$$

Final Answers:

1. **Fixed Cost = ₹2,80,000**
2. **Break-even Sales (Value) = ₹7,00,000**

16. (Margin of Safety Calculation)

Debashish Ltd. has provided the following financial data for the fiscal year ended March 31, 2009:

- ☛ **Sales Volume:** 1,00,000 units
- ☛ **Selling Price per Unit:** ₹10
- ☛ **Variable Cost per Unit:** ₹6
- ☛ **Fixed Costs per annum:** ₹3,00,000

You are required to calculate the Margin of Safety in both units and value.

Solution:**Step 1: Contribution per Unit**

$$\text{Contribution per Unit} = \text{Selling Price} - \text{Variable Cost} = ₹10 - ₹6 = ₹4$$

Step 2: Break-even Sales (in Units)

$$\text{BEP (Units)} = \frac{\text{Fixed Costs}}{\text{Contribution per Unit}} = \frac{₹3,00,000}{₹4} = 75,000 \text{ units}$$

Step 3: Margin of Safety (Units)

$$\text{Margin of Safety (Units)} = \text{Actual Sales} - \text{BEP Units} = 1,00,000 - 75,000 = 25,000 \text{ units}$$

Step 4: Margin of Safety (Value)

$$\text{Margin of Safety (Value)} = \text{MOS Units} \times \text{Selling Price per Unit} = 25,000 \times ₹10 = ₹2,50,000$$

Final Answer:

- ◆ **Margin of Safety = 25,000 units**
- ◆ **Margin of Safety (in) = 2,50,000**

TEST YOURSELF**17. (Break-even Analysis and Profit at 90% Capacity)**

A business manufactures 500 units of a product every month. The variable cost per unit is 200, the selling price per unit is 250, and the monthly fixed costs are 12,000.

You are required to calculate:

1. Break-even Point in Units
2. Break-even Sales (in)
3. Profit when operating at 90% capacity utilization

18. (Comprehensive CVP and Profitability Analysis - Multi-part)

Part A: A business reports the following financial figures:

- ☛ **Sales Revenue:** ₹80,000
- ☛ **Variable Costs:** ₹4,000
- ☛ **Fixed Costs:** ₹4,000

You are required to calculate:

1. Profit/Volume (P/V) Ratio
2. Break-even Sales (₹)
3. Margin of Safety (₹)
4. Profit (₹)
5. Profit Margin (%)

Part B: Based on the following information:

- ☛ **Variable Cost per Unit:** ₹15
- ☛ **Selling Price per Unit:** ₹20
- ☛ **Fixed Costs:** ₹54,000

You are required to calculate:

1. P/V Ratio
2. Break-even Sales (₹)

Part C: (Revised Selling Price to Achieve Target BEP)

The Break-even Point in units has been revised and reduced to 6,000 units.

You are required to determine the new selling price per unit, assuming:

- ☛ Variable Cost per Unit = 15
- ☛ Fixed Costs = ₹ 54,000

19. (Comprehensive CVP Analysis - P/V Ratio, BEP, Margin of Safety)

Using the following financial information of a business, calculate:

1. Profit/Volume (P/V) Ratio
2. Break-even Sales (₹)
3. Margin of Safety (₹)

Given:

- ☛ Sales Revenue = ₹ 1,00,000
- ☛ Total Expenses = ₹ 180,000
- ☛ Fixed Expenses = ₹20,000
- ☛ Net Profit = ₹ 20,000

20. (Break-even Point and Sales Percentage at BEP)

A business has achieved 100% sales amounting to ₹3,00,000.

You are required to:

1. Determine the Break-even Sales (in ₹)
2. Calculate the Percentage of Sales at which the Break-even Point occurs
3. Compute the Profit earned at 80% of full capacity

21. (Composite Break-even Analysis - Multi-product CVP for Tanuja Ltd.)

Tanuja Ltd. manufactures and sells two products – Alpha and Beta. The pricing and cost structure for each product is as follows:

Product Alpha:

- Selling Price per unit = ₹7.00
- Variable Cost per unit = ₹2.94

Product Beta:

- Selling Price per unit = ₹15.00
- Variable Cost per unit = ₹4.50

According to the Marketing Division's forecast, the sales mix is such that for every 5 units of Alpha, 1 unit of Beta is sold.

The total fixed costs incurred by the company are ₹36,000.

You are required to compute:

1. Composite Profit/Volume (P/V) Ratio
2. Break-even Sales in units for both Product Alpha and Product Beta

22. (Break-even Analysis and Net Sales Value - Tanuja Ltd.)

Tanuja Ltd. provides the following cost and pricing data for one of its products:

Particulars	Amount (₹)
Direct Material Cost per unit	₹10
Direct Labour Cost per unit	₹5
Variable Overheads (60% of Labour)	₹3
Fixed Overheads	₹50,000
Selling Price per unit	₹25
Trade Discount	4%

You are required to:

1. Calculate the Break-even Point (in units)
2. Calculate the Net Sales Value at the Break-even Point
3. Determine the Net Profit if sales are 10% and 25% above the break-even volume

23. (Multi-Product CVP Table Completion - Ms. Ridhima Ltd.)

Ms. Ridhima Ltd. produces and sells three independent products - A, B, and C. The following table contains partial information regarding their cost structure and profitability:

You are required to solve and fill in the missing values in the table:

Particulars	Product A	Product B	Product C
No. of Units Sold	?	20,000	10,000
Selling Price per Unit (₹)	30	?	40
Variable Cost (% of Sales)	85%	85%	?
Contribution (₹)	?	50,000	75,000
Fixed Cost (₹)	1,20,000	10,000	?
Profit/Loss (₹)	40,000	?	20,000

Instructions: Use the standard contribution formula and profit equation to compute the missing values:

☛ **Contribution = Sales – Variable Cost**

☛ **Profit/Loss = Contribution – Fixed Cost**

24. (P/V Ratio Based on Changing Average Cost)

For a manufacturing concern:

☛ When the production volume is 8,000 units, the average cost is 18 per unit.

☛ When the production volume is 20,000 units, the average cost reduces to 4.50 per unit.

If the break-even point is reached at 8,000 units of production and sale, you are required to:

Calculate the Profit/Volume (P/V) Ratio

25. (Product Mix Impact on Profit - Marginal Costing Analysis)

The table below provides cost and profit data for three distinct products-X, Y, and Z-manufactured by JABRA Co. Ltd.:

Particulars	X	Y	Z	Total
Sales (₹)	3,00,000	1,80,000	1,20,000	6,00,000
Variable Cost (₹)	2.40,000	1,26,000	72,000	4,38,000
Contribution (₹)	60,000	54,000	48,000	1,62,000
Fixed Cost (₹)	—	—	—	81,000
Profit (₹)	—	—	—	₹8,10,000

Required:

Can JABRA Co. Ltd. improve its profitability by changing the sales mix of Products X, Y, and Z?

Use the marginal costing technique to support your analysis.

26. (Break-even Concept, Chart Construction, and Strategic Use - Tanuja Ltd.)

Tanuja Ltd. manufactures a single product. The following cost and pricing details have been provided:

Particulars	Amount (₹)
Direct Labour Cost per unit	₹ 22
Direct Material Cost per unit	₹12
Variable Overhead per unit	₹6
Fixed Costs	₹4,00,000
Selling Price per unit	₹60

You are required to answer the following:

1. Explain what is meant by the term “break-even”.
2. Why might a company with a wide product range find break-even analysis less useful or reliable?
3. Using the above data, construct a break-even chart showing the minimum number of units that must be sold to break even. Ensure your diagram includes complete labels.
4. List and explain any key factors a company like Tanuja Ltd. should consider before relying on break-even analysis as a basis for decision-making.

27. (Comprehensive Break-even and Profitability Analysis - Sharda Ltd.)

Sharda Ltd. provides the following sales data for two successive years:

Year	Units Sold	Selling Price (₹)	Average Cost (₹)
2020	12,000	₹50	₹30
2021	15,000	₹50	₹38

You are required to calculate:

1. P/V Ratio and Fixed Cost
2. Break-even Sales (in ₹)
3. Sales required to earn a profit of ₹12,000
4. Selling Price per unit to earn a profit of ₹1,50,000 by selling 9,000 units
5. Margin of Safety (in ₹) when profit is ₹30,000

28. (Break-even and Capacity Utilization Analysis - Merged Plant Scenario)

Two plants, Plant X and Plant Y, produce the same product and sell it at the same selling price. The following are the details at current capacity utilization:

Capacity Utilization	Plant X (70%)	Plant Y (60%)
Sales (in lakhs)	₹150	₹90
Variable Cost (in lakhs)	₹105	₹75
Fixed Cost (in lakhs)	₹30	₹20

It is proposed to merge Plant Y with Plant X, resulting in additional fixed costs of ₹2 lakhs.

You are required to:

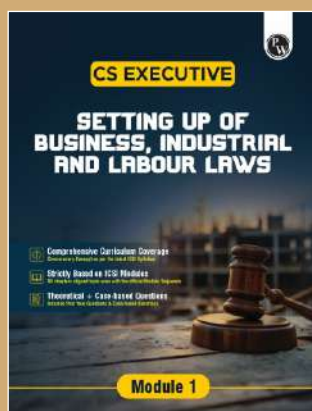
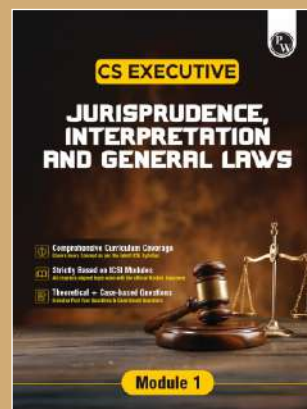
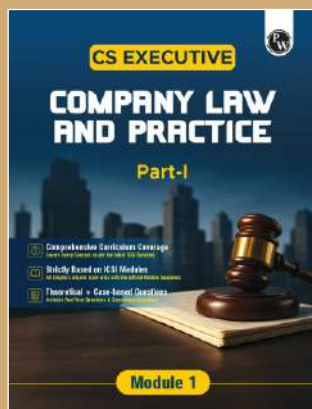
(a) **Calculate:**

1. The break-even point (₹) of Plant X and Plant Y before the merger, and
2. The break-even point of the merged plant after combining the operations and accounting for the additional fixed cost.

(b) **Determine:**

1. The capacity utilization required for the merged plant to achieve a profit of 18 lakhs.

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